

AMATH 731: Applied Functional Analysis Lecture Notes

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November 24, 2014

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Preface

From the book by Reed and Simon:

Mathematics has its roots in numerology, geometry, and physics. Since the time of Newton, the search for mathematical models of physical phenomena has been a source of mathematical problems. In fact, whole branches of mathematics have grown out of attempts to analyse particular physical situations. An example is the development of harmonic analysis from Fourier's work on the heat equation.

Although mathematics and physics have grown apart in this century, physics has continued to stimulate mathematical research. Partially because of this, the influence of physics on mathematics is well understood. However, the contributions of mathematics to physics are not as well understood. It is a common fallacy to suppose that mathematics is important for physics only because it is a useful tool for making computations. Actually, mathematics plays a more subtle role that in the long run is more important. When a successful mathematical model is created for a physical phenomenon, that is, a model that can be used for accurate computations and predictions, the mathematical structure of the model itself provides a new way of thinking about the phenomenon. Put slightly differently, when a model is successful, it is natural to think of the physical quantities in terms of the mathematical objects that represent them and to interpret similar or secondary phenomena in terms of the same model. Because of this, an investigation of the internal mathematical structure of the model can alter and enlarge our understanding of the physical phenomenon. Of course, the outstanding example of this is Newtonian mechanics, which provided such a clear and coherent picture of celestial motions that it was used to interpret practically all physical phenomena. The model itself became central to an understanding of the physical world, and it was difficult to give it up in the late nineteenth century, even in the face of contradictory evidence. A more modern example of this influence of mathematics on physics is the use of group theory to classify elementary particles.

From the book by Kreyzig:

Functional analysis is an abstract branch of mathematics that originated from classical analysis. Its development started about eighty years ago, and nowadays functional analytic methods and results are important in various fields of mathematics and its applications. The impetus came from linear algebra, linear ordinary and partial differential equations, calculus of variations, approximation theory and, in particular, linear integral equations, whose theory had the greatest effect on the development and promotion of the modern ideas. Mathematicians observed that problems from different fields often enjoy related features and properties. This fact was used for an effective unifying approach towards such problems, the unification being obtained by the omission of unessential details. Hence, the advance of such an *abstract approach* is that it concentrates on the essential facts, so that these facts become clearly visible, since the investigator's attention is not disturbed by unimportant details. In this respect, the abstract method is the simplest and most economical method for treating mathematical systems. Since any such abstract system will, in general, have various concrete realisations (concrete *models*), we see that the abstract method is quite versatile in its application to concrete situations. It helps to free the problem from isolation and creates relations and transitions between

fields that have at first no contact with one another.

In the abstract approach, one usually starts from a set of elements satisfying certain axioms. The nature of the elements is left unspecified. This is done on purpose. The theory then consists of logical consequences that result from the axioms and are derived as theorems once and for all. This means that, in this axiomatic fashion, one obtains a mathematical structure whose theory is developed in an abstract way. Those general theorems can then later be applied to various special sets satisfying those axioms.

For example, in algebra, this approach is used in connection with fields, rings, and groups. In functional analysis, we use it in connection with *abstract spaces*; these are of basic importance, and we shall consider some of them (Banach spaces, Hilbert spaces) in great detail. We shall see that in this connection the concept of a “space” is used in a very wide and surprisingly general sense. An *abstract space* will be a set of (unspecified) elements satisfying certain axioms. And by choosing different sets of axioms, we shall obtain different types of abstract spaces.

From Vrscay:

Functional analysis is the study of functions and operators, a kind of higher-level version of basic real analysis. In most, if not all, research areas of applied mathematics, you will be faced with having to perform “operations” that require solid mathematical justification, even if they always appear to work, for example, numerically, since there should always be a mathematical basis for why they work, or, indeed, when they are expected to work and not to work.

In many cases, the “operations” mentioned above are iterative procedures of the form

$$x_{n+1} = T x_n, \quad n = 0, 1, \dots, \quad (1)$$

where the x_n belong to some suitable space or set—call it “ X ”—and T is a mapping from X to itself, i.e.,

$$T : X \rightarrow X. \quad (2)$$

Examples of X could be

- The real numbers $\mathbb{R}$;
- The complex numbers $\mathbb{C}$;
- N -vectors of real numbers $\mathbb{R}^N$;
- N -vectors of complex numbers $\mathbb{C}^N$;
- Functions;
- Vectors of functions;
- Measures;
- Operators themselves!

In all of these iterations procedures, we would definitely like the iteration sequence $\{x_n\}$ to “converge” to a limit $\bar{x} \in X$. It would be even better if the limit $\bar{x}$ was unique, but that may be too much to ask.

A well known example of an iteration procedure is the Newton-Raphson iterations method for finding approximations to the zeros of a real-valued function $f : \mathbb{R} \rightarrow \mathbb{R}$:

$$x_{n+1} = N(x_n) = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, \dots \quad (3)$$

Here we are concerned with the so-called “Newton operator” $N : \mathbb{R} \rightarrow \mathbb{R}$. You have probably seen, but perhaps not analysed in great detail, that if x^* is a simple zero of f , then for x_0 sufficiently close to x^* , the iteration sequence

$$x_{n+1} = N(x_n), \quad n = 0, 1, \dots$$

converges to x^* . (In fact, the convergence is quadratic.)

Another example involves the existence and uniqueness of solutions to the initial value problem (IVP)

$$x' = f(x), \quad x(0) = x_0. \quad (4)$$

Here, for simplicity, we simply consider the scalar case, i.e., $f : \mathbb{R} \rightarrow \mathbb{R}$. The basic proof of existence-uniqueness of the solution to the IVP involves the existence of a *contractive* operator T that maps a suitable *Banach space* (a complete normed linear space) of functions, call it $\mathcal{F}$, to itself. And when you start with any function $f_0 \in \mathcal{F}$ and perform the iteration (the so-called “Picard iteration procedure”),

$$f_{n+1} = T f_n, \quad n = 0, 1, \dots, \quad (5)$$

then the sequence of *functions* $\{f_n\}$ converges to the solution of the IVP above. (Of course, we’ll have to explain what is meant by “convergence” in this example.)

Now, we all know how to deal with convergent sequences of real numbers. In other words, we know what the statement

$$\lim_{n \rightarrow \infty} x_n = \bar{x}, \quad x_n \in \mathbb{R} \quad (6)$$

means. In precise mathematical terms, it means: given an $\epsilon > 0$, there exists an $N_\epsilon > 0$ such that

$$|x_n - \bar{x}| < \epsilon \quad \text{for all } n \geq N_\epsilon. \quad (7)$$

The above limit is often written more informally as

$$x_n \rightarrow \bar{x} \quad \text{as } n \rightarrow \infty, \quad (8)$$

or

$$|x_n - \bar{x}| \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (9)$$

The last expression is simply stating that the *distance* between the points x_n and the limit $\bar{x}$ is going to zero as $n \rightarrow \infty$.

But what about the case of iteration of sequences of *functions*, i.e.,

$$x_n \equiv f_n \in \mathcal{F}? \quad (10)$$

What does it mean to say that

$$\lim_{n \rightarrow \infty} f_n = \bar{f}? \quad (11)$$

As you might already know, we need to work with a *distance function*, or *metric* “ d ” between elements of our function space. In this way, the above statement translates to

$$d(f_n, \bar{f}) \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty. \quad (12)$$

But it’s a little more complicated than that! When working with the real numbers $\mathbb{R}$, we enjoy the benefits of the *completeness* of the real line: any “Cauchy sequence” of real numbers $\{x_n\}$ converges to a real number.

Can we say that same for the particular space $\mathcal{F}$ or any space X , be it of functions, measures, etc.? In other words, is the metric space X *complete*? We’ll have to consider this matter. In fact, you have undoubtedly encountered other situations in which such questions arise—for example, in Fourier series. The partial sums of a Fourier series are functions; therefore, one must necessarily deal with the question of convergence of these partial sums to a function.

In this course, we shall study some standard methods of addressing such problems. In your own research down the road, however, you may well be confronted with the following questions:

- Given a particular problem, what “space” X should I use? And what operator(s) T should I consider? (Perhaps the operator should depend on X .)
- Can I find a “solution” by means of some kind of iteration method?
- Can I find a “solution” by means of some kind of “inversion” method?

Inverse Problems

This is a very important concept in applied mathematics, science, and engineering. Many problems may be posed in the following way:

Given an “observation” y in some space Y , find $x \in X$ (note that X does not necessarily have to be the same as Y) such that

$$Ax = y. \quad (13)$$

Here, A is assumed to be a linear operator $A : X \rightarrow Y$.

In the special finite-dimensional case that $X = Y = \mathbb{R}^n$, under suitable conditions on A , we may simply write

$$x = A^{-1}y. \quad (14)$$

But what happens if X , hence A , is “infinite-dimensional”? We’ll have to discuss what “infinite-dimensional” means, of course.

In fact, problems exist even when X and Y are finite-dimensional. For example, A may represent a “degradation” operator, for example, the blurring of a digital signal or image (which may be represented by an element in $\mathbb{R}^n$). The signal y is what we observe. In other words, we would like to extract the original unblurred signal x . Such problems are called “ill-posed” since there is rarely a unique solution x .

The mathematician Hadamard defined an “ill-posed problem” to be one that violates at least one of the following criteria: with respect to the problem in (13), given a y ,

1. A solution x exists;
2. The solution x is unique;
3. The solution x varies continuously with continuous variations in y .

In general, one cannot hope to find a unique solution x for ill-posed inverse problems. With reference to (13), this means that we shall not be able to find a unique $x \in X$ such that

$$Ax - y = O. \quad (15)$$

(Note that the left-hand side of the above equation is an element in Y . As such, the right-hand side must also be an element in Y . Here O denotes the zero element of Y .) Acknowledging this difficulty, we accept the fact that an exact equality is not achievable and therefore tolerate some deviation, i.e., we let

$$Ax - y = e, \quad (16)$$

where $e \in Y$ is $\|e\|_Y$ is hopefully small. Here $\|\cdot\|_Y$ denotes an appropriate norm in Y .

Of course, we'd like to make the deviation as small as we can, if this is even possible. One possible strategy is to look for an $x \in X$ that minimises this deviation, i.e., look for a solution to the following minimization problem:

$$\bar{x} = \min_{x \in X} \|Ax - y\|_Y^2. \quad (17)$$

(It's always better to square the norm so that we produce a quadratic optimization problem.) This sounds good but, in practise, it may be very difficult, if not impossible, to find such a global minimum. For example, the presence of many local minima can complicate numerical procedures. As well, many of these local minima may correspond to quite poor solutions, i.e., solutions that are too far removed from the original data y .

One way of overcoming this difficulty is to impose additional restrictions on the solution. For example, to keep solutions close to the original data y , we may add a term to the objective function in (17) as follows:

$$\bar{x} = \min_{x \in X} \{ \|Ax - y\|_X^2 + \lambda \|x - y\|_X \}, \quad (18)$$

where $\lambda > 0$ is a constant. (Note that we have now assumed that $X = Y$.) The final term in the above expression is an example of a *regularization term*. In this case, the distance between x and the original data y is viewed as a kind of *penalty term*. Other regularization terms are possible and can be added to the objective function if deemed necessary. For example, in the case of image processing, one may desire that the solution $\bar{x}$ be relatively smooth, or at least piecewise smooth, in which case the regularization term would involve gradients.

1 Review of Real Analysis

Many basic results from real analysis will be important in this course, not only in their own right, but also because of their analogues in metric spaces (e.g., convergence, Cauchy convergence). In what follows, we summarize some of these basic and important results.

1.1 Convergence and Cauchy Sequences

Let's start with one of the simplest results of real analysis, the *triangle inequality*:

$$|x + y| \leq |x| + |y|, \quad x, y \in \mathbb{R}. \quad (1.1)$$

A slight modification produces one of the most fundamental results in analysis (and probably one of the most often employed results when you include its generalisations/analogues in other spaces). First replace y with $-y$,

$$|x - y| \leq |x| + |y|, \quad x, y \in \mathbb{R}, \quad (1.2)$$

(since $|y| = |-y|$) and replace x and y with $x - z$, $y - z$, respectively, for any $z \in \mathbb{R}$ to obtain

$$|(x - z) - (y - z)| \leq |x - z| + |y - z|, \quad x, y, z \in \mathbb{R}, \quad (1.3)$$

which reduces to

$$\boxed{|x - y| \leq |x - z| + |z - y|}. \quad (1.4)$$

Keeping in mind that $|x - y|$ measures the distances between x and y on the real line, the above inequality can be interpreted as follows:

The distance between any two points x and y on the real line is less than the sum of their respective distances to a third point z on the real line.

Of course, we know that this property is true for points $x, y \in \mathbb{R}^n$ in the case of the Euclidean distance in $\mathbb{R}^n$. In general, however, (1.4) expresses one of the fundamental properties of a *metric*, or distance function, between elements of a *metric space*, one of the topics to be seen soon.

There is actually something even deeper here. Equation (1.1) represents a fundamental property of the *norm* $|x|$, which characterises the magnitude of a real number. In a *normed vector space*, for example, the real line $\mathbb{R}$ (and indeed $\mathbb{R}^n$), we can use the norm to define a distance between two elements of the space. We're very much used to this idea because of our acquaintance with the spaces $\mathbb{R}^n$. But it also applies to other normed spaces, for example, spaces of functions, as we'll see soon.

1.2 Convergence of Sequences and Cauchy Sequences

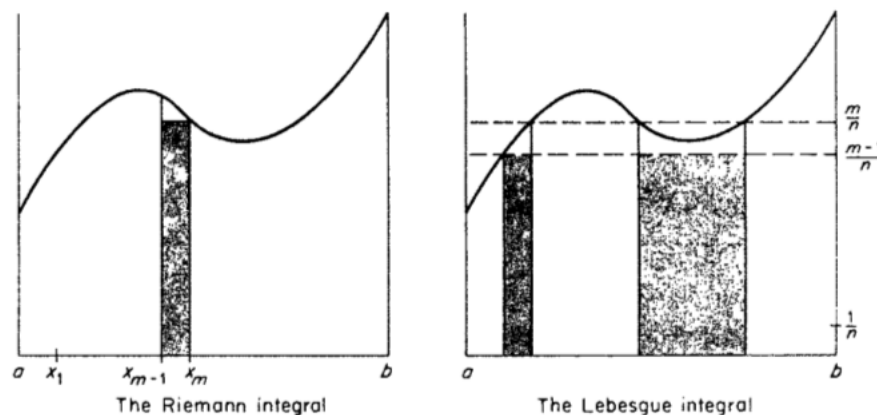
2 Measure Theory

Toward the end of the nineteenth century it became clear to many mathematicians that the Riemann integral, about which one learns in calculus courses, should be replaced by some other type of integral, more general and more flexible, better suited for dealing with limit processes. Among the many attempts made by several mathematicians, it was Lebesgue's construction that turned out to be the most successful.

Here is the main idea: the Riemann integral of a function f over an interval $[a, b]$ can be approximated by sums of the form

$$\sum_{i=1}^n f(t_i)m(E_i),$$

where $E_1, \dots, E_n$ are disjoint intervals whose union is $[a, b]$, $m(E_i)$ denotes the length of E_i , and $t_i \in E_i$ for $i = 1, \dots, n$. In other words, computing the Riemann integral involves dividing the domain of f into finer and finer pieces. For "nasty" functions, this method does not work, and so a different method is needed—the simplest modification is to divide the range into finer and finer pieces, as shown in the figure below.



This method depends more on the function and so has the possibility of working for more types of functions. We are thus interested in sets $f^{-1}([a, b])$ and their size. In other words, the problem has been transferred to one of defining an extended notion of size. We must first decide what sets are to have a size. Why not all sets? Because, for example, it is possible to break up a unit ball into a finite number of wild pieces, move the pieces around by rotation and translation, and reassemble them to get two balls of radius one. This is the Banach-Tarski paradox, and it is a classical example showing that not all sets in $\mathbb{R}^3$ can have a size if we want that size to be invariant under rotations and translations (and not be trivial, such as assigning zero to all sets).

Lebesgue discovered, however, that a completely satisfactory theory of integration results if the sets E_i in the above sum are allowed to belong to a larger class of subsets of the line, the so-called

“measurable sets”, and if the class of functions under consideration is enlarged to what he called “measurable functions”.

The passage from Riemann’s theory of integration to that of Lebesgue is a process of *completion*, the notion of which will be defined later. It is of the same fundamental importance in analysis as is the construction of the real number system from the rationals.

The above mentioned object m , called the *measure*, is intimately related to the geometry of the real line. In this chapter, we shall present an abstract version of the Lebesgue integral, relative to *any* countably additive measure on *any* set. The abstract theory will show that a large part of integration theory is independent of any geometry (or topology) of the underlying space.

2.1 The Concept of Measurability

The class of measurable functions (to be defined later) plays a fundamental role in integration theory. It has some basic properties in common with another most important class of functions, namely, the continuous functions. The material will be presented so that the analogies between the concepts of *topological space*, *open set*, and *continuous function*, and *measurable space*, *measurable set*, and *measurable function*, are strongly emphasized.

Definition 2.1.1 General Topology

1. A collection $\mathcal{T}$ of subsets of a set X is called a **topology** on X if $\mathcal{T}$ has the following three properties:
 - (a) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$;
 - (b) If $V_i \in \mathcal{T}$ for $i = 1, \dots, n$, then $V_1 \cap V_2 \cap \dots \cap V_n \in \mathcal{T}$;
 - (c) If $\{V_\alpha\}$ is an arbitrary collection of members of $\mathcal{T}$ (finite, countable, or uncountable), then $\bigcup_\alpha V_\alpha \in \mathcal{T}$.
2. If $\mathcal{T}$ is a topology on X , then the pair $(X, \mathcal{T})$ (often just X if the topology is unimportant) is called a **topological space**, and the members of $\mathcal{T}$ are called the **open sets** in X .
3. If X and Y are topological spaces and if $f : X \rightarrow Y$ is a mapping, then f is called **continuous** if $f^{-1}(V)$ is an open set in X for every open set V in Y .

Definition 2.1.2 σ -algebra

1. A collection $\mathcal{M}$ of subsets of a set X is called a σ -**algebra** in X if $\mathcal{M}$ has the following properties:
 - (a) $X \in \mathcal{M}$;
 - (b) If $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$, where A^c is the complement of A relative to X .
 - (c) If $A = \bigcup_{n=1}^{\infty} A_n$ and if $A_n \in \mathcal{M}$ for $n = 1, 2, 3, \dots$, then $A \in \mathcal{M}$.
2. If $\mathcal{M}$ is a σ -algebra in X , then the pair $(X, \mathcal{M})$ (often just X if the σ -algebra is unimportant) is called a **measurable space**, and the members of $\mathcal{M}$ are called the **measurable sets** in X .
3. If X is a measurable space, Y a topological space, and $f : X \rightarrow Y$ a mapping, then f is called a **measurable function** if $f^{-1}(V)$ is a measurable set in X for every open set V in Y .

REMARK: The prefix σ refers to the fact that (c) is required to hold for all *countable* unions of members of $\mathcal{M}$. If (c) is required for finite unions only, then $\mathcal{M}$ is called an **algebra** of sets.

We will often use the terms *real measurable function* and *complex measurable function*. These have the obvious meanings of being measurable functions $X \rightarrow \mathbb{R}$ and $X \rightarrow \mathbb{C}$, respectively, where X is a measurable space.

Now, let $\mathcal{M}$ be a σ -algebra in a set X . Referring to the first and third properties of the first part of the definition above, we immediately derive the following facts:

1. Since $\emptyset = X^c$, (a) and (b) imply that $\emptyset \in \mathcal{M}$;
2. Taking $A_{n+1} = A_{n+2} = \dots = \emptyset$ in (c), we see that $A_1 \cup A_2 \cup \dots \cup A_n \in \mathcal{M}$ if $A_i \in \mathcal{M}$ for $i = 1, \dots, n$;
3. Since by set theory

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c,$$

$\mathcal{M}$ is closed under the formation of countable (and also finite) intersections.

4. Since $A - B = B^c \cap A$, we have $A - B \in \mathcal{M}$ if $A \in \mathcal{M}$ and $B \in \mathcal{M}$

Theorem 2.1.1

Let Y and Z be topological spaces and let $g : Y \rightarrow Z$ be continuous.

1. If X is a topological space, if $f : X \rightarrow Y$ is continuous, and if $h = g \circ f$, then $h : X \rightarrow Z$ is continuous.
2. If X is a measurable space, if $f : X \rightarrow Y$ is measurable, and if $h = g \circ f$, then $h : X \rightarrow Z$ is measurable.

Informally, continuous functions of continuous functions are continuous, and continuous functions of measurable functions are measurable.

PROOF: If V is open in Z , then $g^{-1}(V)$ is open in Y , and

$$h^{-1}(V) = f^{-1}(g^{-1}(V)).$$

We now prove each statement in turn.

1. If f is continuous, it follows that $h^{-1}(V)$ is open.
2. If f is measurable, it follows that $h^{-1}(V)$ is measurable. ■

Theorem 2.1.2

Let $u, v : X \rightarrow \mathbb{R}$ be real measurable functions on a measurable space X , let $\Phi : \mathbb{R}^2 \rightarrow Y$ be a continuous mapping of the plane into a topological space Y , and define $h : X \rightarrow Y$ by

$$h(x) = \Phi(u(x), v(x))$$

for all $x \in X$. Then h is measurable.

PROOF: Define $f : X \rightarrow \mathbb{R}^2$ by $f(x) = (u(x), v(x))$. Since $h = \Phi \circ f$, Theorem 2.1.1 shows that it is enough to prove the measurability of f .

If R is any open rectangle in the plane with sides parallel to the axes, then R is a Cartesian product of two segments I_1 and I_2 , and

$$f^{-1}(R) = u^{-1}(I_1) \cap v^{-1}(I_2),$$

which is measurable by our assumption on u and v . Every open set V in the plane is a countable union of such rectangles R_i , and since

$$f^{-1}(V) = f^{-1}\left(\bigcup_{i=1}^{\infty} R_i\right) = \bigcup_{i=1}^{\infty} f^{-1}(R_i),$$

we have that $f^{-1}(V)$ is measurable. ■

Corollary 2.1.1

Let X be a measurable space.

1. If $f = u + iv$, where u and v are real measurable functions on X , then f is a complex measurable function on X .
2. If $f = u + iv$ is a complex measurable function on X , then u , v , and $|f|$ are real measurable functions on X , where $|f| \equiv |f(x)|$ for all $x \in X$.
3. If f and g are complex measurable functions on X , then so are $f + g$ and $f g$.
4. If E is a measurable set in X and if

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases},$$

then χ_E is a measurable function.

5. If f is a complex measurable function on X , there is a complex measurable function α on X such that $|\alpha| = 1$ and $f = \alpha|f|$.

PROOF

1. This follows from Theorem 2.1.2 with $\Phi(z) = z$ (what is z ?)
2. This follows from Theorem 2.1.1 with $g(z) = \operatorname{Re}(z)$, $g(z) = \operatorname{Im}(z)$, and $g(z) = |z|$.
3. For real f and g , this follows from Theorem 2.1.2 with $\Phi(s, t) = s + t$ and $\Phi(s, t) = st$. The complex case then follows from 1 and 2.
4. This is evident. We will call χ_E the **characteristic function** of the set E .
5. Let $E = \{x \mid f(x) = 0\}$, let Y be the complex plane with the origin removed, define $\varphi(z) = \frac{z}{|z|}$ for $z \in Y$, and put

$$\alpha(x) = \varphi(f(x) + \chi_E(x)) \quad (x \in X).$$

If $x \in E$, then $\alpha(x) = 1$, and if $x \notin E$, then $\alpha(x) = \frac{f(x)}{|f(x)|}$. Since φ is continuous on Y and since E is measurable (why?), the measurability of α follows from 3, 4, and Theorem 2.1.1. ■

Theorem 2.1.3

If $\mathcal{F}$ is any collection of subsets of a measurable space X , then there exists a smallest σ -algebra $\mathcal{M}^*$ in X such that $\mathcal{F} \subset \mathcal{M}^*$. This $\mathcal{M}^*$ is sometimes called the **σ -algebra generated by $\mathcal{F}$** .

Definition 2.1.3 Borel Sets

Let X be a topological space. By Theorem ??, there exists a smallest σ -algebra $\mathcal{B}$ in X such that every open set in X belongs to $\mathcal{B}$. The members of $\mathcal{B}$ are called the **Borel sets** of X .

We have that

- all closed sets are Borel sets (being, by definition, the complements of open sets);
- all countable unions of closed sets; and
- all countable intersections of open sets.

Since $\mathcal{B}$ is a σ -algebra, we may now regard X as a measurable space with the Borel sets playing the role of the measurable sets; such measurable spaces are sometimes called **Borel measurable spaces**. Consider, then, the measurable space $(X, \mathcal{B})$. If $f : X \rightarrow Y$ is a continuous mapping of X , where Y is any topological space, then it is clear from definitions that $f^{-1}(V) \in \mathcal{B}$ for every open set V in Y . In other words, *every continuous mapping of X is Borel measurable*, where **Borel measurable function** has the same definition as a measurable function except that the measurable space is Borel. A Borel measurable function is also called a **Borel function**.

Theorem 2.1.4

Suppose $\mathcal{M}$ is a σ -algebra in a set X and Y is a topological space. Let $f : X \rightarrow Y$.

1. If Ω is the collection of all sets $E \subset Y$ such that $f^{-1}(E) \in \mathcal{M}$, then Ω is a σ -algebra in Y .
2. If f is measurable and E is a Borel set in Y , then $f^{-1}(E) \in \mathcal{M}$.
3. If $Y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \in \mathcal{M}$ for every $\alpha \in \mathbb{R}$, then f is measurable.
4. If f is measurable, if Z is a topological space, if $g : Y \rightarrow Z$ is a Borel function, and if $h = g \circ f$, then $h : X \rightarrow Z$ is measurable.

REMARK: Part 3 is a frequently used criterion for the measurability of real-valued functions. Note that 4 generalizes Part 2 of Theorem 2.1.1.

PROOF

1. This follows from the relations

$$\begin{aligned} f^{-1}(Y) &= X, \\ f^{-1}(Y - A) &= X - f^{-1}(A), \\ f^{-1}(A_1 \cup A_2 \cup \cdots) &= f^{-1}(A_1) \cup f^{-1}(A_2) \cup \cdots. \end{aligned}$$

2. Let Ω be as in 1. The measurability of f implies that Ω contains all open sets in Y , and since Ω is a σ -algebra, Ω contains all Borel sets in Y .
3. Let Ω be the collection of all $E \subset [-\infty, \infty]$ such that $f^{-1}(E) \in \mathcal{M}$. Choose a real number α , and choose $\alpha_n < \alpha$ so that $\alpha_n \rightarrow \alpha$ as $n \rightarrow \infty$. Since $(\alpha_n, \infty] \in \Omega$ for each n , since

$$[-\infty, \alpha) = \bigcup_{n=1}^{\infty} [-\infty, \alpha_n] = \bigcup_{n=1}^{\infty} (\alpha_n, \infty]^c,$$

and since 1 shows that Ω is a σ -algebra, we see that $[-\infty, \alpha) \in \Omega$. The same is then true of $(\alpha, \beta) = [-\infty, \beta) \cap (\alpha, \infty]$. Since every open set in $[-\infty, \infty]$ is a countable union of segments of the above types, Ω contains every open set. Thus f is measurable.

4. Let $V \subset Z$ be open. Then $g^{-1}(V)$ is a Borel set of Y , and since $h^{-1}(V) = f^{-1}(g^{-1}(V))$, from 2 we have that $h^{-1}(V) \in \mathcal{M}$. ■

Definition 2.1.4 **lim sup and lim inf**

Let $\{a_n\}$ be a sequence in $[-\infty, \infty]$, and let

$$b_k = \sup\{a_k, a_{k+1}, a_{k+2}, \dots\}, \quad k = 1, 2, 3, \dots, \quad (2.1)$$

and let

$$\beta = \inf\{b_1, b_2, b_3, \dots\}. \quad (2.2)$$

β is called the **upper limit**, or **limit superior**, of $\{a_n\}$, and we write

$$\beta = \limsup_{n \rightarrow \infty} a_n. \quad (2.3)$$

The **lower limit**, or **limit inferior**, is defined by

$$\liminf_{n \rightarrow \infty} a_n = -\limsup_{n \rightarrow \infty} (-a_n). \quad (2.4)$$

If $\{f_n\}$ is a sequence of extended-real functions on a set X , then $\sup_n f_n$ and $\limsup_{n \rightarrow \infty} f_n$ are the functions on X defined by

$$\left(\sup_n f_n \right)(x) := \sup_n (f_n(x)), \quad (2.5)$$

$$\left(\limsup_{n \rightarrow \infty} f_n \right)(x) := \limsup_{n \rightarrow \infty} (f_n(x)). \quad (2.6)$$

If $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, the limit being assumed to exist at every $x \in X$, then we call f the **pointwise limit** of the sequence $\{f_n\}$.

Theorem 2.1.5

Let $\{a_n\}$ be a sequence in $[-\infty, \infty]$, and let

$$b_k = \sup\{a_k, a_{k+1}, a_{k+2}, \dots\}, \quad k = 1, 2, 3, \dots, \quad (2.7)$$

and let $\beta = \limsup_{n \rightarrow \infty} a_n$. Then the following properties hold:

1. $b_1 \geq b_2 \geq b_3 \geq \dots$, so that $b_k \rightarrow \beta$ as $k \rightarrow \infty$.
2. There is a subsequence $\{a_{n_i}\}$ of $\{a_n\}$ such that $a_{n_i} \rightarrow \beta$ as $i \rightarrow \infty$, and β is the largest number with this property.
3. If $\{a_n\}$ converges, then

$$\limsup_{n \rightarrow \infty} a_n = \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_n. \quad (2.8)$$

Theorem 2.1.6

If X is a measurable space, and $f_n : X \rightarrow [-\infty, \infty]$ is measurable for $n = 1, 2, 3, \dots$, and

$$g = \sup_{n \geq 1} f_n, \quad h = \limsup_{n \rightarrow \infty} f_n,$$

then g and h are measurable.

PROOF: $g^{-1}((\alpha, \infty]) = \bigcup_{n=1}^{\infty} f_n^{-1}((\alpha, \infty])$. Hence, Theorem 2.1.4 Part 3 implies that g is a measurable function. The same result holds with $\inf$ in place of $\sup$, and since

$$h = \inf_{k \geq 1} \left\{ \sup_{i \geq k} f_i \right\},$$

it follows that h is also a measurable function. ■

Corollary 2.1.2

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If f and g are measurable (with range in $[-\infty, \infty]$), then so are $\max\{f, g\}$ and $\min\{f, g\}$. In particular, this is true of the functions $f^+ := \max\{f, 0\}$ and $f^- := -\min\{f, 0\}$, which are called, respectively, the *positive part* and *negative part* of f .

REMARK: Note that if f^+ and f^- are the positive and negative parts, respectively, of f , then we have $|f| = f^+ + f^-$ and $f = f^+ - f^-$, a standard representation of f as a difference of two non-negative functions with the following minimality property: if $f = g - h$, $g \geq 0$, and $h \geq 0$, then $f^+ \leq g$ and $f^- \leq h$. This is due to the fact that $f \leq g$ and $0 \leq g$ implies that $\max\{f, 0\} \leq g$.

2.1.1 Simple Functions

Definition 2.1.5 Simple Function

A complex function s on a measurable space X whose range consists of only finitely many points is called a **simple function**.

Among the simple functions is the *non-negative simple functions*, whose range is a finite subset of $[0, \infty)$. Note that we explicitly exclude ∞ from the values of a simple function.

If $\alpha_1, \dots, \alpha_n$ are the distinct values of a simple function s , and if we set $A_i = \{x \mid s(x) = \alpha_i\}$, then

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i},$$

where χ_{A_i} is the characteristic function of A_i as defined earlier.

Theorem 2.1.7

Let X be a measurable space and $f : X \rightarrow [0, \infty]$ a measurable function. Then there exist simple measurable functions s_n on X such that

1. $0 \leq s_1 \leq s_2 \leq \dots \leq f$.
2. $s_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$ for every $x \in X$.

PROOF: Put $\delta_n = 2^{-n}$. To each positive integer n and each real number t corresponds a unique integer $k = k_n(t)$ that satisfies $k\delta_n \leq t \leq (k+1)\delta_n$. Define

$$\varphi_n(t) = \begin{cases} k_n(t)\delta_n & \text{if } 0 \leq t < n \\ n & \text{if } n \leq t \leq \infty \end{cases} \quad (2.9)$$

Each φ_n is then a Borel function on $[0, \infty]$, $t - \delta_n < \varphi_n(t) \leq t$ if $0 \leq t \leq n$, $0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq t$, and $\varphi_n(t) \rightarrow t$ as $n \rightarrow \infty$ for every $t \in [0, \infty]$. It follows that the functions $s_n = \varphi_n \circ f$ satisfy 1 and 2. They are measurable by Theorem 2.1.4 Part 4. ■

2.2 Elementary Properties of Measures

Definition 2.2.1 Measure and Measure Space

1. A **positive measure** is a function μ , defined on a σ -algebra $\mathcal{M}$, whose range is in $[0, \infty]$ and that is *countably additive*, meaning that if $\{A_i\}$ is a *disjoint* countable collection of members of $\mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i). \quad (2.10)$$

Also, $\mu(A) < \infty$ for at least one $A \in \mathcal{M}$.

2. A **measure space** is a measurable space that has a positive measure defined on the σ -algebra of its measurable sets. It can be characterised by the triple $(X, \mathcal{M}, \mu)$, where X is the measurable space on which the σ -algebra is defined.
3. A **complex measure** is a complex-valued countably additive function defined on a σ -algebra.

REMARK: What we have called here a *positive measure* is frequently just called a *measure*; we add the word “positive” for emphasis. If $\mu(E) = 0$ for every $E \in \mathcal{M}$, then μ is a positive measure, by the definition. The value ∞ is admissible for a positive measure, but when we talk of a *complex measure* μ , it is understood that $\mu(E)$ is a complex number for every $E \in \mathcal{M}$. The *real measures* form a subclass of the complex ones, of course.

Theorem 2.2.1 Important Properties of Measures

Let μ be a positive measure on a σ -algebra $\mathcal{M}$. Then

1. $\mu(\emptyset) = 0$;
2. $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$ if $A_1, \dots, A_n$ are pairwise disjoint members of $\mathcal{M}$ (*finite additivity*);
3. $A \subset B$ implies $\mu(A) \leq \mu(B)$ if $A, B \in \mathcal{M}$ (*monotonicity*);
4. $\mu(A_n) \rightarrow \mu(A)$ as $n \rightarrow \infty$ if $A = \bigcup_{n=1}^{\infty} A_n$, $A_n \in \mathcal{M}$ for all n , and $A_1 \subset A_2 \subset A_3 \subset \dots$;
5. $\mu(A_n) \rightarrow \mu(A)$ as $n \rightarrow \infty$ if $A = \bigcap_{n=1}^{\infty} A_n$, $A_n \in \mathcal{M}$ for all n , $A_1 \supset A_2 \supset A_3 \supset \dots$, and $\mu(A_1)$ is finite.

REMARK: As the proof will show, these properties, with the exception of 3, also hold for complex measures.

PROOF

1. Take $A \in \mathcal{M}$ so that $\mu(A) < \infty$, and take $A_1 = A$ and $A_2 = A_3 = \dots = \emptyset$ in (2.10).
2. Take $A_{n+1} = A_{n+2} = \dots = \emptyset$ in (2.10).
3. Since $B = A \cup (B - A)$ and $A \cap (B - A) = \emptyset$, we see that 2 implies $\mu(B) = \mu(A) + \mu(B - A) \geq \mu(A)$.

4. Put $B_1 = A_1$, and put $B_n = A_n - A_{n-1}$ for $n = 2, 3, 4, \dots$. Then $B_n \in \mathcal{M}$, $B_i \cap B_j = \emptyset$ if $i \neq j$, $A_n = B_1 \cup \dots \cup B_n$, and $A = \bigcup_{i=1}^{\infty} B_i$. Hence,

$$\mu(A_n) = \sum_{i=1}^n \mu(B_i) \quad \text{and} \quad \mu(A) = \sum_{i=1}^{\infty} \mu(B_i).$$

Then the result follows by the definition of the sum of an infinite series.

5. Put $C_n = A_1 - A_n$. Then $C_1 \subset C_2 \subset C_3 \subset \dots$, $\mu(C_n) = \mu(A_1) - \mu(A_n)$, $A_1 - A = \bigcup_n C_n$, and so by 4,

$$\mu(A_1) - \mu(A) = \mu(A_1 - A) = \lim_{n \rightarrow \infty} \mu(C_n) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_n),$$

from which the result follows. ■

Example 2.2.1 Here are a few examples of measure spaces.

1. For any $E \subset X$, where X is any set, define $\mu(E) = \infty$ if E is an infinite set, and let $\mu(E)$ be the number of points in E if E is finite. Then μ is called the *counting measure* on X .
2. Fix $x_0 \in X$, define $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$, for any $E \subset X$. Then μ is called the *unit mass measure* concentrated at x_0 .
3. Let μ be the counting measure on the set $\{1, 2, 3, \dots\}$, let $A_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap_n A_n = \emptyset$, but $\mu(A_n) = \infty$ for $n = 1, 2, 3, \dots$. This shows that the hypothesis $\mu(A_1) < \infty$ in Theorem 2.2.1 Part 5 is not superfluous.

2.2.1 Arithmetic in $[0, \infty]$

Throughout integration theory, one inevitably encounters ∞ . One reason is that one wants to be able to integrate over sets of infinite measure; after all, the real line has infinite length. Another reason is that even if one is primarily interested in real-valued functions, the lim sup of a sequence of positive real functions or the sum of a sequence of positive real functions may well be ∞ at some points.

Let us define $a + \infty = \infty + a = \infty$ for $0 \leq a \leq \infty$, and

$$a \cdot \infty = \infty \cdot a = \begin{cases} \infty & \text{if } 0 < a \leq \infty \\ 0 & \text{if } a = 0 \end{cases}.$$

Sums and products of real numbers are defined in the usual way.

The reason for defining $0 \cdot \infty = 0$ is that the commutative, associative, and distributive laws hold in $[0, \infty]$ without any restriction.

The cancellation laws have to be treated with some care: $a + b = a + c$ implies $b = c$ only when $a < \infty$, and $ab = ac$ implies $b = c$ only when $0 < a < \infty$.

Observe that the following useful proposition holds: if $0 \leq a_1 \leq a_2 \leq \cdots$, $0 \leq b_1 \leq b_2 \leq \cdots$, $a_n \rightarrow a$, and $b_n \rightarrow b$, then $a_n b_n \rightarrow ab$.

If we combine this with Theorems 2.1.6 and 2.1.7, we see that sums and products of measurable functions into $[0, \infty]$ are measurable.

2.3 Integration of Positive Functions

Definition 2.3.1 Integral of Simple Functions

Let X be any set, $\mathcal{M}$ a σ -algebra in X , and μ a positive measure on $\mathcal{M}$. If $s : X \rightarrow [0, \infty)$ is a measurable simple function of the form

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i}, \quad (2.11)$$

where $\alpha_1, \dots, \alpha_n$ are the distinct values of s , $A_i = \{x \mid s(x) = \alpha_i\}$, and χ_{A_i} are the characteristic functions of the A_i , and if $E \in \mathcal{M}$, we define

$$\int_E s \, d\mu := \sum_{i=1}^n \alpha_i \mu(A_i \cap E). \quad (2.12)$$

Note that the convention $0 \cdot \infty = 0$ has been used here since it may happen that $\alpha_i = 0$ for some i and that $\mu(A_i \cap E) = \infty$.

Definition 2.3.2 Lebesgue Integral

Let X be any set, $\mathcal{M}$ a σ -algebra in X , and μ a positive measure on $\mathcal{M}$. If $f : X \rightarrow [0, \infty]$ is a measurable function and $E \in \mathcal{M}$, we define

$$\int_E f \, d\mu := \sup \int_E s \, d\mu, \quad (2.13)$$

where the supremum is taken over all simple measurable functions s such that $0 \leq s \leq f$. This is called the *Lebesgue integral* of f over E with respect to the measure μ . It is a number in $[0, \infty]$.

REMARK: Note that we have two definitions for $\int_E f \, d\mu$ if f is a simple function. However, both of these definitions assign the same value to the integral since f is, in this case, the largest of the functions s that occur on the right-hand side of (2.13).

2.4 Integration of Complex Functions

2.5 Sets of Measure Zero

2.6 Positive Borel Measures

2.6.1 Vector Spaces and Topological Preliminaries

2.6.2 The Riesz Representation Theorem

2.6.3 Regularity Properties of Borel Measures

2.6.4 Lebesgue Measure

2.6.5 Continuity Properties of Measurable Functions

3 Metric Spaces

A *metric space* is a set X with a *metric* on it. The metric associates with any pair of elements (*points*) of X a *distance*. The metric is defined axiomatically, the axioms begin suggested by certain simple properties of the familiar distance between points on the real line $\mathbb{R}$ and the complex plane $\mathbb{C}$. Basic examples show that the concept of a metric space is remarkably general. A very important additional property that a metric space may have is *completeness*. Another concept of theoretical and practical interest is *separability* of a metric space. Separable metric spaces are simpler than non-separable ones.

Example 3.0.1 Three Important Inequalities

We first derive three important inequalities, the *Holder inequality*, the *Cauchy-Schwarz inequality*, and the *Minkowski inequality*.

Let $p, q \in \mathbb{R}$, $p > 1$, and define q by

$$\frac{1}{p} + \frac{1}{q} = 1. \quad (3.1)$$

Then we have

$$1 = \frac{p+q}{pq}, \quad pq = p+q, \quad (p-1)(q-1) = 1. \quad (3.2)$$

Hence, $\frac{1}{p-1} = q-1$, so that $u := t^{p-1}$ implies $t := u^{q-1}$.

Now, let α and β be any positive numbers. Since $\alpha\beta$ is the area of the rectangle in the figure below, we thus obtain by integration the inequality

$$\alpha\beta \leq \int_0^\alpha t^{p-1} dt + \int_0^\beta u^{q-1} du = \frac{\alpha^p}{p} + \frac{\beta^q}{q}. \quad (3.3)$$

Note that this inequality is trivially true if $\alpha = 0$ or $\beta = 0$.

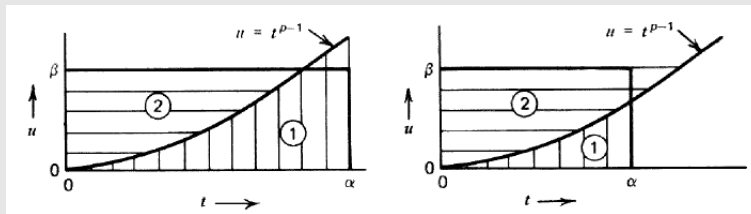


Figure 3.1: Inequality (3.3), where region 1 corresponds to the first integral in (3.3) and region 2 to the second.

Now, let (ξ_i) and (η_i) be two real sequences such that

$$\sum_{i=1}^{\infty} |\xi_i|^p = 1, \quad \text{and} \quad \sum_{i=1}^{\infty} |\eta_i|^q = 1. \quad (3.4)$$

Setting $\alpha = |\tilde{\xi}_i|$ and $\beta = |\tilde{\eta}_i|$, we have from (3.3) the inequality

$$|\tilde{\xi}_i \tilde{\eta}_i| \leq \frac{1}{p} |\tilde{\xi}_i|^p + \frac{1}{q} |\tilde{\eta}_i|^q. \quad (3.5)$$

If we sum over j and use (3.4) and (3.2), we obtain

$$\sum_{i=1}^{\infty} |\tilde{\xi}_i \tilde{\eta}_i| \leq \frac{1}{p} + \frac{1}{q} = 1. \quad (3.6)$$

We now take any non-zero sequences (ξ_i) and (η_i) and set

$$\tilde{\xi}_j = \frac{\xi_j}{\left(\sum_{k=1}^{\infty} |\xi_k|^p\right)^{1/p}}, \quad \tilde{\eta}_j = \frac{\eta_j}{\left(\sum_{m=1}^{\infty} |\eta_m|^q\right)^{1/q}}. \quad (3.7)$$

Then (3.4) is satisfied, so that we may apply (3.6). Substituting (3.7) into (3.6) and multiplying the resulting inequality by the product of the denominators in (3.7), we arrive at the **Holder inequality** for sums

$$\sum_{j=1}^{\infty} |\xi_j \eta_j| \leq \left(\sum_{k=1}^{\infty} |\xi_k|^p \right)^{1/p} \left(\sum_{m=1}^{\infty} |\eta_m|^q \right)^{1/q}, \quad (3.8)$$

where $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$. If $p = 2$, then $q = 2$, and (3.8) gives the **Cauchy-Schwarz Inequality** for sums

$$\sum_{j=1}^{\infty} |\xi_j \eta_j| \leq \sqrt{\sum_{k=1}^{\infty} |\xi_k|^2} \sqrt{\sum_{m=1}^{\infty} |\eta_m|^2}. \quad (3.9)$$

Now, let $p > 1$. To simplify the formulas, we shall write $\xi_j + \eta_j =: \omega_j$. The triangle inequality for numbers gives

$$|\omega_j|^p = |\xi_j + \eta_j| |\omega_j|^{p-1} \leq (|\xi_j| + |\eta_j|) |\omega_j|^{p-1}.$$

Summing over j from 1 to any fixed n , we obtain

$$\sum_{j=1}^n |\omega_j|^p \leq \sum_{j=1}^n |\xi_j| |\omega_j|^{p-1} + \sum_{j=1}^n |\eta_j| |\omega_j|^{p-1}. \quad (3.10)$$

To the first sum on the right we apply the Holder inequality to obtain

$$\sum_{j=1}^n |\xi_j| |\omega_j|^{p-1} \leq \left[\sum_{k=1}^n |\xi_k|^p \right]^{1/p} \left[\sum_{m=1}^n (|\omega_m|^{p-1})^q \right]^{1/q}.$$

On the right, we simply have $(p-1)q = p$ because $pq = p + q$. Treating the last sum in (3.10) in a similar way, we obtain

$$\sum_{j=1}^n |\eta_j| |\omega_j|^{p-1} \leq \left[\sum_{k=1}^n |\eta_k|^q \right]^{1/q} \left[\sum_{m=1}^n |\omega_m|^p \right]^{1/p}.$$

Together,

$$\sum_{j=1}^n |\omega_j|^p \leq \left\{ \left[\sum_{k=1}^n |\xi_k|^p \right]^{1/p} + \left[\sum_{k=1}^n |\eta_k|^p \right]^{1/p} \right\} \left(\sum_{m=1}^n |\omega_m|^p \right)^{1/q}.$$

Dividing by the last factor on the right and noting that $1 - \frac{1}{q} = \frac{1}{p}$, we obtain

$$\left(\sum_{j=1}^n |\xi_j + \eta_j|^p \right)^{1/p} \leq \left(\sum_{k=1}^n |\xi_k|^p \right)^{1/p} + \left(\sum_{m=1}^n |\eta_m|^p \right)^{1/p}.$$

Now, let $n \rightarrow \infty$. On the right-hand side of the above equation, we have two series that converge if we assume that the corresponding sequences do. Hence the series on the left also converges and we arrive at the **Minkowski inequality** for sums

$$\left(\sum_{j=1}^{\infty} |\xi_j + \eta_j|^p \right)^{1/p} \leq \left(\sum_{k=1}^{\infty} |\xi_k|^p \right)^{1/p} + \left(\sum_{m=1}^{\infty} |\eta_m|^p \right)^{1/p}. \quad (3.11)$$

3.1 Definition and Examples

Definition 3.1.1 Metric, Metric Space

A **metric space** is a pair (X, d) , where X is a set and d is a *metric* on X . A **metric** $d : X \times X \rightarrow \mathbb{R}^+$ is a function such that for all $x, y, z \in X$ we have

1. (Positivity) $d(x, y) \geq 0$, $d(x, x) = 0$ for all $x, y \in X$.
2. (Strict Positivity) $d(x, y) = 0$ implies $x = y$.
3. (Symmetry) $d(x, y) = d(y, x)$.
4. (Triangle Inequality) $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

We often simply write X for the metric space if the metric is understood.

Using the fourth axiom above, we obtain by induction the *generalized triangle inequality*

$$d(x_1, x_n) \leq d(x_1, x_2) + d(x_2, x_3) + \cdots + d(x_{n-1}, x_n). \quad (3.12)$$

Example 3.1.1 Using the triangle inequality and the generalized triangle inequality, show that

$$|d(x, y) - d(z, w)| \leq d(x, z) + d(y, w), \quad \text{and} \quad |d(x, z) - d(y, z)| \leq d(x, y).$$

SOLUTION:

Definition 3.1.2 Subspace

Let (X, d) be a metric space and $Y \subset X$. The **subspace** $(Y, \tilde{d})$ is a metric space defined by the metric $\tilde{d} = d|_{Y \times Y}$, called the metric *induced* on Y by d .

Definition 3.1.3 Bounded Set

Let (X, d) be a metric space and consider the non-empty subset $M \subset X$. M is called **bounded** if its *diameter*

$$\delta(M) := \sup_{x, y \in M} d(x, y)$$

is finite.

Example 3.1.2 Examples of Metric Spaces

Here we go through some basic examples of metric space.

1. *The Real Line*, $(\mathbb{R}, d)$: This is the set of all real numbers $\mathbb{R}$ taken with the usual metric d defined as $d(x, y) = |x - y|$ for all $x, y \in \mathbb{R}$.
2. *n-dimensional Euclidean Space*, $(\mathbb{R}^n, d_p)$: This is the set of all n -tuples of real numbers $\underline{x} = (x_1, \dots, x_n)$ which has defined on it several standard metrics. Let $\underline{x} = (x_1, \dots, x_n)$ and $\underline{y} = (y_1, \dots, y_n)$.

- $(p \equiv E)$: $d_E(\underline{x}, \underline{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$, called the **Euclidean metric**.

- $(p \geq 1)$: $d_p(\underline{x}, \underline{y}) = \left[\sum_{i=1}^n |x_i - y_i|^p \right]^{1/p}$.

- $(p \equiv \infty)$: $d_\infty(\underline{x}, \underline{y}) = \max_{1 \leq i \leq n} |x_i - y_i|$.

3. *Space of Continuous Functions*, $(C([a, b]), d_p)$: This is the space of continuous real-valued function on the closed interval $[a, b]$. Let $f, g : [a, b] \rightarrow \mathbb{R} \in C([a, b])$. Then the metrics are defined by

- $(p \geq 1)$: $d_p(f, g) = \left(\int_a^b |f(t) - g(t)|^p dt \right)^{1/p}$.

- $(p \equiv \infty)$: $d_\infty(f, g) = \max_{a \leq t \leq b} |f(t) - g(t)|$. (Note that we do not need to use the supremum here because a continuous function on a closed interval always achieves its maximum.)

Note that one can also use complex-valued functions here, so that we have instead $f, g : [a, b] \rightarrow \mathbb{C}$.

4. *Sequence Space*, (ℓ_p, d_p) : This is the space of all sequences $\underline{x} = (x_1, x_2, \dots)$, $x_i \in \mathbb{R}$, such that $\sum_{i=1}^\infty |x_i|^p \leq \infty$ for all $p \geq 1$, with metric defined by

$$d_p(\underline{x}, \underline{y}) = \left(\sum_{i=1}^\infty |x_i - y_i|^p \right)^{1/p}, \quad p \geq 1.$$

Note that we can also use complex sequences here, so that the $x_i, y_i \in \mathbb{C}$.

5. *Sequence Space, (ℓ_∞, d_∞)* : This is the space of all bounded sequences $\underline{x} = (x_1, x_2, \dots)$, $x_i \in \mathbb{R}$ or $x_i \in \mathbb{C}$, such that $\sup_{i \geq 1} |x_i| \leq \infty$, with metric d_∞ defined by

$$d_\infty(\underline{x}, \underline{y}) = \sup_{i \geq 1} |x_i - y_i|.$$

6. *Space of Continuous Functions with Continuous First Derivative, $(C^1([a, b]), d_p)$* : This is the space of all continuous real- (or complex-) valued functions whose first derivatives are continuous on the closed real interval $[a, b]$. There are two common metrics. Let $f, g \in C^1([a, b])$:

- $d_{1,\infty}(f, g) = \max\{d_\infty(f, g), d_\infty(f', g')\}$, where d_∞ is the metric defined on $C([a, b])$.
- $d_{1,2}(f, g) = \sqrt{d_\infty(f, g)^2 + d_\infty(f', g')^2}$, where again d_∞ is the metric defined on $C([a, b])$.

7. *Discrete Metric Space, (X, d)* : Let X be any non-empty set and define d by

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

for all $x, y \in X$.

Example 3.1.3 Product of Metric Spaces

The Cartesian product $X = X_1 \times X_2$ of two metric spaces (X_1, d_1) and (X_2, d_2) can be made into a metric space (X, d) in many ways. For example, letting $x = (x_1, x_2)$ and $y = (y_1, y_2)$ we can define d in the following ways:

- $d(x, y) = d_1(x_1, y_1) + d_2(x_2, y_2)$;
- $d(x, y) = \sqrt{d_1(x_1, y_1)^2 + d_2(x_2, y_2)^2}$;
- $d(x, y) = \max\{d_1(x_1, y_1), d_2(x_2, y_2)\}$.

(Complete this by proving these are metrics...)

3.2 Covergence, Cauchy Sequence, Completeness

We know that sequences of real numbers play an important role in calculus, and it is the metric $|\cdot|$ on $\mathbb{R}$ that enables us to define the basic concept of convergence of such a sequence. The same holds for sequences of complex numbers; in this case, we have to use the metric on the complex plane. In

an arbitrary metric space (X, d) , the situation is similar.

Definition 3.2.1 Convergence of a Sequence, Limit

A sequence (x_n) in a metric space (X, d) (where each element $x_n \in X$, of course) is said to **converge**, or to **be convergent** if there is an $x \in X$ such that

$$\lim_{n \rightarrow \infty} d(x_n, x) = 0.$$

x is called the **limit** of the sequence (x_n) , and we write

$$\lim_{n \rightarrow \infty} x_n = x,$$

or, simply, $x_n \rightarrow x$. We say that (x_n) *converges to* x , or *has the limit* x . If (x_n) is not convergent, then we call it **divergent**.

REMARK: How is the metric d being used in this definition? We see that d yields the sequence of real numbers $a_n := d(x_n, x)$, whose convergence defines that of (x_n) . And remember that the convergence of a sequence of real numbers is based on the $\epsilon - N_\epsilon$ definition given earlier. We can give a similar $\epsilon - N_\epsilon$ definition of convergence for metric spaces:

Definition 3.2.2 Convergence of a Sequence, Limit-Alternate

A sequence (x_n) in a metric space (X, d) is said to **converge**, or to **be convergent** if there is an $x \in X$ such that for all $\epsilon > 0$ there exists $N_\epsilon > 0$ such that $d(x_n, x) < \epsilon$ for all $n > N_\epsilon$.

REMARK: To avoid trivial misunderstandings, we note that the limit of a convergent sequence must be a point of the space X . For instance, let X be the open interval $(0, 1)$ on $\mathbb{R}$ with the usual metric defined by $d(x, y) = |x - y|$. Then, the sequence $(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$ is not convergent since 0, the point to which the sequence “wants to converge to”, is not in X .

Proposition 3.2.1 Uniqueness of Limits

Let (X, d) be a metric space. If a sequence in X converges, then it is bounded and its limit is unique.

PROOF: Consider the convergent sequence (x_n) with limits x and z , $x \neq z$. Then $d(x, z) > 0$, but also, by the triangle inequality,

$$d(x, z) \leq d(x, x_n) + d(x_n, z),$$

which holds for all n . But as $n \rightarrow \infty$, $x_n \rightarrow x$ and $x_n \rightarrow z$, which gives

$$d(x, z) \leq 0, \tag{3.13}$$

which contradicts the assumption $d(x, z) > 0$. So we must have $x = z$. ■

Definition 3.2.3 Bounded Sequence

Let (X, d) be a metric space and consider the sequence (x_n) in X . It is called a **bounded sequence** if the set $\{x_n\} \subset X$ is bounded, that is, if

$$\delta(\{x_n\}) = \sup_{x_n, x_m \in \{x_n\}} d(x_n, x_m)$$

is finite.

Proposition 3.2.2

Let (X, d) be a metric space. Every convergent sequence in X is bounded.

PROOF: Let (x_n) be a convergent sequence in X with limit x . Then, taking $\epsilon = 1$, we can find N such that $d(x_n, x) < 1$ for all $n > N$. Hence, by the triangle inequality, for all n we have $d(x_n, x) < 1 + a$, where $a = \max\{d(x_1, x), \dots, d(x_N, x)\}$. So (x_n) is bounded since the diameter $\delta(\{x_n\}) = 1 + a$. ■

Proposition 3.2.3

Let (X, d) be a metric space and (x_n) and (y_n) sequences in X converging to x and y , respectively. Then the sequence $(d(x_n, y_n))$ of real numbers converges to $d(x, y)$.

PROOF: We prove this using the ϵ - N_ϵ definition of convergence of real sequences. Let $\epsilon > 0$. By the convergence of (x_n) and (y_n) there exist $N_\epsilon^{(1)} > 0$ and $N_\epsilon^{(2)}$ such that

$$\begin{aligned} d(x_n, x) &< \frac{\epsilon}{2} \text{ for all } n > N_\epsilon^{(1)}, \\ d(y_n, y) &< \frac{\epsilon}{2} \text{ for all } n > N_\epsilon^{(2)}. \end{aligned}$$

Let $N = \max\{N_\epsilon^{(1)}, N_\epsilon^{(2)}\}$. By the generalised triangle inequality, we can write

$$d(x_n, y_n) \leq d(x_n, x) + d(x, y) + d(y, y_n) \Rightarrow d(x_n, y_n) - d(x, y) \leq d(x_n, x) + d(y_n, y),$$

and also

$$\begin{aligned} d(x, y) &\leq d(x, x_n) + d(x_n, y_n) + d(y_n, y) \Rightarrow d(x, y) - d(x_n, y_n) \leq d(x, x_n) + d(y_n, y) \\ &\Rightarrow d(x_n, y_n) - d(x, y) > d(x, x_n) + d(y_n, y). \end{aligned}$$

Combining the two inequalities gives

$$|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y).$$

Therefore, for all $n > N$, we have

$$|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y) \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad \blacksquare$$

REMARK: Observe that by the proof we have shown that the metric $d : X \times X \rightarrow \mathbb{R}$ is a continuous function on $X \times X$.

Definition 3.2.4 Cauchy Sequence

Let (X, d) be a metric space and consider the sequence (x_n) . The sequence is called **Cauchy**, or a **Cauchy sequence**, if for all $\epsilon > 0$ there exists $N_\epsilon > 0$ such that

$$d(x_m, x_n) < \epsilon \text{ for every } m, n > N_\epsilon.$$

Definition 3.2.5 Equivalent Cauchy Sequences

Two sequences (x_n) and (y_n) in a metric space (X, d) are called **equivalent**, and written $(x_n) \sim (y_n)$, if $\lim_{n \rightarrow \infty} d(x_n, y_n) = 0$.

Theorem 3.2.1 Convergent Sequences

Every convergent sequence in a metric space is a Cauchy sequence.

PROOF: Let (x_n) be a convergent sequence in X with limit x . Then, for every $\epsilon > 0$ there exists $N_\epsilon > 0$ such that $d(x_n, x) < \frac{\epsilon}{2}$ for all $n > N_\epsilon$. By the triangle inequality, we get for all $m, n > N_\epsilon$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad \blacksquare$$

Definition 3.2.6 Complete Metric Space

A metric space is called **complete** if every Cauchy sequence in the space converges (that is, has a limit that is an element of the space).

3.3 The Topology of Metric Spaces

Definition 3.3.1 Ball and Sphere

Let (X, d) be a metric space. Given a point $x_0 \in X$ and a real number $r > 0$, we define three types of sets:

1. $B(x_0; r) \equiv B_r(x_0) := \{x \in X \mid d(x, x_0) < r\}$, called the **open ball** of radius r centred at x_0 ;
2. $\overline{B(x_0; r)} \equiv \overline{B_r(x_0)} := \{x \in X \mid d(x, x_0) \leq r\}$, called the **closed ball** of radius r centred at x_0 ;
3. $S(x_0; r) \equiv S_r(x_0) := \{x \in X \mid d(x, x_0) = r\}$, called the **sphere** of radius r centred at x_0 .

REMARK: In working with metric spaces, it is a great advantage that we use a terminology that is analogous to that of Euclidean geometry. However, we should beware of a danger, namely, of assuming that balls and spheres in an arbitrary and *abstract* metric space enjoy the same properties as balls and spheres in $\mathbb{R}^3$, because this is generally not so. An unusual property is that a sphere can be empty. For example, in a discrete metric space, we have $S(x_0; r) = \emptyset$ if $r \neq 1$. (What about spheres of radius one in this case?)

REMARK: The definitions above immediately imply that

$$S(x_0; r) = \overline{B(x_0; r)} - B(x_0; r).$$

Definition 3.3.2 Open Set, Closed Set

Let (X, d) be a metric space and $M \subset X$.

1. M is called **open** if for all points $p \in M$ there exists $r > 0$ such that $B(p; r) \subset M$.
2. M is called **closed** if its complement $M^c = X - M$ is open.

An open ball of radius ϵ centred at x_0 , i.e., $B(x_0; \epsilon)$ is often called an ϵ -**neighbourhood** of x_0 . Then, a **neighbourhood** of x_0 is any subset M of X that contains an ϵ -neighbourhood of x_0 .

It is also possible to define a closed set in the following way:

Definition 3.3.3 Closed Set—Alternate

Let (X, d) be a metric space and $M \subset X$. M is called **closed** if every convergent sequence in M has its limit in M , i.e., if $(x_n) \subset M$, $\lim_{n \rightarrow \infty} x_n = x$, $x \in X \Rightarrow x \in M$.

We can then *prove* using this definition that the complement of an open set is closed.

Proposition 3.3.1

Let (X, d) be a metric space and $M \subset X$.

1. If M is open, then M^c is closed.
2. If M is closed, then M^c is open.

PROOF

1. Suppose (x_n) is a convergent sequence in M^c and that its limit is p . We must show that $p \in M^c$ by the definition above. Assume for a contradiction that $p \in M$. Then there exists $\epsilon > 0$ such that $B(p; \epsilon) \subset M$, and N such that $x_n \in B(p; \epsilon) \subset M$ for $n > N$, meaning that M is closed, a contradiction to the assumption that M is open. So $p \in M^c$ and so M^c is closed.
2. Let $p \in M^c$ and $\epsilon > 0$. We must show that M^c is open, i.e., that there exists an open ball, say $B(p; \epsilon)$, centred at p contained entirely in M^c . Assume for a contradiction that $B(p; \epsilon)$ is not contained entirely in M^c , i.e., that $B(p; \epsilon) \cap M \neq \emptyset$ for all ϵ . Let $\epsilon = \frac{1}{n}$. Then there exists

$x_n \in B(p; \frac{1}{n}) \cap M$, i.e., $x_n \in M$ and $d(x_n, p) < \frac{1}{n}$ for all n . So the sequence (x_n) is in M and has limit p , and since M is closed we must have $p \in M$, a contradiction to the assumption that $p \in M^c$. So $B(p; \epsilon)$ is contained entirely in M^c . As p was arbitrary, the proof is complete. ■

REMARK: This remark concerns the alternate definition of a closet set given above. It is meant to emphasise the fact that the limit x of the sequence $(x_n) \subset M$ has to be in the metric space X .

Let $X = (0, 2) \subset \mathbb{R}$, with subset $M = (1, 2)$, so that $M^c = (0, 1]$. From the definition of an open set, it is clear that M satisfies the requirements of an open set. But what about the set M^c ? From the second part of 3.3.1, M^c is closed. But $(0, 1]$ doesn't look like a closed set, at least from what we have seen in the past! One might immediately come up with the convergent sequence $x_n = \frac{1}{n}$, with limit $0 \notin (0, 1]$, implying according to the alternate definition that $(0, 1]$ is not closed. It appears we have a contradiction between the two definitions of closet set!

But of course, we don't. The interval $(0, 1]$ is neither closed nor open when viewed as a **subset of $\mathbb{R}$** . But in this example, we are considering it as a particular subset of the given metric space, $X = (0, 2)$. We have to look carefully at the alternate definition of a closet set and keep in mind the additional requirement that the limit point x of the sequence is assumed to be in the metric space X . Therefore, the sequence $x_n = \frac{1}{n}$ does not qualify for use in the definition since its limit, 0, is not an element of X . So there is no contradiction, and the subset $(0, 1]$ is indeed closed.

But perhaps the subset M itself is closed. After all, we are not allowed to consider sequences in M such as $x_n = 2 - \frac{1}{n}$ that converge to 2, since 2 is not an element of X . But if we consider the sequence $x_n = 1 + \frac{1}{n}$, which is a sequence in A , we see that it converges to the limit 1, which is an element of X but not an element of A . So the requirement of the alternate definition is not satisfied, and A is not closed, as expected. (Note that the requirement in the definition is for *all* convergent sequences.)

But now what about X itself. Is it closed, open, neither, or both? The answer to this is not quite clear—see the Appendix at the end of this section.

Finally, is X a complete metric space. No, because the Cauchy sequence $x_n = \frac{1}{n}$ does not converge (to an element in X).

Proposition 3.3.2

Let (X, d) be a metric space and $p \in X$, $r > 0$. Then $B(p; r)$ is an open subset of X .

PROOF: We want to show that $B(p; r)$ is open, i.e., that there exists an open ball about each point in $B(p; r)$ that is contained entirely in $B(p; r)$. So let $q \in B(p; r)$. Then clearly $d(q, p) < r$. Now, let $r_1 = r - d(q, p)$ and $x \in B(q; r_1)$. Then $d(x, q) < r_1$, and by the triangle inequality,

$$d(x, p) \leq d(x, q) + d(q, p) < r_1 + d(q, p) = r.$$

Thus, $x \in B(p; r)$, which means that $B(q; r_1) \subset B(p; r)$, concluding the proof. ■

It follows immediately, then, from our definitions that the closed ball $\overline{B(p; r)}$ is a closed subset of a metric space.

Definition 3.3.4 Interior and Interior Point

Let (X, d) be a metric space and $M \subset X$. A point x_0 is called an **interior point** of M if M is a neighbourhood of x_0 . The **interior** of M is the set of all interior points of M and is often denoted $\text{Int}(M)$.

REMARK: $\text{Int}(M)$ is open and is the largest open set contained in M .

Definition 3.3.5 Accumulation Point

Let (X, d) be a metric space and $M \subset X$. A point $x_0 \in X$ (which may or may not be a point in M) is called an **accumulation point**, or **limit point**, if every neighbourhood of x_0 contains at least one point $y \in M$ distinct from x_0 .

Definition 3.3.6 Closure of a Set

Let (X, d) be a metric space and $M \subset X$. The set consisting of the point of M and the accumulation points of M is called the **closure** of M and is denoted $\overline{M}$.

REMARK: $\overline{M}$ is the smallest closed set containing M .

Proposition 3.3.3

Let (X, d) be a metric space and $M \subset X$. M is closed if and only if $\overline{M} = M$, i.e., if and only if M contains all of its accumulation points.

PROOF: To be completed. ■

Theorem 3.3.1

Let M be a non-empty subset of a metric space (X, d) and $\overline{M}$ its closure. Then $x \in \overline{M}$ if and only if there is a sequence (x_n) in M such that $\lim_{n \rightarrow \infty} x_n = x$.

PROOF: $(\Rightarrow)$ Let $x \in \overline{M}$. If $x \in M$, a sequence of the required type is $(x, x, x, x, \dots)$. If $x \notin M$, it is a point of accumulation of M . Hence, for each $n = 1, 2, \dots$, the ball $B(x; 1/n)$ contains an $x_n \in M$, and $\lim_{n \rightarrow \infty} x_n = x$ because $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

$(\Leftarrow)$ Conversely, if (x_n) is in M and $\lim_{n \rightarrow \infty} x_n = x$, then $x \in M$ or every neighbourhood of x contains points $x_n \neq x$, so that x is a point of accumulation of M . Hence, $x \in \overline{M}$. ■

Definition 3.3.7 Dense Set, Separable Space

Let (X, d) be a metric space and $M \subset X$. M is called **dense in X** if $\overline{M} = X$. X is called **separable** if it has a countable subset that is dense in X .

3.3.1 Continuity

Definition 3.3.8 Continuous Mapping

Let (X_1, d_1) and (X_2, d_2) be metric spaces. A mapping $T : X_1 \rightarrow X_2$ is called **continuous at** $x_0 \in X_1$ if for every $\epsilon > 0$ there exists $\delta > 0$ such that $d_2(T(x), T(x_0)) < \epsilon$ for all $x \in X$ satisfying $d_1(x, x_0) < \delta$. T is called **continuous** on $M \subset X$ if T is continuous at every point in M .

Proposition 3.3.4

Let (X_1, d_1) and (X_2, d_2) be metric spaces and $T : X_1 \rightarrow X_2$. T is continuous at $x_0 \in X$ if and only if the sequence $(T(x_n)) \subset X_2$ converges (under the d_2 metric) to $T(x_0)$ for every sequence $(x_n) \subset X_1$ that converges (under the d_1 metric) to x_0 .

PROOF: ($\Rightarrow$) Assume T is continuous at x_0 . Then, for a given $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_1(x, x_0) < \delta \implies d_2(T(x), T(x_0)) < \epsilon.$$

Let $\lim_{n \rightarrow \infty} x_n = x_0$. Then there exists $N > 0$ such that for all $n > N$ we have

$$d_1(x_n, x_0) < \delta.$$

Hence, for all $n > N$,

$$d_2(T(x_n), T(x_0)) < \epsilon.$$

By definition, this means that $\lim_{n \rightarrow \infty} T(x_n) = T(x_0)$.

($\Leftarrow$) Conversely, assume that $\lim_{n \rightarrow \infty} x_n = x_0$ implies $\lim_{n \rightarrow \infty} T(x_n) = T(x_0)$. We must prove that T is continuous at x_0 . Suppose this is false. Then there is $\epsilon > 0$ such that for every $\delta > 0$ there is an $x \neq x_0$ satisfying $d_1(x, x_0) < \delta$ but $d_2(T(x), T(x_0)) \geq \epsilon$. In particular, for $\delta = \frac{1}{n}$, there is an x_n satisfying $d_1(x_n, x_0) < \frac{1}{n}$ but $d_2(T(x_n), T(x_0)) \geq \epsilon$. Clearly, $\lim_{n \rightarrow \infty} x_n = x_0$, but $(T(x_n))$ does not converge to $T(x_0)$. This contradicts the assumption that $\lim_{n \rightarrow \infty} T(x_n) = T(x_0)$, proving the theorem. ■

Theorem 3.3.2

A mapping T between metric spaces (X_1, d_1) and (X_2, d_2) is continuous if and only if the inverse image of any open subset of X_2 is an open subset of X_1 .

PROOF

1. Suppose that T is continuous. Let $S \subset X_2$ be open and S_0 the inverse image of S . If $S_0 = \emptyset$, then it is open. Let $S_0 \neq \emptyset$. For any $x_0 \in S_0$, let $y_0 = T(x_0)$. Since S is open, it contains an ϵ -neighbourhood N of y_0 . Since T is continuous, x_0 has a δ -neighbourhood N_0 that is mapped into N . Since $N \subset S$, we have $N_0 \subset S_0$, so that S_0 is open because $x_0 \in S_0$ was arbitrary.

2. Conversely, assume that the inverse image of every open set in X_2 is an open set in X_1 . Then, for every $x_0 \in X$ and any ϵ -neighbourhood N of $T(x_0)$, the inverse image N_0 of N is open, since N is open and N_0 contains x_0 . Hence, N_0 also contains a δ -neighbourhood of x_0 , which is mapped into N because N_0 is mapped into N . Consequently, by definition, T is continuous at x_0 . Since $x_0 \in X_1$ was arbitrary, T is continuous. ■

Theorem 3.3.3

Let X, Y, Z be metric spaces and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, $a \in X$, $b = f(a) \in Y$. If f is continuous at a and g is continuous at b , then $g \circ f$ is continuous at a .

PROOF: To be completed. ■

Definition 3.3.9 Lipschitz Continuity

Let (X, d_X) and (Y, d_Y) be two metric spaces and $T : X \rightarrow Y$. T is called **Lipschitz continuous** if there exists a real constant $K \geq 0$ such that, for all $x_1, x_2 \in X$,

$$d_Y(T(x_1), T(x_2)) \leq K d_X(x_1, x_2).$$

K is called a **Lipschitz constant**.

Proposition 3.3.5

Let $I \subset \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$ a differentiable function with $|f'(x)| \leq \lambda$ for all $x \in I$. Then f is Lipschitz continuous with Lipschitz constant λ .

PROOF: By the mean value theorem, for any two points $x, y \in I$, there exists a point c between x and y such that

$$|f(x) - f(y)| = |f'(c)(x - y)| = |f'(c)||x - y| \leq \lambda|x - y|. \quad \blacksquare$$

Example 3.3.1 Here we go through some important examples of Lipschitz continuous functions.

- The function $f(x) = \sqrt{x^2 + 5}$ defined on $\mathbb{R}$ is Lipschitz continuous with Lipschitz constant $K = 1$ because it is everywhere differentiable and the absolute value of the derivative is bounded above by 1.
- Likewise, the function $f(x) = \sin(x)$ defined on $\mathbb{R}$ is Lipschitz continuous with $K = 1$ because its derivative, $\cos$, is bounded above by 1 in absolute value.
- The function $f(x) = |x|$ defined on $\mathbb{R}$ is Lipschitz continuous with $K = 1$. This is an example of a Lipschitz continuous function that is not differentiable.

3.3.2 Equicontinuity

Definition 3.3.10 Equicontinuity

Let $\mathcal{F}$ be a family of functions from a metric space (X, d_X) to (Y, d_Y) . $\mathcal{F}$ is called an **equicontinuous family** if for all $\epsilon > 0$ and all $x \in X$ there exists $\delta > 0$ such that for all $f \in \mathcal{F}$ $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \epsilon$.

$\mathcal{F}$ is called **uniformly equicontinuous** if for all $\epsilon > 0$ there exists $\delta > 0$ such that for all $x \in X$ and all $f \in \mathcal{F}$ $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \epsilon$.

For comparison's sake, note that to say all $f \in \mathcal{F}$ are continuous (i.e., each $f \in \mathcal{F}$ is continuous) means that for all $\epsilon > 0$ and for all $x \in X$ and for all $f \in \mathcal{F}$ there exists $\delta > 0$ such that $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) < \epsilon$. Thus, for mere continuity, δ can depend on f and on x and on ϵ , while equicontinuity says that δ is independent of f . Uniform continuity says that δ is dependent only on ϵ .

Theorem 3.3.4

Let (f_n) be a sequence of functions from one metric space to another with the property that the family $\{f_n\}$ is equicontinuous. Suppose also that $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x (i.e., the sequence of functions converges pointwise to a function f). Then f is continuous.

PROOF: Let $f_n : (X, d_X) \rightarrow (Y, d_Y)$. Given $\epsilon > 0$ and x , choose $\delta > 0$ such that $d_X(x, x') < \delta$ implies $d_Y(f_n(x), f_n(x')) < \frac{\epsilon}{2}$ for all n . Since d is continuous, we have $d(f(x), f(x')) = \lim_{n \rightarrow \infty} d(f_n(x), f_n(x'))$, so that $d_X(x, x') < \delta$ implies $d_Y(f(x), f(x')) \leq \frac{\epsilon}{2} < \epsilon$. ■

Theorem 3.3.5

Let $\{f_n\}$ be an equicontinuous family of functions from one metric space (X, d_X) to (Y, d_Y) with Y complete. Suppose that for a dense set $D \subset X$, we know that $f_n(x)$ converges for all $x \in D$. Then $f_n(x)$ converges for all $x \in X$.

PROOF: To be completed. ■

The above theorem tells us that, in general, pointwise convergence on a dense set combined with equicontinuity implies pointwise convergence everywhere. More spectacularly, for a sequence of functions on $[0, 1]$, uniform equicontinuity and pointwise convergence imply *uniform convergence*.

Theorem 3.3.6

Let $\{f_n\}$ be a uniformly equicontinuous family of functions on $[0, 1]$. Suppose that $f_n(x) \rightarrow f(x)$ for each $x \in [0, 1]$. Then $f_n(x) \rightarrow f(x)$ uniformly in x .

PROOF: Let $\epsilon > 0$ be given. Choose $\delta > 0$ such that $|x - y| < \delta$ implies $|f_n(x) - f_n(y)| < \frac{\epsilon}{3}$ for all n . Now, choose $y_1, \dots, y_m$ such that every point of $[0, 1]$ is within δ of some y_i . Since $y_1, \dots, y_m$ is a finite set, we can find n such that $n > N$ implies $|f_n(y_i) - f(y_i)| < \frac{\epsilon}{3}$ for all $i = 1, 2, \dots, m$. By an $\frac{\epsilon}{3}$ argument, $d_\infty(f_n, f) < \epsilon$ for all $n > N$. ■

Theorem 3.3.7 Arzela-Ascoli

Let (f_n) be a sequence of uniformly bounded equicontinuous functions on $[a, b]$. Then, some subsequence (f_{n_k}) converges uniformly on $[a, b]$.

PROOF: Let $\{r_i\}_{i=1}^\infty$ be a countable dense set in $[a, b]$. There exist successive subsequences of (f_n) :

$$\begin{aligned} f_{11}, f_{12}, \dots &\text{ converges at } r_1, \\ f_{21}, f_{22}, \dots &\text{ converges at } r_1, r_2. \end{aligned}$$

Consider the diagonal sequence (f_{nn}) . (f_{nn}) converges at all $\{r_i\}$. Now, let $\bar{f}_n = f_{nn}$. Then $(\bar{f}_n)$ converges uniformly on $[a, b]$. To prove this, let $\epsilon > 0$ be given. Then there exists $\delta > 0$ such that $|\bar{f}_n(x) - \bar{f}_n(y)| < \frac{\epsilon}{3}$ for $|x - y| < \delta$ for all n . Divide $[a, b]$ into N_0 subintervals $[x_{i-1}, x_i]$, $i = 1, \dots, N_0$, each of length less than δ . There exists $\bar{r}_i \in [x_{i-1}, x_i]$ such that $\bar{r}_i = r_k$ for some k , for all i . Then there exists N such that

$$|\bar{f}_n(\bar{r}_i) - \bar{f}_m(\bar{r}_i)| < \frac{\epsilon}{3}$$

for all $n, m > N$, $i = 1, \dots, N_0$. For $x \in [a, b]$, $x \in [x_{i-1}, x_i]$ for some i . This means that

$$\begin{aligned} |\bar{f}_n(x) - \bar{f}_m(x)| &= |(\bar{f}_n(x) - \bar{f}_n(\bar{r}_i)) + (\bar{f}_n(\bar{r}_i) - \bar{f}_m(\bar{r}_i)) + (\bar{f}_m(\bar{r}_i) - \bar{f}_m(x))| \\ &\leq |\bar{f}_n(x) - \bar{f}_n(\bar{r}_i)| + |\bar{f}_n(\bar{r}_i) - \bar{f}_m(\bar{r}_i)| + |\bar{f}_m(\bar{r}_i) - \bar{f}_m(x)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

Thus, $(\bar{f}_n)$ is a uniformly Cauchy sequence by the Weierstrass theorem (see real analysis review notes). Therefore, $(\bar{f}_n)$ converges uniformly on $[a, b]$. ■

3.3.3 Appendix: Topological Spaces

It is not difficult to show that the collection of all open subsets of a metric space (X, d) , call it $\mathcal{T}$, has the following properties:

1. $\emptyset \in \mathcal{T}; X \in \mathcal{T}$;
2. The union of any members of $\mathcal{T}$ is a member of $\mathcal{T}$;
3. The intersection of finitely-many members of $\mathcal{T}$ is a member of $\mathcal{T}$.

PROOF: 1 follows by noting that $\emptyset$ is open since $\emptyset$ has no elements and, obviously, X is open. As for 2, any point x of the union U of open sets belongs to (at least) one of these sets, call it M , and M contains a ball $B \subset M$ about x since M is open. Then $B \subseteq U$, by the definition of a union. Finally,

for 3, if y is any point of the intersection of open sets $M_1, \dots, M_n$, then each M_j contains a ball about $y \in X$ and a smallest of these balls is contained in that intersection. ■

The three properties above form the basis of the area of *general topology*. In general topology, any set X that has a set of its subsets $\mathcal{T}$ satisfying the three properties is called a **topological space**, with $\mathcal{T}$ being called a **topology** for X .

Because the set of all open sets satisfies the three properties of a topological space, we have that *all metric spaces are topological spaces*. (Of course, the converse isn't necessarily true because topological spaces need not have a metric defined on them.)

3.4 Equivalent Metrics

Definition 3.4.1 Equivalent Metrics

Two metrics d_1 and d_2 on a set X are called **equivalent** if they have the same convergent sequences.

REMARK: Note that another way of stating this definition is to say that d_1 and d_2 are equivalent if they *generate* the same topology, i.e., they have the same open and closed sets.

This means that if a sequence converges under d_1 then it also converges (to the same limit) under d_2 , and vice versa.

Example 3.4.1 Prove that the equivalence of metrics is an equivalence relation.

SOLUTION:

Example 3.4.2 Show that the metrics

$$d_1(x, y) = |x - y|, \quad d_2(x, y) = \frac{|x - y|}{1 + |x - y|}, \quad d_3(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

are possible metrics on $\mathbb{R}$, but no two of them are equivalent.

SOLUTION:

Example 3.4.3 d_p and d_∞ on $\mathbb{R}^\ell$ are equivalent metrics for all $p \geq 1$. To see this, use the fact that

$$d_\infty(x, y) \leq d_p(x, y) \leq \ell^{1/p} d_\infty(x, y).$$

(prove this!) Therefore, if the sequence (x_n) converges to x under d_∞ then it also converges to x under d_p . Conversely, if a sequence converges under d_p then it also does do under d_∞ . So the two metrics are equivalent.

Also, d_{p_1} and d_{p_2} are equivalent for all $p_1, p_2 \geq 1$.

Example 3.4.4 The metrics d_1 and d_∞ are *not* equivalent on $C([a, b])$. To see this, let $[a, b] = [0, 1]$, and define the sequence of functions (f_n) by

$$f_n(x) = \begin{cases} 1 - nx, & 0 \leq x \leq \frac{1}{n} \\ 0, & \frac{1}{n} \leq x \leq 1 \end{cases}.$$

Then (f_n) converges to the zero function $f_0(x) \equiv 0$ for all $x \in [0, 1]$ under the metric d_1 since $d_1(f_n, 0) = \frac{1}{2n}$. But, $d_\infty(f_n, 0) = 1$, and so under d_∞ (f_n) does not converge to f_0 .

Proposition 3.4.1

Let (X, d) be a metric space and define $\rho := \frac{d}{1+d}$. Then ρ is a metric that is equivalent to d .

PROOF: It is clear that ρ satisfies the first three conditions of a metric. Let us verify the triangle inequality. First, note that the function f defined by $f(t) = \frac{t}{1+t}$, $t \geq 0$, is an increasing functions since $f'(t) = \frac{1}{(1+t)^2} > 0$. Now, using this, we have

$$\begin{aligned} \rho(x, y) &= \frac{d(x, y)}{1 + d(x, y)} \leq \frac{d(x, z) + d(z, y)}{1 + d(x, z) + d(z, y)} \\ &= \frac{d(x, z)}{1 + d(x, z) + d(z, y)} + \frac{d(z, y)}{1 + d(x, z) + d(z, y)} \leq \rho(x, z) + \rho(z, y), \end{aligned}$$

proving the triangle inequality, and thus proving that ρ is indeed a metric. Note that for all $x, y \in X$, $0 \leq \rho < 1$. Now, suppose the sequence (x_n) in X converges to x under d . So $\lim_{n \rightarrow \infty} d(x_n, x) = 0$. Then,

$$\lim_{n \rightarrow \infty} \rho(x_n, x) = \frac{\lim_{n \rightarrow \infty} d(x_n, x)}{1 + \lim_{n \rightarrow \infty} d(x_n, x)} = 0.$$

Conversely, suppose the sequence (x_n) in X converges to x under ρ , so that $\lim_{n \rightarrow \infty} \rho(x_n, x) = 0$. Then,

$$\lim_{n \rightarrow \infty} d(x_n, x) = \frac{\lim_{n \rightarrow \infty} \rho(x_n, x)}{1 - \lim_{n \rightarrow \infty} \rho(x_n, x)} = 0.$$

Therefore, ρ and d are equivalent. ■

REMARK: A metric d such that $0 \leq d(x, y) < 1$ for all $x, y \in X$ is sometimes called a *normalized metric*.

3.5 Examples of Complete Metric Spaces

In various applications, a set X is given (for instance, a set of sequences or a set of functions), and X is made into a metric space by choosing a metric d on it. The remaining task is then to find out whether (X, d) has the desirable property of being complete. To prove completeness, we take an arbitrary Cauchy sequence (x_n) in X and show that it converges in X (i.e., the limit to which the sequence converges is a point in X). For different spaces, such proofs may vary in complexity, but they have approximately the same general pattern (in roughly this order):

1. Construct an element x (to be used as a limit);
2. Prove that x is in the space considered;
3. Prove that $\lim_{n \rightarrow \infty} x_n = x$ under the metric.

We now present completeness proofs for some metric spaces that occur quite frequently in theoretical and practical investigations. We will frequently use the completeness of the real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$ in the proofs. The following fact will also be useful.

Theorem 3.5.1 Complete Subspace

A subspace $(M, \tilde{d})$ of a complete metric space (X, d) (where $\tilde{d}$ is the metric on M induced by d) is complete if and only if M is closed in X .

PROOF: ($\Rightarrow$) Let $(M, \tilde{d})$ be a complete metric space under the metric $\tilde{d}$ induced by d . By 3.3.1, for every $x \in \overline{M}$, there is a sequence (x_n) in M that converges to x . Since (x_n) is Cauchy (every convergent sequence is a Cauchy sequence, remember) and M is complete, (x_n) converges in M , the limit being unique. Hence, $x \in M$. This proves that M is closed because $x \in \overline{M}$ was arbitrary.

($\Leftarrow$) Conversely, let M be closed and (x_n) Cauchy and convergent in M . Then, letting $\lim_{n \rightarrow \infty} x_n = x \in X$, we have $x \in \overline{M}$ by 3.3.1, and $x \in M$ since $M = \overline{M}$ (M being closed), by assumption. Hence, the arbitrary Cauchy sequence (x_n) converges in M , which proves the completeness of M . ■

Theorem 3.5.2 Completeness of $\mathbb{R}^\ell$ and $\mathbb{C}^\ell$

The metric spaces $(\mathbb{R}^\ell, d_p)$ and $(\mathbb{C}^\ell, d_p)$ are complete for $p \geq 1$ and $p = \infty$.

PROOF: We focus on $\mathbb{R}^\ell$ first.

- *Case 1: $p = \infty$* Let $x_1 = (x_1^1, x_1^2, \dots, x_1^\ell)$ and $x_2 = (x_2^1, x_2^2, \dots, x_2^\ell)$ and recall that the d_∞ norm is defined by

$$d_\infty(x_1, x_2) = \max_{1 \leq i \leq \ell} |x_1^i - x_2^i|.$$

Therefore,

$$|x_1^i - x_2^i| \leq d_\infty(x_1, x_2) \quad \text{for } 1 \leq i \leq \ell. \quad (3.14)$$

Now, let $(x_n) \subseteq \mathbb{R}^\ell$ be a Cauchy sequence under d_∞ , where $x_n = (x_n^1, x_n^2, \dots, x_n^\ell)$. By definition, therefore, for every $\epsilon > 0$ there exists N_ϵ such that for all $m, n > N_\epsilon$

$$d_\infty(x_n, x_m) < \epsilon.$$

Consider the sequences $(x_n^1), (x_n^2), (x_n^3), \dots$, i.e., the sequences comprising the first component, second component, third component, ... of each element in the main sequence (x_n) . For each j , $1 \leq j \leq \ell$, we have for $m, n > N_\epsilon$

$$|x_m^j - x_n^j| \leq d_\infty(x_m, x_n) < \epsilon,$$

where the first inequality is from (3.14) and the second due to the fact that (x_n) is a Cauchy sequence. Therefore (x_n^j) are Cauchy sequences for all $1 \leq j \leq \ell$. Therefore, by the completeness of $\mathbb{R}$, there exists $q = (q^1, q^2, \dots, q^\ell)$ such that

$$\lim_{n \rightarrow \infty} x_n^j = q^j \quad \text{for all } 1 \leq j \leq \ell.$$

Therefore,

$$\lim_{n \rightarrow \infty} d_\infty(x_n, q) = \lim_{n \rightarrow \infty} \left(\max_{1 \leq j \leq \ell} |x_n^j - q^j| \right) = 0,$$

which proves that $(\mathbb{R}^\ell, d_\infty)$ is complete.

- *Case 2:* $1 \leq p < \infty$ Again, let $x_1 = (x_1^1, x_1^2, \dots, x_1^\ell)$ and $x_2 = (x_2^1, x_2^2, \dots, x_2^\ell)$ and recall that the d_p norm is defined by

$$d_p(x_1, x_2) = \left(\sum_{i=1}^{\ell} |x_1^i - x_2^i|^p \right)^{1/p}.$$

Therefore,

$$\max_{1 \leq i \leq \ell} |x_1^i - x_2^i| \leq \left(\sum_{i=1}^{\ell} |x_1^i - x_2^i|^p \right)^{1/p} \leq \left(\ell \max_{1 \leq i \leq \ell} |x_1^i - x_2^i|^p \right)^{1/p},$$

i.e.,

$$d_\infty(x_1, x_2) \leq d_p(x_1, x_2) \leq \ell^{1/p} d_\infty(x_1, x_2). \quad (3.15)$$

Now, let (x_n) be a Cauchy sequence in $\mathbb{R}^\ell$, so that for every $\epsilon > 0$ there exists N_ϵ such that for all $m, n > N_\epsilon$

$$d_\infty(x_n, x_m) < \epsilon.$$

Then, as before, consider the sequences $(x_n^1), (x_n^2), (x_n^3), \dots$. For every $1 \leq j \leq \ell$, we have for all $m, n > N_\epsilon$,

$$\begin{aligned} |x_m^j - x_n^j| &\leq d_\infty(x_m, x_n) \quad \text{by (3.14)} \\ &\leq d_p(x_m, x_n) \quad \text{by (3.15)} \\ &< \epsilon \end{aligned}$$

proving that (x_n^j) is a Cauchy sequence for all $1 \leq j \leq \ell$. Therefore, there exists $q = (q^1, q^2, \dots, q^\ell)$ such that

$$\lim_{n \rightarrow \infty} x_n^j = q^j \quad \text{for all } 1 \leq j \leq \ell.$$

Finally, using (3.15) and the fact, as we've shown above, that $\lim_{n \rightarrow \infty} d_\infty(x_n, q) = 0$, so that also $\lim_{n \rightarrow \infty} \ell^{1/p} d_\infty(x_n, q) = 0$, by the *squeeze theorem*, we have

$$\lim_{n \rightarrow \infty} d_p(x_n, q) = 0,$$

proving that $(\mathbb{R}^\ell, d_p)$ is complete for $1 \leq p < \infty$.

An analogous proof applies for the cases $(\mathbb{C}^\ell, d_\infty)$ and $(\mathbb{C}^\ell, d_p)$, $1 \leq p < \infty$. ■

Example 3.5.1 From the above theorem, we have that $(\mathbb{R}, |\cdot|)$ is a complete metric space. The interval $[a, b] \subset \mathbb{R}$ is closed, and so by Theorem 3.5.1 $([a, b], |\cdot|)$ is a complete metric space.

Theorem 3.5.3 Completeness of ℓ_∞

The metric space (ℓ_∞, d_∞) is complete.

PROOF: Let (x_m) be any Cauchy sequence in the space ℓ_∞ , where $x_m = (\xi_m^1, \xi_m^2, \dots)$. Recall that the metric d_∞ on ℓ_∞ is given by

$$d_\infty(x_1, x_2) = \sup_j |\xi_1^j - \xi_2^j|,$$

and the sequence (x_m) is Cauchy, for any $\epsilon > 0$ there exists N_ϵ such that for all $m, n > N_\epsilon$,

$$d_\infty(x_m, x_n) = \sup_j |x_m^j - x_n^j| < \epsilon.$$

Additionally, for every j , we have for all $m, n > N_\epsilon$,

$$|\xi_m^j - \xi_n^j| < \epsilon. \quad (3.16)$$

Hence, for every j , the sequence (ξ_n^j) is a Cauchy sequence of real numbers. Therefore, by the completeness of $\mathbb{R}$, there exists $q := (q^1, q^2, \dots)$ such that $\lim_{n \rightarrow \infty} (\xi_n^j) = q^j$ for all j . Using these infinitely many limits, we now show that $q \in \ell_\infty$ and that $\lim_{n \rightarrow \infty} x_n = q$.

From (3.16), as $n \rightarrow \infty$,

$$|\xi_m^j - q^j| \leq \epsilon. \quad (3.17)$$

Since $x_m \in \ell_\infty$, there is a real number k_m such that $|\xi_m^j| \leq k_m$ for all j . Hence, by the triangle inequality,

$$|q^j| \leq |q^j - \xi_m^j| + |\xi_m^j| \leq \epsilon + k_m.$$

This inequality holds for every j , and the right-hand side does not involve j . Hence (q^j) is a bounded sequence of real numbers. This implies that $q \in \ell_\infty$. Also, from (3.17), we obtain

$$d_\infty(x_m, q) = \sup_j |\xi_m^j - q^j| \leq \epsilon.$$

This shows that $\lim_{n \rightarrow \infty} x_m = q$, and since (x_m) was arbitrary, we conclude that (ℓ_∞, d_∞) is complete. ■

Theorem 3.5.4 **Completeness of $(\ell_{c,\infty}, d_\infty)$**

The metric space $(\ell_\infty(\mathbb{C}), d_\infty)$ of all convergent sequences of real numbers, with the metric induced by the one from ℓ_∞ , is complete.

PROOF: $(\ell_{c,\infty}, d_\infty)$ is a subspace of (ℓ_∞, d_∞) . If we can show that $(\ell_{c,\infty}, d_\infty)$ is closed in (ℓ_∞, d_∞) , then by 3.5.1 we will have proven that $(\ell_{c,\infty}, d_\infty)$ is complete. For brevity, let $c \equiv (\ell_{c,\infty}, d_\infty)$.

Consider any $x = (\xi^j) \in \bar{c}$, the closure of c . By Theorem 3.3.1, there are $x_n = (\xi_n^j) \in c$ such that $\lim_{n \rightarrow \infty} x_n = x$, where $x := (\xi^1, \xi^2, \dots)$. Hence, given any $\epsilon > 0$ there is N_ϵ such that for $n \geq N_\epsilon$ and all j we have

$$|\xi_n^j - \xi^j| \leq d_\infty(x_n, x) < \frac{\epsilon}{3},$$

in particular, for $n = N_\epsilon$ and all j . Since $x_{N_\epsilon} \in c$, its terms $\xi_{N_\epsilon}^j$ form a convergent sequence. Such a sequence is Cauchy. Hence, there is a N_1 such that

$$|\xi_{N_\epsilon}^j - \xi_{N_\epsilon}^k| < \frac{\epsilon}{3}.$$

The triangle inequality now yields for all $j, k \geq N_1$ the following inequality:

$$|\xi^j - \xi^k| \leq |\xi^j - \xi_{N_\epsilon}^j| + |\xi_{N_\epsilon}^j - \xi_{N_\epsilon}^k| + |\xi_{N_\epsilon}^k - \xi^k| < \epsilon.$$

This shows that the sequence $x = (\xi^i)$ is convergent. Hence, $x \in c$. Since $x \in \bar{c}$ was arbitrary, we have that c is closed in ℓ_∞ , and so c is complete. ■

Theorem 3.5.5 **Completeness of (ℓ_p, d_p)**

The metric space (ℓ_p, d_p) , for $1 \leq p < \infty$, is complete.

PROOF: Recall first that the metric d_p in this metric space is defined as

$$d_p(x_1, x_2) = \left(\sum_{j=1}^{\infty} |\xi_1^j - \xi_2^j|^p \right)^{1/p}.$$

Now, let (x_n) be a Cauchy sequence in ℓ_p , where $x_m = (\xi_m^1, \xi_m^2, \dots)$. Then, for every $\epsilon > 0$ there exists $N_\epsilon > 0$ such that for all $m, n > N_\epsilon$,

$$d_p(x_m, x_n) = \left(\sum_{j=1}^{\infty} |\xi_m^j - \xi_n^j|^p \right)^{1/p} < \epsilon. \quad (3.18)$$

Using exactly the same arguments as in the proof of the completeness of $\mathbb{R}^\ell$, in particular, using (3.15), we have for all $m, n > N_\epsilon$,

$$|\xi_m^j - \xi_n^j| < \epsilon.$$

We choose a fixed j . From the above equation, we see that $(\xi_1^j, \xi_2^j, \dots)$ is a Cauchy sequence of real numbers. It converges by the completeness of $\mathbb{R}$, say $\lim_{n \rightarrow \infty} \xi_m^j = \xi^j$. Using these limits, we define $x = (\xi^1, \xi^2, \dots)$ and show that $x \in \ell_p$ and that $\lim_{n \rightarrow \infty} x_m = x$.

From (3.18), we have that for all $m, n > N_\epsilon$,

$$\sum_{j=1}^k |\xi_m^j - \xi_n^j|^p < \epsilon^p,$$

where $k = 1, 2, \dots$. Letting $n \rightarrow \infty$, we obtain for $m > N_\epsilon$,

$$\sum_{j=1}^k |\xi_m^k - \xi^j|^p \leq \epsilon^p, \quad k = 1, 2, \dots$$

Now let $k \rightarrow \infty$. For $m > N_\epsilon$,

$$\sum_{j=1}^{\infty} |\xi_m^j - \xi^j|^p \leq \epsilon^p. \quad (3.19)$$

This shows that $x_m - x = (\xi_m^j - \xi^j) \in \ell_p$. Since $x_m \in \ell_p$, it follows by the Minkowski inequality that

$$x = x_m + (x - x_m) \in \ell_p.$$

Furthermore, the series in (3.19) represents $[d_p(x_m, x)]^p$, so that (3.19) implies that $\lim_{m \rightarrow \infty} x_m = x$. Since (x_m) was arbitrary, this proves the completeness of (ℓ_p, d_p) , $1 \leq p < \infty$. ■

Theorem 3.5.6

The metric space $(C([a, b]), d_\infty)$ is complete for all closed intervals $[a, b] \subset \mathbb{R}$.

PROOF: Recall that the d_∞ metric on $C([a, b])$ is defined as

$$d_\infty(f, g) = \max_{a \leq x \leq b} |f(x) - g(x)|, \quad f, g \in C([a, b]).$$

Now, suppose that $(f_n) \subset C([a, b])$ is a Cauchy sequence, which means that for all $\epsilon > 0$ there exists $N_\epsilon > 0$ such that $d_\infty(f_n, f_m) < \epsilon$ for all $m, n > N_\epsilon$, i.e.,

$$\max_{a \leq x \leq b} |f_n(x) - f_m(x)| < \epsilon. \quad (3.20)$$

We must prove that the sequence converges to an element, call it f , of $C([a, b])$. Now, (3.20) implies that

$$|f_n(x) - f_m(x)| < \epsilon \quad \text{for all } x \in [a, b].$$

For a fixed $x \in [a, b]$, the above equation implies that the sequence $(f_n(x)) \subset \mathbb{R}$ is a Cauchy sequence. By the completeness of $\mathbb{R}$, it follows that the sequence $(f_n(x))$ converges to a limit, call it $f(x)$, in $\mathbb{R}$. This set of limit points $f(x)$ defines a function $f : [a, b] \rightarrow \mathbb{R}$. The convergence to this function is *pointwise*. We must now show that the convergence is *uniform*, i.e., the convergence is in the d_∞ metric (clarify this point...).

This is easily done by taking the limit $m \rightarrow \infty$ on both sides of (3.20) to obtain

$$|f_n(x) - f(x)| < \epsilon \quad \text{for all } x \in [a, b],$$

which implies that for all $n > N_\epsilon$,

$$\max_{a \leq x \leq b} |f(x) - f_n(x)| < \epsilon \Rightarrow d_\infty(f_n, f) < \epsilon \quad \text{for all } n > N_\epsilon.$$

This proves that the sequence of functions (f_n) converges in the d_∞ metric, i.e., uniformly to f . It remains to show that $f \in C([a, b])$, i.e., that f is a continuous function in the interval $[a, b]$.

Now, for $x, y \in [a, b]$ and $n \leq 1$ (?)

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \quad (3.21)$$

by the triangle inequality. Our goal is to make the entire right-hand side of (3.21) less than ϵ for x and y sufficiently close, thereby proving that f is continuous.

From the (uniform) convergence of the sequence (f_n) to f , it follows that for $\epsilon > 0$ (different from the one used above), there exists $N_1 > 0$ such that for any $n > N_1$,

$$|f(x) - f_n(x)| < \frac{\epsilon}{3} \quad \text{and} \quad |f(y) - f_n(y)| < \frac{\epsilon}{3}.$$

This takes care of the first and third terms on the right-hand side of (3.21). As for the middle term, recall that all functions f_n were assumed to be continuous. Therefore, for a fixed $n > N_1$ and $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f_n(x) - f_n(y)| < \frac{\epsilon}{3} \quad \text{for all } x, y \text{ such that } |x - y| < \delta.$$

(Note that δ will depend on n , but this does not affect the proof.) Putting all of these results together, we have that

$$|f(x) - f(y)| < \epsilon \quad \text{for all } x, y \in [a, b] \text{ such that } |x - y| < \delta.$$

Therefore, $f \in C([a, b])$ and the proof is complete. The proof is almost the same as this if instead we take the space $C([a, b])$ of *complex-valued* functions on $[a, b]$. ■

REMARK: In the proof, we mention that convergence in the d_∞ metric corresponds to uniform convergence. Formally, we have (clarify this)

Theorem 3.5.7 Uniform Convergence

A sequence of functions in the metric space $(C([a, b]), d_\infty)$ is convergent if and only if the sequence converges uniformly.

PROOF: To be completed. ■

Example 3.5.2 Examples of Incomplete Metric Spaces

To gain a good understanding of completeness and related concepts, let us finally look at some examples of incomplete metric spaces.

1. *The rationals* $\mathbb{Q}$: This is the set of all rational numbers with the usual metric given by $d(x, y) = |x - y|$, where $x, y \in \mathbb{Q}$. It is not complete. To see this, consider the Cauchy sequence defined by

$$y_1 = 1, \quad y_{n+1} = \frac{y_n}{2} + \frac{1}{y_n}, \quad n \geq 1,$$

which has the limiting value $\sqrt{2} \notin \mathbb{Q}$.

2. *Polynomials*: Let X be the set of all polynomials considered as functions of t on some finite closed interval $J = [a, b] \subset \mathbb{R}$, and define a metric d on X by

$$d(x, y) = \max_{t \in J} |x(t) - y(t)|.$$

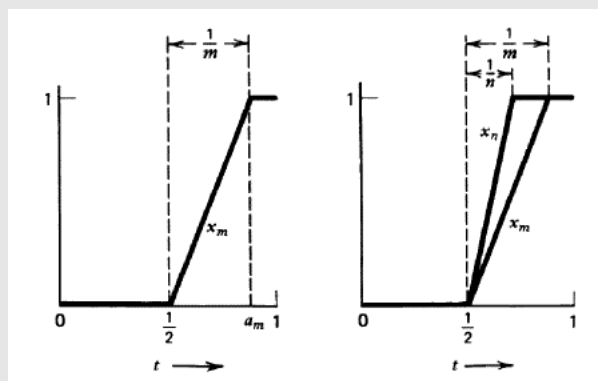
This metric space (X, d) is not complete. In fact, an example of a Cauchy sequence without limit in X is given by any sequence of polynomials that converges uniformly on J to a continuous function that is not a polynomial.

There is one last important example of a metric space that is not complete.

Example 3.5.3 The metric space $(C([a, b]), d_1)$, with the metric d_1 defined as

$$d_1(x, y) = \int_a^b |x(t) - y(t)| dt,$$

is not complete. For a proof, it suffices to give a counterexample. Without loss of generality, let $[a, b] = [0, 1]$. Consider the functions x_m in the figure below.



These functions form a Cauchy sequence because $d(x_m, x_n)$ is the area of the triangle in the right-hand side of the above figure, and for every $\epsilon > 0$,

$$d(x_m, x_n) < \epsilon \quad \text{when } m, n > \frac{1}{\epsilon}.$$

We now show that this Cauchy sequence does not converge (to a continuous function). We have

$$x_m(t) = \begin{cases} 0 & \text{if } t \in [0, \frac{1}{2}] \\ 1 & \text{if } t \in [a_m, 1], \end{cases}$$

where $a_m = \frac{1}{2} + \frac{1}{m}$. Hence, for every $x \in C([0, 1])$,

$$\begin{aligned} d(x_m, x) &= \int_0^1 |x_m(t) - x(t)| \, dt \\ &= \int_0^{\frac{1}{2}} |x(t)| \, dt + \int_{\frac{1}{2}}^{a_m} |x_m(t) - x(t)| \, dt + \int_{a_m}^1 |1 - x(t)| \, dt. \end{aligned}$$

Since the integrands are non-negative, so is each integral on the right-hand side. Hence, $\lim_{n \rightarrow \infty} d(x_n, x) = 0$ would imply that each integral approaches zero and, since x is continuous, we should have

$$x(t) = \begin{cases} 0 & \text{if } t \in [0, \frac{1}{2}) \\ 1 & \text{if } t \in (\frac{1}{2}, 1] \end{cases}.$$

But this is impossible for a continuous function. Hence, (x_m) does not converge. This is enough to prove that $(C([a, b], d_1))$ is not complete.

3.6 Completion of Metric Spaces

Recall the definition of a dense set and a separable metric space.

Definition 3.6.1 Dense Set, Separable Space

Let (X, d) be a metric space and $M \subset X$. M is called **dense in X** if $\overline{M} = X$. X is called **separable** if it has a countable subset that is dense in X .

Example 3.6.1 Here are some examples of separable and non-separable spaces.

1. *The Real Line, $\mathbb{R}$* : The real line $\mathbb{R}$ is separable.

PROOF: The set $\mathbb{Q}$ of all rational numbers is countable and is dense in $\mathbb{R}$. Informally, the latter is expressed as, “every element of $\mathbb{R}$ is a limit of a sequence of rational numbers”. ■

2. *The Complex Plane, $\mathbb{C}$* : The complex plane $\mathbb{C}$ is separable.

PROOF: A countable dense subset of $\mathbb{C}$ is the set of all complex numbers whose real and imaginary parts are both rational. ■

3. *Discrete Metric Spaces*: A discrete metric space is separable if and only if it is countable.

PROOF: Let X be a discrete metric space. Then no proper subset of X can be dense in X (how?). Hence, the only dense set in X is X itself, and the statement follows. ■

4. *The Space ℓ_∞* : The metric space ℓ_∞ is not separable.

PROOF: Let $y = (\eta_1, \eta_2, \dots)$ be a sequence of zeros and ones. Then $y \in \ell_\infty$. To y we associate the real number $\hat{y}$ whose binary representation is

$$\frac{\eta_1}{2^1} + \frac{\eta_2}{2^2} + \dots.$$

We now use the facts that the set of points in the interval $[0, 1]$ is uncountable, each $\hat{y} \in [0, 1]$ has a binary representation, and different $\hat{y}$ s have different binary representations. Hence, there are uncountably many sequences of zeros and ones. The d_∞ metric on ℓ_∞ shows that any two of them that are not equal must be of distance 1 apart. If we let each of these sequences be the center of a small ball, say, of radius $\frac{1}{3}$, these balls do not intersect and we have uncountable many of them. If M is any dense set in ℓ_∞ , each of these non-intersecting balls must contain an element of M . Hence M cannot be countable. Since M was an arbitrary dense set, this shows that ℓ_∞ cannot have dense subsets that are countable. Consequently, ℓ_∞ is not separable. ■

5. *The Space ℓ_p* : The metric space ℓ_p , for $1 \leq p < \infty$, is separable.

PROOF: Let M be the set of all sequences y of the form

$$y = (\eta_1, \eta_2, \dots, \eta_n, 0, 0, \dots),$$

where n is any positive integer and the η_j s are rational numbers. M is countable. We show that M is dense in ℓ_p . Let $x = (\xi_j) \in \ell_p$ be arbitrary. Then, for every $\epsilon > 0$ there is an n_ϵ such that

$$\sum_{j=n_\epsilon+1}^{\infty} |\xi_j|^p < \frac{\epsilon^p}{2}$$

because on the left we have the remainder of a converging series. Since the rationals are dense in $\mathbb{R}$, for each ξ_j there is a rational η_j close to it. Hence, we can find a $y \in M$ satisfying

$$\sum_{j=1}^n |\xi_j - \eta_j|^p < \frac{\epsilon^p}{2}.$$

It follows that

$$[d_p(x, y)]^p = \sum_{j=1}^{n_\epsilon} |\xi_j - \eta_j|^p + \sum_{j=n_\epsilon+1}^{\infty} |\xi_j|^p < \epsilon^p.$$

We thus have $d_p(x, y) < \epsilon$ and see that M is dense in ℓ_p . ■

Theorem 3.6.1 Weierstrass Approximation Theorem

Let P be the set of polynomials with real coefficients. Given $f \in C([a, b])$, for any $\epsilon > 0$, there exists $p \in P$ such that $|f(t) - p(t)| < \epsilon$ for all $t \in [a, b]$.

Example 3.6.2 Letting P be the set of polynomials with real coefficients, as a subset of $C([a, b])$ like in the previous theorem, the condition $|f(t) - p(t)| < \epsilon$ is equivalent to $d_\infty(f, p) < \epsilon$, where d_∞ is the metric on $C([a, b])$. So P is dense in $(C([a, b]), d_\infty)$. Now, letting P' be the set of all polynomials with rational coefficients, we have that P' is countable and P' is also dense in $(C([a, b]), d_\infty)$. Therefore, $(C([a, b]), d_\infty)$ is separable.

Now, we know that the rational line $\mathbb{Q}$ is not complete, but it can be “enlarged” to the real line $\mathbb{R}$, which is complete. And this “completion” $\mathbb{R}$ of $\mathbb{Q}$ is such that $\mathbb{Q}$ is dense in $\mathbb{R}$. It is quite important that an arbitrary incomplete metric space can be “completed” in a similar fashion.

Definition 3.6.2 Isometric Mapping, Isometric Spaces

Let (X, d) and $(\tilde{X}, \tilde{d})$ be metric spaces.

1. A mapping $T : X \rightarrow \tilde{X}$ is called an **isometry** if T preserves distance, that is, if for all $x, y \in X$ $\tilde{d}(T(x), T(y)) = d(x, y)$, where $T(x)$ and $T(y)$ are the images of x and y , respectively.
2. (X, d) is said to be **isometric** to the space $(\tilde{X}, \tilde{d})$ if there exists a bijective isometry $X \rightarrow \tilde{X}$. The spaces (X, d) and $(\tilde{X}, \tilde{d})$ are then called **isometric spaces**.

Hence, isometric spaces may differ at most by the nature of their points but are indistinguishable from the viewpoint of the metric. And in any study in which the nature of the points does not matter, we may regard the two spaces as identical—as two copies of the same “abstract” space.

Theorem 3.6.2 Completion

For metric space (X, d) there exists a complete metric space $(\hat{X}, \hat{d})$ that has a subspace W that is isometric to X and is dense in $\hat{X}$. This space $\hat{X}$ is unique up to isometry, that is, if $\tilde{X}$ is any complete metric space having a dense subspace $\tilde{W}$ isometric to X , then $\tilde{X}$ and $\hat{X}$ are isometric.

PROOF: The proof is lengthy but straightforward. We sub-divide it into four steps. They are:

1. Construct $(\hat{X}, \hat{d})$;
2. Construct an isometry $T : (X, d) \rightarrow (W, \hat{d})$ with W dense in $\hat{X}$;
3. Prove the completeness of $(\hat{X}, \hat{d})$; and
4. Prove the uniqueness of $\hat{X}$ up to isometry.

Roughly speaking, the task is to assign suitable limits to Cauchy sequences in X that do not converge. However, we should not introduce “too many” limits, but take into account that certain sequences “may want to converge with the same limit” since the terms of those sequences “ultimately come arbitrarily close to each other”. This intuitive idea can be expressed mathematically in terms of a

suitable equivalence relation. This is not artificial but is suggested by the process of completion of the rational line mentioned above.

1. *Construction of $(\hat{X}, \hat{d})$:* Let $\hat{X}$ be the set of all equivalence classes $\hat{x}, \hat{y}, \dots$ of Cauchy sequences, where the equivalent relation is $\sim$ denoting equivalent Cauchy sequences, as defined in 3.2.5. We write $(x_n) \in \hat{x}$ to mean that (x_n) is a member, i.e., a *representative*, of the equivalent class $\hat{x}$. Now, let

$$\hat{d}(\hat{x}, \hat{y}) := \lim_{n \rightarrow \infty} d(x_n, y_n), \quad (3.22)$$

where $(x_n) \in \hat{x}$ and $(y_n) \in \hat{y}$. Let us show that this limit exists. We have

$$d(x_n, y_n) \leq d(x_n, x_m) + d(x_m, y_m) + d(y_m, y_n);$$

hence, we obtain

$$d(x_n, y_n) - d(x_m, y_m) \leq d(x_n, x_m) + d(y_m, y_n),$$

and a similar inequality with m and n interchanged. Together,

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_m, y_n).$$

Since (x_n) and (y_n) are Cauchy (by construction), we can make the right-hand side as small as we please. This means that $(d(x_n, y_n))$ is a Cauchy sequence of real numbers, which converges, implying that the limit in (3.22) exists.

We must also show that the limit in (3.22) is independent of the particular choice of representatives. In fact, if $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, then by definition of equivalent Cauchy sequences,

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

since by definition $d(x_n, x'_n) \rightarrow 0$ and $d(y_n, y'_n) \rightarrow 0$ as $n \rightarrow \infty$. This implies that

$$\lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(x'_n, y'_n),$$

showing that $\hat{d}(\hat{x}, \hat{y})$ can be calculated using any member of the equivalence classes $\hat{x}$ and $\hat{y}$, as we sought to do.

We now prove that the metric $\hat{d}$ is a metric on $\hat{X}$. Positivity is clear. For strict positivity:

$$\hat{d}(\hat{x}, \hat{y}) = 0 \Rightarrow \lim_{n \rightarrow \infty} d(x_n, y_n) = 0 \Rightarrow (x_n) \sim (y_n) \Rightarrow \hat{x} = \hat{y}.$$

For symmetry:

$$\hat{d}(\hat{x}, \hat{y}) = \lim_{n \rightarrow \infty} d(x_n, y_n) = \lim_{n \rightarrow \infty} d(y_n, x_n) = \hat{d}(\hat{y}, \hat{x}),$$

and for the triangle inequality, let $\hat{x}, \hat{y}, \hat{z} \in \hat{X}$, $(x_n) \in \hat{x}$, $(y_n) \in \hat{y}$, and $(z_n) \in \hat{z}$. Then, since d is a metric, $d(x_n, y_n) \leq d(x_n, z_n) + d(z_n, y_n)$, and letting $n \rightarrow \infty$ on both sides gives

$$\hat{d}(\hat{x}, \hat{y}) \leq \hat{d}(\hat{x}, \hat{z}) + \hat{d}(\hat{z}, \hat{y}).$$

This proves that $\hat{d}$ is a metric.

2. *Construction of an isometry $T : (X, d) \rightarrow (W, \hat{d})$:* To each $b \in X$ we associate the class $\hat{b} \in \hat{X}$ that contains the constant Cauchy sequence $(b, b, \dots)$. This defines a mapping $T : (X, d) \rightarrow (W, \hat{d})$ onto the subspace $W = T(X) \subset \hat{X}$. The mapping T is given by $b \mapsto \hat{b} = T(b)$, where $(b, b, \dots) \in \hat{b}$. We see that T is an isometry since (3.22) becomes simply

$$\hat{d}(\hat{b}, \hat{c}) = d(b, c),$$

where $\hat{c}$ is the class of (y_n) , where $y_n = c$ for all n . Any isometry is injective, and $T : X \rightarrow W$ is surjective since $T(X) = W$. Therefore, T is a bijection, and $(W, \hat{d})$ and (X, d) are isometric.

We now show that W is dense in $\hat{X}$. Consider any $\hat{x} \in \hat{X}$. Let $(x_n) \in \hat{x}$. For every $\epsilon > 0$ there exists N such that

$$d(x_n, x_N) < \frac{\epsilon}{2} \quad \text{for all } n > N.$$

Let $(x_N, x_N, \dots) \in \hat{x}_N$. Then $\hat{x}_N \in W$. By (3.22),

$$\hat{d}(\hat{x}, \hat{x}_N) = \lim_{n \rightarrow \infty} d(x_n, x_N) \leq \frac{\epsilon}{2} < \epsilon.$$

This shows that every ϵ -neighbourhood of the arbitrary $\hat{x} \in \hat{X}$ contains an element of W . Hence, W is dense in $\hat{X}$.

3. *Completeness of $\hat{X}$:* Let $(\hat{x}_n)$ be any Cauchy sequence in $\hat{X}$. Since W is dense in $\hat{X}$, for every $\hat{x}_n$ there is a $\hat{z}_n \in W$ such that

$$\hat{d}(\hat{x}_n, \hat{z}_n) < \frac{1}{n}. \quad (3.23)$$

Hence, by the triangle inequality,

$$\hat{d}(\hat{z}_m, \hat{z}_n) \leq \hat{d}(\hat{z}_m, \hat{x}_m) + \hat{d}(\hat{x}_m, \hat{x}_n) + \hat{d}(\hat{x}_n, \hat{z}_n) < \frac{1}{m} + \hat{d}(\hat{x}_m, \hat{x}_n) + \frac{1}{n},$$

and this is less than any given $\epsilon > 0$ for sufficiently large m and n because $(\hat{x}_m)$ is Cauchy. Hence, $(\hat{z}_m)$ is Cauchy. Since $T : (X, d) \rightarrow (W, \hat{d})$ is an isometry and $\hat{z}_m \in W$, the sequence (z_m) , where $z_m = T^{-1}(\hat{z}_m)$, is Cauchy in X . Let $\hat{x} \in \hat{X}$ be the class to which (z_m) belongs. We show that $\hat{x}$ is the limit of $(\hat{x}_n)$. By (3.23),

$$\hat{d}(\hat{x}_n, \hat{x}) \leq \hat{d}(\hat{x}_n, \hat{z}_n) + \hat{d}(\hat{z}_n, \hat{x}) < \frac{1}{n} + \hat{d}(\hat{z}_n, \hat{x}). \quad (3.24)$$

Since $(z_m) \in \hat{x}$ and $\hat{z}_n \in W$, so that $(z_n, z_n, z_n, \dots) \in \hat{z}_n$, in inequality (3.24) becomes

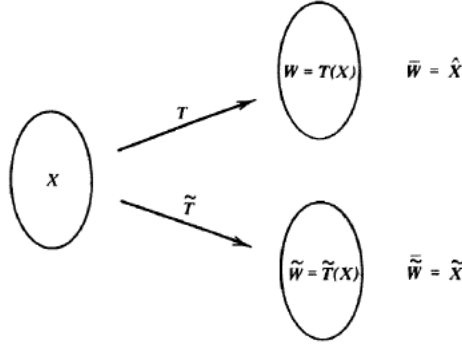
$$\hat{d}(\hat{x}_n, \hat{x}) < \frac{1}{n} + \lim_{m \rightarrow \infty} d(z_n, z_m),$$

and the right-hand side is smaller than any given $\epsilon > 0$ for sufficiently large n . Hence, the arbitrary Cauchy sequence $(\hat{x}_n)$ in $\hat{X}$ has the limit $\hat{x} \in \hat{X}$, and so $\hat{X}$ is complete.

4. *Uniqueness of $\hat{X}$ up to isometry:* Suppose there are two completions:

$$\begin{aligned} (\hat{X}, \hat{d}), \quad W \subset \hat{X}, \quad W \text{ isometric to } X, \quad W \text{ dense in } \hat{X}, \\ (\tilde{X}, \tilde{d}), \quad \tilde{W} \subset \tilde{X}, \quad \tilde{W} \text{ isometric to } X, \quad \tilde{W} \text{ dense in } \tilde{X}. \end{aligned}$$

The isometries are T and $\tilde{T}$, respectively, as shown in the figure below.



This shows that $T \circ \tilde{T}^{-1} : \tilde{W} \rightarrow W$ is an isometry. Extend this to $S : \tilde{X} \rightarrow \hat{X}$. Then, for any $\tilde{x}, \tilde{y} \in \tilde{X}$, we have sequences $(\tilde{x}_n), (\tilde{y}_n)$ in $\tilde{W}$ such that $\lim_{n \rightarrow \infty} \tilde{x}_n = \tilde{x}$ and $\lim_{n \rightarrow \infty} \tilde{y}_n = \tilde{y}$; hence,

$$\tilde{d}(\tilde{x}, \tilde{y}) = \lim_{n \rightarrow \infty} \tilde{d}(\tilde{x}_n, \tilde{y}_n)$$

follows from

$$|\tilde{d}(\tilde{x}, \tilde{y}) - \tilde{d}(\tilde{x}_n, \tilde{y}_n)| \leq \tilde{d}(\tilde{x}, \tilde{x}_n) + \tilde{d}(\tilde{y}, \tilde{y}_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

(the inequality being similar to the one used in Part 1). Since $\tilde{W}$ is isometric to $W \subset \hat{X}$ and $\tilde{W} = \hat{X}$, the distances on $\tilde{X}$ and $\hat{X}$ must be the same. Hence, $\tilde{X}$ and $\hat{X}$ are isometric. ■

3.7 L_p Spaces

Definition 3.7.1 L_p Space

The L_p space on the closed interval $[a, b] \subset \mathbb{R}$, denoted $L_p[a, b]$ for $1 \leq p < \infty$ is the set of equivalence classes of measurable functions $f : [a, b] \rightarrow \mathbb{R}$ such that

$$\int_a^b |f(t)|^p dt < \infty.$$

Theorem 3.7.1 Riesz-Fischer

For any $[a, b] \subset \mathbb{R}$ and $1 \leq p < \infty$, define a function $d_p : L_p[a, b] \times L_p[a, b] \rightarrow \mathbb{R}$ by

$$d_p(f, g) = \left(\int_a^b |f(t) - g(t)|^p dt \right)^{1/p} \quad \text{for all } f, g \in L_p[a, b].$$

Then $(L_p[a, b], d_p)$ is a complete metric space.

Our reason for looking at L_p spaces is the following fact:

$$C[a, b] \text{ is a dense subspace of } (L_p[a, b], d_p).$$

By the completion theorem, therefore,

$$(L_p[a, b], d_p) \text{ is the completion of } (C[a, b], d_p).$$

The reason for defining $L_p[a, b]$ as the *equivalence class* of measurable functions is that for functions in L_p spaces the notion of equivalence must be defined as follows: two functions f and g in $L_p[a, b]$ are considered the same if $f(t) = g(t)$ “almost everywhere”, i.e., f and g are equal for all $t \in [a, b]/S$, where S is a set of measure zero (“for all t except possibly on a set of measure zero”). We need such a definition because it is possible for $d_p(f, g) = 0$ but $f(t) \neq g(t)$ for some $t \in [a, b]$, contradicting the usual notion of equivalent functions and hence the definition of the metric.

3.8 Appendix: Additional Topics

3.8.1 Pseudometrics

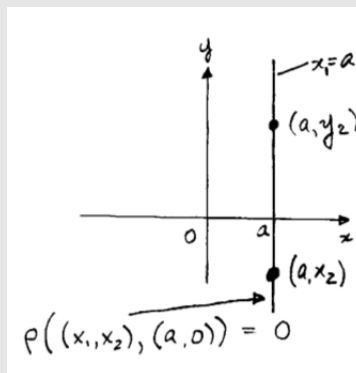
Definition 3.8.1 Pseudometric

A real-valued function on a set X $\rho : X \times X \rightarrow \mathbb{R}$ is called a **pseudometric** if it satisfies conditions 1, 3, and 4 for a metric but not necessarily condition 2, i.e.,

1. (Positivity) $\rho(x, y) \geq 0$ and $\rho(x, x) = 0$ for all $x, y \in X$.
2. (Strict Positivity) $\rho(x, y) = 0 \Rightarrow x = y$.
3. (Symmetric) $\rho(x, y) = \rho(y, x)$ for all $x, y \in X$.
4. (Triangle Inequality) For all $x, y, z \in X$, $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$.

Example 3.8.1 Here are a couple of examples of pseudometrics.

1. $X = \mathbb{R}^2$, and for $x = (x_1, x_2)$ and $y = (y_1, y_2)$, define $\rho(x, y) = |x_1 - y_1|$. This is a pseudometric on the plane $\mathbb{R}^2$.



2. The d_p metric on the space $L_p[a, b]$ is a pseudometric, as was alluded to earlier. This is due to the fact that $d_p(f, g) = 0$ does not necessarily imply that $f(x) = g(x)$ for all $x \in [a, b]$.

Recall that the equivalence of functions for this space was modified so that f and g were equivalent as long as they didn't differ on a set of (Lebesgue) measure zero. But in terms of the usual pointwise equivalence, f and g can differ on a set of measure zero.

Example 3.8.2 Does

$$d(x, y) = \int_a^b |x(t) - y(t)| \, dt$$

define a metric or a pseudometric on X if X is

1. The set of all real-valued continuous functions on $[a, b]$;
2. The set of all real-valued Riemann integrable functions on $[a, b]$?

SOLUTION:

3.8.2 A Metric Space for Sets

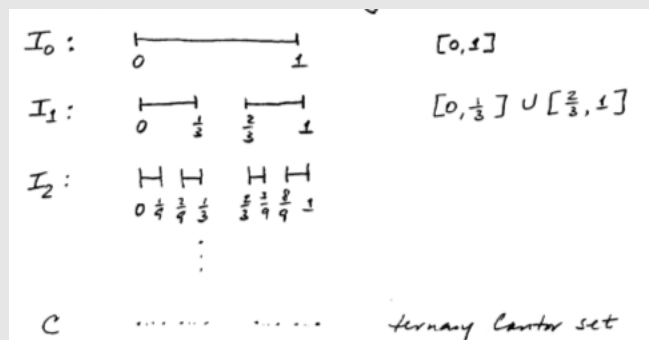
Example 3.8.3 Consider the closed sets I_n on $\mathbb{R}$ defined by

$$I_n = \left[-\frac{1}{2^n}, \frac{1}{2^n} \right], \quad n = 0, 1, 2, \dots$$

As $n \rightarrow \infty$, the intervals I_n are shrinking in size and approaching the limit set $I := \{0\}$. But how do we make sense of the statement $\lim_{n \rightarrow \infty} I_n = I$?

The question at the end of the above example can be restated as: can we define a metric d between sets so that $\lim_{n \rightarrow \infty} d(I_n, I) = 0$?

Example 3.8.4 Consider the classical “middle-thirds” bisection procedure that produces the ternary Cantor set $\mathcal{C}$ on $[0, 1]$:

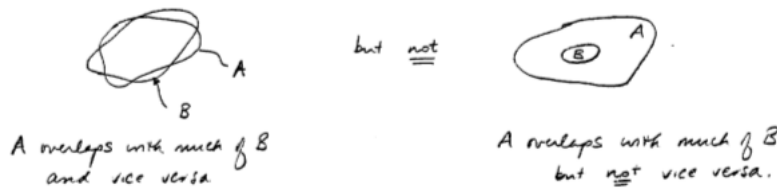


Again, how do we make sense of the statement $\lim_{n \rightarrow \infty} I_n = \mathcal{C}$? Once again, can we define a metric d between sets so that $\lim_{n \rightarrow \infty} d(I_n, \mathcal{C}) = 0$?

Let (X, d) be a complete metric space, for example $\mathbb{R}^n$. For two sets $A, B \subset X$ to be “close” to each other, it is obviously not sufficient that they be close to each other in the sense of the figure below.



Rather, they must “overlap” each other well, for example,



Let the distance from a point $x \in X$ to a set $A \subseteq X$ be written as

$$d(x, A) := \inf_{y \in A} d(x, y),$$

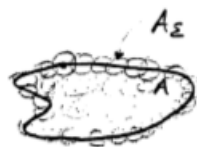
where remember d is the metric on X .



Now, define the ϵ -neighbourhood of a set $A \subset X$ as

$$A_\epsilon := \{x \in X \mid d(x, A) < \epsilon\}.$$

A_ϵ is obtained from A by constructing an (open) ϵ -ball around each point $x \in A$, as shown in the figure below.

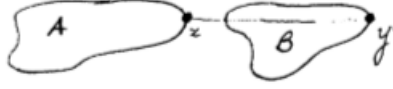


For A and B (both subsets of X) to be “ ϵ -close”, let us demand that

$$B \subset A_\epsilon \quad \text{and} \quad A \subset B_\epsilon$$

This is the starting point for developing the *Hausdorff metric* between sets.

Now, what does $B \subset A_\epsilon$ mean? It means that all points $y \in B$ lie within ϵ of some point in A . To express this mathematically, find the point $y' \in B$ that lies *farthest* away from A , as shown in the figure below.



The distance from this point y' to A is

$$d(y', A) = \sup_{y \in B} d(y, A) = \sup_{y \in B} \inf_{x \in A} d(y, x).$$

In other words, for B to be contained in a δ -neighbourhood A_δ of A , we would require $\delta > d(y', A)$. We shall refer to this quantity as the distance from the B to the set A :

$$d(B, A) := \sup_{y \in B} d(y, A).$$

For $B \subset A_\epsilon$, we shall demand that $d(B, A) < \epsilon$.

However, we also demand that $A \subset B_\epsilon$, which means that any point $x \in A$ lies within ϵ of some point y of B . Again, to express this mathematically, find the point x' that lies farthest away from B . Then define

$$d(A, B) := d(x', B) = \sup_{x \in A} d(x, B) = \sup_{x \in A} \inf_{y \in B} d(x, y),$$

which is the distance from the set A to the set B .

Definition 3.8.2 Hausdorff Distance

The **Hausdorff distance** between two subsets A and B of a complete metric space (X, d) is defined as

$$h(A, B) := \max\{d(A, B), d(B, A)\} = \max\{\sup_{x \in A} d(x, B), \sup_{y \in B} d(y, A)\}.$$

Thus, $h(A, B) < \epsilon$ implies that $d(A, B) < \epsilon$ and $d(B, A) < \epsilon$, or equivalently, that $A \subset B_\epsilon$ and $B \subset A_\epsilon$.

Note that $d(A, B)$ is *not* necessarily equal to $d(B, A)$.

Example 3.8.5 Let $X = [0, 1]$ under d , the Euclidean metric. Let

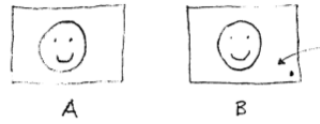
$$A = \left[0, \frac{1}{3}\right], \quad \text{and} \quad B = [0, 1].$$

Then $d(A, B) = 0$ because we have to draw ϵ -balls of radius more than $\frac{2}{3}$ around points in A in order to cover B . On the other hand, $d(B, A) = \frac{2}{3} \neq d(A, B)$.

Therefore, $h(A, B) = \max\{0, \frac{2}{3}\} = \frac{2}{3}$.

REMARK: The Hausdorff distance looks like an excellent distance function for sets. However, it can be “too excellent” from practical perspectives, for example, from a visual perspective.

For example, take two photographs that are almost identical, except that photo B has an extra small dot.



Even though the photos look almost identical, the Hausdorff distance $h(A, B)$ between them could be large. This could plague practical calculations that use h to approximate target images.

Now, we are going to want the Hausdorff distance h to be a metric over an appropriate space. From the previous discussion on, for example, the cantor set in $[0, 1]$, it would appear that this space would consist of all non-empty subsets of X . However, it is desirable that h be a metric and *not* a pseudo-metric. For example, $h([0, 1], [0, 1]) = h([0, 1], (0, 1)) = h([0, 1], [0, 1))$, etc. For this reason, as well as the fact that the usual “fractal” sets are closed, it would seem desirable to consider only closed subsets. However, questions of convergence of sets are also involved. For this reason additionally, the sets should be *compact*.

Let (X, d) be a compact metric space. Let $\mathcal{H}(X)$ denote the set of all non-empty compact subsets of X . Then $(\mathcal{H}(X), h)$ is a complete metric space. Note that the “points” in $\mathcal{H}(X)$ are non-empty compact subsets of X .

4 The Contraction Mapping Theorem

The *contraction mapping theorem*, also called the *Banach fixed point theorem*, concerns *contraction mappings* of a complete metric space onto itself. It states conditions sufficient for the existence and uniqueness of a *fixed point* (a point that is mapped to itself). The theorem also gives an iterative process by which we can obtain approximations to the fixed point and error bounds. We consider three important fields of application of the theorem, namely, linear algebraic equations, ordinary differential equations, and integral equations. Other applications, like to partial differential equations, also exist, and will be discussed in later chapters.

4.1 The Theorem

Definition 4.1.1 Fixed Point

A **fixed point** of a mapping $T : X \rightarrow X$ of a set X into itself is a point $x \in X$ that is mapped onto itself, that is, a point x such that

$$T(x) = x.$$

As a couple of quick examples, a translation has no fixed points, a rotation of the plane has a single fixed point (the centre of rotation), the mapping $x \mapsto x^2$ of $\mathbb{R}$ into itself has two fixed points (0 and 1), and the projection $(\xi_1, \xi_2) \mapsto \xi_1$ of $\mathbb{R}^2$ onto the ξ_1 -axis has infinitely many fixed points (all points of the ξ_1 -axis).

The contraction mapping theorem to be stated below is an existence and uniqueness theorem for fixed points of certain mappings, and it also gives a constructive procedure for obtaining better and better approximations to the fixed point (the solution of the practical problem). This procedure is called an **iteration**. By definition, this is a method such that we choose an arbitrary starting point x_0 in a given set and calculate recursively a sequence $x_0, x_1, x_2, \dots$ from a relation of the form

$$x_{n+1} = T(x_n), \quad n = 0, 1, 2, \dots;$$

that is, we choose an arbitrary x_0 and determine successively $x_1 = T(x_0)$, $x_2 = T(x_1)$, $\dots$.

Iteration procedures are used in nearly every branch of applied mathematics, and convergence proofs and error estimates are very often obtained by an application of Banach's fixed point theorem (or more difficult fixed point theorems). Banach's theorem gives sufficient conditions for the existence (and uniqueness) of a fixed point for a class of mappings, called contractions.

Definition 4.1.2 Contraction Mapping

Let (X, d) be a metric space. A mapping $T : X \rightarrow X$ is called a **contraction** on X if there is a real number α satisfying $0 \leq \alpha < 1$ such that for all $x, y \in X$,

$$d(T(x), T(y)) \leq \alpha d(x, y). \quad (4.1)$$

REMARK: Note that a contraction mapping is a special case of a Lipschitz continuous map (definition 3.3.9) in which: (1) the codomain metric space Y is the same as the domain; and (2) the Lipschitz constant K is restricted to the interval $[0, 1)$.

REMARK: Note that a contraction mapping is continuous at any point in $x_0 \in X$ since given any $\epsilon > 0$ we may let $\delta = \frac{\epsilon}{1+\alpha}$. Then $d(T(x), T(x_0)) < Kd(x, x_0) \leq K \frac{\epsilon}{1+\alpha} < \epsilon$ for all x satisfying $d(x, x_0) < \delta$.

Geometrically, this means that any points x and y have images that are closer together than those points x and y ; more precisely, the ratio $\frac{d(T(x), T(y))}{d(x, y)}$ does not exceed a constant α that is strictly less than one.

Theorem 4.1.1 Contraction Mapping/Banach Fixed Point Theorem

Let (X, d) be a complete metric space ($X \neq \emptyset$) with $T : X \rightarrow X$ a contraction on X . Then T has precisely one fixed point.

PROOF: The idea is to construct a sequence (x_n) and show that it is Cauchy, so that by the completeness of X it will converge to a point (the fixed point of T) in X . We then show that this fixed point is unique.

Let $x_0 \in X$ and define the iterative sequence (x_n) by

$$x_0, \quad x_1 = T(x_0), \quad x_2 = T(x_1) = T^2(x_0), \quad \dots, \quad x_n = T^n(x_0). \quad (4.2)$$

This is the sequence of the images of x_0 under repeated application of T . Now we show that (x_n) is Cauchy. By (4.1) and (4.2),

$$\begin{aligned} d(x_{n+1}, x_n) &= d(T(x_n), T(x_{n-1})) \\ &\leq \alpha d(x_n, x_{n-1}) = \alpha d(T(x_{n-1}), T(x_{n-2})) \\ &\leq \alpha^2 d(x_{n-1}, x_{n-2}) \\ &\dots \leq \alpha^n d(x_1, x_0). \end{aligned} \quad (4.3)$$

Hence, by the triangle inequality and the formula for the sum of a finite geometric series, we obtain for $m > n$:

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + \dots + d(x_{n+1}, x_n) \\ &\leq (\alpha^{m-1} + \dots + \alpha^n) d(x_1, x_0) \\ &= \alpha^n \frac{1 - \alpha^{m-n}}{1 - \alpha} d(x_0, x_1). \end{aligned}$$

Since $0 \leq \alpha < 1$, in the numerator we have $1 - \alpha^{m-n} < 1$. Consequently,

$$d(x_m, x_n) \leq \frac{\alpha^n}{1 - \alpha} d(x_0, x_1). \quad (4.4)$$

Now, on the right-hand side, $0 \leq \alpha < 1$, and $d(x_0, x_1)$ is fixed, so that we can make the right-hand side as small as we want by taking n sufficiently large. Specifically, we have for $m, n > N$,

$$d(x_m, x_n) \leq \frac{\alpha^{N+1}}{1 - \alpha} d(x_1, x_0).$$

Then, given $\epsilon > 0$, there exists N such that

$$\frac{\alpha^{N+1}}{1 - \alpha} d(x_1, x_0) < \epsilon.$$

(because $0 \leq \alpha < 1$) Thus, $d(x_m, x_n) < \epsilon$ for $n, m > N$, so that (x_n) is a Cauchy sequence. Since X is complete (x_n) converges to a point $x \in X$. We now show that this limit x is a fixed point of the mapping T .

From the triangle inequality and (4.1) we have

$$d(x, T(x)) \leq d(x, x_m) + d(x_m, T(x)) \leq d(x, x_m) + \alpha d(x_{m-1}, x),$$

and we can make the sum at the end smaller than any pre-assigned $\epsilon > 0$ because of the convergence of (x_n) . So in the limit $m \rightarrow \infty$, we have that $d(x, T(x)) = 0$, so that $x = T(x)$, showing that x is a fixed point of T .

Finally, x is the only fixed point of T because from $T(x) = x$ and $T(\tilde{x}) = \tilde{x}$ we obtain by (4.1)

$$d(x, \tilde{x}) = d(T(x), T(\tilde{x})) \leq \alpha d(x, \tilde{x}),$$

which implies that $d(x, \tilde{x}) = 0$ since $\alpha < 1$. Hence $x = \tilde{x}$. ■

Corollary 4.1.1

Under the conditions of the contraction mapping theorem, the iterative sequence (4.2) with arbitrary $x_0 \in X$ converges to the unique fixed point x of T , i.e.,

$$\lim_{n \rightarrow \infty} T^{on}(x_0) = x \text{ for all } x_0 \in X.$$

Corollary 4.1.2 Error Bounds

Under the conditions of the contraction mapping theorem, we have the following error estimates: the **prior estimate**

$$d(x_m, x) \leq \frac{\alpha^m}{1 - \alpha} d(x_0, x_1) \quad (4.5)$$

and the **posterior estimate**

$$d(x_m, x) \leq \frac{\alpha}{1 - \alpha} d(x_{m-1}, x_m). \quad (4.6)$$

PROOF: The first statement of the theorem is clear from the proof of the theorem. Inequality (4.5) follows from (4.4) by letting $n \rightarrow \infty$. We now derive (4.6). Taking $m = 1$ and writing y_0 for x_0 and y_1 for x_1 , we have from (4.5),

$$d(y_1, x) \leq \frac{\alpha}{1-\alpha} d(y_0, y_1).$$

Setting $y_0 = x_{m-1}$, we have $y_1 = T(y_0) = x_m$, from which one obtains (4.6). ■

The prior error bound (4.5) can be used at the beginning of a calculation for estimating the number of steps necessary to obtain a given accuracy. (4.6) can be used at intermediate stages or at the end of a calculation. It is at least as accurate as (4.5) and may be better.

Theorem 4.1.2 Contraction on a Ball

Let $T : X \rightarrow X$ be a mapping from a complete metric space (X, d) to itself that is a contraction on a closed ball $Y = \{x \mid d(x, x_0) \leq r\}$, that is, T satisfies (4.1) for all $x, y \in Y$. Moreover, assume that

$$d(x_0, T(x_0)) < (1 - \alpha)r. \quad (4.7)$$

Then the iterative sequence $x_{n+1} = T(x_n)$ converges to an $x \in Y$. This x is a fixed point of T and is the only fixed point of T in Y .

PROOF: We merely have to show that all the x_m s as well as x lie in Y . We put $m = 0$ in (4.4), change n to m and use (4.7) to get

$$d(x_0, x_m) \leq \frac{1}{1-\alpha} d(x_0, x_1) < r.$$

Hence, all x_m s are in Y . Also, $x \in Y$ since (x_m) converges to x and Y is closed (so that by Theorem 3.5.1 the subspace Y is complete). The result then follows from the contraction mapping theorem. ■

Definition 4.1.3 Eventually Contractive Mapping

Let (X, d) be a metric space and $T : X \rightarrow X$. T is called **eventually contractive** if for some integer $p \geq 1$ the function f^{op} is a contraction map.

Proposition 4.1.1 Eventually Contractive Mapping

Let $T : X \rightarrow X$ be a mapping on a complete metric space (X, d) , and suppose that T^{om} is a contraction on X for some positive integer m . Then T has a unique fixed point.

PROOF: By assumption, $B := T^{om}$ is a contraction on X . By the contraction mapping theorem, therefore, B has a unique fixed point, call it $\hat{x}$, so that $B(\hat{x}) = \hat{x}$. Hence $B^n(\hat{x}) = \hat{x}$. We also know from the contraction mapping theorem that

$$\lim_{n \rightarrow \infty} B^n(x) = \hat{x} \text{ for all } x \in X.$$

For the particular $x = T(\hat{x})$, since $B^{\circ n} = T^{\circ nm}$, we thus obtain

$$\begin{aligned}\hat{x} &= \lim_{n \rightarrow \infty} B^n(x) = \lim_{n \rightarrow \infty} B^n(T(\hat{x})) = \lim_{n \rightarrow \infty} T(B^n(\hat{x})) \\ &= \lim_{n \rightarrow \infty} T(\hat{x}) \\ &= T(\hat{x}).\end{aligned}$$

This shows that $\hat{x}$ is a fixed point of T . Since every fixed point of T is also a fixed point of B , we see that T cannot have more than one fixed point (since B doesn't have more than one fixed point). This completes the proof. ■

4.2 Application to Linear Equations

4.3 Application to Ordinary Differential Equations

The most interesting applications of the contraction mapping theorem arise in connection with function spaces. The theorem then yields existence and uniqueness theorems for differential equations.

Here we deal with the following *initial value problem (IVP)* for the *first-order ordinary differential equation (ODE)*

$$x' = f(t, x), \quad x(t_0) = x_0, \quad (4.8)$$

where t_0 and x_0 are real numbers.

We shall use the Banach fixed point theorem to prove the famous Picard's theorem that, while not the strongest of its type that is known, plays a vital role in the theory of ODEs. The idea is simple: (4.8) will be converted to an integral equation, which will define a mapping T , and the conditions of the theorem will imply that T is a contraction such that its fixed point becomes the (unique) solution to the problem.

Theorem 4.3.1 Picard's Existence and Uniqueness for ODEs

Let f be a continuous function on a rectangle

$$R = \{(t, x) \mid |t - t_0| \leq a, |x - x_0| \leq b\}$$

and thus bounded on R , say

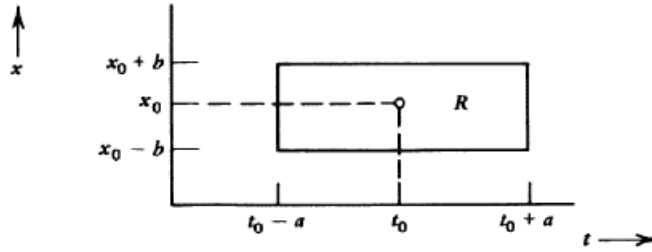
$$|f(t, x)| \leq c \quad \text{for all } (t, x) \in R. \quad (4.9)$$

Suppose also that f satisfies a *Lipschitz condition* on R with respect with its second argument, i.e., there is a constant k (the Lipschitz constant) such that for $(t, x), (t, v) \in R$,

$$|f(t, x) - f(t, v)| \leq k|x - v|. \quad (4.10)$$

Then the IVP (4.8) has a unique solution. This solution exists on an interval $[t_0 - \beta, t_0 + \beta]$, where

$$\beta < \min \left\{ a, \frac{b}{c}, \frac{1}{k} \right\}. \quad (4.11)$$

Figure 4.1: The Rectangle R .Figure 4.2: Geomeric illustration of inequality (4.10) for: (A) relatively small c ; (B) relatively large c . The solution curve must remain in the shaded region bounded by straight lines with slopes $\pm c$.

PROOF: Let $C(J)$ be the metric space of all real-valued continuous functions defined on the interval $J = [t_0 - \beta, t_0 + \beta]$ with metric d defined by

$$d(x, y) = \max_{t \in J} |x(t) - y(t)|.$$

We have seen that $C(J)$ is complete. Let $\tilde{C}$ be the subspace of $C(J)$ consisting of all those functions $x \in C(J)$ that satisfy

$$|x(t) - x_0| \leq c\beta. \quad (4.12)$$

It can be shown that $\tilde{C}$ is closed in $C(J)$ (show it!), so that $\tilde{C}$ is complete by Theorem 3.5.1.

By integration, we see that (4.8) can be written as $x = T(x)$, where $T : \tilde{C} \rightarrow \tilde{C}$, sometimes called the *Picard operator*, is defined by

$$T(x(t)) = x_0 + \int_{t_0}^t f(\tau, x(\tau)) d\tau. \quad (4.13)$$

Indeed, T is defined for all $x \in \tilde{C}$ because $c\beta < b$ by (4.11), so that if $x \in \tilde{C}$, then $\tau \in J$ and $(\tau, x(\tau)) \in R$, and the integral in (4.13) exists since f is continuous on R . To see that T maps $\tilde{C}$ into itself (something that is required if we want T to be a contraction), we can use (4.13) and (4.9), obtaining

$$|T(x(t)) - x_0| = \left| \int_{t_0}^t f(\tau, x(\tau)) d\tau \right| \leq c|t - t_0| \leq c\beta.$$

We now show that T is a contraction on $\tilde{C}$. By the Lipschitz condition (4.10),

$$\begin{aligned} |T(x(t)) - T(v(t))| &= \left| \int_{t_0}^t f(\tau, x(\tau)) - f(\tau, v(\tau)) \, d\tau \right| \\ &\leq |t - t_0| \max_{\tau \in J} k |x(\tau) - v(\tau)| \\ &\leq k\beta d(x, v). \end{aligned}$$

Since the last expression does not depend on t , we can take the maximum on the left and have

$$d(T(x), T(v)) \leq \alpha d(x, v) \quad \text{where} \quad \alpha = k\beta.$$

From (4.11), we see that $\alpha = k\beta < 1$, so that T is indeed a contraction on $\tilde{C}$. The contraction mapping theorem then implies that T has a unique fixed point $x \in \tilde{C}$, that is, a continuous function x on J satisfying $x = T(x)$. Writing $x = T(x)$ out, we have by (4.13),

$$x(t) = x_0 + \int_{t_0}^t f(\tau, x(\tau)) \, d\tau. \quad (4.14)$$

Since $(\tau, x(\tau)) \in R$, where f is continuous, (4.17) may be differentiated. Hence, x is also differentiable and satisfies (4.8). Conversely, every solution of (4.8) must satisfy (4.17). This completes the proof. ■

We can give an alternate, yet equivalent, formulation of this theorem and its proof.

Theorem 4.3.2 Picard Existence and Uniqueness for ODEs—Alternate

If f is continuous on a region

$$R = \{(t, y) \mid t_0 \leq t \leq t_1, |y - y_0| \leq b\},$$

satisfies a Lipschitz condition with respect to y on R , then there exists $t_0 < a \leq t_1$ such that the IVP (4.8) has a unique solution for $t_0 \leq t \leq a$.

PROOF: Let $M := \max_R |f|$. Now, $y \in C^1[t_0, a]$ is a solution to the IVP if and only if $y \in C[t_0, a]$ is a solution to the integral equation

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds, \quad t \in [t_0, a].$$

As we have seen, y is a solution to the IVP implies that y is a solution to the integral equation. If y is a solution to the integral equation, then $y \in C^1[t_0, a]$ and $y' = f(t, y)$ by the fundamental theorem of calculus. Also, $y(t_0) = y_0$. So y is a solution to the IVP. Let

$$T(g)(t) := y_0 + \int_{t_0}^t f(s, g(s)) \, ds,$$

and

$$\begin{aligned} S_a &= \{g \in C[t_0, a] \mid |g(t) - y_0| \leq b \, \forall t \in [t_0, a]\}, \\ X &= C[t_0, a] \quad d_\infty(g, h) = \max_{t_0 \leq t \leq a} |g(t) - h(t)|. \end{aligned}$$

Note that

$$S_a = \overline{B_b(y_0)} = \{g \in X \mid d_\infty(g, y_0) \leq b\}.$$

Now, we know that (X, d_∞) is a complete metric space, and it can be shown that S_a is closed in X , which means that by 3.5.1 (S_a, d_∞) is a complete metric space.

Because the idea is to use the contraction mapping theorem on T , which requires us to have T act on a complete metric space, we have completed the first step, which is identifying the complete metric space (S_a, d_∞) on which T acts. Now we must show that $T : S_a \rightarrow S_a$, i.e., that T maps S_a to itself. We have

$$|T(g)(t) - y_0| = \left| \int_{t_0}^t f(s, g(s)) \, ds \right| \leq \int_{t_0}^t |f(s, g(s))| \, ds \leq M(t - t_0) \leq M(a - t_0),$$

which means that $d_\infty(T(g), y_0) \leq M(a - t_0)$, proving that T maps S_a to itself.

Now, $M(a - t_0) \leq b$ is equivalent to $a \leq t_0 + \frac{b}{M}$ (assuming $M \neq 0$). We also require T to be a contraction on S_a . We have

$$\begin{aligned} |T(g)(t) - T(h)(t)| &= \left| \int_{t_0}^t f(s, g(s)) - f(s, h(s)) \, ds \right| \\ &\leq \int_{t_0}^t |f(s, g(s)) - f(s, h(s))| \, ds. \end{aligned}$$

Because by assumption f satisfies a Lipschitz condition with respect to y on R , there exists $L \geq 0$ such that $|f(t, y_2) - f(t, y_1)| \leq L|y_2 - y_1|$ for all $(t, y_1), (t, y_2) \in R$. Thus,

$$\begin{aligned} |T(g)(t) - T(h)(t)| &\leq \int_{t_0}^t L|g(s) - h(s)| \, ds \leq \int_{t_0}^a L|g(s) - h(s)| \, ds \\ &\leq \int_{t_0}^a L d_\infty(g, h) \, ds = L(a - t_0) d_\infty(g, h) \\ &\Rightarrow d_\infty(T(g), T(h)) \leq L(a - t_0) d_\infty(g, h). \end{aligned}$$

Now, $L(a - t_0) < 1$ is equivalent to $a < t_0 + \frac{1}{L}$ (assuming $L \neq 0$). For a satisfying $t_0 < a \leq t_1$, $a \leq t_0 + \frac{b}{M}$, and $a < t_0 + \frac{1}{L}$, we have that $T : S_a \rightarrow S_a$ is a contraction mapping. By the contraction mapping theorem, T has a unique fixed point $y^* \in S_a$. Hence y^* satisfies the IVP. ■

Example 4.3.1 Consider the IVP

$$y' = 1 + y^2, \quad y(0) = 0.$$

This has the solution $y(t) = \tan(t)$ on $[0, \frac{\pi}{2})$ (or $(-\frac{\pi}{2}, \frac{\pi}{2})$). Let's see what interval the theory provides. $M := \max_R |f| = 1 + b^2$, and $\frac{\partial f}{\partial y} = 2y$, so that by Proposition 3.3.5, the Lipschitz constant L is $2b$. The conditions on a are $a > 0$, $a \leq \frac{b}{1+b^2}$, and $a < \frac{1}{2b}$. Let

$$F(b) := \min \left\{ \frac{b}{1+b^2}, \frac{1}{2b} \right\} = \begin{cases} \frac{b}{1+b^2}, & b \leq 1 \\ \frac{1}{2b}, & b \geq 1 \end{cases}.$$

The maximum of F is $\frac{1}{2}$ and it occurs at $b = 1$. Thus, the theory gives a solution on $[0, a]$ for any $0 < a < \frac{1}{2}$.

In the proof above, we have seen that under certain conditions of f , the operator T is contractive on a complete metric space (S_a, d_∞) of functions supported on $[t_0, a]$. We also had to obtain an estimate of a based on the properties of f :

1. First, we established that $a \leq t_0 + \frac{b}{M}$, where b can be prescribed and $M = \max_R |f|$.
2. Then, we established that $a < t_0 + \frac{1}{L}$, where L is the Lipschitz constant for the second argument of f .

In what follows, we show that these restriction can often be “softened” so that the existence of a unique solution to the IVP can be established over a larger interval. This is done by showing that the operator T is eventually contraction instead of contractive.

Let us return to the following fundamental set of identities involving the operator T :

$$|T(g)(t) - T(h)(t)| = \left| \int_{t_0}^t f(s, g(s)) - f(s, h(s)) \, ds \right| \leq \int_{t_0}^t |f(s, g(s)) - f(s, h(s))| \, ds \quad (4.15)$$

$$\leq L \int_{t_0}^t |g(s) - h(s)| \, ds \quad (4.16)$$

$$\leq L d_\infty(g, h) \int_{t_0}^t ds = L d_\infty(g, h)(t - t_0). \quad (4.17)$$

Note that we have not integrated out of the value a , but rather are keeping the right-hand side as a function of t . This will be useful below.

We replace g and h in the above relation with $T(g)$ and $T(h)$, respectively, to obtain:

$$|T^2(g)(t) - T^2(h)(t)| \leq L \int_{t_0}^t |T(g)(s) - T(h)(s)| \, ds.$$

Now insert (4.17):

$$|T^2(g)(t) - T^2(h)(t)| \leq L^2 d_\infty(g, h) \int_{t_0}^t (s - t_0) \, ds = \frac{1}{2} L^2 d_\infty(g, h)(t - t_0)^2.$$

We can repeat this procedure for $T^2(g)$ and $T^2(h)$, etc., to arrive at the following result, which can be proved by induction:

$$|T^n(g)(t) - T^n(h)(t)| \leq \frac{1}{n!} L^n (t - t_0)^n d_\infty(g, h), \quad t \in [t_0, a].$$

Taking the supremum over $t \in [t_0, a]$ on both sides, we obtain the important result

$$d_\infty(T^n(g), T^n(h)) \leq \frac{1}{n!} L^n (a - t_0)^n d_\infty(g, h).$$

For sufficiently large n , say $n = p$,

$$\frac{1}{p!} L^p (a - t_0)^p < 1, \quad (4.18)$$

which implies that $U := T^p$ for some $p > 1$ is a contraction, i.e., that T is eventually contractive. From the contraction mapping theorem, it follows that U has a unique fixed point. But we also know, from

Proposition 4.1.1 that the unique fixed point, call it u^* , of U , is also the unique fixed point of T . This implies that u^* is the unique solution to the IVP.

Note that the above analysis can also be extended over to the “other side” of t_0 , i.e., to an interval $[c, t_0]$, provided that suitable conditions on f are met.

A final comment: from (4.18), one might be tempted to conclude that the outer endpoint a of the interval $[t_0, a]$ over which the unique solution exists can be made as large as possible, i.e., given any $a > 0$, we can find a $p > 0$ that guarantees that the inequality (4.18) is true. This could pose a problem, since we know that some solutions “blow up” in finite time. Consider the following IVP:

$$\frac{dy}{dt} = y^2, \quad y(0) = y_0 > 0. \quad (4.19)$$

The function $f(t, y) = y^2$ is Lipschitz continuous in the variable y , so a unique solution exists. It is given by

$$y(t) = \frac{y_0}{1 - y_0 t}, \quad 0 \leq t < \frac{1}{y_0}.$$

Nevertheless, the solution y “blows up” at $t = \frac{1}{y_0}$.

If we return to the original proof using the contraction mapping theorem, we see that, in fact, no such problem exists. The proof rests on the assumption that the solution is an element of a closed ball of continuous functions—the space S_a . These functions are necessarily bounded. As a result, the endpoint a may not be arbitrarily large—it depends on the function $f(t, y)$ on the right-hand side of the IVP. a probably won’t have to be as small as the value determined in the proof, but finding larger values could be a tricky procedure, involving some kind of “juggling”, along with the knowledge that the operator T is eventually contractive.

Example 4.3.2 Consider the IVP

$$\frac{\partial x}{\partial t} = x^{1/3}, \quad x(0) = x_0.$$

Integrating gives

$$x(t) = \left(x_0^{2/3} + \frac{2}{3}t \right)^{3/2}.$$

For $x_0 \neq 0$, there is a unique solution x . When $x_0 = 0$, the above method gives $x(t) = \left(\frac{2}{3}t \right)^{3/2}$. But this solution is not unique! Indeed, $x(t) = 0$ satisfies the ODE and the initial condition. This means that when $x_0 = 0$, the solution is not unique. The reason for this is that $x^{1/3}$ is not Lipschitz continuous at $x = 0$.

4.3.1 Picard’s Method of Successive Approximations

The contractivity of the T (or T^p) operator is the basis for the *Picard method of successive substitutions/approximations*, a method that provides estimates to the solution of the IVP (4.8). Often, these estimates are in the form of power series about the point t_0 (which is often zero).

The idea is to start with a function $u_0(t)$ that will be the “seed” of the following iteration procedure:

$$u_{n+1} = T(u_n).$$

It is often most convenient to start with the constant function $u_0(t) = y_0$. Substitution into the integral equation gives

$$u_{n+1}(t) = y_0 + \int_{t_0}^t f(s, u_n(s)) \, ds, \quad n = 0, 1, 2, \dots$$

From the contractivity (or eventual contractivity) of the operator T (over an appropriate interval), it follows that the sequence of functions (u_n) will converge uniformly to the solution y of the IVP (over an appropriate interval).

Let us now illustrate this method with a simple example. Consider the IVP

$$\frac{dy}{dt} = ay, \quad y(0) = y_0, \quad (4.20)$$

where a and y_0 are arbitrary, non-zero real numbers. For convenience, we have set $t_0 = 0$. Of course, we know that the solution to this IVP is

$$y(t) = y_0 e^{at}.$$

We can confirm this using the Picard method. The solution of the IVP must satisfy the equivalent integral equation

$$y(t) = y_0 + \int_0^t ay(s) \, ds,$$

which is the fixed-point equation $y = T(y)$. Starting with the constant function $u_0(t) = y_0$ as the “seed” for the iteration procedure, we have

$$u_1(t) = y_0 + \int_0^t au_0(s) \, ds = y_0 + \int_0^t ay_0 \, ds = y_0 + ay_0 t = y_0(1 + at).$$

Also,

$$u_2(t) = y_0 + \int_0^t au_1(s) \, ds = y_0 + \int_0^t ay_0(1 + as) \, ds = y_0 \left(1 + at + \frac{1}{2}(at)^2 \right).$$

One can conjecture, and in fact prove by induction, that

$$u_n(t) = y_0 \left(1 + at + \dots + \frac{1}{n!}(at)^n \right), \quad n \geq 0,$$

which is the n th degree Taylor polynomial $P_n(t)$ to the solution $y(t) = y_0 e^{at}$. For each $t \in \mathbb{R}$, these Taylor polynomials are partial sums of the infinite Taylor series expansion of the function $y(t)$. As such, we see that the sequence of functions (u_n) converges to the solution. A little more work will show that the convergence is uniform over closed subintervals that include the point $t_0 = 0$.

Earlier, we commented that it was convenient to start the Picard iteration with the constant function $u_0(t) = y_0$; but we don't have to. We can, in fact, start with any function that satisfies the initial condition $u_0(t) = y_0$. For example, let us consider

$$u_0(t) = y_0 \cos(t).$$

Then,

$$u_1(t) = y_0 + \int_0^t a u_0 \, ds = y_0 + \int_0^t a y_0 \cos(s) \, ds = y_0(1 + a \sin(t)).$$

Once again:

$$u_2(t) = y_0 + \int_0^t a u_1(s) \, ds = y_0 + \int_0^t a y_0(1 + a \sin(s)) \, ds = y_0(1 + at - a^2 \cos(t) + a^2).$$

It is perhaps not obvious that these functions are “getting closer” to the solution $y(t) = y_0 e^{at}$. But it is not too hard to show that the Taylor series expansions of $u_1(t)$ and $u_2(t)$ (i.e., expanding the sin and cos appearing the iterates) agree, respectively, with the first two and three terms of the Taylor series expansion of the solution y .

4.4 Application to Integral Equations

5 Normed Linear Spaces and Banach Spaces

Particularly useful and important metric spaces are obtained if we take a vector space and define on it metric by means of a *norm*. The resulting space is called a *normed space*. If it is a complete metric space, it is called a *Banach space*. The theory of normed spaces, in particular Banach spaces, and the theory of linear operators defined on them, are the most highly developed parts of functional analysis.

5.1 Quick Review of Vector Spaces

Definition 5.1.1 Vector Space

A **vector space over a field** $\mathbb{F}$, the elements of which are called **scalars**, is a set $\mathcal{V}$ of objects called **vectors** together with two operations,

$$\begin{aligned} + : \mathcal{V} \times \mathcal{V} &\rightarrow \mathcal{V} \quad \text{such that } +(\underline{v}, \underline{w}) = \underline{v} + \underline{w} \text{ for all } \underline{v}, \underline{w} \in \mathcal{V} && \text{(Addition);} \\ \cdot : \mathbb{F} \times \mathcal{V} &\rightarrow \mathcal{V} \quad \text{such that } \cdot(\lambda, \underline{v}) = \lambda \cdot \underline{v} \equiv \lambda \underline{v} && \text{(Scalar Multiplication)} \end{aligned}$$

satisfying the following axioms:

1. Associativity of addition and scalar multiplication

$$\begin{aligned} (\underline{u} + \underline{v}) + \underline{w} &= \underline{u} + (\underline{v} + \underline{w}) && \text{for all } \underline{u}, \underline{v}, \underline{w} \in \mathcal{V} \\ \lambda(\mu \underline{v}) &= (\lambda\mu) \underline{v} && \text{for all } \lambda, \mu \in \mathbb{F} \text{ and } \underline{v} \in \mathcal{V} \end{aligned}$$

2. Distributivity of vectors and scalars

$$\begin{aligned} \lambda(\underline{v} + \underline{w}) &= \lambda \underline{v} + \lambda \underline{w} \\ (\lambda + \mu) \underline{v} &= \lambda \underline{v} + \mu \underline{v} \end{aligned} \quad \text{for all } \lambda, \mu \in \mathbb{F} \text{ and } \underline{v}, \underline{w} \in \mathcal{V}$$

3. The pair $(\mathcal{V}, +)$ is an Abelian group, i.e., along with the associativity of $+$ written above, $\mathcal{V}$ contains an identity element, called the **zero vector** and denoted $\underline{0}$ such that

$$\underline{0} + \underline{v} = \underline{v} + \underline{0} = \underline{v} \quad \text{for all } \underline{v} \in \mathcal{V}.$$

Also for every $\underline{v} \in \mathcal{V}$, there exists an inverse element, denoted $-\underline{v}$ such that $\underline{v} + (-\underline{v}) = -\underline{v} + \underline{v} = \underline{0}$. (We will write $\underline{v} + (-\underline{v}) \equiv \underline{v} - \underline{v}$.) Finally, the operation $+$ is commutative, i.e.,

$$\underline{v} + \underline{w} = \underline{w} + \underline{v} \quad \text{for all } \underline{v}, \underline{w} \in \mathcal{V}.$$

4. $1 \cdot \underline{v} = \underline{v}$ for all $\underline{v} \in \mathcal{V}$, where 1 is the multiplicative identity of $\mathbb{F}$.

Example 5.1.1 Examples of Vector Spaces

Here we go through some examples of vector spaces.

1. *The Euclidean Space* $\mathbb{R}^n$: This is the set of all (ordered) n -tuples of real numbers with the scalar field being $\mathbb{R}$.
2. *The Complex Space* $\mathbb{C}^n$: This is the set of all (ordered) n -tuples of complex numbers with the scalar field being $\mathbb{C}$.
3. *The Space Continuous Functions* $C([a, b])$: This is, as we have seen, the space of all continuous real-valued (or complex-valued) function defined on the closed interval $[a, b] \subset \mathbb{R}$.

Depending on context, this can be a vector space over $\mathbb{R}$ or $\mathbb{C}$.

4. *The Space of Sequences ℓ_2* : This is the set of all square-summable sequences of real (or complex) numbers, with the scalar field either $\mathbb{R}$ or $\mathbb{C}$.

Definition 5.1.2 Linear Dependence, Linear Independence

Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$ and $\mathcal{W} \subset \mathcal{V}$. $\mathcal{W}$ is called a **linearly dependent** set if there are $\lambda_j \in \mathbb{F}$ with $\lambda_j \neq 0$ and $\underline{v}_j \in \mathcal{W} - \{\underline{0}\}$ such that

$$\sum_{j=1}^n \lambda_j \underline{v}_j = \underline{0}.$$

Equivalently, $\mathcal{W}$ is linearly dependent if there is a vector $\underline{w} \in \mathcal{W}$ such that $\underline{w} \in \text{span}(\mathcal{W} - \{\underline{w}\})$, i.e., there is some $\underline{w} \in \mathcal{W}$ that can be written as a linear combination of the other vectors in $\mathcal{W}$. $\mathcal{W}$ is called **linearly independent** if it is not linearly dependent. For a finite subset $\{\underline{w}_j\}_{j=1}^n \subset \mathcal{V}$, linearly independence can be defined as

$$\sum_{j=1}^n \lambda_j \underline{w}_j = \underline{0} \quad \Rightarrow \quad \lambda_1 = \lambda_2 = \cdots = \lambda_n = 0.$$

Definition 5.1.3 Finite and Infinite Dimensional Vector Space

A vector space $\mathcal{V}$ is called **finite dimensional** if there is a positive integer n such that $\mathcal{V}$ contains a linear independent set of n vectors whereas any set of $n + 1$ or more vectors of $\mathcal{V}$ is linearly dependent. n is called the *dimension* of $\mathcal{V}$, and we write $n = \dim(\mathcal{V})$. By definition, $\mathcal{V} = \{0\}$ is finite dimensional and $\dim(\mathcal{V}) = 0$. If $\mathcal{V}$ is not finite dimensional, it is called **infinite dimensional**.

Definition 5.1.4 Basis

Let $\mathcal{V}$ be a vector space. $\mathcal{W} \subseteq \mathcal{V}$ is called a **basis** of $\mathcal{V}$ if $\mathcal{W}$ is a linearly independent spanning set, i.e., if $\mathcal{W}$ is linearly independent and $\text{span}(\mathcal{W}) = \mathcal{V}$.

If there is a finite basis $\mathcal{W} = \{\underline{w}_j\}_{j=1}^n$, then $\mathcal{V}$ is a *finite-dimensional* vector space with dimension $\dim(\mathcal{V}) = n$. In this case, every $\underline{v} \in \mathcal{V}$ has the form

$$\underline{v} = \sum_{j=1}^n \lambda_j \underline{w}_j,$$

where the *coordinates*, or *coefficients* of $\underline{v}$ with respect to the basis $\mathcal{W}$, $\{\lambda_j\}_{j=1}^n \subset \mathbb{C}$, are uniquely determined.

If $\mathcal{V}$ is *infinite-dimensional*, i.e., there is no finite basis, then the above formula needs to be modified. We will only encounter spaces in which there is a countably infinite basis $\{\underline{v}_j\}_{j=1}^\infty$ such that any element $\underline{v} \in \mathcal{V}$ can be expressed as the linear combination

$$\underline{v} = \sum_{j=1}^{\infty} \lambda_j \underline{v}_j. \quad (5.1)$$

5.2 Norms and Normed Spaces; Banach Spaces

Definition 5.2.1 Norm, Normed Space

A **normed (linear) space** is a pair $(X, \|\cdot\|)$, where X is a vector space (over a field $\mathbb{F}$) and $\|\cdot\| : X \times X \rightarrow \mathbb{R}$ is a real-valued function called the **norm** and is defined to have the following properties:

1. (Positivity) $\|x\| \geq 0$ for all $x \in X$;
2. (Strict Positivity) $\|x\| = 0$ if and only if $x = 0$;
3. (Triangle Inequality) $\|x + y\| \leq \|x\| + \|y\|$;
4. (Homogeneity) $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{F}$ and $x \in X$.

We will write only X if the norm is understood.

An easy consequence of the last property of the norm is

$$|\|y\| - \|x\|| \leq \|y - x\|, \quad (5.2)$$

from which we obtain the following:

Proposition 5.2.1 Continuity of the Norm

Let $(X, \|\cdot\|)$ be a normed space. Then $\|\cdot\| : X \times X \rightarrow \mathbb{R}$ is a continuous mapping.

PROOF: To be completed. ■

Theorem 5.2.1 Induced Metric

Let $(X, \|\cdot\|)$ be a normed space. Define a function $d : X \times X \rightarrow \mathbb{R}$ by

$$d(x, y) := \|x - y\| \quad \text{for all } x, y \in X.$$

Then d is a metric on X and (X, d) is metric space. d is called the metric **induced** by the norm $\|\cdot\|$.

PROOF: It is easy to check the axioms of a metric for $\|x - y\|$:

1. $d(x, y) = \|x - y\| \geq 0$ by definition of the norm;
2. $d(x, y) = 0 \Leftrightarrow x - y = 0$ from the definition of the norm, which means that $x = y$;
3. $d(x, y) = \|x - y\| = \|y - x\| = d(y, x)$;
4. $d(x, y) = \|x - y\| = \|(x - z) + (z - y)\| \leq \|x - z\| + \|z - y\| = d(x, z) + d(z, y)$, using the triangle inequality for norms. ■

Proposition 5.2.2 Translation Invariance

A metric d induced by a norm $\|\cdot\|$ on a normed space $(X, \|\cdot\|)$ satisfies

$$d(x + a, y + a) = d(x, y) \quad \text{and} \quad d(\alpha x, \alpha y) = |\alpha|d(x, y)$$

for all $x, y, a \in X$ and all scalars α .

PROOF: We have

$$d(x + a, y + a) = \|x + a - (y + a)\| = \|x - y\| = d(x, y)$$

and

$$d(\alpha x, \alpha y) = \|\alpha x - \alpha y\| = |\alpha| \|x - y\| = |\alpha|d(x, y). \quad \blacksquare$$

The axioms of a norm are very similar in look to those of a metric. And from this point of view, normed spaces and metric spaces are quite similar. They are not, however, the same (or else why would we define them!); but do note that, by using the induced norm,

All normed spaces are metric spaces.

But the converse isn't necessarily true: not all metric spaces are normed linear spaces. For example, on $\mathbb{R}$, we have seen the metric

$$\rho(x, y) = \frac{|x - y|}{1 + |x - y|}.$$

This is a metric not generated by a norm since if we define $\|x\| = \frac{|x|}{1+|x|}$, then this mapping isn't a norm since it does not satisfy the homogeneity condition.

Whereas for a metric space (X, d) X is allowed to be *any* set, in a normed space (X, d) X must be a *vector space*. We often say that metric spaces give us *geometric structure* because they allow us to talk about distances between points. So, on top of the structure provided to us by a metric space, normed spaces have the *algebraic structure* that comes from it being a vector space.

Definition 5.2.2 Banach Space

A complete normed linear space is called a **Banach space**.

REMARK: Note that completeness is meant with respect to the metric induced by the norm.

Example 5.2.1 Examples of Normed Linear Spaces

Here we go through some simple examples of normed linear spaces. The most important ones are the ones we've seen already.

1. *Euclidean Space $\mathbb{R}^n$ and Complex Space $\mathbb{C}^n$* : Both $\mathbb{R}^n$ and $\mathbb{C}^n$ are normed spaces with norms $\|\cdot\|_p$ (called the *p-norm*) and $\|\cdot\|_\infty$ (called the *infinity norm*) defined by

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad p \geq 1 \quad \text{and} \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

Note that the metrics generated by these norms are precisely the p -metric d_p and the metric d_∞ that we looked at earlier. There is also the Euclidean norm $\|\cdot\|_E$ defined by

$$\|x\|_E = \sqrt{\sum_{i=1}^n x_i^2} = \|x\|_2.$$

Both $(\mathbb{R}^n, \|\cdot\|_p)$ and $(\mathbb{R}^n, \|\cdot\|_\infty)$ (along with the complex counterparts) are Banach spaces.

2. *The Space $C[a, b]$* : A norm on $C[a, b]$ is $\|\cdot\|_\infty$ defined by

$$\|f\|_\infty = \max_{a \leq t \leq b} |f(t)|.$$

This norm generates the infinity metric $d_\infty(f, g) = \|f - g\|_\infty$ that we have already seen. $(C[a, b], \|\cdot\|_\infty)$ is a Banach space.

3. *The Space ℓ_p* : A norm on ℓ_p , for $1 \leq p < \infty$, is the p -norm $\|\cdot\|_p$ defined by

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p \right)^{1/p}.$$

This norm induces the metric d_p that was already defined for this space. We have also shown that (ℓ_p, d_p) is a complete metric space. Hence, $(\ell_p, \|\cdot\|_p)$ is a Banach space.

4. *The Space ℓ_∞* : A norm on ℓ_∞ is the *infinity norm* $\|\cdot\|_\infty$ defined by

$$\|x\|_\infty = \sup_i |x_i|.$$

As with the other examples, this norm induces the metric d_∞ that we previously defined for this space.

Example 5.2.2 The Holder Spaces

Let α satisfy $0 < \alpha \leq 1$ and define $C^\alpha[0,1]$ to be the space of all real-valued functions x that satisfy

$$|x(t) - x(s)| \leq K|t - s|^\alpha, \quad 0 \leq t, s \leq 1$$

for some finite $K > 0$. Any x satisfying this relation is continuous. Now, let

$$N_\alpha(x) = \inf\{K \mid |x(t) - x(s)| \leq K|t - s|^\alpha, \quad 0 \leq t, s \leq 1\}.$$

For example, if $x(t) = \cos(\pi t)$, then $N_1(x) = \pi$. Define a norm on $C^\alpha[0,1]$ by

$$\|x\|_\alpha := \|x\|_\infty + N_\alpha(x),$$

where $\|x\|_\infty = \max_{0 \leq t \leq 1} |x(t)|$. (Is $(C^\alpha[0,1], \|\cdot\|_\alpha)$ a Banach space?)

Example 5.2.3 Continuous Functions

Let (T, d) be a metric space and let $X = C(T, \mathbb{R})$ denote the space of all continuous real-valued functions defined on T . Define the norm on this space by

$$\|x\|_\infty = \sup\{|x(t)| \mid t \in T\}.$$

When T is a compact metric space, $(C(T, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space.

Example 5.2.4 Bounded Functions

Let (T, d) be a metric space and let $X = B(T, \mathbb{R})$ denote the space of all bounded real-valued functions defined on T . Define the norm on this space by

$$\|x\|_\infty = \sup\{|x(t)| \mid t \in T\}.$$

$(B(T, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space.

Example 5.2.5 Bounded Continuous Functions

Let (T, d) be a metric space and let $X = BC(T, \mathbb{R})$ denote the space of all real-valued, bounded, continuous functions defined on T . The norm on this space is defined as

$$\|x\|_\infty = \sup\{|x(t)| \mid t \in T\}.$$

$(BC(T, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space.

Some commonly employed cases are T being an interval on the real line, such as $T = [0, 1]$, or when $T = [0, \infty)$ or $T = (-\infty, \infty)$.

Note that if T is a compact metric space, for example $T = [a, b] \subset \mathbb{R}$, then every real-valued continuous function defined on T is bounded, implying that $BC(T, \mathbb{R}) = C(T, \mathbb{R})$.

Definition 5.2.3 Subspace of a Normed Space

Let $(X, \|\cdot\|)$ be a normed linear space and $Y \subset X$. Then $(Y, \|\cdot\|_Y)$ is called a **subspace** of $(X, \|\cdot\|)$. The norm $\|\cdot\|_Y$ is the norm $\|\cdot\|$ restricted to the subset Y , and is called the norm *induced* by X . If Y is closed in X , then Y is called a **closed subspace** of X .

Definition 5.2.4 Subspace of a Banach Space

Let $(X, \|\cdot\|)$ be a Banach space and $Y \subset X$. Then $(Y, \|\cdot\|_Y)$ is called a **subspace** of X . Note that $(Y, \|\cdot\|_Y)$ does *not* have to be complete.

Theorem 5.2.2 Subspace of a Banach Space

A subspace Y of a Banach space X is complete if and only if Y is closed in X .

PROOF: Immediate from Theorem 3.5.1. ■

Definition 5.2.5 **Isometrically Isomorphic**

Two normed linear spaces $(X_1, \|\cdot\|_1)$ and $(X_2, \|\cdot\|_2)$ are called **isometrically isomorphic** if there exists a linear bijection ϕ from X_1 to X_2 such that $\|\phi(x)\|_2 = \|x\|_1$ for all $x \in X_1$.

Theorem 5.2.3

$L_2[0, 1]$ and ℓ_2 are isometrically isomorphic.

PROOF: (Idea) From the theory of Fourier series,

$$f(x) = \sum_{n=1}^{\infty} a_n \sqrt{2} \sin(n\pi x)$$

with convergence in L_2 . Thus, there is a correspondence

$$f \leftrightarrow (a_1, a_2, \dots),$$

and by the Parseval formula,

$$\|f\|_{L_2} = \|(a_n)\|_{\ell_2}. \quad \blacksquare$$

5.2.1 Sequences and Convergence; Bases

The convergence of sequences and related concepts in normed spaces follows readily from the corresponding definitions for metric spaces and from the fact that all normed spaces are metric spaces.

Definition 5.2.6 **Convergence of a Sequence, Limit**

A sequence (x_n) in a normed space $(X, \|\cdot\|)$ is called **convergent** if there exists an $x \in X$ such that

$$\lim_{n \rightarrow \infty} \|x_n - x\| = 0,$$

that is, if for all $\epsilon > 0$ there exists $N_\epsilon > 0$ such that $\|x_n - x\| < \epsilon$ for all $n > N_\epsilon$. We then sometimes write $x_n \rightarrow x$ and call x the **limit** of (x_n) .

Definition 5.2.7 **Cauchy Sequence**

A sequence (x_n) in a normed space $(X, \|\cdot\|)$ is called a **Cauchy sequence** if for every $\epsilon > 0$ there exists N_ϵ such that

$$\|x_m - x_n\| < \epsilon \text{ for all } m, n > N_\epsilon. \quad (5.3)$$

The algebraic structure of normed linear spaces allows us to define infinite series in a way similar to that in calculus. If (x_k) is a sequence in a normed space $(X, \|\cdot\|)$, we can associate with it the sequence (s_n) of *partial sums*

$$s_n := x_1 + x_2 + \dots + x_n,$$

where $n = 1, 2, \dots$

Definition 5.2.8 Infinite Series, Convergence

Let (s_n) be the sequence of partial sums associated with a sequence (x_k) in a normed linear space $(X, \|\cdot\|)$. If (s_n) is convergent with limit s , i.e., $\lim_{n \rightarrow \infty} \|s_n - s\| = 0$, then the **infinite series**, or just **series**

$$\sum_{k=1}^{\infty} x_k = x_1 + x_2 + \cdots$$

is said to **converge** or to be **convergent**. s is then called the **sum** of the series, and we write

$$\sum_{k=1}^{\infty} x_k = x_1 + x_2 + \cdots = s.$$

Note that

$$\|s_n\| = \|x_1 + x_2 + \cdots + x_n\| \leq \|x_1\| + \|x_2\| + \cdots + \|x_n\|.$$

Definition 5.2.9 Absolute Convergence of Series

Consider the infinite series

$$\sum_{k=1}^{\infty} x_k \tag{5.4}$$

of elements x_k from a normed linear space $(X, \|\cdot\|)$. If the series

$$\|x_1\| + \|x_2\| + \cdots$$

converges, then (5.4) is said to be **absolutely convergent**.

It is easy to see that if the original series (5.4) converges then so does the series of norms, so that convergence implies absolute convergence. However, the converse is not always true; see Theorem 5.2.5 below.

In the special case of Banach spaces, we may use the following *Cauchy test* for convergence of partial sums without knowing the limit of the sequence (just as in the case of real numbers).

Theorem 5.2.4 The Cauchy Test

Let $(X, \|\cdot\|)$ be a Banach space. An infinite series $\sum_{k=1}^{\infty} x_k$ converges in X if and only if for every $\epsilon > 0$ there exists an integer N such that

$$\|s_n - s_m\| = \left\| \sum_{k=m}^n x_k \right\| \leq \epsilon \text{ for all } n \geq m > N.$$

Theorem 5.2.5 Absolute Convergence

For an infinite series in a normed space X , absolute convergence implies convergence if and only if X is complete, i.e., X is a Banach space.

PROOF: We prove only the forward ($\Rightarrow$) direction. From the triangle inequality, we have that

$$\left\| \sum_{k=m}^n x_k \right\| \leq \sum_{k=m}^n \|x_k\|.$$

Using the Cauchy test, it follows that the series $\sum_{k=1}^{\infty} x_k$ is convergent. ■

The concept of convergence of a series can be used to define a “basis” of a normed space as follows.

Definition 5.2.10 Schauder Basis

Let $(X, \|\cdot\|)$ be a normed linear space. If X contains a sequence (e_n) with the property that for every $x \in X$ there is a unique sequence of scalars (α_n) such that

$$\|x - (\alpha_1 e_1 + \cdots + \alpha_n e_n)\| \rightarrow 0 \text{ as } n \rightarrow \infty, \quad (5.5)$$

then (e_n) is called a **Schauder basis** for X . The series

$$\sum_{k=1}^{\infty} \alpha_k e_k,$$

which has the sum x is then called the *expansion* of x with respect to the basis (e_n) , and we write

$$x = \sum_{k=1}^{\infty} \alpha_k e_k.$$

For example, recall the space ℓ_p :

$$\ell_p = \left\{ (x_1, x_2, \dots) \mid \sum_{k=1}^{\infty} |x_k| < \infty \right\}, \quad \|x\|_p = \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{1/p}.$$

This space has a Schauder basis (e_n) given as follows:

$$\begin{aligned} e_1 &= (1, 0, 0, 0, \dots) \\ e_2 &= (0, 1, 0, 0, \dots) \\ e_3 &= (0, 0, 1, 0, \dots) \\ &\vdots \end{aligned}$$

Theorem 5.2.6

If a normed space X , then X is separable.

5.2.2 Completeness

Theorem 5.2.7 Completion

Let $(X, \|\cdot\|)$ be a normed space. Then there is a Banach space $\hat{X}$ and an isometry A from X onto a subspace W of $\hat{X}$ that is dense in $\hat{X}$. The space $\hat{X}$ is unique up to isometries.

PROOF: The Completion Theorem for metric spaces, Theorem 3.6.2, implies the existence of a complete metric space $(\hat{X}, \hat{d})$ and an isometry $A : X \rightarrow W = A(X)$, where W is dense in $\hat{X}$ and $\hat{X}$ is unique up to isometries. Consequently, to prove the present theorem, we must make $\hat{X}$ into a vector space and then introduce on $\hat{X}$ a suitable norm.

To define on $\hat{X}$ the two algebraic operations of a vector space, we consider any $\hat{x}, \hat{y} \in \hat{X}$ and any representatives $(x_n) \in \hat{x}$ and $(y_n) \in \hat{y}$. Remember that $\hat{x}$ and $\hat{y}$ are equivalence classes of Cauchy sequences in X . We set $z_n := x_n + y_n$. Then (z_n) is Cauchy in X since

$$\|z_n - z_m\| = \|x_n + y_n - (x_m + y_m)\| \leq \|x_n - x_m\| + \|y_n - y_m\|.$$

We define the sum $\hat{z} := \hat{x} + \hat{y}$ of $\hat{x}$ and $\hat{y}$ to be the equivalence class for which (z_n) is a representative; thus, $(z_n) \in \hat{z}$. This definition is independent of the particular choice of Cauchy sequences belonging to $\hat{x}$ and $\hat{y}$. In fact, we have from (3.22) in the proof of Theorem 3.6.2 that if $(x_n) \sim (x'_n)$ and $(y_n) \sim (y'_n)$, then $(x_n + y_n) \sim (x'_n + y'_n)$ because

$$\|x_n + y_n - (x'_n + y'_n)\| \leq \|x_n - x'_n\| + \|y_n - y'_n\|.$$

Similarly, we define the product $\alpha\hat{x} \in \hat{X}$ of a scalar α and $\hat{x}$ to be the equivalence class for which (αx_n) is a representative. Again, this definition is independent of the particular choice of a representative of $\hat{x}$. The zero element of $\hat{X}$ is the equivalence class containing all Cauchy sequences that converge to zero. It is not difficult to see that those two algebraic operations have all the properties required by the definition, so that $\hat{X}$ is a vector space. From the definition, it follows that on W the operations of vector space induced from $\hat{X}$ agree with those induced from X by means of A .

Furthermore, A induces on W a norm $\|\cdot\|_1$, whose value at every $\hat{y} = Ax \in W$ is $\|\hat{y}\|_1 = \|x\|$. The corresponding metric on W is the restriction of $\hat{d}$ to W since A is isometric. We can extend the norm $\|\cdot\|_1$ to $\hat{X}$ by setting $\|\hat{x}\|_2 := \hat{d}(\hat{0}, \hat{x})$ for every $\hat{x} \in \hat{X}$. In fact, it should be clear that $\|\cdot\|_2$ satisfies the first two axioms of a norm, and the other two axioms follow from those of $\|\cdot\|_1$ by a limit process. ■

Theorem 5.2.8 Completeness

Every finite-dimensional subspace Y of a normed space X is complete. In particular, every finite-dimensional normed space is complete, i.e., every finite-dimensional normed space is a Banach space.

PROOF: We consider an arbitrary Cauchy sequence (y_m) in Y and show that it is convergent in Y ; let the limit be y . Let $\dim(Y) = n$ and $\{e_1, \dots, e_n\}$ any basis for Y . Then each y_m has a unique

representation of the form

$$y_m = \alpha_1^{(m)} e_1 + \cdots + \alpha_n^{(m)} e_n.$$

Since (y_m) is a Cauchy sequence, for every $\epsilon > 0$ there exists $N > 0$ such that $\|y_m - y_r\| < \epsilon$ for all $m, r > N$. From this and Lemma 5.2.3 below, we have for some $c > 0$

$$\epsilon > \|y_m - y_r\| = \left\| \sum_{j=1}^n (\alpha_j^{(m)} - \alpha_j^{(r)}) e_j \right\| \geq c \sum_{j=1}^n |\alpha_j^{(m)} - \alpha_j^{(r)}|,$$

where $m, r > N$. Division by $c > 0$ gives

$$|\alpha_j^{(m)} - \alpha_j^{(r)}| \leq \sum_{j=1}^n |\alpha_j^{(m)} - \alpha_j^{(r)}| < \frac{\epsilon}{c} \text{ for all } m, r > N.$$

This shows that each of the n sequences

$$(\alpha_j^{(m)}) = (\alpha_j^{(1)}, \alpha_j^{(2)}, \dots), \quad j = 1, 2, \dots, n,$$

is Cauchy in $\mathbb{R}$ or $\mathbb{C}$. Hence, it converges. Let α_j denote the limits. Using these n limits $\alpha_1, \dots, \alpha_n$, we define

$$y = \alpha_1 e_1 + \cdots + \alpha_n e_n.$$

Clearly, $y \in Y$. Furthermore,

$$\|y_m - y\| = \left\| \sum_{j=1}^n (\alpha_j^{(m)} - \alpha_j) e_j \right\| \leq \sum_{j=1}^n |\alpha_j^{(m)} - \alpha_j| \|e_j\|.$$

On the right, $\alpha_j^{(m)} \rightarrow \alpha_j$. Hence, $\|y_m - y\| \rightarrow 0$, that is $y_m \rightarrow y$. This shows that (y_m) is a convergence sequence in Y . Since (y_m) was an arbitrary Cauchy sequence in Y , we have that Y is complete. ■

From this and Theorem 3.5.1, we have the following.

Theorem 5.2.9 Closedness

Every finite-dimensional subspace Y of a normed space X is closed in X .

Note that infinite-dimensional subspaces need not be closed. For example, let $X = C[0, 1]$ and $Y = \text{span}(x_0, x_1, \dots)$, where $x_j(t) = t^j$, so that Y is the set of all polynomials. Y is not closed in X . Why?

5.2.3 Compactness

Definition 5.2.11 Compactness

A metric space (X, d) is called **compact** if every sequence in X has a convergent subsequence.

A subspace $(M, d|_M)$ of X is called **compact** if every (generally infinite) sequence in M has a convergent subsequence whose limit is an element of M .

REMARK: Compare the definition of a compact subspace to that of a closed set (in particular, the alternate definition). For a closed set, we required every *convergent* sequence to have a limit in M , whereas here the sequence does not have to be convergent—we just need a convergent *subsequence*

Lemma 5.2.1 Compactness

A compact subset of a metric space is closed and bounded.

PROOF: Let M be a subset of a metric space. Since the closure $\overline{M}$ is closed, by definition (see remark above, or Definition 3.3.3) there is a sequence $(x_n) \subset M$ such that $\lim_{n \rightarrow \infty} x_n = x$. Since M is compact (by assumption), $x \in M$. Hence, M is closed because $x \in \overline{M}$ was arbitrary. We now prove that M is bounded. If M were unbounded, it would contain an unbounded sequence (y_n) such that $d(y_n, b) > n$, where b is any fixed element of M and d is the metric on M . This sequence could not have a convergent subsequence since a convergent subsequence must be bounded by Proposition 3.2.1. ■

Note that the converse of this lemma is in general false. For an example, consider the space (ℓ_2, d_2) and the set

$$\overline{B_1(0)} := \{(x_1, x_2, \dots) \mid \sum x_i^2 \leq 1\}.$$

$\overline{B_1(0)}$ is closed and bounded but it is *not* compact. To see why, let $e_i := (0, \dots, 0, 1, 0, \dots)$, the sequence with 1 in the i th position and 0 elsewhere. So $d_2(e_i, e_j) = \sqrt{2}$ for all $i \neq j$. So the sequence $(e_i) \subset \overline{B_1(0)}$ has no convergent subsequence since it is not possible for any subsequence to be Cauchy (the distance between distinct points is a constant $\sqrt{2}$). Therefore, $\overline{B_1(0)}$ is not compact.

Proposition 5.2.3

A closed subset of a compact set is compact.

PROOF: To be completed. ■

Theorem 5.2.10 Compactness

In a *finite-dimensional* normed space $(X, \|\cdot\|)$, any subset $M \subset X$ is compact if and only if it is closed and bounded.

PROOF: Compactness implies closedness and boundedness by Lemma 5.2.1, which gives the forward direction ($\Rightarrow$) of the proof. For the converse ($\Leftarrow$), let M be closed and bounded. Let $\dim(X) = n$ and $\{e_1, \dots, e_n\}$ a basis for X . We consider any sequence $(x_m) \subset M$. Each x_m can be written as

$$x_m = \xi_1^{(m)} e_1 + \dots + \xi_n^{(m)} e_n.$$

Since M is bounded, so is (x_m) , say $\|x_m\| \leq k$ for all m . By 5.2.3,

$$k \geq \|x_m\| = \left\| \sum_{j=1}^n \xi_j^{(m)} e_j \right\| \geq c \sum_{j=1}^n |\xi_j^{(m)}|,$$

where $c > 0$. Hence, the sequence of numbers $(\xi_j^{(m)})$ (j fixed) is bounded, and, by the Bolzano-Weierstrass theorem, has a limit, call it ξ_j (here, $1 \leq j \leq n$). As in the proof of Lemma 5.2.3, we conclude that (x_m) has a subsequence (z_m) that converges to $z := \sum \xi_j e_j$. Since M is closed, $z \in M$. This shows that the arbitrary sequence (x_m) in M has a subsequence that converges in M . Hence M is compact. ■

We see that in $\mathbb{R}^n$ (or in any other finite-dimensional normed space), the compact subsets are precisely the closed and bounded subsets, so that this property (closedness and boundedness) can be used to *define* compactness. However, this can no longer be done for infinite-dimensional normed spaces.

Lemma 5.2.2 F. Riesz

Let Y and Z be subspaces of a normed space X of any dimension, and suppose that Y is closed and is a proper subset of Z . Then, for every real number θ in the interval $(0, 1)$ there is a $z \in Z$ such that

$$\|z\| = 1, \quad \|z - y\| \geq \theta \text{ for all } y \in Y.$$

Theorem 5.2.11 Finite Dimension

In normed space $(X, \|\cdot\|)$, the closed unit ball $M = \{x \mid \|x\| \leq 1\}$ is compact if and only if X is finite-dimensional.

Theorem 5.2.12 Continuous Mappings

Let X and Y be metric spaces and $T : X \rightarrow Y$ a continuous mapping. Then the image of a compact subset M of X under T is compact.

PROOF: By the definition of compactness, it suffices to show that every sequence (y_n) in the image $T(M) \subset Y$ contains a subsequence that converges in $T(M)$. Since $y_n \in T(M)$, we have $y_n = T(x_n)$ for some $x_n \in M$. Since M is compact, (x_n) contains a subsequence (x_{n_k}) that converges in M . The image of (x_{n_k}) is a subsequence of (y_n) , which converges in $T(M)$ by Proposition 3.3.4 because T is continuous. Hence, $T(M)$ is compact. ■

From this result, we conclude that the following property, well-known from calculus for continuous functions, carries over to metric spaces.

Theorem 5.2.13 **Extreme Value/Weierstrass**

A continuous mapping T of a compact subset M of a metric space X into $\mathbb{R}$, i.e., $T : M \rightarrow \mathbb{R}$, assumes a maximum and a minimum at some points of M .

PROOF: $T(M) \subset \mathbb{R}$ is compact by the previous theorem, and closed and bounded by 5.2.1 (applied to $T(M)$), so that $\inf T(M) \in T(M)$, $\sup T(M) \in T(M)$, and the inverse images of these two points consist of points of M at which $T(x)$ is minimum or maximum, respectively. ■

5.2.4 Equivalent Norms**Definition 5.2.12** **Equivalent Norms**

A norm $\|\cdot\|_1$ on a vector space X is called **equivalent** to a norm $\|\cdot\|_2$ on X if there are positive numbers a, b such that for all $x \in X$ we have

$$a \|x\|_2 \leq \|x\|_1 \leq b \|x\|_2. \quad (5.6)$$

REMARK: Equivalence of norms is an equivalence relation.

Proposition 5.2.4

Equivalent norms on a linear space X generate equivalent metrics on X , that is, they define the same topology on X .

PROOF: (Idea) This follows from the definition (5.6) and the fact that every non-empty open set is a union of open balls. One can also show that the Cauchy sequences in $(X, \|\cdot\|_1)$ and $(X, \|\cdot\|_2)$ are equivalent (recall definition of equivalent Cauchy sequences in Definition 3.2.5). ■

Example 5.2.6 In $\mathbb{R}^n$, $\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty$ for all $x \in \mathbb{R}^n$ and $p \geq 1$. Thus, $\|\cdot\|_p$ is equivalent to $\|\cdot\|_\infty$ for all $p \geq 1$. This shows that $\|\cdot\|_{p'}$ is equivalent to $\|\cdot\|_p$ for any $p, p' \geq 1$.

Proposition 5.2.5

All norms on $\mathbb{R}^n$ are equivalent.

PROOF: Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. By the previous example, it is sufficient to show that $\|\cdot\|$ is equivalent to the Euclidean norm $\|\cdot\|_2$. Let $d(x, y) = \|x - y\|$ and $d_2(x, y) = \|x - y\|_2$. Then, letting $\{e_1, \dots, e_n\}$ be a basis for $\mathbb{R}^n$, we have by the Cauchy-Schwarz inequality

$$\|x\| = \|x_1 e_1 + \dots + x_n e_n\| \leq |x_1| \|e_1\| + \dots + |x_n| \|e_n\| \leq \beta \|x\|_2,$$

where

$$\beta = \sqrt{\sum_{i=1}^n \|e_i\|^2}.$$

Now we show that there exists $\alpha > 0$ such that $\alpha \|x\|_2 \leq \|x\|$ for all $x \in \mathbb{R}^n$. For $x \neq 0$, this is equivalent to $\alpha \leq \left\| \frac{x}{\|x\|_2} \right\|$. Note that $\left\| \frac{x}{\|x\|_2} \right\| = 1$, so it suffices to show that there exists $\alpha > 0$ such that

$$\alpha \leq \|y\| \text{ for all } y \text{ such that } y \in S = \{x \mid \|x\|_2 = 1\}. \quad (5.7)$$

S is a closed and bounded subset of $\mathbb{R}^n$, so by the Heine-Borel theorem it is compact. Suppose for a contradiction that (5.7) is not true. Then there exists a sequence (y_n) such that $\|y_n\|_2 = 1$, $\|y_n\| \rightarrow 0$. By the compactness of S , there exists a subsequence (y_{n_k}) such that $y_{n_k} \rightarrow y$ under the d_2 metric and $y \in S$, i.e., $\|y\|_2 = 1 \Rightarrow y \neq 0$. Since $\|y_{n_k} - y\| \leq \beta \|y_{n_k} - y\|_2$, we have $y_{n_k} \rightarrow y$ under the d metric, which means that $\|y_{n_k}\| \rightarrow \|y\|$ by the continuity of the norm $\|\cdot\|$. But $y \neq 0 \Rightarrow \|y\| \neq 0$, so that the last limit contradicts $\|y_{n_k}\| \rightarrow 0$. Therefore, there exists $\alpha > 0$, $\beta > 0$ such that $\alpha \|x\|_2 \leq \|x\| \leq \beta \|x\|_2$ for all $x \in \mathbb{R}^n$. ■

PROOF: (Alternate) Let $N(x) = \|x\|$ be a norm on $\mathbb{R}^n$. As before, it is sufficient to show that it is equivalent to the Euclidean norm $\|\cdot\|_2$.

We know that N is continuous on $\mathbb{R}^n$. Consider the sphere $S_1 = \{x \in \mathbb{R}^n \mid \|x\|_2 = 1\}$. S_1 is closed and bounded, so by the Heine-Borel theorem it is compact. Therefore, N attains a maximum value B and a minimum value A for all $x \in S_1$. Note that $x = 0$ is not an element of S_1 , so that $A > 0$.

Now, let $x \in \mathbb{R}^n$ and define $y = \frac{x}{\|x\|_2} \in S_1$. Then,

$$A \leq N(y) \leq B.$$

But

$$N(y) = N\left(\frac{x}{\|x\|_2}\right) = \frac{1}{\|x\|_2} N(x) \Rightarrow A \leq \frac{1}{\|x\|_2} N(x) \leq B,$$

or,

$$A \|x\|_2 \leq N(x) \leq B \|x\|_2.$$

Therefore, $N(x) = \|x\|$ is equivalent to the Euclidean norm $\|\cdot\|_2$. ■

Lemma 5.2.3 Linear Combinations

Let $\{x_1, \dots, x_n\}$ be a linearly independent set of vectors in a normed space X of any dimension. Then there is a number $c > 0$ such that for every choice of scalars $\alpha_1, \dots, \alpha_n$ we have

$$\|\alpha_1 x_1 + \dots + \alpha_n x_n\| \geq c(|\alpha_1| + \dots + |\alpha_n|). \quad (5.8)$$

PROOF: Let $s := |\alpha_1| + \dots + |\alpha_n|$. If $s = 0$, then all α_j are zero, so that (5.8) holds for any c . Let $s > 0$. Then (5.8) is equivalent to the inequality that we obtain from (5.8) by dividing by s and letting $\beta_j := \frac{\alpha_j}{s}$, that is,

$$\|\beta_1 x_1 + \dots + \beta_n x_n\| \geq c, \quad \sum_{j=1}^n |\beta_j| = 1. \quad (5.9)$$

Hence, it suffices to prove the existence of a $c > 0$ such that (5.9) holds for every n -tuple of scalars $\beta_1, \dots, \beta_n$ with $\sum_{j=1}^n |\beta_j| = 1$.

Suppose for a contradiction that this is false. Then there exists a sequence (y_m) of vectors

$$y_m = \beta_1^{(m)} x_1 + \dots + \beta_n^{(m)} x_n, \quad \sum_{j=1}^n |\beta_j^{(m)}| = 1,$$

such that $\lim_{m \rightarrow \infty} \|y_m\| = 0$.

Now we reason as follows. Since $\sum_{j=1}^n |\beta_j^{(m)}| = 1$, we have $|\beta_j^{(m)}| \leq 1$ for all j . Hence, for each fixed j , the sequence

$$(\beta_j^{(m)}) = (\beta_j^{(1)}, \beta_j^{(2)}, \dots)$$

is bounded. Consequently, by the Bolzano-Weierstrass theorem, $(\beta_1^{(m)})$ has a convergent subsequence. Let β_1 denote the limit of that subsequence, and let $(y_{1,m})$ denote the corresponding subsequence of (y_m) . By the same argument, $(y_{1,m})$ has a subsequence $(y_{2,m})$ for which the corresponding subsequence of scalars $\beta_2^{(m)}$ converges. Let β_2 denote the limit of that sequence. Continuing in this way, after n steps, we obtain a subsequence $(y_{n,m}) = (y_{n,1}, y_{n,2}, \dots)$ of (y_m) whose terms are of the form

$$y_{n,m} = \sum_{j=1}^n \gamma_j^{(m)} x_j, \quad \sum_{j=1}^n |\gamma_j^{(m)}| = 1,$$

with scalars $\gamma_j^{(m)}$ satisfying $\lim_{m \rightarrow \infty} \gamma_j^{(m)} = \beta_j$. Hence,

$$\lim_{m \rightarrow \infty} y_{n,m} = y := \sum_{j=1}^n \beta_j x_j,$$

where $\sum_{j=1}^n |\beta_j| = 1$, so that not all β_j can be zero. Since $\{x_1, \dots, x_n\}$ is a linearly independent set, we thus have $y \neq 0$. On the other hand, $\lim_{m \rightarrow \infty} y_{n,m} = y$ implies $\lim_{m \rightarrow \infty} \|y_{n,m}\| = \|y\|$ by the continuity of the norm. Since $\|y_m\| \rightarrow 0$ by assumption and $(y_{n,m})$ is a subsequence of (y_m) , we must have $\|y_{n,m}\| \rightarrow 0$. Hence, $\|y\| = 0$, so that $y = 0$, a contradiction to the assumption that $y \neq 0$. So the result holds. ■

Using this, we can prove a more generic version of Proposition 5.2.5, which does *not* hold for infinite-dimensional spaces.

Theorem 5.2.14 Equivalent Norms

On a finite-dimensional vector space, all norms are equivalent to each other.

PROOF: Let X be an n -dimensional vector space with basis $\{e_1, \dots, e_n\}$. Any $x \in X$ can be written as

$$x = \alpha_1 e_1 + \dots + \alpha_n e_n,$$

where $\alpha_1, \dots, \alpha_n$ are scalars. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on X . By Lemma 5.2.3, there is a positive constant c such that

$$\|x\|_1 \geq c(|\alpha_1| + \dots + |\alpha_n|).$$

On the other hand, the triangle inequality gives

$$\|x\|_2 \leq \sum_{j=1}^n |\alpha_j| \|e_j\|_2 \leq k \sum_{j=1}^n |\alpha_j|, \quad k = \max_j \|e_j\|_2.$$

Together, $a \|x\|_2 \leq \|x\|_1$, where $a = \frac{c}{k} > 0$. The other inequality in (5.6) is now obtained by an interchanvge of the roles of $\|\cdot\|_1$ and $\|\cdot\|_2$ in the preceding argument. ■

5.2.5 Convexity

Definition 5.2.13 Convex Set and Convex Function

A set M in a linear space is called **convex** if for all $u, v \in M$ and $0 \leq \alpha \leq 1$ $\alpha u + (1 - \alpha)v \in M$.

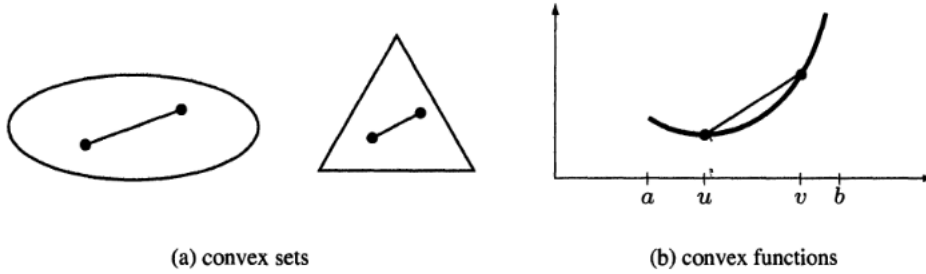
A function $f : M \rightarrow \mathbb{R}$ is called **convex** if M is convex and

$$f(\alpha u + (1 - \alpha)v) \leq \alpha f(u) + (1 - \alpha)f(v)$$

for all $u, v \in M$ and all $0 \leq \alpha \leq 1$.

Intuitively, the convexity of a set M means that if the two points u and v belong to M , then the segment joining them also belongs to M .

The convexity of the real function $f : [a, b] \rightarrow \mathbb{R}$, for example, means that the chords always lie above the graph of f .



Example 5.2.7 Let $(X, \|\cdot\|)$ be a normed space and let $u_0 \in X$ and $r \geq 0$. Then, the ball

$$B = \{u \in X \mid \|u - u_0\| \leq r\}$$

is a convex set. To prove this, suppose $u, v \in B$ and $0 \leq \alpha \leq 1$. Then,

$$\begin{aligned} \|\alpha u + (1 - \alpha)v - u_0\| &= \|\alpha(u - u_0) + (1 - \alpha)(v - u_0)\| \\ &\leq \|\alpha(u - u_0)\| + \|(1 - \alpha)(v - u_0)\| \\ &= \alpha \|u - u_0\| + (1 - \alpha) \|v - u_0\| \\ &\leq \alpha r + (1 - \alpha)r \\ &= r. \end{aligned}$$

Hence, $\alpha u + (1 - \alpha)v \in B$, so B is convex.

Example 5.2.8 Let $(X, \|\cdot\|)$ be a normed space. Let $f(u) := \|u\|$. Then $f : X \rightarrow \mathbb{R}$ is convex.

To prove this, let $u, v \in X$ and $0 \leq \alpha \leq 1$. Then,

$$\|\alpha u + (1 - \alpha)v\| \leq \|\alpha u\| + \|(1 - \alpha)v\| = \alpha \|u\| + (1 - \alpha) \|v\|.$$

This proves the convexity of f .

Definition 5.2.14 Convex Hull

Let M be a subset of a linear space X over $\mathbb{F}$. Then, define the sets

$\text{co}(M) :=$ smallest convex set of X containing M ;

$\overline{\text{co}}(M) :=$ smallest closed convex set of X containing M .

$\text{co}(M)$ is called the **convex hull** of M and $\overline{\text{co}}(M)$ is called the **closed convex hull** of M .

Proposition 5.2.6

Let M be a non-empty subset of the normed space $(X, \|\cdot\|)$ over the field $\mathbb{F}$. The $u \in \text{co}(M)$ if and only if for some fixed $n = 1, 2, \dots$,

$$u = \alpha_1 u_1 + \dots + \alpha_n u_n,$$

where $u_1, \dots, u_n \in M$ and $0 \leq \alpha_1, \dots, \alpha_n \leq 1$ with $\alpha_1 + \dots + \alpha_n = 1$.

PROOF: Observe that it follows from

$$0 \leq \alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_m \leq 1,$$

as well as $\alpha_1 + \dots + \alpha_n = 1$, $\beta_1 + \dots + \beta_m = 1$, and $\alpha + \beta = 1$, that

$$\alpha \alpha_1 + \dots + \alpha \alpha_n + \beta \beta_1 + \dots + \beta \beta_m = \alpha + \beta = 1.$$

■

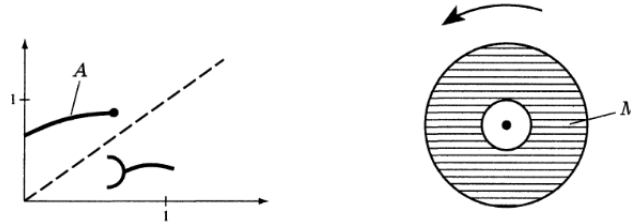
5.3 The Schauder Fixed Point Theorem

Theorem 5.3.1 Brouwer Fixed-Point

Let M be a compact, convex, non-empty subset of a finite-dimensional normed space and $A : M \rightarrow M$ a continuous mapping. Then A has a fixed point.

REMARK: We want to show through some counterexamples that each of the assumptions of the Brouwer fixed-point theorem is essential.

1. Let $M := [0, 1]$. The function $A : M \rightarrow M$ pictured below (left-hand side) has *no* fixed point. The set M is compact and convex, but A is *not* continuous.
2. Let $M := \mathbb{R}$. The continuous function $A : M \rightarrow M$ defined through $A(u) := u + 1$ has *no* fixed point. The set M is convex, but not compact.
3. Let M be a closed annulus as pictured below (right-hand side). Then, a proper rotation $A : M \rightarrow M$ of the annulus around the center is *fixed-point free*. Here, the operator A is continuous and M is compact, but M is *not* convex.



Corollary 5.3.1

The continuous operator $B : K \rightarrow K$ has a fixed point provided K is a subset of a normed space that is homeomorphic to a set M as considered in Theorem 5.3.1.

Corollary 5.3.2

A continuous map of a closed ball in $\mathbb{R}^n$ into itself must have a fixed point.

Example 5.3.1 Let $M = [a, b]$, where $-\infty < a < b < \infty$. Then, each continuous function $A : [a, b] \rightarrow [a, b]$ has a fixed point.

This is the simplest special case of the Brouwer fixed-point theorem. Let us give a direct proof. To this end, we set

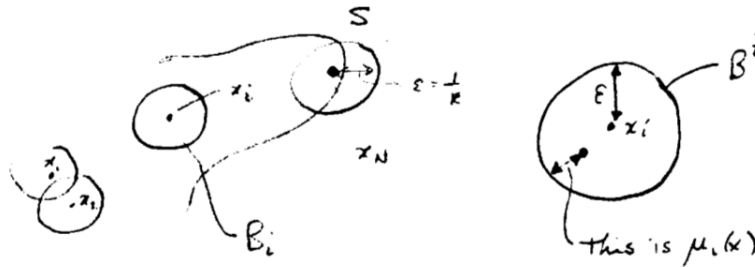
$$B(u) := A(u) - u \quad \text{for all } u \in [a, b].$$

Since $A(a), A(b) \in [a, b]$, we get $A(a) \geq a$ and $A(b) \geq b$. Hence,

$$B(a) \geq 0 \quad \text{and} \quad B(b) \leq 0.$$

By the *intermediate-value theorem*, the continuous real-valued function B has a zero $u \in [a, b]$, i.e., $B(u) = 0$. Hence $A(u) = u$.

Let $(X, \|\cdot\|)$ be a Banach space over a field $\mathbb{F}$, $S \subset X$ such that S is compact, convex, and non-empty. Let $T : S \rightarrow S$ be continuous on S . Then T has a fixed point.

$$S \subset \bigcup_{i=1}^N B_\epsilon(x_i), \quad \epsilon = \frac{1}{R}, \quad N = N_\epsilon.$$


1. If $x \in B^i$, then $\mu_i(x) = \epsilon - \|x - x_i\|$;
2. If $x \notin B^i$, then $\mu_i(x) = 0$.

$$\mu_i(x) = \max\{0, \epsilon - \|x - x_i\|\}.$$

We now form the convex combination

$$J_\epsilon(x) := \frac{\sum_{i=1}^N \mu_i(x) x_i}{\sum_{j=1}^N \mu_j(x)} = \sum_{i=1}^N p_i(x) x_i,$$

$$p_i(x) = \frac{\mu_i(x)}{\sum_{j=1}^N \mu_j(x)}, \quad 0 \leq p_i(x) \leq 1, \quad \sum_{i=1}^N p_i(x) = 1 \quad \forall x \in S.$$
$$U(x^{(\epsilon)}) = (J_\epsilon \circ T)(x^{(\epsilon)}) = x^{(\epsilon)}.$$

We now investigate if $x^{(\epsilon)}$ is a fixed point of T . To do this, examine $\|x^{(\epsilon)} - T(x^{(\epsilon)})\|$:

$$\|x^{(\epsilon)} - T(x^{(\epsilon)})\| = \|J_\epsilon \circ T(x^{(\epsilon)}) - T(x^{(\epsilon)})\|.$$

Now, for $x \in S$, examine $\|J_\epsilon(x) - x\|$:

$$\begin{aligned} \|J_\epsilon(x) - x\| &= \left\| \sum_{i=1}^N p_i(x) x_i - x \right\| \\ &= \left\| \sum_{i=1}^N p_i(x) (x_i - x) \right\| \quad (\text{since } \sum p_i(x) = 1) \\ &\leq \sum_{i=1}^N |p_i(x)| \|x_i - x\| \\ &< \epsilon, \end{aligned}$$

with the last step following from the fact that if $x \notin B^i$, then $p_i(x) = 0$ and if $x \in B^i$ then $\|x_i - x\| < \epsilon$. So we have

$$\|x^{(\epsilon)} - T(x^{(\epsilon)})\| < \epsilon,$$

i.e., $x^{(\epsilon)}$ is an “ ϵ -approximate” fixed point of T . Finally, let $\epsilon = \frac{1}{R} \rightarrow 0$. $(x^{(\epsilon)})$ is a sequence of ϵ -approximate fixed points in S . By the compactness of S , there exists a converging subsequence $(x^{(\epsilon_k)})$, say with limit $\bar{x} \in S$. Therefore, $T(\bar{x}) = \bar{x}$. ■

The following example shows that the compactness of S cannot be lessened to closed and bounded.

Example 5.3.2 Kakutani

In the space $(\ell_2, \|\cdot\|_2)$, let

$$B = \{x = (x_i) \mid \|x\|_2 \leq 1\}, \quad \partial B = \{x \mid \|x\|_2 = 1\},$$

and

$$T : x = (x_i) \rightarrow \left(\sqrt{1 - \|x\|_2^2}, x_1, x_2, \dots \right).$$

Then $T : B \rightarrow \partial B$ since

$$\left\| \left(\sqrt{1 - \|x\|_2^2}, x_1, x_2, \dots \right) \right\|_2^2 = 1 - \|x\|_2^2 + x_1^2 + x_2^2 + \dots = 1.$$

T is continuous since

$$\begin{aligned} \|T(x) - T(y)\| &= \left\| \left(\sqrt{1 - \|x\|_2^2} - \sqrt{1 - \|y\|_2^2}, x_1 - y_1, x_2 - y_2, \dots \right) \right\|_2 \\ &= \left\| \left(\sqrt{1 - \|x\|_2^2} - \sqrt{1 - \|y\|_2^2}, 0, 0, \dots \right) + (0, x_1 - y_1, x_2 - y_2, \dots) \right\|_2 \\ &\leq \left| \sqrt{1 - \|x\|_2^2} - \sqrt{1 - \|y\|_2^2} \right| + \|x - y\|_2. \end{aligned}$$

Thus, if $x^{(n)} \rightarrow x$, using

$$\|T(x) - T(x^{(n)})\|_2 \leq \left| \sqrt{1 - \|x\|_2^2} - \sqrt{1 - \|x^{(n)}\|_2^2} \right| + \|x - x^{(n)}\|_2,$$

we see that $T(x^{(n)}) \rightarrow T(x)$. But T has *no* fixed point. To see this, suppose $T(x) = x$. Then $x \in \partial B$, which means that

$$(0, x_1, x_2, \dots) = (x_1, x_2, \dots) \rightarrow 0 = x_1, x_2, x_3 = \dots \Rightarrow x = 0.$$

This contradicts $x \in \partial B$.

5.3.1 Application to Ordinary Differential Equations

Let us consider again the initial value problem (IVP) (4.8)

$$y' = f(t, y) \text{ on } [t_0, a], \quad y(0) = y_0.$$

Recall that this is equivalent to the integral equation

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds, \quad y \in C[t_0, a].$$

(Equivalent in that the solution the latter implies a solution to the former.) Recall also the Arzela-Ascoli theorem, Theorem 3.3.7, which states that every bounded equicontinuous sequence of functions on $[a, b]$ has a subsequence that converges uniformly on $[a, b]$.

We now state another existence theorem for the IVP above, one that is more general than the one stated before because some of the assumptions of the previous theorem are relaxed. Whereas before we needed the Banach fixed point theorem (contraction mapping theorem), here we'll need the Schauder fixed point theorem. Note that the following is just an *existence* theorem: unlike the previous case, the theorem does *not* establish *uniqueness*.

Theorem 5.3.3 **Peano**

Consider the rectangular region R defined by

$$R = \{(t, y) \mid t_0 \leq t \leq \tau, |y - y_0| \leq b\}.$$

Assume that f is continuous on R and let $M = \max_R |f|$. Then, the initial value problem

$$y' = f(t, y) \text{ on } [t_0, a], \quad y(0) = y_0.$$

has a solution on $[t_0, a]$, where $a = \min \left\{ \tau, t_0 + \frac{b}{M} \right\}$.

REMARK: The aforementioned relaxed assumption in this theorem, which is what disallows us from establishing uniqueness, is that f is merely *continuous*, not *Lipschitz continuous* as was assumed in the previous existence-uniqueness theorem.

PROOF: Let

$$T(y)(t) = y_0 + \int_{t_0}^t f(s, y(s)) \, ds$$

and

$$B = \{g \in C[t_0, a] \mid |g(t) - y_0| \leq b \, \forall t \in [t_0, a]\}.$$

Consider the norm $\|\cdot\|_\infty$. Then $T : B \rightarrow B$, where $B := \overline{B_b(y_0)}$ is closed and convex but *not* compact (why?). Consider instead

$$S = \{g \in C[t_0, a] \mid |g(t) - y_0| \leq b \, \forall t \in [t_0, a], |g(t_2) - g(t_1)| \leq M|t_2 - t_1| \, \forall t_1, t_2 \in [t_0, a]\}.$$

Then, S is convex. Indeed, for $g, h \in S$ and $0 \leq \lambda \leq 1$, we have

$$\begin{aligned} & |\lambda g(t_2) + (1 - \lambda)h(t_2) - (\lambda g(t_1) + (1 - \lambda)h(t_1))| \\ &= |\lambda(g(t_2) - g(t_1)) + (1 - \lambda)(h(t_2) - h(t_1))| \\ &\leq \lambda|g(t_2) - g(t_1)| + (1 - \lambda)|h(t_2) - h(t_1)| \\ &\leq \lambda M|t_2 - t_1| + (1 - \lambda)M|t_2 - t_1| \\ &= M|t_2 - t_1|. \end{aligned}$$

S is also compact. Indeed, for a sequence $(g_n) \subset S$,

$$|g_n(t)| \leq |y_0| + b \text{ for all } t \in [t_0, a] \text{ for all } n \geq 1.$$

Then, given $\epsilon > 0$, let $\delta = \frac{\epsilon}{M}$. Then,

$$|g_n(t_2) - g_n(t_1)| \leq M|t_2 - t_1| < M \frac{\epsilon}{M} = \epsilon$$

for $|t_2 - t_1| < \delta$. By the Arzela-Ascoli theorem, there exists a subsequence (g_{n_k}) converging uniformly on $[t_0, a]$, say, to g . This means that $g \in C[t_0, a]$ and $|g(t_2) - g(t_1)| \leq M|t_2 - t_1|$ for all $t_1, t_2 \in [t_0, a]$, so that $g \in S$. Therefore, S is compact.

Now,

$$|T(y)(t_2) - T(y)(t_1)| = \left| \int_{t_1}^{t_2} f(s, y(s)) \, ds \right| \Rightarrow |T(y)(t_2) - T(y)(t_1)| \leq M|t_2 - t_1|$$

for all $t_1, t_2 \in [t_0, a]$ and

$$|T(y) - y_0| = \left| \int_{t_0}^t f(s, y(s)) \, ds \right| \leq M(t - t_0) \leq M \frac{b}{M} = b,$$

so that T maps S to itself.

Also, T is continuous on S . Indeed, if $(y_n) \subset S$ and $y_n \xrightarrow{d_\infty} y$, then $y_n \rightarrow y$ uniformly, so that $f(s, y_n(s)) \rightarrow f(s, y(s))$ uniformly since f is uniformly continuous on R . Therefore, $T(y_n) \rightarrow T(y)$ uniformly, which implies that $T(y_n) \xrightarrow{d_\infty} T(y)$.

Finally, by the Schauder fixed point theorem, there exists $y \in S$ such that $T(y) = y$. This completes the proof. ■

5.4 Linear Operators

In calculus, we consider the real line $\mathbb{R}$ and real-valued functions on $\mathbb{R}$ (or on a subset of $\mathbb{R}$). Obviously, any such function is a mapping of its domain into $\mathbb{R}$. In functional analysis, we consider more general spaces, such as metric spaces and normed spaces, and mappings of these spaces.

In the case of vector spaces and, in particular, normed spaces, a mapping is called an **operator**.

Of special interest are operators that “preserve” the two algebraic operations of a vector space in the sense of the following definition.

Definition 5.4.1 Linear Operator

A **linear operator** T is an operator such that

1. The domain, denoted $\mathcal{D}(T)$, of T is a vector space and the range, denoted $\mathcal{R}(T)$, lies in a vector space over the same field; and
2. for all $x, y \in \mathcal{D}(T)$ and scalars α ,

$$T(x + y) = T(x) + T(y), \quad T(\alpha x) = \alpha T(x). \quad (5.10)$$

Definition 5.4.2 Null Space

The **null space** of a linear operator T , denoted $\mathcal{N}(T)$, is the set of all $x \in \mathcal{D}(T)$ such that $T(x) = 0$.

Note that (5.10) is equivalent to

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y) \text{ for all } x, y \in \mathcal{D}(T). \quad (5.11)$$

By taking $\alpha = 0$ in (5.10), we obtain the following formula:

$$T(0) = 0. \quad (5.12)$$

(5.10) expresses the fact that a linear operator T is a **homomorphism** of a vector space (its domain) into another vector space, that is, T preserves the two operations of a vector space in the following sense. In (5.10), on the left, we first apply a vector space operator (addition or multiplication by scalars) and then map the resulting vector into the range, call it Y , whereas on the right-hand side we first map x and y into Y and then perform the vector space operations in Y , the outcome being the same. This property makes linear operators important. In turn, vector spaces are important in functional analysis mainly because of the linear operators defined on them.

Example 5.4.1 Here are some basic examples of linear operators.

1. *Identity Operator*: The identity operator $I_X : X \rightarrow X$ is defined by $I_X(x) = x$ for all $x \in X$. We sometimes write simply I for I_X if the underlying set is understood.

2. *Zero Operator*: The zero operator $0 : X \rightarrow Y$ is defined by $0(x) = 0$ for all $x \in X$.
3. *Differentiation*: Let X be the vector space of all polynomials on $[a, b] \subset \mathbb{R}$. We may define a linear operator T on X by setting $T(x)(t) = x'(t)$ for every $x \in X$, where the prime denotes differentiation with respect to t . This operator maps T maps X onto itself.
4. *Integration*: A linear operator T from $C[a, b]$ into itself can be defined by

$$T(x)(t) = \int_a^t x(s) \, ds \text{ for all } t \in [a, b].$$

5. *Multiplication by t* : Another linear operator from $C[a, b]$ into itself is defined by

$$T(x)(t) = tx(t).$$

T plays a role in quantum theory, as we will see.

6. *Elementary Vector Algebra*: The *cross product* with one factor kept fixed defines a linear operator $T_1 : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Similarly, the *dot product* with one fixed factor defines a linear operator $T_2 : \mathbb{R}^3 \rightarrow \mathbb{R}$, say

$$T_2(x) = x \cdot a.$$

7. *Matrices*: A real-valued matrix $A = (\alpha_{jk})$ with r rows and n columns defines an operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^r$ by means of $y = T(x) = Ax$, where $x = (\xi_1, \dots, \xi_n)$ has n components and $y = (\eta_1, \dots, \eta_r)$ has r components, and both vectors are written as column vectors because of the usual convention of matrix multiplication. Writing $y = Ax$ out, we have

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_r \end{pmatrix} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ \vdots & \vdots & \dots & \vdots \\ \alpha_{r1} & \alpha_{r2} & \dots & \alpha_{rn} \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \vdots \\ \xi_n \end{pmatrix}.$$

T is linear because matrix multiplication is a linear operation. If A were a complex-valued matrix, it would define a linear operator from $\mathbb{C}^n$ to $\mathbb{C}^r$.

Theorem 5.4.1 Range and Null Space

Let T be a linear operator. Then

1. The range $\mathcal{R}(T)$ is a vector space.
2. If $\dim(T) = n < \infty$, then $\dim(\mathcal{R}(T)) \leq n$.
3. The null space $\mathcal{N}(T)$ is a vector space.

Corollary 5.4.1

Linear operators preserve linear dependence.

Let us turn to the inverse of a linear operator. We first remember that a mapping $T : \mathcal{D}(T) \rightarrow Y$ is called **injective** or **one-to-one** if different points in the domain have different images, that is, if for any $x_1, x_2 \in \mathcal{D}(T)$,

$$T(x_1) = T(x_2) \rightarrow x_1 = x_2. \quad (5.13)$$

Also, T is called **surjective**, or **onto**, if $\mathcal{R}(T) = Y$, or equivalently, if for all points $y \in Y$ there exists a point $x \in X$ such that $T(x) = y$.

T is called **bijective** if it is injective and surjective (one-to-one and onto).

Definition 5.4.3 Inverse Operator

Let X and Y be vector spaces and $T : X \rightarrow Y$ be a bijective linear operator. The mapping $T^{-1} : Y \rightarrow X$ defined by $T^{-1}(y) = x$ for all $y \in Y$ and all $x \in X$, is called the **inverse** of T .

REMARK: Note that the inverse operator is only defined for bijective linear operators. If an operator has an inverse, it is sometimes called **invertible**. So all bijective linear operators are invertible.

It is clear from the definition of an inverse operator that for an invertible operator $T : X \rightarrow Y$

$$\begin{aligned} T^{-1}(T(x)) &= x \text{ for all } x \in X, \\ T(T^{-1}(y)) &= y \text{ for all } y \in Y. \end{aligned}$$

Theorem 5.4.2 Inverse Operator

Let X and Y be vector spaces, both real or both complex. Let $T : X \rightarrow Y$ be a linear operator. Then:

1. The inverse $T^{-1} : Y \rightarrow X$ exists if and only if $\mathcal{N}(T) = \{0\}$, i.e., if and only if $T(x) = 0 \Rightarrow x = 0$.
2. If T^{-1} exists, it is a linear operator.
3. If $\dim(X) = n < \infty$ and T^{-1} exists, then $\dim(X) = \dim(Y)$.

PROOF:

1. Suppose that $T(x) = 0$ implies $x = 0$. Let $T(x_1) = T(x_2)$. Since T is linear,

$$T(x_1 - x_2) = T(x_1) - T(x_2) = 0,$$

so that $x_1 - x_2 = 0$ by the hypothesis. Hence, $T(x_1) = T(x_2)$ implies $x_1 = x_2$, and T^{-1} exists by (5.13). Conversely, if T^{-1} exists, then (5.13) holds. From (5.13) with $x_2 = 0$ and the fact that $T(0) = 0$, we obtain $T(x_1) = T(0) = 0 \Rightarrow x_1 = 0$.

2. We assume that T^{-1} exists and show that T^{-1} is linear. The domain of T^{-1} is Y and is a vector space by Theorem 5.4.1. We consider any $x_1, x_2 \in X$ and their images

$$y_1 = T(x_1) \quad \text{and} \quad y_2 = T(x_2).$$

Then

$$x_1 = T^{-1}(y_1) \quad \text{and} \quad x_2 = T^{-1}(y_2).$$

T is linear, so that for any scalars α and β , we have

$$\alpha y_1 + \beta y_2 = \alpha T(x_1) + \beta T(x_2) = T(\alpha x_1 + \beta x_2).$$

Since $x_j = T^{-1}(y_j)$ for $j = 1, 2$, this implies

$$T^{-1}(\alpha y_1 + \beta y_2) = \alpha x_1 + \beta x_2 = \alpha T^{-1}(y_1) + \beta T^{-1}(y_2),$$

and proves that T^{-1} is linear.

3. We have $\dim(Y) \leq \dim(X)$ by Theorem 5.4.1, and $\dim(X) \leq \dim(Y)$ by the same theorem applied to T^{-1} . ■

Let us now consider the *product* of linear operators. Let $T : X \rightarrow Y$ and $S : Y \rightarrow Z$ be linear operators, where X, Y, Z are vector spaces. Then the product $ST : X \rightarrow Z$ is defined as

$$(ST)(x) := (S \circ T)(x) = S(T(x)) \text{ for all } x \in X.$$

Definition 5.4.4 Commuting Operators

Let X be any vector space and $S : X \rightarrow X$ and $T : X \rightarrow X$ any two operators on X . S and T are said to **commute** if $ST = TS$, that is, if $(ST)(x) = (TS)(x)$ for all $x \in X$.

Lemma 5.4.1

Inverse of Product Let $T : X \rightarrow Y$ and $S : Y \rightarrow Z$ be bijective linear operators, where X, Y, Z are vector spaces. Then the inverse $(ST)^{-1} : Z \rightarrow X$ of the product (i.e., the composition) ST exists and

$$(ST)^{-1} = T^{-1}S^{-1}. \quad (5.14)$$

PROOF: The operator $ST : X \rightarrow Z$ is bijective, so that its inverse $(ST)^{-1}$ exists. We thus have

$$(ST)(ST)^{-1} = I_Z,$$

where I_Z is the identity operator on Z . Applying S^{-1} and using $S^{-1}S = I_Y$ (the identity operator on Y), we obtain

$$S^{-1}ST(ST)^{-1} = T(ST)^{-1} = S^{-1}I_Z = S^{-1}.$$

Applying T^{-1} and using $T^{-1}T = I_X$, we obtain the desired result

$$T^{-1}T(ST)^{-1} = (ST)^{-1} = T^{-1}S^{-1}. \quad \blacksquare$$

Definition 5.4.5 Bounded Below Linear Operator

Let $L : X \rightarrow Y$ be a linear operator, where $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are normed linear spaces. L is called **bounded below** if there exists $m > 0$ such that $\|L(x)\|_Y \geq m \|x\|_X$ for all $x \in X$.

5.5 Bounded and Continuous Linear Operators

Let us now take norms into account when considering linear operators.

Definition 5.5.1 Bounded Linear Operator

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces and $T : X \rightarrow Y$ a linear operator. T is called **bounded** if there is a $c \in \mathbb{R}$ such that

$$\|T(x)\|_Y \leq c \|x\|_X \text{ for all } x \in X. \quad (5.15)$$

REMARK: (5.15) shows that a bounded linear operator maps bounded sets in X onto bounded sets in Y . This is what motivates the term “bounded operator”.

Also, note that the present use of the word “bounded” is different from that in calculus, where a bounded function is one whose range is a bounded set.

Now, what is the smallest possible c such that (5.15) still holds for all non-zero $x \in X$? (We can leave out $x = 0$ since $T(x) = 0$ for $x = 0$.) By division,

$$\frac{\|T(x)\|_Y}{\|x\|_X} \leq c, \quad x \neq 0,$$

and this shows that c must be at least as big as the supremum of the expression on the left-hand side taken over $X - \{0\}$. Hence, the answer to our question is that the smallest possible c in (5.15) is that supremum.

Definition 5.5.2 Operator Norm

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed linear spaces and $T : X \rightarrow Y$. The quantity

$$\|T\| := \sup_{x \neq 0} \frac{\|T(x)\|_Y}{\|x\|_X} \quad (5.16)$$

is called the **norm** of T . If $X = \{0\}$, we define $\|T\| = 0$.

So we *define* the smallest possible c in (5.15) to be the *operator norm*.

Note that taking $c = \|T\|$ in (5.15) gives

$$\|T(x)\|_Y \leq \|T\| \|x\|_X. \quad (5.17)$$

Lemma 5.5.1 **Norm**

Let $T : X \rightarrow Y$ be a bounded linear operator. Then

1. The operator norm is a norm.
2. An alternative formula for the norm of T is

$$\|T\| = \sup_{\|x\|_X=1} \|T(x)\|_Y. \quad (5.18)$$

PROOF:

1. (Pending...use the definition, not the second part of this lemma!)
2. We write $\|x\|_X = a$ and set $y = \frac{1}{a}x$, where $x \neq 0$. Then $\|y\|_X = \frac{\|x\|_X}{a} = 1$, and since T is linear, (5.16) gives

$$\|T\| = \sup_{x \neq 0} \frac{1}{a} \|T(x)\|_Y = \sup_{x \neq 0} \left\| T\left(\frac{1}{a}x\right) \right\|_Y = \sup_{\|y\|_X=1} \|T(y)\|_Y. \quad \blacksquare$$

Example 5.5.1 Let us look at some typical examples of bounded linear operators.

1. *Identity Operator*: The identity operator $I : X \rightarrow X$ on a non-empty normed space $(X, \|\cdot\|_X)$ is bounded and has a norm $\|I\| = 1$.
2. *Zero Operator*: The zero operator $0 : X \rightarrow Y$ on a normed space $(X, \|\cdot\|_X)$ is bounded and has norm $\|0\| = 0$.
3. *Differentiation Operator*: Let $(X, \|\cdot\|)$ be the normed space of all polynomials on $[0, 1] \subset \mathbb{R}$ with norm $\|x\| = \max_{0 \leq t \leq 1} |x(t)|$. A differentiation operator T is defined on X by

$$T(x(t)) = x'(t),$$

where the prime denotes differentiation with respect to t . This operator is linear but not bounded. Indeed, let $x_n(t) = t^n$, where $n \in \mathbb{N}$. Then $\|x_n\| = 1$ and

$$T(x_n(t)) = x'_n(t) = nt^{n-1},$$

so that $\|T(x_n)\| = n$ and $\frac{\|T(x_n)\|}{\|x_n\|} = n$. Since $n \in \mathbb{N}$ is arbitrary, this shows that there is no *fixed* number c such that $\frac{\|T(x_n)\|}{\|x_n\|} \leq c$. From this, and (5.15), we conclude that T is not bounded.

4. *Integral Operator*: We can define an integral operator $T : C[0, 1] \rightarrow C[0, 1]$ (with norm on both copies of $C[0, 1]$ being $\|\cdot\|_\infty$) by

$$y = T(x) \quad \text{where} \quad y(t) = \int_0^1 k(t, s)x(s) \, ds.$$

Here, k is a given function, which is called the *kernel* of T , and is assumed to be continuous on the closed square $[0, 1] \times [0, 1]$. This operator is linear and it is bounded. To prove the

latter, we first note that the continuity of k on the closed square implies that k is bounded, say $|k(t, s)| \leq k_0$ for all $(t, s) \in [0, 1] \times [0, 1]$, where $k_0 \in \mathbb{R}$. Furthermore,

$$|x(t)| \leq \max_{0 \leq t \leq 1} |x(t)| = \|x\|_\infty.$$

Hence,

$$|y(t)| = |T(x)(t)| = \left| \int_0^1 k(t, s)x(s) \, ds \right| \leq \max_{0 \leq t \leq 1} \int_0^1 |k(t, s)||x(s)| \, ds \leq \int_0^1 |k(t, s)| \|x\|_\infty \, ds,$$

so that

$$\|T(x)\|_\infty \leq \left(\max_{0 \leq t \leq 1} \int_0^1 k(t, s) \, ds \right) \|x\|_\infty.$$

Therefore, T is bounded.

5. *Matrices:* A real-valued matrix $A = (\alpha_{jk})$ with r rows and n columns defines an operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^r$ by means of

$$T(x) = y = Ax, \quad (5.19)$$

where $x = (\xi_j)$ and $y = (\eta_j)$ are columns vectors with n and r components, respectively, and we used matrix multiplication. In terms of components, (5.19) becomes

$$\eta_j = \sum_{k=1}^n \alpha_{jk} \xi_k \quad (5.20)$$

T is linear because matrix multiplication is a linear operation, and it is also bounded. To prove the latter, let us take the Euclidean norm on $\mathbb{R}^n$ and $\mathbb{R}^r$. From (5.20) and the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} \|T(x)\|^2 &= \sum_{j=1}^r \eta_j^2 = \sum_{j=1}^r \left(\sum_{k=1}^n \alpha_{jk} \xi_k \right)^2 \leq \sum_{j=1}^r \left[\left(\sum_{k=1}^n \alpha_{jk}^2 \right)^{1/2} \left(\sum_{k=1}^n \xi_k^2 \right)^{1/2} \right]^2 \\ &= \|x\|^2 \sum_{j=1}^r \sum_{k=1}^n \alpha_{jk}^2. \end{aligned}$$

Noting that the double sum in the last line does not depend on x , we can write our result in the form

$$\|T(x)\|^2 \leq c^2 \|x\|^2 \quad \text{where} \quad c^2 = \sum_{j=1}^r \sum_{k=1}^n \alpha_{jk}^2.$$

This proves that T is bounded.

Example 5.5.2 Integral Operator

Let us continue the fourth example above about the integral operator. We had

$$\|T(f)\|_\infty \leq \left(\max_{0 \leq t \leq 1} \int_0^1 |k(t,s)| \, ds \right) \|f\|_\infty.$$

By definition of the operator norm, we have

$$\|T\| \leq \max_{0 \leq t \leq 1} \int_0^1 |k(t,s)| \, ds.$$

Let us now prove that in fact

$$\|T\| = \max_{0 \leq t \leq 1} \int_0^1 |k(t,s)| \, ds.$$

The right-hand side of the above equation can be written as $\int_0^1 |k(t_0,s)| \, ds$ for some t_0 . Let's now show that $k(t_0,s)$ is uniformly continuous on $[0,1]$. Given $\epsilon > 0$, there exists $\delta > 0$ such that

$$|k(t_0,s_2) - k(t_0,s_1)| < \epsilon \quad \text{for } |s_2 - s_1| < \delta.$$

Let $A_\epsilon = \{s \mid |k(t_0,s)| \leq \epsilon\}$. A_ϵ is compact since

$$A_\epsilon \subset \bigcup_{s \in A_\epsilon} (s - \delta, s + \delta) \Rightarrow A_\epsilon \subset V_\epsilon := \bigcup_{i=1}^N (s_i - \delta, s_i + \delta).$$

(Every open cover of A_ϵ has a finite subcover.) Let $\tilde{V}_\epsilon = V_\epsilon \cap [0,1]$, $\tilde{U}_\epsilon = [0,1] - \tilde{V}_\epsilon \Rightarrow A_\epsilon \subset \tilde{V}_\epsilon$. Let

$$f_\epsilon(s) := \frac{k(t_0,s)}{|k(t_0,s)|} \text{ for } s \in \tilde{U}_\epsilon$$

and extend f_ϵ linearly so that $|f_\epsilon(s)| \leq 1 \Rightarrow \|f_\epsilon\|_\infty = 1$. Now,

$$\begin{aligned} T(f_\epsilon)(t_0) &= \int_{\tilde{U}_\epsilon} k(t_0,s) f_\epsilon(s) \, ds + \int_{\tilde{V}_\epsilon} k(t_0,s) f_\epsilon(s) \, ds \\ &= \int_{\tilde{U}_\epsilon} |k(t_0,s)| \, ds + \int_{\tilde{V}_\epsilon} k(t_0,s) f_\epsilon(s) \, ds. \end{aligned}$$

For $s \in \tilde{V}_\epsilon$, $|s - s_i| < \delta$ for some $s_i \in A_\epsilon \Rightarrow |k(t_0,s) - k(t_0,s_i)| < \epsilon$, which implies that

$$|k(t_0,s)| < |k(t_0,s_i)| + \epsilon < 2\epsilon \Rightarrow \int_{\tilde{V}_\epsilon} |k(t_0,s)| \, ds < 2\epsilon.$$

Thus,

$$\begin{aligned} T(f_\epsilon)(t_0) &\geq \int_{\tilde{U}_\epsilon} |k(t_0,s)| \, ds - \int_{\tilde{V}_\epsilon} |k(t_0,s)| \, ds \\ &= \int_0^1 |k(t_0,s)| \, ds - 2 \int_{\tilde{V}_\epsilon} |k(t_0,s)| \, ds \geq \int_0^1 |k(t_0,s)| \, ds - 4\epsilon. \end{aligned}$$

For ϵ sufficiently small, the right-hand side is positive, which means that

$$|T(f_\epsilon)(t_0)| \geq \int_0^1 |k(t_0,s)| \, ds - 4\epsilon \Rightarrow \frac{\|T(f_\epsilon)\|_\infty}{\|f_\epsilon\|_\infty} \geq \int_0^1 |k(t_0,s)| \, ds - 4\epsilon.$$

Therefore,

$$\|T\| \geq \int_0^1 |k(t_0, s)| \, ds - 4\epsilon.$$

Now, let $\epsilon \rightarrow 0$. Then $\|T\| \geq \int_0^1 |k(t_0, s)| \, ds$. Therefore, $\|T\| = \max_{0 \leq t \leq 1} \int_0^1 |k(t, s)| \, ds$.

Example 5.5.3 Matrices

Let us again look at the matrix operators $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, this time with the infinity norm $\|\cdot\|_\infty$ on both $\mathbb{R}^n$ and $\mathbb{R}^m$. Let $L(x) = Ax$, $A = (a_{ij})$. Then,

$$|(L(x))_i| \leq \sum_{j=1}^n |a_{ij}| |x_j| \leq \sum_{j=1}^n |a_{ij}| \|x\|_\infty \Rightarrow \|L(x)\|_\infty \leq \left(\max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}| \right) \|x\|_\infty.$$

Similarly to the previous example with the integral operator, let us now show that

$$\|L\| = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|.$$

The right-hand side of the above equation is $\sum_{j=1}^n |a_{i_0 j}|$ for some i_0 . Let

$$\hat{x}_j = \begin{cases} 1 & \text{if } a_{i_0 j} \geq 0 \\ -1 & \text{otherwise} \end{cases}.$$

Then $\|\hat{x}\|_\infty = 1$, and

$$\|L(\hat{x})\|_\infty = \max_i \left| \sum_j a_{ij} \hat{x}_j \right| \geq \left| \sum_j a_{i_0 j} \hat{x}_j \right| = \sum_j |a_{i_0 j}|.$$

Thus, $\frac{\|L(\hat{x})\|_\infty}{\|\hat{x}\|_\infty} = \max_i \sum_j |a_{ij}|$. Therefore, $\|L\| = \max_i \sum_j |a_{ij}|$.

Theorem 5.5.1 Finite Dimension

If a normed space $(X, \|\cdot\|)$ is finite dimensional, then every linear operator on X is bounded.

PROOF: Let $\dim(X) = n$ and $\{e_1, \dots, e_n\}$ a basis for X . We take any $x = \sum_j \xi_j e_j$ and consider any linear operator T on X . Since T is linear,

$$\|T(x)\| = \left\| \sum_{j=1}^n \xi_j T(e_j) \right\| \leq \sum_{j=1}^n |\xi_j| \|T(e_j)\| \leq \max_k \|T(e_k)\| \sum_{j=1}^n |\xi_j|.$$

To the last sum we apply Lemma 5.2.3 with $\alpha_j = \xi_j$ and $x_j = e_j$. Then we obtain

$$\sum_{j=1}^n |\xi_j| \leq \frac{1}{c} \left\| \sum_{j=1}^n \xi_j e_j \right\| = \frac{1}{c} \|x\|.$$

Together,

$$\|T(x)\| \leq \gamma \|x\| \quad \text{where} \quad \gamma = \frac{1}{c} \max_k \|T(e_k)\|.$$

Therefore, T is bounded. ■

Operators are mappings, so that the definition of continuity in Definition 3.3.8 applies to them.

Definition 5.5.3 Continuous Mapping

Let $T : X \rightarrow Y$ be an operator (not necessarily linear) between two normed space $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$. T is called *continuous at x_0* if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$\|T(x) - T(x_0)\|_Y < \epsilon \text{ for all } x \in X \text{ such that } \|x - x_0\|_X < \delta.$$

T is called **continuous** if T is continuous at every $x \in X$.

Theorem 5.5.2 Continuity and Boundedness

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be two normed spaces and $T : X \rightarrow Y$ a linear operator. Then

1. T is continuous if and only if T is bounded.
2. If T is continuous at a single point, it is continuous. In particular, T is continuous if and only if T is continuous at 0.

PROOF:

1. For $T = 0$, the statement is trivial. Let $T \neq 0$. Then $\|T\| \neq 0$. We assume T to be bounded and consider any $x_0 \in X$. Let any $\epsilon > 0$ be given. Then, since T is linear, for every $x \in X$ such that

$$\|x - x_0\|_X \text{ such that } \delta = \frac{\epsilon}{\|T\|}$$

we obtain

$$\|T(x) - T(x_0)\|_Y = \|T(x - x_0)\|_Y \leq \|T\| \|x - x_0\|_X < \|T\| \delta = \epsilon.$$

Since $x_0 \in X$ was arbitrary, this shows that T is continuous.

Conversely, assume that T is continuous at an arbitrary $x_0 \in X$. Then, given any $\epsilon > 0$, there is a $\delta > 0$ such that

$$\|T(x) - T(x_0)\|_Y \leq \epsilon \text{ for all } x \in X \text{ satisfying } \|x - x_0\|_X \leq \delta. \quad (5.21)$$

We now take any $y \neq 0$ in X and set

$$x = x_0 + \frac{\delta}{\|y\|_X} y \Rightarrow x - x_0 = \frac{\delta}{\|y\|_X} y.$$

Hence, $\|x - x_0\|_X = \delta$, so that we may use (5.21). Since T is linear, we have

$$\|T(x) - T(x_0)\|_Y = \|T(x - x_0)\|_Y = \left\| T\left(\frac{\delta}{\|y\|_X} y\right) \right\|_Y = \frac{\delta}{\|y\|_X} \|T(y)\|_Y,$$

and (5.21) implies

$$\frac{\delta}{\|y\|_X} \|T(y)\|_Y \leq \epsilon \Rightarrow \|T(y)\|_Y \leq \frac{\epsilon}{\delta} \|y\|_X.$$

This can be written as $\|T(y)\|_Y \leq c \|y\|_X$, where $c = \frac{\epsilon}{\delta}$, and shows that T is bounded.

2. Continuity of T at a point implies boundedness of T by the second part of the proof of 1, which in turn implies continuity of T by the first part as well.

In particular, assuming T is continuous at 0, we can show that T is continuous at x as follows: suppose the sequence $(x_n) \subset X$ converges to x , i.e., $x_n \rightarrow x$. Then

$$\|T(x_n) - T(x)\|_Y = \|T(x_n - x)\|_Y.$$

Since $x_n - x \rightarrow 0$, we have $T(x_n - x) \rightarrow 0$, which implies that $T(x_n) - T(x) \rightarrow 0$, which implies that $T(x_n) \rightarrow T(x)$. So T is continuous at $x \in X$. Since x was arbitrary, we have that T is continuous (on X). The converse is proven by reversing these arguments. ■

Corollary 5.5.1 Continuity, Null Space

Let $T : X \rightarrow Y$ be a bounded linear operator and X and Y normed linear spaces. Then

1. For any convergent sequence $(x_n) \subset X$, say $\lim_{n \rightarrow \infty} x_n = x$, we have that the sequence $(T(x_n)) \subset Y$ converges to $T(x)$, i.e., $\lim_{n \rightarrow \infty} T(x_n) = T(x)$.
2. The null space $\mathcal{N}(T)$ is closed.

PROOF:

1. As $n \rightarrow \infty$,

$$\|T(x_n) - T(x)\| = \|T(x_n - x)\| \leq \|T\| \|x_n - x\| \rightarrow 0.$$

2. For every $x \in \overline{\mathcal{N}(T)}$, there is a sequence $(x_n) \in \mathcal{N}(T)$ such that $x_n \rightarrow x$; recall 3.3.1. Hence, $T(x_n) \rightarrow T(x)$ by the first part of this corollary. Also, $T(x) = 0$ since $T(x_n) = 0$, so that $x \in \mathcal{N}(T)$. Since $x \in \overline{\mathcal{N}(T)}$ was arbitrary, $\mathcal{N}(T)$ is closed. ■

Example 5.5.4 Continuity of the Differentiation Operator

Let $D : (C^1[0, 1], \|\cdot\|_\infty) \rightarrow (C[0, 1], \|\cdot\|_\infty)$ be the differentiation operator, i.e., $D = \frac{d}{dt}$. D is not continuous at 0. To prove this, let $f_n(t) = \frac{1}{n} \sin(n\pi t)$. Then

$$\|f_n\| = \frac{1}{n} \Rightarrow f_n \rightarrow 0.$$

Then $D(f_n)(t) = f'_n(t) = \pi \cos(n\pi t) \Rightarrow \|D(f_n)\| = \pi$, which implies that the sequence $(D(f_n))$ does not converge to 0. This proves that D is not continuous at 0. Therefore D is not continuous anywhere. By the previous theorem, this means that D is an unbounded operator. Indeed, letting $g_n(t) = \sin(n\pi t)$, we have $\|g_n\| = 1$, and

$$\|D(g_n)\| = \|n\pi \cos(n\pi t)\| = n\pi \rightarrow \infty \text{ as } n \rightarrow \infty.$$

So $\|D\| = \infty$, and hence D is unbounded.

Now, we proved that D is not continuous anywhere under the $\|\cdot\|_\infty$ norm. If, instead, we consider the differentiation operator $D : C^1[0, 1] \rightarrow C[0, 1]$ such that

$$\begin{aligned} \|f\|_X &= \max\{\|f\|_\infty, \|f'\|_\infty\} \text{ and} \\ \|g\|_Y &= \|g\|_\infty, \end{aligned}$$

then we can show that D is continuous. Consider $f_n(t) = \frac{1}{\pi n^2} \sin(n\pi t)$. Then

$$\|f_n\|_X = \max\left\{\frac{1}{\pi n^2}, \frac{1}{n}\right\} = \frac{1}{n}.$$

As $n \rightarrow \infty$, we therefore have $\|f_n\|_X \rightarrow 0 \Rightarrow f_n \rightarrow 0$. Now,

$$D(f_n) = \frac{1}{n} \cos(n\pi t) \Rightarrow \|D(f_n)\|_Y = \left\|\frac{1}{n} \cos(n\pi t)\right\|_\infty = \frac{1}{n},$$

which means that $\|D(f_n)\|_Y \rightarrow 0$ as $n \rightarrow \infty$, i.e., $D(f_n) \rightarrow 0$. This proves that D is continuous at 0, which proves that D is continuous on X . This proves that D is bounded. Indeed, let $g_n(t) = \frac{1}{n\pi} \sin(n\pi t)$. Then

$$\|g_n\|_X = \max\left\{\frac{1}{n\pi}, 1\right\} = 1,$$

so that $D(g_n)(t) = \cos(n\pi t) \Rightarrow \|D(g_n)\|_Y = \|\cos(n\pi)\|_\infty = 1$, which implies that $\|D\| \leq 1$.

REMARK: The last part of the above example leads to the following natural infinity norm on the space $C^n[a, b]$ of k -times continuously differentiable functions on $[a, b]$, i.e., on the space

$$C^n[a, b] = \{f : [a, b] \rightarrow \mathbb{R} \mid f^{(n)} \in C[a, b]\}$$

(Note that $f^{(n)} \in C[a, b] \Rightarrow f^{(n-1)} \in C[a, b]$):

$$\|f\|_{n,\infty} := \max_{0 \leq k \leq n} \{\|f^{(k)}\|_\infty\}.$$

For example, if $f(t) = \sin(5t)$ on $[-\pi, \pi]$, then $\|f^{(k)}\|_\infty = 5^k$, $k = 0, 1, 2, \dots$, so that $\|f\|_{n,\infty} = 5^n$, $n = 0, 1, 2, \dots$. This norm defines the following metric on $C^n[a, b]$:

$$d_{n,\infty}(f, g) := \|f - g\|_{n,\infty} = \max_{1 \leq k \leq n} \{\|f^{(k)} - g^{(k)}\|_\infty\}.$$

It is easy to prove the following formulas,

$$\|T_1 T_2\| \leq \|T_1\| \|T_2\| \quad \text{and} \quad \|T^n\| \leq \|T\|^n, \quad n \in \mathbb{N}, \quad (5.22)$$

where $T_2 : X \rightarrow Y$, $T_1 : Y \rightarrow Z$, and $T : X \rightarrow X$, where X, Y, Z are normed spaces.

We now state some further definitions.

Definition 5.5.4

Two operators T_1 and T_2 are called **equal**, written $T_1 = T_2$ if they have the same domain and if $T_1(x) = T_2(x)$ for all x in the domain.

The **restriction** of an operator $T : X \rightarrow Y$ to a subset $B \subset X$ is denoted $T|_B : B \rightarrow Y$ and it is defined by

$$T|_B(x) = T(x) \quad \forall x \in B.$$

An **extension** of T to a set $M \supset X$ is an operator $\tilde{T} : M \rightarrow Y$ such that $\tilde{T}|_X = T$, that is, $\tilde{T}(x) = T(x)$ for all $x \in X$. Hence, T is the restriction of $\tilde{T}$ to X .

Theorem 5.5.3 Bounded Linear Extensions

Let $T : X \rightarrow Y$ be a bounded linear operator such that Y is a Banach space. Then T has an extension

$$\tilde{T} : \bar{X} \rightarrow Y$$

such that $\tilde{T}$ is a bounded linear operator with norm $\|\tilde{T}\| = \|T\|$.

PROOF: We consider any $x \in X$. By Theorem 3.3.1, there is a sequence $(x_n) \subset X$ such that $\lim_{n \rightarrow \infty} x_n = x$. Since T is linear and bounded, we have

$$\|T(x_n) - T(x_m)\| = \|T(x_n - x_m)\| \leq \|T\| \|x_n - x_m\|.$$

This shows that $(T(x_n))$ is a Cauchy sequence in Y because (x_n) converges. By assumption, Y is complete (being a Banach space), so that $(T(x_n))$ converges (being a Cauchy sequence), so that $\lim_{n \rightarrow \infty} T(x_n) = y$ for some $y \in Y$. Now, define $\tilde{T}$ by

$$\tilde{T}(x) = y.$$

We show that this definition is independent of the particular choice of a sequence in X converging to x . Suppose that $x_n \rightarrow x$ and $z_n \rightarrow x$. Then $v_m \rightarrow x$, where (v_m) is the sequence $(x_1, z_1, x_2, z_2, \dots)$. Hence, $(T(v_m))$ converges by Corollary 5.5.1, and the two subsequences $(T(x_n))$ and $(T(z_n))$ of $(T(v_m))$ have the same limit. This proves that $\tilde{T}$ is uniquely defined at every $x \in \bar{X}$.

Clearly, $\tilde{T}$ is linear and $\tilde{T}(x) = T(x)$ for all $x \in X$, so that $\tilde{T}$ is an extension of T . We now use

$$\|T(x_n)\| \leq \|T\| \|x_n\|$$

and let $n \rightarrow \infty$. Then $T(x_n) \rightarrow y = \tilde{T}(x)$. Since $x \mapsto \|x\|$ is a continuous mapping, we thus obtain

$$\|\tilde{T}(x)\| \leq \|T\| \|x\|.$$

Hence, $\tilde{T}$ is bounded and $\|\tilde{T}\| \leq \|T\|$. Of course, $\|\tilde{T}\| \geq \|T\|$ because the norm, being defined by a supremum, cannot decrease in an extension. Together, we have $\|\tilde{T}\| = \|T\|$. ■

5.5.1 Inverse of Linear Operators

Theorem 5.5.4 Norm of the Inverse

A linear operator $L : X \rightarrow Y$ on normed linear spaces $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ has a bounded inverse if and only if L is bounded below. In this case,

$$\|L^{-1}\| = \frac{1}{\inf_{\|x\|=1} \|L(x)\|_Y}.$$

PROOF: Suppose L^{-1} is bounded. Then there exists $M > 0$ such that $\|L^{-1}(y)\|_X \leq M \|y\|_Y$ for all $y \in \mathcal{R}(L)$, where remember $\mathcal{R}(L)$ is a subspace of Y . Let $y = L(x)$. Then

$$\|x\|_X \leq \|L(x)\|_Y \Rightarrow \|L(x)\|_Y \geq \frac{1}{M} \|x\|_X \text{ for all } x \in X.$$

So L is bounded below.

Conversely, suppose that L is bounded below. Then there exists $m > 0$ such that $\|L(x)\|_Y \geq m \|x\|_X$ for all $x \in X$. L is one-to-one because its kernel consists of only the zero vector: $L(x) = 0 \Rightarrow \|x\|_X = 0 \Rightarrow x = 0$. Thus $L^{-1} : \mathcal{R}(L) \rightarrow X$ exists (by construction L^{-1} is onto). L^{-1} is linear because

$$L^{-1}(\beta_1 y_1 + \beta_2 y_2) = z \Rightarrow \beta_1 y_1 + \beta_2 y_2 = L(z) \text{ for all } y_1, y_2 \in \mathcal{R}(L).$$

Let $L^{-1}(y_1) = x_1$, $L^{-1}(y_2) = x_2$, so that $y_1 = L(x_1)$ and $y_2 = L(x_2)$. Then,

$$L(\beta_1 x_1 + \beta_2 x_2) = \beta_1 L(x_1) + \beta_2 L(x_2) = \beta_1 y_1 + \beta_2 y_2 = L(z).$$

Therefore,

$$L^{-1}(\beta_1 y_1 + \beta_2 y_2) = \beta_1 L^{-1}(y_1) + \beta_2 L^{-1}(y_2),$$

so that L^{-1} is linear. Now, in the expression $\|L(x)\|_Y \geq m \|x\|_X$, let $x = L^{-1}(y)$, so that $\|y\|_Y \geq m \|L^{-1}y\|_X \Rightarrow \|L^{-1}(y)\|_X \leq \frac{1}{m} \|y\|_Y$. Therefore, L^{-1} is bounded. Now,

$$\|L^{-1}\| = \sup_{y \in \mathcal{R}(L)} \frac{\|L^{-1}(y)\|_X}{\|y\|_Y} = \sup_{x \neq 0} \frac{\|x\|_X}{\|L(x)\|_Y} = \frac{1}{\inf_{x \neq 0} \frac{\|L(x)\|_Y}{\|x\|_X}} = \frac{1}{\inf_{\|x\|_X=1} \|L(x)\|_Y}. \quad \blacksquare$$

Example 5.5.5 Inverse of the Differentiation Operator

Let $D : X \rightarrow Y$, where $D = \frac{d}{dt}$, $X = \{f \in C^1[0, 1] \mid f(0) = 0\}$ and $Y = C[0, 1]$, with the norm being $\|\cdot\|_\infty$ on X and Y . Note that $\mathcal{N}(D) = \{0\}$, so that D^{-1} exists. Now, we can write any $f \in X$ as

$$f(t) = \int_0^t f'(s) \, ds.$$

Indeed, we have $f(0) = 0$, and f is certainly C^1 . Therefore, $D^{-1}(g)(t) = \int_0^t g(s) \, ds$. Indeed, we have $D\left(\int_0^t g(s) \, ds\right) = \frac{d}{dt} \int_0^t g(s) \, ds = g(t)$, so that $(DD^{-1})(g) = g$. Now,

$$|D^{-1}(g)(t)| \leq \int_0^t |g(s)| \, ds \leq \|g\|_\infty t \Rightarrow \|D^{-1}(g)\|_\infty \leq \|g\|_\infty \Rightarrow \|D^{-1}\| = \sup_{g \neq 0} \frac{\|D^{-1}(g)\|_\infty}{\|g\|_\infty} \leq 1.$$

Equality in the above expression holds if we let $g(t) = 1$, so that the supremum exists (i.e., it is achieved). Therefore, $\|D^{-1}\| = 1$. Alternatively,

$$|f(t)| \leq \int_0^t |f'(s)| \, ds \Rightarrow \|f\|_\infty \leq \|f'\|_\infty = \|D(f)\|.$$

So D is bounded from below and $\inf_{\|f\|=1} \|D(f)\|_\infty \geq 1$, with equality holding for $f(t) = t$, so that $\inf_{\|f\|_\infty=1} \|D(f)\|_\infty = 1$. Therefore, by the previous theorem, $\|D^{-1}\| = \frac{1}{1} = 1$.

As a side note, observe that if we let $g(t) = a$, for $a > 0$, then

$$f(t) = \int_0^t a \, ds = at \Rightarrow \|f\|_\infty = a \rightarrow 0 \text{ as } a \rightarrow 0^+.$$

This shows that D^{-1} is not bounded below.

Example 5.5.6 Let $L : \ell_\infty \rightarrow \ell_\infty$ with the norm $\|\cdot\|_\infty$ on both sides defined by

$$L(x_1, x_2, \dots) = \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots\right).$$

L is linear and one-to-one. Now, let $e_n := (0, \dots, 0, 1, 0, \dots)$, with 1 in the i th position. For $\|x\|_\infty = 1$, we have

$$\|L(x)\|_\infty = \sup_i \frac{|x_i|}{i} \leq 1 \quad \text{and} \quad \|L(e_1)\|_\infty = 1 \Rightarrow \|L\| = 1.$$

Let $\|L(e_n)\|_\infty = \frac{1}{n}$. Since $\|e_n\|_\infty = 1$, we have that L is not bounded from below. In particular, since

$$L^{-1}(y_1, y_2, \dots) = (y_1, 2y_2, 3y_3, \dots),$$

we have that

$$L^{-1}(e_m) = me_m \Rightarrow \|L^{-1}(e_m)\|_\infty = m.$$

This shows that L^{-1} is not bounded.

Definition 5.5.5 Condition Number

If X is a normed linear space and $T : X \rightarrow X$ is a bounded linear operator with a bounded inverse, then the **condition number** of T is defined as

$$k(T) := \|T\| \|T^{-1}\|.$$

Theorem 5.5.5

Let X be a normed linear space and $T : X \rightarrow X$ a bounded linear operator with a bounded inverse. If x^* is the unique solution to $T(x) = b$ and $x^* + \Delta x^*$ is the unique solution to $T(x) = b + \Delta b$, then

$$\frac{\|\Delta x^*\|}{\|x^*\|} \leq k(T) \frac{\|\Delta b\|}{\|b\|}.$$

In words, the relative error in x is less than or equal to the condition number times the relative error in b .

PROOF: $T(x^* + \Delta x^*) = b + \Delta b$ and $T(x^*) = b$ implies that $T(\Delta x^*) = \Delta b$. Then $\Delta x^* = T^{-1}(\Delta b)$, which implies that $\|\Delta x^*\| = \|T^{-1}(\Delta b)\| \leq \|T^{-1}\| \|\Delta b\|$, and since $b = T(x^*)$, we have $\|b\| = \|T(x^*)\| \leq \|T\| \|x^*\|$, so that $\|x^*\| \geq \frac{\|b\|}{\|T\|} \Rightarrow \frac{1}{\|x^*\|} \leq \frac{\|T\|}{\|b\|}$. Therefore,

$$\frac{\|\Delta x^*\|}{\|x^*\|} \leq \|T^{-1}\| \|T\| \frac{\|\Delta b\|}{\|b\|} = k(T) \frac{\|\Delta b\|}{\|b\|}. \quad \blacksquare$$

Example 5.5.7 Let

$$T = \begin{pmatrix} 4.1 & 2.8 \\ 9.7 & 6.6 \end{pmatrix}, \quad b = \begin{pmatrix} 4.1 \\ 9.7 \end{pmatrix}, \quad T(x) = b.$$

This system has solution

$$x^* = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Let

$$b + \Delta b = \begin{pmatrix} 4.11 \\ 9.70 \end{pmatrix} \Rightarrow \Delta b = \begin{pmatrix} 0.01 \\ 0.00 \end{pmatrix}$$

Then $T(x) = b + \Delta b$ has solution

$$x^* + \Delta x^* = \begin{pmatrix} 0.34 \\ 0.97 \end{pmatrix}.$$

Now, let $\|x\| = \|x\|_1 = |x_1| + |x_2|$. Then $\|A\| = \max_j \sum_i |a_{ij}|$, where $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. So

$$\frac{\|\Delta b\|}{\|b\|} = \frac{0.01}{13.8} = \frac{1}{1380}, \quad \frac{\|\Delta x^*\|}{\|x^*\|} = 1.63.$$

Also, $\|T\| = 13.8$, and

$$T^{-1} = \begin{pmatrix} -66 & 28 \\ 97 & -41 \end{pmatrix} \Rightarrow \|T^{-1}\| = 163.$$

So the condition number is $k(T) = 13.8 \cdot 163 = 2249.4$. In this case, because T is linear, we have equality in the expression from the previous theorem:

$$\frac{\|\Delta x^*\|}{\|x^*\|} = k(T) \frac{\|\Delta b\|}{\|b\|}.$$

5.5.2 Linear Functionals

A *functional* is an operator whose range lies on the real line $\mathbb{R}$ or in the complex plane $\mathbb{C}$. And *functional analysis* was initially the study of functionals.

Functionals are operators, so that previous definitions apply. In particular, we have the following.

Definition 5.5.6 Linear Functional

A **linear functional** f is a linear operator with domain a vector space X and range in the scalar field $\mathbb{F}$ of X , i.e., $f : X \rightarrow \mathbb{F}$.

Definition 5.5.7 Bounded Linear Functional

A **bounded linear functional** f is a bounded linear operator on a normed space $(X, \|\cdot\|)$ with range in the scalar field $\mathbb{F}$ of X . In other words, for all $x \in X$, there exists a real number c such that

$$|f(x)| \leq c \|x\|. \quad (5.23)$$

Furthermore, the **norm** of f is

$$\|f\| = \sup_{x \in X} \frac{|f(x)|}{\|x\|} \Leftrightarrow \|f\| = \sup_{\|x\|=1} |f(x)|. \quad (5.24)$$

Note then that

$$|f(x)| \leq \|f\| \|x\|. \quad (5.25)$$

Theorem 5.5.2 also applies to functionals.

Theorem 5.5.6 Continuity and Boundedness

A linear functional in a normed linear space is continuous if and only if it is bounded.

Example 5.5.8 Here are some typical examples of functionals.

1. *The Norm*: The norm $\|\cdot\| : X \rightarrow \mathbb{R}$ on a normed space $(X, \|\cdot\|)$ is a functional on X that is not linear.
2. *Dot Product*: The familiar dot product with one factor kept fixed defines a functional $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by means that

$$f(x) = x \cdot a = \xi_1 \alpha_1 + \xi_2 \alpha_2 + \xi_3 \alpha_3,$$

where $a = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{R}^3$ is fixed. Note that f is linear and bounded. In fact,

$$|f(x)| = |x \cdot a| \leq \|x\| \|a\|,$$

so that $\|f\| \leq \|a\|$, which follows from (5.24) if we take the supremum over all x of norm one. On the other hand, by taking $x = a$ and using (5.25) we obtain

$$\|f\| \geq \frac{|f(a)|}{\|a\|} = \frac{\|a\|^2}{\|a\|} = \|a\|.$$

Hence, the norm of f is $\|f\| = \|a\|$.

3. *Definite Integral*: The definite integral is a number if we consider it for a single function, as we do in calculus most of the time. However, the situation changes completely if we consider that integral for all functions in a certain function space. Then the integral becomes a functional on that space, call it f . As a space, let us choose $C[a, b]$. Then f is defined by

$$f(x) = \int_a^b x(t) dt \text{ for all } x \in C[a, b].$$

f is linear by linearity of the integral. We prove that f is bounded and has norm $\|f\| = b - a$. Now, using the norm $\|\cdot\|_\infty$ on $C[a, b]$, we obtain

$$|f(x)| = \left| \int_a^b x(t) dt \right| \leq (b - a) \max_{a \leq t \leq b} |x(t)| = (b - a) \|x\|.$$

Taking the supremum over all x of norm 1, we obtain $\|f\| \leq b - a$. To get $\|f\| \geq b - a$, we choose the particular case $x(t) = x_0 \equiv 1$. Then, noting that $\|x_0\| = 1$, and using (5.25):

$$\|f\| \geq \frac{|f(x_0)|}{\|x_0\|} = |f(x_0)| = \int_a^b dt = b - a.$$

4. *The Space $C[a, b]$* : Another practically important functional on $C[a, b]$ is obtained if we choose a fixed $t_0 \in [a, b]$ and set

$$f_1(x) = x(t_0) \text{ for all } x \in C[a, b].$$

f_1 is linear. f_1 is bounded and has norm $\|f_1\| = 1$. In fact, we have

$$|f_1(x)| = |x(t_0)| \leq \|x\|,$$

and this implies that $\|f_1\| \leq 1$ by (5.24). On the other hand, for $x_0 = 1$, we have $\|x_0\| = 1$ and we obtain from (5.25)

$$\|f_1\| \geq |f_1(x_0)| = 1.$$

5. *The Space ℓ_2* : We can obtain a linear functional f on the (Hilbert) space ℓ_2 by choosing a fixed $a = (\alpha_j) \in \ell_2$ and setting

$$f(x) = \sum_{j=1}^{\infty} \xi_j \alpha_j,$$

where $x = (\xi_j) \in \ell_2$. This series converges absolutely and f is bounded, since the Cauchy-Schwarz inequality gives

$$|f(x)| = \left| \sum_{j=1}^{\infty} \xi_j \alpha_j \right| \leq \sum_{j=1}^{\infty} |\xi_j \alpha_j| \leq \sqrt{\sum_{j=1}^{\infty} |\xi_j|^2} \sqrt{\sum_{j=1}^{\infty} |\alpha_j|^2} = \|x\| \|a\|.$$

5.6 Representing Linear Operators and Functionals on Finite-Dimensional Spaces

Finite-dimensional vector spaces are simpler than infinite-dimensional ones, and it is natural to ask what simplification this entails with respect to linear operators and functionals defined on such a space.

Linear operators on finite-dimensional vector spaces can be represented in terms of matrices. In this way, matrices become the most important tool for studying linear operators in the finite-dimensional case.

Let X and Y be finite-dimensional vector spaces over the same field, and $T : X \rightarrow Y$ a linear operator. We choose a basis $E = \{e_1, \dots, e_n\}$ for X and a basis $B = \{b_1, \dots, b_r\}$ for Y , with the vectors arranged in a definite order that we keep fixed. Then, every $x \in X$ has a unique representation

$$x = \xi_1 e_1 + \dots + \xi_n e_n. \quad (5.26)$$

Since T is linear, x has the image

$$y = T(x) = T\left(\sum_{k=1}^n \xi_k e_k\right) = \sum_{k=1}^n \xi_k T(e_k). \quad (5.27)$$

Since the representation (5.26) is unique, we have our first result:

T is uniquely determined if the images $y_k := T(e_k)$ of the n basis vectors $e_1, \dots, e_n$ are prescribed.

Since y and $y_k = T(e_k)$ are in Y , they have a unique representations of the form

$$y = \sum_{j=1}^r \eta_j b_j \Rightarrow T(e_k) = \sum_{j=1}^r \tau_{jk} b_j. \quad (5.28)$$

Substitution into (5.27) gives

$$y = \sum_{j=1}^r \eta_j b_j = \sum_{k=1}^n \xi_k T(e_k) = \sum_{k=1}^n \xi_k \sum_{j=1}^r \tau_{jk} b_j = \sum_{j=1}^r \left(\sum_{k=1}^n \tau_{jk} \xi_k \right) b_j.$$

Since the b_j s form a linearly independent set, the coefficients of each b_j on the left and right must be the same, that is,

$$\eta_j = \sum_{k=1}^n \tau_{jk} \xi_k, \quad j = 1, \dots, r. \quad (5.29)$$

This gives the next result.

The image $y = T(x) = \sum_{j=1}^r \eta_j b_j$ of $x = \sum_{k=1}^n \xi_k e_k$ can be obtained from (5.29).

Note the unusual position of the summation index j of τ_{jk} in (5.28), which is necessary in order to arrive at the usual position of the summation index in (5.29).

Now, the coefficients in (5.29) for a matrix

$$T_{EB} := (\tau_{jk})$$

with r rows and n columns. If a basis E for X and a basis B for Y are given, with the elements of E and B arranged in a definite order (which is arbitrary but fixed), then the matrix T_{EB} is uniquely determined by the linear operator T . We say that the matrix T_{EB} **represents** the operator T with respect to those bases.

By introducing the column vectors $\underline{x} = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}$ and $\underline{y} = \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_r \end{pmatrix}$, we can write (5.29) in matrix notation:

$$\underline{y} = T_{EB} \underline{x}. \quad (5.30)$$

So we have that a linear operator T determines a unique matrix representing T with respect to given bases for X and Y , where the vectors of each of the bases are assumed to be arranged in a fixed order. Conversely, any matrix with r rows and n columns determines a linear operator that it represents with respect to given bases for X and Y .

5.7 Normed Spaces of Operators

Consider any two normed spaces X and Y (both real or both complex) and consider the set

$$B(X, Y)$$

of all bounded linear operators from X into Y , that is, each such operator is defined on all X and its range lies in Y . We let $B(X, X) =: B(X)$ if $Y = X$.

Theorem 5.7.1 The Space $B(X, Y)$

The set $B(X, Y)$ is a normed linear space with the operator norm.

PROOF: The first step is to prove that $B(X, Y)$ is a linear space, i.e., a vector space. This is easy to prove if we define the sum $T_1 + T_2$ of two operators $T_1, T_2 \in B(X, Y)$ in a natural way by

$$(T_1 + T_2)(x) := T_1(x) + T_2(x)$$

and the product αT of $T \in B(X, Y)$ and a scalar α by

$$(\alpha T)(x) := \alpha T(x).$$

It remains to show that the operator norm $\|T\|$ for all $T \in B(X, Y)$ is a norm. But this was done in Lemma 5.5.1. So we are done. ■

Proposition 5.7.1

If $T \in B(X, Y)$ and $S \in B(Y, Z)$ for X, Y, Z normed linear spaces, then the composition $ST \in B(X, Z)$ and $\|ST\| \leq \|S\| \|T\|$.

PROOF: We have

$$\|(ST)(x)\| \leq \|S\| \|T(x)\| \leq \|S\| \|T\| \|x\|,$$

so that ST is bounded. Also,

$$\|ST\| = \sup_{\|x\|=1} \|(ST)(x)\| \leq \|S\| \|T\|,$$

proving the second result. ■

Corollary 5.7.1

If $L \in B(X)$, then $L^n \in B(X)$ and $\|L^n\| \leq \|L\|^n$.

5.7.1 Convergence of Sequences of Operators and Functionals**Definition 5.7.1 Convergence of Sequences in $B(X, Y)$**

Let $(T_n) \subset B(X, Y)$ be a sequence of bounded linear operators. The sequence is said to **converge** to an operator $T \in B(X, Y)$ in **operator norm** if $\lim_{n \rightarrow \infty} \|T_n - T\| = 0$.

This notion of convergence is also sometimes called “convergence in the operator norm topology” and “convergence in the uniform topology”.

Definition 5.7.2 Strong Convergence

A sequence of bounded linear operators $(T_n) \subset B(X, Y)$ is said to **converge strongly** to $T \in B(X, Y)$ if

$$\lim_{n \rightarrow \infty} \|(T_n - T)x\| = 0 \text{ for all } x \in X. \quad (5.31)$$

The notions of convergence in operator norm/uniform topology and strong convergence are somewhat analogous to uniform convergence and pointwise convergence, respectively, for sequences of functions on an interval $[a, b]$. The term “strong” is a bit of a misnomer because in fact operator norm convergence is “stronger” than strong convergence.

Theorem 5.7.2

If the sequence $(T_n) \subset B(X, Y)$ converges uniformly to $T \in B(X, Y)$, then it converges strongly to T .

PROOF: Uniform convergence implies that $\lim_{n \rightarrow \infty} \|T - T_n\| = 0$. But $\|(T - T_n)x\| \leq \|T - T_n\| \|x\|$ for any $x \in X$. Thus, uniform convergence implies that $\|(T - T_n)(x)\| \rightarrow 0$ for any $x \in X$, i.e., the sequence converges strongly. ■

The converse of this statement does not exist, i.e., a sequence may converge strongly but not uniformly. An example can be found in Example 2 p. 250 of Naylor and Sell.

Proposition 5.7.2 Convergence of Sequences in $B(X, Y)$

1. Consider a sequence $(T_n) \subset B(X, Y)$ such that $\lim_{n \rightarrow \infty} T_n = T \in B(X, Y)$. If $S \in B(Y, Z)$, then $\lim_{n \rightarrow \infty} ST_n = ST$.
2. If $T \in B(X, Y)$ and $(S_n) \subset B(X, Y)$ such that $\lim_{n \rightarrow \infty} S_n = S \in B(X, Y)$, then $\lim_{n \rightarrow \infty} S_n T = ST$.

In what case(s) will $B(X, Y)$ be a Banach space? This is a central question, which we answer in the following theorem.

Theorem 5.7.3 Completeness

If Y is a Banach space, then $B(X, Y)$ is a Banach space.

PROOF: We consider an arbitrary Cauchy sequence $(T_n) \subset B(X, Y)$ and show that (T_n) converges to an operator $T \in B(X, Y)$. Since (T_n) is Cauchy, for every $\epsilon > 0$ there exists $N > 0$ such that

$$\|T_n - T_m\| < \epsilon \text{ for all } m, n > N.$$

For all $x \in X$ and $m, n > N$, we thus obtain

$$\|T_n(x) - T_m(x)\| = \|(T_n - T_m)(x)\| \leq \|T_n - T_m\| \|x\| < \epsilon \|x\|. \quad (5.32)$$

Now, for any fixed x and given $\tilde{\epsilon}$, we may choose $\epsilon = \epsilon_x$ such that $\epsilon_x \|x\| < \tilde{\epsilon}$. Then, from (5.32), we have $\|T_n(x) - T_m(x)\| < \tilde{\epsilon}$, and see that $(T_n(x))$ is Cauchy in Y . Since Y is complete by assumption, $(T_n(x))$ converges, say, to y , i.e., $\lim_{n \rightarrow \infty} T_n(x) = y$. Clearly, the limit $y \in Y$ depends on the choice $x \in X$. This defines an operator $T : X \rightarrow Y$, where $y = T(x)$. The operator T is linear since

$$\lim_{n \rightarrow \infty} T_n(\alpha x + \beta z) = \lim_{n \rightarrow \infty} (\alpha T_n(x) + \beta T_n(z)) = \alpha \lim_{n \rightarrow \infty} T_n(x) + \beta \lim_{n \rightarrow \infty} T_n(z).$$

We now prove that T is bounded and that $\lim_{n \rightarrow \infty} T_n = T$, i.e., that $\|T_n - T\| \rightarrow 0$. Since (5.32) holds for every $m > N$ and $\lim_{m \rightarrow \infty} T_m(x) = T(x)$, we may let $m \rightarrow \infty$. Using the continuity of the norm, we then obtain from (5.32) that for every $n > N$ and all $x \in X$

$$\|T_n(x) - T(x)\| = \left\| T_n(x) - \lim_{m \rightarrow \infty} T_m(x) \right\| = \lim_{m \rightarrow \infty} \|T_n(x) - T_m(x)\| \leq \epsilon \|x\|. \quad (5.33)$$

This shows that $T_n - T$ with $n > N$ is a bounded linear operator. Since T_n is bounded, $T = T_n - (T_n - T)$ is bounded, that is, $T \in B(X, Y)$. Furthermore, if in (5.33) we take the supremum over all x of norm one, we obtain

$$\|T_n - T\| \leq \epsilon \text{ for all } n > N.$$

Hence, $\|T_n - T\| \rightarrow 0$. ■

5.7.2 The Dual Space

Let us return to the linear functionals on a vector space X . It is of basic importance that the set of all these linear functionals can itself be made into a vector space. This space is denoted X^* and is called the **algebraic dual space** of X . Its algebraic operations of a vector space are defined in a natural way as follows. The *sum* $f_1 + f_2$ of two functionals f_1 and f_2 is the functional whose value at every $x \in X$ is

$$(f_1 + f_2)(x) := f_1(x) + f_2(x).$$

The *product* αf of a scalar α and a functional f is the functional whose value at $x \in X$ is

$$(\alpha f)(x) := \alpha f(x).$$

Note that this agrees with the usual way of adding functions and multiplying them by constants.

Now, let $\dim(X) = n < \infty$ and $\{e_1, \dots, e_n\}$ be a basis for X . For every functional $f \in X^*$, and every $x = \sum_{j=1}^n \xi_j e_j$, by the definitions above we have

$$f(x) = f\left(\sum_{j=1}^n \xi_j e_j\right) = \sum_{j=1}^n \xi_j f(e_j) = \sum_{j=1}^n \xi_j \alpha_j, \quad (5.34)$$

where

$$\alpha_j = f(e_j), \quad j = 1, 2, \dots, n, \quad (5.35)$$

and f is uniquely determined by its values α_j at the n basis vectors of X .

Conversely, every n -tuple of scalars $\alpha_1, \dots, \alpha_n$ determines a linear functional on X by (5.34) and (5.35). In particular, let us take the n -tuples

$$\begin{aligned} &(1, 0, 0, \dots, 0, 0) \\ &(0, 1, 0, \dots, 0, 0) \\ &\vdots \\ &(0, 0, 0, \dots, 0, 1). \end{aligned}$$

By (5.34) and (5.35), this gives n functionals, which we denote by $f_1, \dots, f_n$, with values

$$f_k(e_j) = \delta_{jk}. \quad (5.36)$$

The set $\{f_1, \dots, f_n\}$ is called the **dual basis** of the basis $\{e_1, \dots, e_n\}$ for X .

Lemma 5.7.1

Let X be a finite-dimensional vector space. If $x_0 \in X$ has the property that $f(x_0) = 0$ for all $f \in X^*$, then $x_0 = 0$.

PROOF: Let $\{e_1, \dots, e_n\}$ be a basis for X and $x_0 = \sum_{j=1}^n \xi_{0j} e_j$. Then (5.34) and (5.35) becomes

$$f(x_0) = \sum_{j=1}^n \xi_{0j} \alpha_j.$$

By assumption, this is zero for every $f \in X^*$, that is, for every choice of $\alpha_1, \dots, \alpha_n$. Hence, all ξ_{0j} must be zero, i.e., $x_0 = 0$. ■

Theorem 5.7.4 Dimension of X^*

Let X be an n -dimensional vector space and $E = \{e_1, \dots, e_n\}$ a basis for X . Then $F = \{f_1, \dots, f_n\}$ given by $f_k(e_j) = \delta_{jk}$ is a basis for the dual X^* of X and $\dim(X^*) = \dim(X) = n$.

We now consider the dual space of a *normed* linear space.

Definition 5.7.3 Dual Space X'

Let $(X, \|\cdot\|_X)$ be a normed space. Then the set of all bounded linear functionals on X constitutes a normed space with norm defined by

$$\|f\| = \sup_{x \neq 0} \frac{|f(x)|}{\|x\|_X} = \sup_{\|x\|=1} |f(x)|, \quad (5.37)$$

which is called the **dual space** of $(X, \|\cdot\|_X)$ and is denoted $(X', \|\cdot\|)$.

Since a linear functional on X maps X into $\mathbb{R}$ or $\mathbb{C}$ (the scalar field of X), and since $\mathbb{R}$ or $\mathbb{C}$, taken with the usual metric, is complete, we see that $(X', \|\cdot\|) = B(X, \mathbb{R})$ or $B(X, \mathbb{C})$. Therefore, applying Theorem 5.7.3, we get:

Theorem 5.7.5 Completeness of Dual Space

The dual space X' of a normed space X is a Banach space (whether or not X is).

Example 5.7.1

1. The dual space of $\mathbb{R}^n$ is $\mathbb{R}^n$.

PROOF: We have by Theorem 5.5.1 that $(\mathbb{R}^n)' = (\mathbb{R}^n)^*$, and every $f \in (\mathbb{R}^n)^*$ has a representation (5.34), i.e.,

$$f(x) = \sum_{k=1}^n \xi_k \gamma_k, \quad \gamma_k = f(e_k).$$

By the Cauchy-Schwarz inequality,

$$|f(x)| \leq \sum_{k=1}^n |\xi_k \gamma_k| \leq \sqrt{\sum_{j=1}^n \xi_j^2} \sqrt{\sum_{k=1}^n \gamma_k^2} = \|x\| \sqrt{\sum_{k=1}^n \gamma_k^2}.$$

Taking the supremum over all x of norm one, we obtain

$$\|f\| \leq \sqrt{\sum_{k=1}^n \gamma_k^2}.$$

However, since for $x = (\gamma_1, \dots, \gamma_n)$ equality is achieved in the Cauchy-Schwarz inequality, we must in fact have

$$\|f\| = \sqrt{\sum_{k=1}^n \gamma_k^2}.$$

This proves that the norm of f is the Euclidean norm, and $\|f\| = \|c\|$, where $c = (\gamma_k) \in \mathbb{R}^n$. Hence, the mapping $(\mathbb{R}^n)' \rightarrow \mathbb{R}^n$ can be defined by $f \mapsto c = (f(e_k))$. This is norm-preserving and, since it is linear and bijective, it is an isomorphism. ■

2. The dual space of ℓ_1 is ℓ_∞ .

PROOF: To be completed. ■

3. The dual space of ℓ_p is ℓ_q , where $1 < p < \infty$ and q satisfies $\frac{1}{p} + \frac{1}{q} = 1$.

PROOF: To be completed. ■

5.7.3 Series Expansions of Bounded Linear Operators

In the same way that we can write down series expansions (like Taylor expansions) of real-valued functions, we can write down series expansions of bounded linear operators.

The Exponential of a Bounded Linear Operator

Definition 5.7.4 Operator Exponential

Given an $A \in B(X)$, we define the **exponential** of A , denoted $\exp(A)$ or e^A , as

$$\exp(A) := \lim_{n \rightarrow \infty} S_n,$$

where $S_n \in B(X)$ is the partial sum

$$S_n = I + A + \frac{1}{2}A^2 + \cdots + \frac{1}{n!}A^n,$$

where I is the identity operator on X .

The convergence of the sequence (S_n) to $\exp(A)$ is in the operator norm. This follows from the fact that the series $\sum_k \frac{1}{k!}a^k$ converges absolutely for any $a \in \mathbb{R}$ and

$$\|S_n\| = \left\| \sum_{k=0}^n \frac{1}{k!}A^k \right\| \leq \sum_{k=0}^n \frac{1}{k!} \|A\|^k, \quad n = 1, 2, \dots$$

Because of the completeness of X , it follows that

1. $\exp(A) \in B(X)$;
2. $\|\exp(A)\| \leq \exp(\|A\|)$.

This result has special importance in the case $X = \mathbb{R}^n$, in which case $B(X)$ becomes the space of $n \times n$ matrices. Given an $n \times n$ matrix A , interpreted as a linear operator $A \in B(X)$, we may define its exponential $\exp(A)$ as

$$\exp(A) \equiv e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!},$$

where $A^0 = I$, the $n \times n$ identity matrix.

Now, let $t \in \mathbb{R}$ and define

$$e^{tA} = \sum_{k=0}^{\infty} \frac{1}{k!} t^k A^k.$$

It is a well-known that $x(t) = x_0 e^{tA}$ is the unique solution of the linear system of ODEs $\frac{dx}{dt} = Ax$, $x(0) = x_0$.

Other Linear Operators Defined by Series Expansion

We can use series expansions to define other well-known functions from calculus. For example,

$$\cos(A) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} A^{2n}, \quad A \in B(X); \quad (5.38)$$

$$\sin(A) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} A^{2n+1}, \quad A \in B(X); \quad (5.39)$$

$$\log(I - A) = \sum_{n=1}^{\infty} \frac{1}{n} A^n, \quad \|A\| < 1. \quad (5.40)$$

Finally, we have the very important *geometric series*.

Definition 5.7.5 Geometric Series

Let $A \in B(X)$ satisfying $\|A\| < 1$. Then we define the **geometric series** as

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n, \quad \|A\| < 1. \quad (5.41)$$

5.7.4 Application: The Neumann Series

Consider the following *Fredholm integral equation*:

$$f(t) - \int_0^1 k(t, s) f(s) \, ds = g(t), \quad 0 \leq t \leq 1. \quad (5.42)$$

If we let

$$L(f)(t) := \int_0^1 k(t, s) f(s) \, ds,$$

then we can write (5.42) as

$$f - L(f) = g \Leftrightarrow (I - L)(f) = g \Leftrightarrow f = L(f) + g.$$

Many problems in applied mathematics can be written as such an operator equation. The problem is essentially: given a function g (in some prescribed space), can we solve for f ? In this section, we develop one method of finding such a solution.

Theorem 5.7.6

If $L \in B(X)$, where X is a Banach space, with $\|L\| < 1$, then $(I - L)^{-1}$ exists, $(I - L)^{-1} \in B(X)$, $(I - L)^{-1} = \sum_{n=0}^{\infty} L^n$, and $\|(I - L)^{-1}\| \leq \frac{1}{1 - \|L\|}$.

PROOF: $\|L\| < 1$ implies that the series $\sum_{n=0}^{\infty} \|L\|^n$ converges. We also know that $\|L^n\| \leq \|L\|^n$, so that the series $\sum_{n=0}^{\infty} \|L^n\|$ converges by the comparison test. Therefore, $\sum_{n=0}^{\infty} L^n$ converges, so that $M := \sum_{n=0}^{\infty} L^n \in B(X)$.

Now, let $S_n = \sum_{k=0}^n L^k$. Then, we have $S_n \rightarrow M$ as $n \rightarrow \infty$. Then $LS_n = \sum_{k=0}^n L^{k+1} = S_n L$. Then, by Proposition 5.7.2, $S_n L \rightarrow ML$ and $LS_n \rightarrow LM$. Also,

$$\sum_{k=0}^n L^{k+1} \rightarrow \sum_{k=0}^{\infty} L^{k+1} = M - I.$$

Thus,

$$\begin{cases} LM = M - I \\ ML = M - I \end{cases} \Rightarrow \begin{cases} (I - L)M = I \\ M(I - L) = I, \end{cases}$$

showing that M is both a right and left inverse of $I - L$. So

$$(I - L)^{-1} = \sum_{n=0}^{\infty} L^n.$$

Finally,

$$\|S_n\| = \left\| \sum_{k=0}^n L^k \right\| \leq \sum_{k=0}^n \|L^k\| \leq \sum_{k=0}^n \|L\|^k.$$

Letting $n \rightarrow \infty$ in the above equation gives

$$\|(I - L)^{-1}\| \leq \sum_{k=0}^{\infty} \|L\|^k = \frac{1}{1 - \|L\|}. \quad \blacksquare$$

The above theorem tells us that when $\|L\| < 1$ we can solve the equation $f - L(f) = g$ as

$$f = (I - L)^{-1}g = (I + L + L^2 + \cdots)(g) = g + L(g) + L^2(g) + \cdots. \quad (5.43)$$

This is called the **Neumann series**. We can truncate the Neumann series to obtain an approximate solution:

$$f - (g + L(g) + \cdots + L^{n-1}(g)) = L^n(g) + L^{n+1}(g) + \cdots = L^n(I + L + L^2 + \cdots)(g) = L^n(I - L)^{-1}(g),$$

so that

$$\|f - (g + L(g) + \cdots + L^{n-1}(g))\| \leq \|L^n\| \|(I - L)^{-1}\| \|g\| \leq \frac{\|L\|^n}{1 - \|L\|} \|g\|.$$

This expression gives an upper bound on the error of the approximate solution $g + L(g) + \cdots + L^{n-1}(g)$.

There is a connection here with the contraction mapping theorem. Letting $L(f) := L(f) + g$, we see that the solution to $f - L(f) = g$ is equivalent to $T(f) = f$. (Note that T is *not* a linear operator, but an *affine* operator.) Note that

$$\|T(f_1) - T(f_2)\| = \|L(f_1) - L(f_2)\| = \|L(f_1 - f_2)\| \leq \|L\| \|f_1 - f_2\|.$$

So $\|L\| < 1$ implies that T is a contraction mapping. We know by the contraction mapping theorem, therefore, that the iteration sequence $f_0 = h$, $f_{n+1} = T(f_n)$ converges to the unique fixed point of T . In particular, if we take $f_0 = 0$, then $f_1 = g$, $f_2 = g + L(g)$, $f_3 = g + L(g + L(g)) = g + L(g) + L^2(g)$. In general, $f_n = g + L(g) + \cdots + L^{n-1}(g)$, the n th partial sum of the Neumann series. So the limit of the iteration sequence is the function to which the Neumann series converges. Note that this result holds regardless of the starting function f_0 . (Show this!)

Example 5.7.2 Consider the following linear Fredholm equation on the function space $C[0, 1]$ with $\|\cdot\|_\infty$ norm.

$$f(t) = t + \int_0^1 stf(s) ds, \quad 0 \leq t \leq 1.$$

This equation has the form $f = g + L(f)$, where $g(t) = t$ and $L(f)(t) = \int_0^1 stf(s) ds$. The kernel of this integral operator is $k(t, s) = st$. Note that the action of the linear operator L defined by this kernel is actually quite simple in form,

$$L(f)(t) = t \int_0^1 sf(s) ds = C_f t,$$

where C_f is a scalar that depends on the function f . In other words, the linear operator L maps the set of continuous functions $C[0, 1]$ onto the very simple space of functions which we'll call

$$\mathcal{F}[0, 1] := \{h : [0, 1] \rightarrow \mathbb{R} \mid h(t) = at \text{ for some } a \in \mathbb{R}\}.$$

Note that this space is a subset of the space of first-degree polynomials defined on $[0, 1]$. In general when the kernel $k(s, t)$ is a multinomial function of s and t (i.e., a sum of powers $s^k t^\ell$), the associated integral linear operator will map functions to an appropriate space (or subspace) of polynomials in t .

Let us now see if our problem admits a solution in terms of Neumann series. We must estimate the norm of L in a rather straightforward manner:

$$|L(f)(t)| = |t| \left| \int_0^1 sf(s) ds \right| \leq |t| \int_0^1 |sf(s)| ds \leq |t| \|f\|_\infty \int_0^1 s ds \leq \frac{1}{2} |t| \|f\|_\infty. \quad (5.44)$$

Taking the supremum on both sides over $[0, 1]$ gives

$$\|L(f)\|_\infty \leq \frac{1}{2} \|f\|_\infty.$$

This implies that

$$\|L\| \leq \frac{1}{2}, \quad (5.45)$$

from which we conclude that the Neumann series approach to this problem is applicable.

Note that in (5.44) we did not maximise the integration variable prematurely, i.e., we did not write

$$|L(f)(t)| = |t| \left| \int_0^t sf(s) ds \right| \leq |t| \int_0^1 |sf(s)| ds \leq |t| \max_{0 \leq s \leq 1} |s| \|f\|_\infty \int_0^1 ds \leq |t| \|f\|_\infty,$$

because then taking the supremum on both sides gives

$$\|L(f)\|_\infty \leq \|f\|_\infty \Rightarrow \|L\| \leq 1,$$

which is poorer than the result obtained in (5.44), firstly, because $\frac{1}{2}$ is "better" than 1 since it is lower in value, and secondly, because the result $\|L\| \leq 1$ does not guarantee convergence of the Neumann series. (We can still try, but there is no guarantee of a solution.)

The result in (5.45) is sufficient to let us continue with the Neumann series approach. But if, for some reason (and there will be reasons, as we'll see later), one wanted to improve the upper bound to $\|L\|$, we could try to find a function that would do so. In this case, since we have seen that L maps continuous functions to functions of the form $h(t) := at$, let's examine what L does to the function $f(t) := t$:

$$L(f)(t) = t \int_0^t s f(s) ds = t \int_0^1 s^2 ds = \frac{1}{3}t. \quad (5.46)$$

In other words, we have found a function f such that $L(f) = \frac{1}{3}f$. This implies that $f(t) = t$ is an *eigenfunction* of the linear operator L . In fact, any multiple of this function, i.e., $h(t) = at$, is an eigenfunction of L .

More importantly for the matter at hand, we have found a function for which $\|L(f)\| = \frac{1}{3}\|f\|$. This implies that we can improve our estimate of the operator norm of L to $\|L\| \leq \frac{1}{3}$.

Let us now return to the Fredholm integral equation and solve it with the Neumann series. Recall that $(I - L)(f) = g$, which yields

$$f = (I - L)^{-1}(g) = (I + L + L^2 + \cdots)(g) = g + L(g) + L^2(g) + \cdots.$$

Here, $g(t) = t$, so that from (5.46),

$$L(g)(t) = L(t) = \frac{1}{3}t.$$

We may then iterate this result, i.e.,

$$L^2(g)(t) = L(L(g))(t) = L\left(\frac{1}{3}t\right) = \frac{1}{3}L(t) = \frac{1}{9}t,$$

and so on. The net result is

$$f(t) = t + \frac{1}{3}t + \frac{1}{9}t^2 + \cdots = \frac{1}{1 - \frac{1}{3}}t = \frac{3}{2}t.$$

We can check this result by substitution into the original integral equation:

$$\text{LHS} = \frac{3}{2}t, \quad \text{RHS} = t + L\left(\frac{3}{2}t\right) = t + \frac{3}{2} \cdot \frac{1}{3}t = \frac{3}{2}t = \text{LHS},$$

so $f(t) = \frac{3}{2}t$ is the unique solution to the Fredholm integral equation.

5.8 The Hahn-Banach Theorem

The Hahn-Banach theorem is an extension theorem for linear functionals. We shall see that the theorem guarantees that a normed space is richly supplied with bounded linear functionals and makes possible an adequate theory of dual spaces, which is an essential part of the general theory of normed spaces. In this way, the Hahn-Banach theorem becomes one of the most important theorems in connection with bounded linear operators.

Generally speaking, in an *extension problem*, one considers a mathematical object (for example, a mapping) defined on a subset Z of a given set X , and one wants to extend the object from Z to the entire set X in such a way that certain basic properties of the object continue to hold for the extended object.

Definition 5.8.1 Subadditivity and Positive-Homogeneity

A mapping $p : X \rightarrow Y$ on a set X into a set Y is called **subadditive** if

$$p(x + y) \leq p(x) + p(y) \text{ for all } x, y \in X. \quad (5.47)$$

p is called **positive-homogeneous** if

$$p(\alpha x) = \alpha p(x) \text{ for all } \alpha \geq 0 \text{ in } \mathbb{R} \text{ and all } x \in X. \quad (5.48)$$

Definition 5.8.2 Sublinear Functional

A **sublinear functional** is a functional on a normed linear space that is subadditive and positive-homogeneous.

In the Hahn-Banach theorem, the object to be extended is a linear functional f that is defined on a subspace Z of a vector space X and has a certain boundedness property that will be formulated in terms of a sublinear functional.

Theorem 5.8.1 Hahn-Banach

Let X be a real vector space and p a sublinear functional on X . Furthermore, let f be a linear functional that is defined on a subspace Z of X and satisfies

$$f(x) \leq p(x) \text{ for all } x \in Z. \quad (5.49)$$

Then f has a linear extension $\tilde{f}$ from Z to X satisfying

$$\tilde{f}(x) \leq p(x) \text{ for all } x \in X, \quad (5.50)$$

that is, $\tilde{f}$ is a linear functional on X , satisfies (5.50) on X and $\tilde{f}(x) = f(x)$ for every $x \in Z$.

Theorem 5.8.2 Hahn-Banach (Generalised)

Let X be a real or complex vector space and p a real-valued functional on X that is subadditive and for every scalar α satisfies

$$p(\alpha x) = |\alpha|p(x). \quad (5.51)$$

Furthermore, let f be a linear function that is defined on a subspace Z of X and satisfies

$$|f(x)| \leq p(x) \text{ for all } x \in Z. \quad (5.52)$$

Then, f has a linear extension $\tilde{f}$ from Z to X satisfying

$$|\tilde{f}(x)| \leq p(x) \text{ for all } x \in X. \quad (5.53)$$

Although the Hahn-Banach theorem says nothing directly about continuity, a principal application of the theorem deals with bounded linear functionals. This brings us back to normed spaces, which is our main concern.

Theorem 5.8.3 Hahn-Banach (Normed Spaces)

Let f be a bounded linear functional on a subspace Z of a normed space $(X, \|\cdot\|)$. Then, there exists a bounded linear functional $\tilde{f}$ on X that is an extension of f to X and has the same norm,

$$\|\tilde{f}\|_X = \|f\|_Z, \quad (5.54)$$

where

$$\|\tilde{f}\|_X = \sup_{x \neq 0} \frac{|\tilde{f}(x)|}{\|x\|}, \quad \|f\|_Z = \sup_{x \neq 0} \frac{|f(x)|}{\|x\|}$$

are operator norms.

From this theorem, we shall now derive another useful result that, roughly speaking, shows that the dual space X' of a normed space X consists of sufficiently many bounded linear functionals to distinguish between the points of X . This will become essential in connection with adjoint operators.

Theorem 5.8.4 Bounded Linear Functionals

Let $(X, \|\cdot\|_X)$ be a normed space and let $x_0 \neq 0$ be any element of X . Then there exists a bounded linear functional $\tilde{f}$ on X such that

$$\|\tilde{f}\| = 1 \text{ and } \tilde{f}(x_0) = \|x_0\|_X.$$

PROOF: We consider the subspace Z of X consisting of all elements $x = \alpha x_0$, where α is a scalar. On Z , we define a linear functional f by

$$f(x) = f(\alpha x_0) = \alpha \|x_0\|. \quad (5.55)$$

f is bounded and has norm $\|f\| = 1$ because for all $x \in X$

$$|f(x)| = |f(\alpha x_0)| = |\alpha| \|x_0\| = \|\alpha x_0\| = \|x\|. \quad (5.56)$$

Then the Hahn-Banach theorem for normed spaces implies that f has a linear extension $\tilde{f}$ from Z to X , with norm $\|\tilde{f}\| = \|f\| = 1$. From (5.55), we see that $\tilde{f}(x_0) = f(x_0) = \|x_0\|$. ■

Example 5.8.1 For the space $(\mathbb{R}^n, \|\cdot\|_2)$, $a \in \mathbb{R}^n$, and $a \neq 0$, the functional f of the theorem above is $f(x) = \frac{x \cdot a}{\|a\|}$. Indeed, we have

$$f(a) = \frac{a \cdot a}{\|a\|} = \|a\| \text{ and } |f(x)| = \frac{|x \cdot a|}{\|a\|} \leq \frac{\|x\| \|a\|}{\|a\|} = \|x\|$$

by the Cauchy-Schwarz inequality. Thus,

$$\frac{|f(x)|}{\|x\|} \leq 1 \text{ for all } x,$$

and since $\frac{|f(a)|}{\|a\|} = 1$, we have that $\|f\| = 1$.

5.8.1 Application to Bounded Linear Functions on $C[a, b]$

The Hahn-Banach theorem for normed spaces has many important applications. One of these will be considered in this section. We will use that theorem for obtaining a general representation formula for bounded linear functionals on $C[a, b]$, where $[a, b] \subset \mathbb{R}$ is a fixed compact interval. The representation will be in terms of a Riemann-Stieltjes integral, which is a generalization of the familiar Riemann integral.

Definition 5.8.3 Bounded Variation

A function w on $[a, b]$ is said to be of **bounded variation** on $[a, b]$ if its *total variation*, denoted $\text{Var}(w)$, on $[a, b]$ is finite, where

$$\text{Var}(w) = \sup \sum_{j=1}^n |w(t_j) - w(t_{j-1})|, \quad (5.57)$$

the supremum being taken over all *partitions*

$$a = t_0 < t_1 < \cdots < t_n = b \quad (5.58)$$

of the interval $[a, b]$; here, $n \in \mathbb{N}$ is arbitrary and so is the choice of values $t_1, \dots, t_{n-1}$ in $[a, b]$, which, however, must satisfy (5.58).

All functions of bounded variation on $[a, b]$ form a vector space. A norm on this space is given by

$$\|w\| = |w(a)| + \text{Var}(w). \quad (5.59)$$

The normed space thus defined is denoted by $BV[a, b]$, where BV is short for “bounded variation”.

We now obtain the concept of the Riemann-Stieltjes integral as follows. Let $x \in C[a, b]$ and $w \in BV[a, b]$. Let P_n be any partition of $[a, b]$ given by (5.58) and denote by $\eta(P_n)$ the length of a largest

interval $[t_{j-1}, t_j]$, that is,

$$\eta(P_n) = \max(t_1 - t_0, \dots, t_n - t_{n-1}).$$

For every partition P_n of $[a, b]$, we consider the sum

$$s(P_n) = \sum_{j=1}^n x(t_j)[w(t_j) - w(t_{j-1})]. \quad (5.60)$$

Now, there exists a number $\mathcal{J}$ with the property that for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$\eta(P_n) < \delta \quad (5.61)$$

implies

$$|\mathcal{J} - s(P_n)| < \epsilon. \quad (5.62)$$

$\mathcal{J}$ is called the **Riemann-Stieltjes integral** of x over $[a, b]$ with respect to w and is denoted by

$$\int_a^b x(t) dw(t). \quad (5.63)$$

Hence, we can obtain (5.63) as the limit of the sums (5.60) for a sequence (P_n) of partitions of $[a, b]$ satisfying $\eta(P_n) \rightarrow 0$ as $n \rightarrow \infty$.

Note that for $w(t) = t$, the integral (5.63) is the familiar Riemann integral of x over $[a, b]$.

Also, if x is continuous on $[a, b]$ and w has a derivative that is integrable on $[a, b]$, then

$$\int_a^b x(t) dw(t) = \int_a^b x(t)w'(t) dt, \quad (5.64)$$

where the prime denotes differentiation with respect to t .

The integral (5.63) depends linearly on $x \in C[a, b]$, that is, for all $x_1, x_2 \in C[a, b]$ and scalars α and β , we have

$$\int_a^b [\alpha x_1(t) + \beta x_2(t)] dw(t) = \alpha \int_a^b x_1(t) dw(t) + \beta \int_a^b x_2(t) dw(t).$$

The integral also depends linearly on $w \in BV[a, b]$; that is, for all $w_1, w_2 \in BV[a, b]$ and scalars γ and δ , we have

$$\int_a^b x(t) d(\gamma w_1 + \delta w_2)(t) = \gamma \int_a^b x(t) dw_1(t) + \delta \int_a^b x(t) dw_2(t).$$

We will also need the inequality

$$\left| \int_a^b x(t) dw(t) \right| \leq \max_{t \in [a, b]} |x(t)| \text{Var}(w). \quad (5.65)$$

We note that this generalises a familiar formula from calculus. In fact, if $w(t) = t$, then $\text{Var}(w) = b - a$ and (5.65) takes the form

$$\left| \int_a^b x(t) dt \right| \leq \max_{t \in [a, b]} |x(t)|(b - a).$$

The representation theorem for bounded linear functionals on $C[a, b]$ by F. Riesz can now be stated as follows.

Theorem 5.8.5 Riesz (Functionals)

Every bounded linear functional f on $C[a, b]$ can be represented by a Riemann-Stieltjes integral

$$f(x) = \int_a^b x(t) \, dw(t), \quad (5.66)$$

where w is a bounded variation on $[a, b]$ and has the total variation

$$\text{Var}(w) = \|f\|. \quad (5.67)$$

REMARK: Note that the w in the theorem is not unique, but can be made unique by imposing the normalising conditions that w be zero at a and continuous from the right:

$$w(a) = 0 \text{ and } w(t+0) = w(t) \text{ for all } a < t < b.$$

5.8.2 The Adjoint Operator

With a bounded linear operator $T : X \rightarrow Y$ on a normed space X we can associate the so-called *adjoint operator* $T^\times$ of T . A motivation for $T^\times$ comes from its usefulness in the solution of equations involving operators; such equations arise, for instance, in physics and other applications.

We consider a bounded linear operator $T : X \rightarrow Y$, where X and Y are normed spaces, and want to define the adjoint operator $T^\times$ of T . For this purpose, we start from any bounded linear functional g on Y . Clearly, g is defined for all $y \in Y$. Setting $y = T(x)$, we obtain a functional on X , call it f :

$$f(x) = g(T(x)) \text{ for all } x \in X. \quad (5.68)$$

f is linear since g and T are linear. f is bounded because

$$|f(x)| = |g(T(x))| \leq \|g\| \|T(x)\| \leq \|g\| \|T\| \|x\|.$$

Taking the supremum over all $x \in X$ of norm one, we obtain the inequality

$$\|f\| \leq \|g\| \|T\|. \quad (5.69)$$

This shows that $f \in X'$, where X' is the dual space of X . By assumption, $g \in Y'$. Consequently, for variable $g \in Y'$, (6.56) defines an operator from Y' into X' , which we call the *adjoint operator* of T and is denoted by $T^\times$. Thus, we have

$$T : X \rightarrow Y \quad \text{and} \quad T^\times : Y' \rightarrow X'. \quad (5.70)$$

Definition 5.8.4 Adjoint Operator

Let $T : X \rightarrow Y$ be a bounded linear operator on normed spaces X and Y . Then the **adjoint operator** $T^\times : Y' \rightarrow X'$ of T is defined by

$$f(x) = (T^\times(g))(x) = g(T(x)) \text{ for all } g \in Y', \quad (5.71)$$

where X' and Y' are the dual spaces of X and Y , respectively.

Theorem 5.8.6

Norm of the Adjoint The adjoint operator $T^\times$ of a bounded linear operator $T : X \rightarrow Y$ on normed spaces X and Y is linear and bounded, and

$$\|T^\times\| = \|T\|. \quad (5.72)$$

PROOF: The operator $T^\times$ is linear since its domain Y' is a vector space and we readily obtain

$$(T^\times(\alpha g_1 + \beta g_2))(x) = (\alpha g_1 + \beta g_2)(T(x)) = \alpha g_1(T(x)) + \beta g_2(T(x)) = \alpha (T^\times(g_1))(x) + \beta (T^\times(g_2))(x).$$

We prove (6.60). From (6.59), we have $f = T^\times(g)$, and by (6.57) it follows that

$$\|T^\times(g)\| = \|f\| \leq \|g\| \|T\|.$$

Taking the supremum over all $g \in Y'$ of norm one, we obtain the inequality

$$\|T^\times\| \leq \|T\|. \quad (5.73)$$

Hence, to get (6.60), we must now prove $\|T^\times\| \geq \|T\|$. Theorem (5.8.4) implies that for every non-zero $x_0 \in X$ there is a $g_0 \in Y'$ such that

$$\|g_0\| = 1 \quad \text{and} \quad g_0(T(x_0)) = \|T(x_0)\|.$$

Here, $g_0(T(x_0)) = (T^\times(g_0))(x_0)$ by the definition of the adjoint $T^\times$. Writing $f_0 = T^\times(g_0)$, we thus obtain

$$\|T(x_0)\| = g_0(T(x_0)) = f_0(x_0) \leq \|f_0\| \|x_0\| = \|T^\times(g_0)\| \|x_0\| \leq \|T^\times\| \|g_0\| \|x_0\|.$$

Since $\|g_0\| = 1$, we thus have for every $x_0 \in X$

$$\|T(x_0)\| \leq \|T^\times\| \|x_0\|.$$

(This includes $x_0 = 0$ since $T(0) = 0$.) But we always have

$$\|T(x_0)\| \leq \|T\| \|x_0\|,$$

and here $\|T\|$ is the *smallest* constant c such that $\|T(x_0)\| \leq c \|x_0\|$ holds for all $x_0 \in X$. Hence, $\|T^\times\|$ cannot be smaller than $\|T\|$, that is, we must have $\|T^\times\| \geq \|T\|$. This and (5.73) imply (6.60), the desired result. ■

Example 5.8.2 Matrices

In n -dimensional Euclidean space $\mathbb{R}^n$, a linear operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be represented by matrices, where such a matrix $T_E = (\tau_{jk})$ depends on the choice of a basis $E = \{e_1, \dots, e_n\}$ for $\mathbb{R}^n$, whose elements are arranged in some order that is kept fixed. We choose a basis E , regard $x = (\xi_1, \dots, \xi_n)$ and $y = (\eta_1, \dots, \eta_n)$ as column vectors and employ the usual notation for matrix multiplication. Then,

$$y = T_E x \Rightarrow \eta_j = \sum_{k=1}^n \tau_{jk} \xi_k, \quad (5.74)$$

where $j = 1, \dots, n$. Let $F = \{f_1, \dots, f_n\}$ be the dual basis of E . This is a basis for $(\mathbb{R}^n)'$ (which is isomorphic to $\mathbb{R}^n$). Then, every $g \in (\mathbb{R}^n)'$ has a representation

$$g = \alpha_1 f_1 + \dots + \alpha_n f_n.$$

Now, by the definition of the dual basis, we have

$$f_j(y) = f_j\left(\sum_{k=1}^n \eta_k e_k\right) = \eta_j.$$

Hence, by (5.74) we obtain

$$g(y) = g(T_E x) = \sum_{j=1}^n \alpha_j \eta_j = \sum_{j=1}^n \sum_{k=1}^n \alpha_j \tau_{jk} \xi_k.$$

Interchanging the order of summation, we can write this in the form

$$g(T_E x) = \sum_{k=1}^n \beta_k \xi_k, \quad \text{where} \quad \beta_k = \sum_{j=1}^n \tau_{jk} \alpha_j. \quad (5.75)$$

We may regard this as the definition of a functional f on X in terms of g , that is,

$$f(x) := g(T_E x) = \sum_{k=1}^n \beta_k \xi_k.$$

Remembering the definition of the adjoint operator, we can write this as

$$f = T^\times g \Rightarrow \beta_k = \sum_{j=1}^n \tau_{jk} \alpha_j.$$

Noting that in β_k we sum with respect to the first subscript (so that we sum over all elements of a column of T_E), we have the following result.

If T is represented by a matrix T_E , then the adjoint operator $T^\times$ is represented by the transpose T_E^T of T_E .

Note that this whole discussion holds if T is a linear operator from $\mathbb{C}^n$ to $\mathbb{C}^n$.

Theorem 5.8.7 Useful Formulas

Let X , Y and Z be normed linear spaces and $S, T \in B(X, Y)$. Then

$$(S + T)^\times = S^\times + T^\times \quad (5.76)$$

$$(\alpha T)^\times = \alpha T^\times. \quad (5.77)$$

Now let $T \in B(X, Y)$ and $S \in B(Y, Z)$. Then

$$(ST)^\times = T^\times S^\times. \quad (5.78)$$

Finally, if $T \in B(X, Y)$ and T^{-1} exists and $T^{-1} \in B(Y, X)$, then $(T^\times)^{-1}$ also exists, $(T^\times)^{-1} \in B(X', Y')$, and

$$(T^\times)^{-1} = (T^{-1})^\times. \quad (5.79)$$

PROOF: To be completed. ■

5.9 The Fréchet Derivative

Definition 5.9.1 Fréchet Derivative

Let X and Y be normed spaces. An operator (usually nonlinear) $F : X \rightarrow Y$ is called **Fréchet differentiable** at a if there exists a bounded linear operator $DF(a) : X \rightarrow Y$, called the **Fréchet derivative** of F , such that

$$\lim_{h \rightarrow 0} \frac{\|F(a+h) - F(a) - DF(a)h\|}{\|h\|} = 0. \quad (5.80)$$

The Fréchet derivative is a generalization of the derivative of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ encountered in first-year calculus and the *Jacobian* (matrix) of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ studied in advanced calculus.

Indeed, for functions $f : \mathbb{R} \rightarrow \mathbb{R}$, the connection is clear if we go back to the definition of $f'(a)$:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

We can write this relation as

$$\lim_{h \rightarrow 0} \frac{|f(a+h) - f(a) - f'(a)h|}{|h|} = 0.$$

The Fréchet derivative of f is the *scalar* $f'(a)$, which multiplies the scalar $a \in \mathbb{R}$ —as such, $f'(a)$ is a linear operator in $\mathbb{R}$.

For functions $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ the Fréchet derivative $DF(a)$ is the Jacobian matrix of F , a linear operator

that is represented by an $m \times n$ matrix,

$$DF(a) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(a) & \cdots & \frac{\partial F_1}{\partial x_m}(a) \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1}(a) & \cdots & \frac{\partial F_m}{\partial x_m}(a) \end{pmatrix}.$$

Here, the rate of change of $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ in the direction $h \in \mathbb{R}^m$ is measured at the point $a \in \mathbb{R}^m$. In fact, the term

$$\frac{DF(a)h}{\|h\|} = DF(a)\hat{h}$$

is, by definition, the *directional derivative* of F at a .

The Fréchet derivative, as defined in (5.80), extends the above concepts of the derivative to operators in general normed spaces, for example, infinite-dimensional function spaces. This is of great importance to computational methods for solving non-linear operator equations.

We consider a few examples below. In all cases, to calculate the Fréchet derivative, it is best to employ the formal definition (5.80). In the analysis of an operator $F : X \rightarrow Y$, the usual procedure is to examine the difference $F(a+h) - F(a)$. All terms that are linear in h (and possibly its derivatives) will comprise the Fréchet derivative. Higher-order terms in h (and its derivatives) will comprise a remainder term, i.e.,

$$F(a+h) - F(a) = Lh + R(a, h),$$

where L is a linear operator. (It may be, for example, an integral operator or a differential operator, or an expression involving both.) From (5.80), it then remains to show that

$$\lim_{h \rightarrow 0} \frac{\|R(a, h)\|}{\|h\|} = 0.$$

If this can be done, then the linear operator L is identified with the Fréchet derivative, i.e., $L \equiv DF(a)$.

Example 5.9.1 Let $X = Y = C[a, b]$ with the $\|\cdot\|_\infty$ norm and let $T : X \rightarrow Y$ be the linear integral operator defined by

$$T(u)(x) = \int_a^b K(x, s)u(s) \, ds,$$

where $K(x, s)$ is continuous on $[a, b] \times [a, b]$. The task is to calculate the Fréchet derivative $DL(u)$.

We first calculate $T(u+h) - T(u)$ for an arbitrary $h \in X$:

$$\begin{aligned} [T(u+h) - T(u)](x) &= \int_a^b K(x, s)[u(s) + h(s)] \, ds - \int_a^b K(x, s)u(s) \, ds \\ &= \int_a^b K(x, s)[u(s) + h(s) - u(s)] \, ds \\ &= \int_a^b K(x, s)h(s) \, ds. \end{aligned}$$

Note that the final term is a linear operator on h , which may not have been unexpected—after all, T is a linear operator. But let us go through the formalities. We may rearrange the above result to read

$$\frac{1}{\|h\|} \left[T(u+h) - T(u) - \int_a^b K(x,s)h(s) \, ds \right] = 0.$$

Since this equation is true for all $h \neq 0$, it follows that the definition (5.80) is satisfied. Therefore, the Fréchet derivative is

$$DT(u) = \int_a^b K(x,s)h(s) \, ds = T(h),$$

which is independent of u , i.e., the bounded linear operator T itself!

Example 5.9.2 As before, let $X = Y = C[a, b]$ with the $\|\cdot\|_\infty$ norm. Now let $T : X \rightarrow Y$ be the *non-linear* integral operator

$$T(u)(x) = u(x) \int_a^b K(x,s)u(s) \, ds,$$

where $K(x,s)$ is continuous on $[a, b] \times [a, b]$. Again, we'd like to find the Fréchet derivative of T .

As before, we start by calculating $T(u+h) - T(u)$ for an arbitrary $h \in X$:

$$[T(u+h) - T(u)](x) = [u(x) + h(x)] \int_a^b K(x,s)[u(s) + h(s)] \, ds - u(x) \int_a^b K(x,s)u(s) \, ds \quad (5.81)$$

$$= u(x) \int_a^b K(x,s)h(s) \, ds + h(x) \int_a^b K(x,s)u(s) \, ds + R(u,h)(x), \quad (5.82)$$

where

$$R(u,h)(x) = h(x) \int_a^b K(x,s)h(s) \, ds.$$

Note that the remainder term $R(u,h)$ is non-linear in h . If $\frac{\|R(u,h)\|}{\|h\|} \rightarrow 0$ as $h \rightarrow 0$, then the first two terms in (5.82) will define the Fréchet derivative of T . We have

$$\|R(u,h)\| = \max_{x \in [a,b]} \left| h(x) \int_a^b K(x,s)h(s) \, ds \right| \leq M \|h\|^2,$$

where $M = (b-a) \max_{[a,b] \times [a,b]} |K(x,s)|$. Thus, the Fréchet derivative of T is given by

$$(DT(u))(h)(x) = u(x) \int_a^b K(x,s)h(s) \, ds + h(x) \int_a^b K(x,s)u(s) \, ds.$$

Note that it is a linear operator on h . It is also bounded. Why?

Example 5.9.3 Let $X = C_0^1[0, 1]$ be the space of all C^1 functions on $[0, 1] \subset \mathbb{R}$ that vanish at the endpoints. We define a norm on this space by

$$\|u\| := \sqrt{\int_0^1 u^2 (u')^2 \, dx}. \quad (5.83)$$

This norm is called the *energy norm*.

Now, consider the functional $K : X \rightarrow \mathbb{R}$ defined by

$$K(u) = \int_0^1 [u^3 + (u')^2] \, dx.$$

The goal is to compute the Fréchet derivative of K .

After a little calculation, one finds that

$$K(u+h) - K(u) = \int_0^1 [3u^2h + 2u'h'] \, dx + R(u, h), \quad R(u, h) = \int_0^1 [3uh^2 + h^3 + (h')^2] \, dx.$$

Note that, once again, the right-hand side of $K(u+h) - K(u)$ has been arranged so that the first term includes all terms that are linear in h , whereas the remainder $R(u, h)$ includes all terms that are non-linear in h . We suspect that the first term represents the Fréchet derivative, but in order to prove this we must show that $\frac{\|R(u, h)\|}{\|h\|} \rightarrow 0$ as $\|h\| \rightarrow 0$. This is, however, somewhat complicated with the energy norm selected for this problem.

In an effort to express $\|R(u, h)\|$ in terms of $\|h\|$, we try the following:

$$\|R(u, h)\| = |R(u, h)| \leq 3 \max_{[0,1]} |u(x)| \int_0^1 h^2 \, dx + \max_{[0,1]} |h(x)| \int_0^1 h^2 \, dx + \int_0^1 (h')^2 \, dx. \quad (5.84)$$

Now note from the definition of the energy norm that

$$\int_0^1 h^2 \, dx \leq \|h\|^2, \quad \text{and} \quad \int_0^1 (h')^2 \, dx \leq \|h\|^2. \quad (5.85)$$

We use this in (5.84):

$$\|R(u, h)\| \leq (3 \|u(x)\|_\infty + \|h(x)\|_\infty + 1) \|h\|^2 \Rightarrow \frac{\|R(u, h)\|}{\|h\|} \leq (2 \|u(x)\|_\infty + \|h(x)\|_\infty + 1) \|h\|. \quad (5.86)$$

It is now tempting to let $\|h\| \rightarrow 0$ and conclude that the ratio on the left-hand side vanishes in this limit, but there is one complication: can we guarantee that h is bounded, so that the middle term on the right-hand side does not “blow up”?

In fact, h must be bounded since it is a C^1 function on $[a, b]$, i.e., there exists $M > 0$ such that $|h(x)| \leq M$ for all x . But for each h , there is an M —what is necessary is to connect M with $\|h\|$. This is made possible with the following result.

Lemma 5.9.1

If $h \in C^1[0, 1]$ and $h(0) = 0$, then

$$\|h\|_\infty = \max_{x \in [0, 1]} |h(x)| \leq 2 \sqrt{\int_0^1 (h')^2 dx}. \quad (5.87)$$

PROOF: If $h = 0$ on $[0, 1]$, then the result holds. We now consider the case that h does not vanish identically over $[0, 1]$. From the fundamental theorem of calculus,

$$\int_0^x h(s)h'(s) ds = \frac{1}{2}h(x)^2 - \frac{1}{2}h(0)^2 = \frac{1}{2}h(x)^2.$$

Applying the Cauchy-Schwarz inequality to the integral on the left yields

$$\frac{1}{2}h(x)^2 \leq \sqrt{\int_0^x h(s)^2 ds} \sqrt{\int_0^x h'(s)^2 ds}.$$

Thus,

$$h(x)^2 \leq 2 \sqrt{\int_0^1 h(s)^2 ds} \sqrt{\int_0^1 h'(s)^2 ds} \leq 2 \max_{[0, 1]} |h(x)| \sqrt{\int_0^1 h'(s)^2 ds} = 2 \|h\|_\infty \sqrt{\int_0^1 h'(s)^2 ds}.$$

Since this inequality holds for all $x \in [0, 1]$, it follows that

$$\max_{[0, 1]} h(x)^2 = \|h\|_\infty^2 \leq 2 \|h\|_\infty \sqrt{\int_0^1 h'(s)^2 ds}.$$

Division on both sides by $\|h\|_\infty > 0$ gives the result. ■

From the lemma and the second inequality in (5.85), it follows that

$$\|h\|_\infty \leq 2 \|h\|.$$

Using the result in (5.86) yields

$$\frac{\|R(u, h)\|}{\|h\|} \leq (2 \|u\|_\infty + 2 \|h\| + 1) \|h\|.$$

Now it follows that

$$\frac{\|R(u, h)\|}{\|h\|} \rightarrow 0 \text{ as } h \rightarrow 0.$$

Therefore, the Fréchet derivative of the non-linear functional K is

$$(DK(u))(h) = \int_0^1 [3u^2h + 2u'h'] dx.$$

Proposition 5.9.1

Let X and Y be normed spaces and $F : X \rightarrow Y$. If F is (Fréchet) differentiable at $a \in X$, then F is continuous at a .

PROOF: We have that $F(a+h) - F(a) - DF(a)h = o(h)$, which means that for any $\epsilon > 0$ there exists $\delta > 0$ such that $\|o(h)\| \leq \epsilon \|h\|$ for all $\|h\| < \delta$. This means that

$$\|F(a+h) - F(a)\| = \|DF(a)h + o(h)\| \leq \|DF(a)h\| + \|o(h)\| \leq \|DF(a)\| \|h\| + \epsilon \|h\|.$$

As $h \rightarrow 0$, the right-hand side tends to zero. Thus,

$$\lim_{h \rightarrow 0} \|F(a+h) - F(a)\| = 0,$$

which implies that

$$\lim_{h \rightarrow 0} F(a+h) = F(a).$$

Therefore, F is continuous at a . ■

Proposition 5.9.2

Let X and Y be normed spaces and $F : X \rightarrow Y$. Then $DF(a)$ is unique for all $a \in X$.

PROOF: Consider two operators L_1 and L_2 satisfying the definition of a Fréchet derivative. Then

$$\frac{1}{\|h\|} (F(a+h) - F(a) - L_1 h) \rightarrow 0 \text{ and } \frac{1}{\|h\|} (F(a+h) - F(a) - L_2 h) \rightarrow 0$$

implies (upon subtracting the two expressions) that

$$\frac{\|(L_1 - L_2)(h)\|}{\|h\|} = \left\| (L_1 - L_2) \left(\frac{h}{\|h\|} \right) \right\| \rightarrow 0 \text{ as } h \rightarrow 0.$$

Let $L := L_1 - L_2$. We now show that $L = 0$. For $x \neq 0$, $tx \neq 0$, $t \neq 0$, since $tx \rightarrow 0$ as $t \rightarrow 0$, we have that $\frac{\|L(tx)\|}{\|tx\|} \rightarrow 0$ as $t \rightarrow 0$. But, since t is a scalar,

$$\frac{\|L(tx)\|}{\|tx\|} = \frac{\|tL(x)\|}{|t| \|x\|} = \frac{|t| \|L(x)\|}{|t| \|x\|} = \frac{\|L(x)\|}{\|x\|}.$$

Thus, $\|L(x)\| = 0$, and hence $L(x) = L_1(x) - L_2(x) = 0 \Rightarrow L_1(x) = L_2(x)$ for all $x \in X$. ■

Theorem 5.9.1 Fréchet Derivative for Bounded Operators

Let X and Y be normed spaces and $F : X \rightarrow Y$. Then $DF(a) = F$ for all a .

PROOF: F satisfies the definition of the Fréchet derivative (verify this!) and it is the unique one by the previous proposition. ■

Theorem 5.9.2 Chain Rule for Fréchet Derivatives

Let $G : X \rightarrow Y$ and $F : Y \rightarrow Z$, where X, Y, Z are normed linear spaces and G is Fréchet differentiable at a and F is Fréchet differentiable at $G(a)$. Then the composition $FG : X \rightarrow Z$ is Fréchet differentiable at a and $D(FG)(a) = DF(G(a))DG(a)$.

PROOF: Let $b := G(a)$. Then,

$$G(a+h) = G(a) + DG(a)h + o(h) = b + k,$$

where $k := DG(a)h + o(h)$. Also,

$$\begin{aligned} F(b+k) &= F(b) + DF(b)k + o(k) \\ \Rightarrow F(G(a+h)) &= F(b+k) = F(b) + DF(b)[DG(a)h + o(h)] + o(DG(a)h + o(h)) \\ &= F(G(a)) + DF(b)DG(a)h + o(h). \end{aligned}$$

Therefore, FG is Fréchet differentiable at a and $D(FG)(a) = DF(b)DG(a)$, as required. ■

5.9.1 The Generalised Mean Value Theorem

The generalised mean value theorem, to be stated below, is useful in showing that an operator is a contraction mapping.

Theorem 5.9.3 Generalised Mean Value

Let X and Y be normed linear spaces $F : X \rightarrow Y$ continuous on the closed segment $\{a + t(b-a) \mid 0 \leq t \leq 1\}$ and Fréchet differentiable on the open segment $\{a + t(b-a) \mid 0 < t < 1\}$ for $a, b \in X$ and $a \neq b$. Then

$$\|F(b) - F(a)\| \leq \left(\sup_{0 < t < 1} \|F'(a + t(b-a))\| \right) \|b - a\|. \quad (5.88)$$

PROOF: Let $\phi(t) = g(F(a + t(b-a)))$, $g \in Y'$, $t \in [0, 1]$. So $\phi : [0, 1] \rightarrow \mathbb{R}$. Now, by the chain rule, and using the fact that

$$\frac{\Delta g}{\Delta t} = \frac{g(F(t + \Delta t)) - g(F(t))}{\Delta t} = g \left[\frac{F(t + \Delta t) - F(t)}{\Delta t} \right] \rightarrow g(F'(t)) \text{ as } \Delta t \rightarrow 0,$$

we have that

$$\phi'(t) = g(F'(a + t(b-a)))(b-a) \text{ for all } 0 < t < 1.$$

Then, by the mean value theorem (for real-valued functions),

$$\phi(1) - \phi(0) = \phi'(\alpha)$$

for some $0 < \alpha < 1$. Thus,

$$g(F(b)) - g(F(a)) = g(F(b) - F(a)) = \overbrace{g(F'(a + \alpha(b-a)))}^{\phi'(\alpha)}(b-a),$$

which implies that

$$|g(F(b) - F(a))| \leq \|g\| \|F'(a + \alpha(b - a))\| \|b - a\|.$$

Then, by Theorem 5.8.4, we can choose g so that $g(F(b) - F(a)) = \|F(b) - F(a)\|$ and $\|g\| = 1$. This gives

$$\|F(b) - F(a)\| \leq \|F'(a + \alpha(b - a))\| \|b - a\| \leq \sup_{0 < t < 1} \|F'(a + t(b - a))\| \|b - a\|. \quad \blacksquare$$

Example 5.9.4 Fréchet Derivative of Integral Operator

Let

$$F(u)(t) = \int_0^1 k(t, s)h(s, u(s)) \, ds,$$

where h is C^1 on $[0, 1] \times \mathbb{R}$ and k is C^0 (i.e., continuous) on $[0, 1] \times [0, 1]$. Let $X = (C[0, 1], \|\cdot\|_\infty)$. We can write F as the composition

$$F(u) = (KH)(u), \quad \text{where } H(u(t)) = h(t, u(t)), \, u \in X \text{ and } K(v) = \int_0^1 k(t, s)v(s) \, ds, \, v \in X.$$

Since K is a linear operator, we have $DK(v) = K(v)$ for all v . Note that both K and H map X into X .

Let us now show that

$$DH(u(t)) = \frac{\partial h}{\partial t}(t, u(t)).$$

For any $u \in X$, we will show that $DH(u)(z) = \frac{\partial h}{\partial u}(t, u)z$ for all $z \in X$. Now, because h is C^1 we can use the mean value theorem to give us

$$H(u + z) - H(u) = h(t, u + z) - h(t, u) = \frac{\partial h}{\partial u}(t, \tilde{u})z,$$

where $\tilde{u}(t)$ is between $u(t)$ and $u(t) + z(t)$ for all $t \in [0, 1]$. This implies that

$$H(u + z) - H(u) - \frac{\partial h}{\partial u}(t, u)z = \left(\frac{\partial h}{\partial u}(t, \tilde{u}) - \frac{\partial h}{\partial u}(t, u) \right) z \quad (5.89)$$

$$\Rightarrow \left\| H(u + z) - H(u) - \frac{\partial h}{\partial u}(t, u)z \right\| \leq \sup_{0 \leq t \leq 1} \left| \frac{\partial h}{\partial u}(t, \tilde{u}) - \frac{\partial h}{\partial u}(t, u) \right| \|z\| \quad (5.90)$$

$$\Rightarrow \frac{\left\| H(u + z) - H(u) - \frac{\partial h}{\partial u}(t, u)z \right\|}{\|z\|} \leq \sup_{0 \leq t \leq 1} \left| \frac{\partial h}{\partial u}(t, \tilde{u}) - \frac{\partial h}{\partial u}(t, u) \right|. \quad (5.91)$$

Note that $|\tilde{u} - u| \leq |z| \leq \|z\|$. The reason for this is that, if $z(t) \geq 0$ for all t , then $u(t) \leq \tilde{u}(t) \leq u(t) + z(t)$, so that $0 \leq \tilde{u}(t) - u(t) \leq z(t) \Rightarrow \|\tilde{u} - u\| \leq \|z\|$, with an analogous argument if $z(t) \leq 0$ for all t . Now, $\frac{\partial h}{\partial u}$ is continuous, hence $\frac{\partial h}{\partial u}$ is uniformly continuous on the compact set $S = \{(t, v) \mid |v - u(t)| \leq 1, 0 \leq t \leq 1\}$. It follows that

$$\sup_{0 \leq t \leq 1} \left| \frac{\partial h}{\partial u}(t, \tilde{u}) - \frac{\partial h}{\partial u}(t, u) \right| \rightarrow 0 \text{ as } \|z\| \rightarrow 0.$$

Therefore, the left-hand side of (5.91) tends to zero as $\|z\| \rightarrow 0$, establishing that $DH(u(t)) = \frac{\partial h}{\partial u}(t, u(t))$.

By the chain rule, we can conclude that

$$DF(u)(z) = \int_0^1 k(t, s) \frac{\partial h}{\partial u}(s, u(s)) z(s) \, ds. \quad (5.92)$$

Example 5.9.5 A Boundary Value Problem

Consider the boundary value problem

$$u'' = f(t, u), \quad u(0) = u(1) = 0.$$

We have that this equivalent to the integral equation $T(u) = u$, where

$$T(u) = \int_0^1 g(t, s) f(s, u(s)) \, ds, \quad \text{where } g(x, s) = \begin{cases} s(x-1) & \text{for } 0 \leq s \leq x < 1 \\ x(s-1) & \text{for } 0 \leq x \leq s \leq 1 \end{cases}.$$

Now, suppose

$$\sup_{0 \leq t \leq 1, u \in \mathbb{R}} \left| \frac{\partial f}{\partial u}(t, u) \right| = L_0 < 8.$$

Then, using the boxed result of the previous example,

$$\begin{aligned} DT(u)h &= \int_0^1 g(t, s) \frac{\partial f}{\partial u}(s, u(s)) h(s) \, ds \\ \Rightarrow \|DT(u)\| &= \max_{0 \leq t \leq 1} \int_0^1 |g(t, s)| \left| \frac{\partial f}{\partial u}(s, u(s)) \right| \, ds \leq L_0 \max_{0 \leq t \leq 1} \int_0^1 |g(t, s)| \, ds = \frac{L_0}{8} < 1, \end{aligned}$$

where we used the fact that $\max_{0 \leq t \leq 1} \int_0^1 |g(t, s)| \, ds = \frac{1}{8}$, which is straightforward to show. So, by the generalised mean value theorem, T is a contraction mapping on $(C[0, 1], \|\cdot\|_\infty)$. By the contraction mapping theorem, there exists a unique fixed point of T that solves the boundary value problem.

Iteration Dynamics Near Locally Attractive Fixed Points

Recall the result for C^1 functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that if p is a fixed point of f , i.e., $f(p) = p$, and $|f'(p)| < 1$, then p is called locally attractive, i.e., there exists an interval I containing p such that $f^n(x) \rightarrow p$ for all $x \in I$. A simple proof is provided by the mean value theorem. In other words, if x_0 is close enough to the attractive fixed point p , then we may find better and better approximations to p by means of the iteration procedure $x_{n+1} = f(x_n)$ (and probably reaching it, to finite accuracy, given the geometric convergence of the iterates).

This procedure may be extended to multi-dimensional (non-linear) mappings using the generalised mean value theorem. Consider mappings $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Suppose that $\mathbf{p} \in \mathbb{R}^n$ is a fixed point of F .

Further suppose that the Fréchet derivative $DF(\mathbf{x})$ exists and is continuous on a neighbourhood N of $\mathbf{p}$ and that $\|DF(\mathbf{p})\| < 1$. Then $\mathbf{p}$ is called *locally attractive*, i.e., there exists a ball $B(\mathbf{p})$ (centred at $\mathbf{p}$) such that $F^n(\mathbf{x}) \rightarrow \mathbf{p}$ for all $\mathbf{x} \in B(\mathbf{p})$. Once again, if $\mathbf{x}_0$ is close enough to $\mathbf{p}$, we may approach $\mathbf{p}$ by means of the iteration sequence $\mathbf{x}_{n+1} = F(\mathbf{x}_n)$. (Prove this!)

Note that these existence results are local—they do not say anything about the structure of the *basin of attraction* of a fixed point $\mathbf{p}$, i.e., about the set of points $\mathbf{x} \in \mathbb{R}^n$ for which the sequence $(F^n(\mathbf{x}))$ converges to $\mathbf{p}$.

Example 5.9.6 Consider the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as

$$F(x, y) = \begin{pmatrix} \frac{1}{2}x + xy + \frac{1}{2}y^2 \\ x^2y \end{pmatrix}.$$

The fixed points of F are

$$(\bar{x}_1, \bar{y}_1) = (0, 0), \quad (\bar{x}_2, \bar{y}_2) = (-1, 1), \quad (\bar{x}_3, \bar{y}_3) = (1, -1 + \sqrt{2}), \quad (\bar{x}_4, \bar{y}_4) = (1, -1 - \sqrt{2}).$$

The Fréchet derivative (i.e., Jacobian, remember) of F is

$$DF(x, y) = \begin{pmatrix} \frac{1}{2} + y & x + y \\ 2xy & x^2 \end{pmatrix}.$$

An examination of $DF(x, y)$ at each of the fixed points (by looking at the eigenvalues) shows that $(\bar{x}_1, \bar{y}_1) = (0, 0)$ is the only attractive fixed point:

$$DF(0, 0) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{pmatrix}, \quad \text{with eigenvalues } \lambda_1 = \frac{1}{2}, \lambda_2 = 0.$$

Therefore, if we start with a point (x_0, y_0) sufficiently close to $(0, 0)$, we expect the iteration sequence $((x_n, y_n))$ to approach $(0, 0)$. We do not expect this to be the case around the other fixed points. This is confirmed with numerical calculations.

In the figure below is plotted an approximation to the basin of attraction of the locally attractive fixed point $(0, 0)$, which again is the set of points $\mathbf{x}_0 = (x_0, y_0) \in \mathbb{R}^2$ for which the sequence $\mathbf{x}_{n+1} = F(\mathbf{x}_n)$ converges to $(0, 0)$. Note the complicated “fractal” structure of the basin boundary.

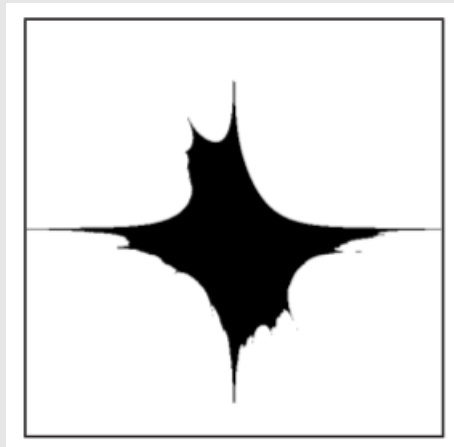


Figure 5.1: Basin of attraction of the fixed point $(0, 0)$. The region pictured above is $-5 \leq x, y \leq 5$.

Note that each of the other fixed points of F will also have, by definition, basins of attraction. The basin of attraction of a repulsive fixed point $\mathbf{p}$ will include $\mathbf{p}$ itself but no points in its neighbourhood. (There may also be other points that are mapped to $\mathbf{p}$.) The “adherence” of a repulsive fixed points will probably not be detected numerically because of roundoff errors.

We can actually state one global result: the x -axis is part of the basin of attraction of $(0, 0)$. To see this, note that

$$F(x, 0) = \begin{pmatrix} \frac{1}{2}x \\ 0 \end{pmatrix}.$$

The x -axis is an invariant set with respect to F . In other words, if we start on the x -axis, i.e., at $y = 0$, we remain on the x -axis. And the iterates x_n are contracted toward $x = 0$ geometrically.

Example 5.9.7 Now let

$$F(x, y) = \begin{pmatrix} x^2 - y^2 - \frac{1}{2} \\ 2xy \end{pmatrix}.$$

The fixed points of F are

$$(\bar{x}_1, \bar{y}_1) = \left(\frac{1}{2}(1 - \sqrt{3}), 0 \right), \quad (\bar{x}_2, \bar{y}_2) = \left(\frac{1}{2}(1 + \sqrt{3}), 0 \right).$$

The Fréchet derivative is

$$DF(x, y) = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix},$$

and $(\bar{x}_1, \bar{y}_1) \approx (-0.366, 0)$ is the only attractive fixed point:

$$DF(\bar{x}_1, \bar{y}_1) = \begin{pmatrix} 1 - \sqrt{3} & 0 \\ 0 & 1 - \sqrt{3} \end{pmatrix}, \quad \text{with eigenvalues } \lambda_1 = \lambda_2 = 1 - \sqrt{3} \approx -0.732.$$

The figure below displays the basin of attraction of this fixed point.

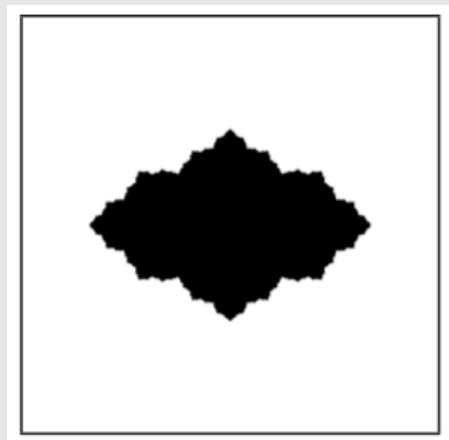


Figure 5.2: Basin of attraction of the fixed point $(-0.366, 0)$. The region pictured above is $-2 \leq x, y \leq 2$.

This example represents the iteration of the complex-valued function $g(z) = z^2 - \frac{1}{2}$ in the complex plane. (Indeed, $f_1(x, y)$ and $f_2(x, y)$ are, respectively, the real and imaginary parts of the complex mapping g .) The boundary of the basin of attraction in the above figure is the so-called *Julia set* of $f(z)$. If we removed the term $-\frac{1}{2}$ from $f_1(x, y)$, the fixed point of F is $(0, 0)$, corresponding to the fixed point 0 of the complex-valued function $f(z) = z^2$. The basin of attraction of this fixed point—hence the Julia set of $f(z) = z^2$ —is the unit circle $|z| = 1$, which can be derived quite easily analytically.

5.9.2 Application: The Newton-Kantorovich Method

Recall Newton's method, also referred to as the Newton-Raphson method, as applied to functions $f : \mathbb{R} \rightarrow \mathbb{R}$. The goal of this method is to provide approximations to the zeros of f . In what follows, we let $\bar{x}$ denote a zero of f , i.e., $\bar{x}$ satisfies $f(\bar{x}) = 0$. The Newton-Raphson function N associated with f is given by

$$N(x) = x - \frac{f(x)}{f'(x)}, \quad (5.93)$$

assuming that f is differentiable at least over a neighbourhood of $\bar{x}$. There are possible complications at critical points of f , i.e., at points satisfying $f'(x) = 0$, and also if the zeros of f are *simple*, but we avoid these details here.

From the definition of N , it is clear that $N(\bar{x}) = \bar{x}$, i.e., $\bar{x}$ is a *fixed point* of N . Our goal is to analyse the iteration procedure

$$x_{n+1} = N(x_n). \quad (5.94)$$

It is well known that if the seed x_0 of this sequence is sufficiently close to $\bar{x}$, then $x_n \rightarrow \bar{x}$. In fact, it can be shown (do it!) that if f is twice-differentiable, then

$$|N(x) - \bar{x}| \leq K|x - \bar{x}|^2. \quad (5.95)$$

This is referred to as *quadratic convergence*: the error in approximating $\bar{x}$ with $N(x)$ is proportional to the square of the error in approximating $\bar{x}$ with x . Repeated application of this result yields

$$|x_n - \bar{x}| \leq K^n |x_0 - \bar{x}|^{2^n} \quad (5.96)$$

This rate of convergence is much faster than the rate $K^n |x_0 - \bar{x}|^n$ that would result from linear convergence, where the exponent “2” is replaced with “1” in (5.95).

Let us backtrack now and recall how the quadratic convergence of the Newton method was established. Taking derivatives of both sides of (5.93) gives

$$N'(x) = \frac{f(x)f''(x)}{|f'(x)|^2}, \quad (5.97)$$

assuming, of course, that f'' exists. From this comes the important result

$$N'(\bar{x}) = 0. \quad (5.98)$$

We now apply Taylor's theorem about the point $\bar{x}$: for x sufficiently close to $\bar{x}$,

$$N(x) = N(\bar{x}) + N'(\bar{x})(x - \bar{x}) + \frac{1}{2}N''(c)(x - \bar{x})^2, \quad (5.99)$$

where c lies between $\bar{x}$ and x . Since $N(\bar{x}) = \bar{x}$ and $N'(\bar{x}) = 0$, we have

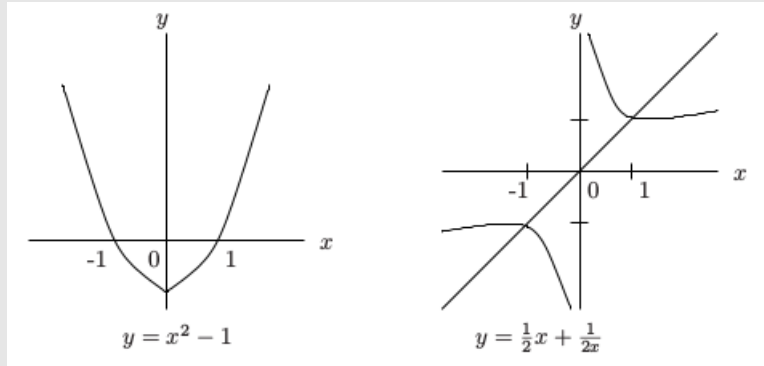
$$N(x) - \bar{x} = \frac{1}{2}N''(c)(x - \bar{x})^2. \quad (5.100)$$

Restricting x to a δ -neighbourhood of $\bar{x}$, taking absolute values, and assuming that $N''(x)$ is continuous/bounded over this set, we arrive at (5.95).

Example 5.9.8 Let's apply Newton's method to the function $f(x) = x^2 - 1$. The zeros of f are $\bar{x}_1 = 1$ and $\bar{x}_2 = -1$. A simple calculation yields

$$N(x) = \frac{1}{2}x + \frac{1}{2x}.$$

The graphs of $f(x)$ and $N(x)$ are sketched below. The sketch of the graph of $N(x)$ shows that $N'(x)$ at its fixed points $\bar{x}_i$, which are the zeros of f .



The next step is to provide an estimate of the radius δ of a ball $B_\delta(\bar{x})$ within which such quadratic convergence to $\bar{x}$ is guaranteed. Indeed, for any $0 < \delta < \frac{1}{K}$, say $\delta = \frac{1}{K}$, where K is given in (5.95), the Newton function N is a contraction over $B_\delta(\bar{x})$. So we now have the existence of a unique fixed point $\bar{x}$, hence zero of f .

We now want to analyse the Newton method as applied to Banach spaces, i.e., we want to solve the equation $F(x) = 0$, where $F : X \rightarrow X$ is a function on a Banach space X . The simple geometric picture employing tangents to the curve $y = f(x)$ for functions $f : \mathbb{R} \rightarrow \mathbb{R}$ may not apply in this case, but we can come up with a Newton-like function, the so-called *Newton-Kantorovich* function, associated with F as follows. First of all, we approximate F in a neighbourhood of a point $x_0 \in X$ in terms of its Fréchet derivative,

$$F(x) \approx F(x_0) + DF(x_0)(x - x_0). \quad (5.101)$$

This follows from the formula definition of the Fréchet derivative, where we have ignored the remainder term $R(x_0, x) = o(h)$, where $h = x - x_0$. (Remember that $DF(x_0)$ is a *operator*, i.e.,

$DF(x_0) : X \rightarrow X$, *not* an element of X !) Ideally, we would like $F(x)$ to be zero, i.e.,

$$F(x_0) + DF(x_0)(x - x_0) = 0. \quad (5.102)$$

We now solve for x in terms of x_0 :

$$DF(x_0)(x - x_0) = -F(x_0) \Rightarrow x - x_0 = -DF(x_0)^{-1}(F(x_0)), \quad (5.103)$$

assuming that the Fréchet derivative $DF(x_0) \in B(X)$, a linear operator, is invertible. A rearrangement yields

$$x = x_0 - DF(x_0)^{-1}F(x_0). \quad (5.104)$$

In other words, given an estimate of x_0 of a zero of F , we produce a new estimate

$$x_1 = N(x_0), \quad (5.105)$$

where $N : X \rightarrow X$ denotes the Newton-Kantorovich (NK) function associated with F :

$$N(x) = x - F'(x)^{-1}F(x) \quad \text{for all } x \in X, \quad (5.106)$$

where, for simplicity, $F'(x) \equiv DF(x)$. Once again, $F(\bar{x}) = 0$ implies that $N(\bar{x}) = \bar{x}$, i.e., that $\bar{x}$ is a fixed point of N .

Theorem 5.9.4

Let X be a Banach space, $F : X \rightarrow X$, $N(x) = x - F'(x)^{-1}F(x)$ the Newton-Kantorovich function associated with F , and $\bar{x}$ a zero of F . Then

$$DN(\bar{x}) = 0. \quad (5.107)$$

REMARK: This is the Banach space analogy to (5.98).

PROOF: Let's write the NK mapping in (5.106) as

$$N(x) = I(x) - DF(x)^{-1}F(x). \quad (5.108)$$

We now compute the Fréchet derivative of N on the right-hand side of the above equation:

$$DN(x) = DI(x) - D[DF(x)^{-1}F(x)] = I(x) - [D[DF(x)^{-1}]]F(x) - DF(x)^{-1}DF(x) \quad (5.109)$$

$$= I(x) - DF(x)^{-1}F(x) - I(x) \quad (5.110)$$

$$= -DF(x)^{-1}F(x), \quad (5.111)$$

where in the second step we used the product rule for Fréchet derivatives (make sure you know exactly what this is...). Then, substituting $F(\bar{x}) = 0$ into (5.111) gives the result. ■

It can be easily shown that $\|DN(\bar{x})\| \leq 1$, so that an argument involving the generalised mean value theorem may be invoked to deduce the existence of a ball $B_\delta(\bar{x})$ within which convergence of the iteration sequence $x_{n+1} = N(x_n)$ to $\bar{x}$ is guaranteed (show this!).

Recalling (5.98), the fact that $\|DX(\bar{x})\| = 0$ suggests that iteration of the NK function might exhibit quadratic convergence near $\bar{x}$, keeping in mind the scalar case. The proof of quadratic convergence is contained in the following theorem.

Theorem 5.9.5

Let X be a Banach space, $F : X \rightarrow X$, $\bar{x}$ a zero of F , and suppose F is Fréchet differentiable in $\overline{B_{\delta_0}(\bar{x})}$, with $F'(\bar{x})^{-1} \in B(X)$ and

$$\|F'(x) - F'(y)\| \leq L \|x - y\| \quad \forall x, y \in \overline{B_{\delta_0}(\bar{x})}. \quad (5.112)$$

Then there exists $\delta \leq \delta_0$ such that if $\|x_0 - \bar{x}\| < \delta$, then the iteration sequence $x_{n+1} = N(x_n)$ converges quadratically to $\bar{x}$.

The proof of this theorem is quite complicated because it does not make any assumptions on the existence of the higher-order Fréchet derivative $F''(x)$. The following discussion is intended to provide a gentle introduction to this proof by applying some of its strategy to the Newton-Raphson function for functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

First of all, we make no assumptions on the existence of $f''(x)$, which means that we cannot use (5.97). For the moment, we simply assume that $f'(x)$ is continuous over a neighbourhood containing $\bar{x}$, with $f'(\bar{x}) \neq 0$. This implies the existence of a neighbourhood $B_\delta(\bar{x})$ over which $f'(x) \neq 0$, so that the Newton function N does not “blow up”. Then, consider the following manipulations.

$$N(x) - \bar{x} = x - \frac{f(x)}{f'(x)} - \bar{x} \quad (5.113)$$

$$= \frac{1}{f'(x)} [f'(x)(x - \bar{x}) - f(x)] \quad (5.114)$$

$$= \frac{1}{f'(x)} [f'(x)(x - \bar{x}) - f(x) + f(\bar{x})] \quad (\text{since } f(\bar{x}) = 0) \quad (5.115)$$

$$= \frac{1}{f'(x)} [f(\bar{x}) - f(x) - f'(x)(\bar{x} - x)]. \quad (5.116)$$

Eventually, we shall want to take absolute values, i.e.,

$$|N(x) - \bar{x}| = \frac{1}{|f'(x)|} |f(\bar{x}) - f(x) - f'(x)(\bar{x} - x)|. \quad (5.117)$$

Notice that the term of the right-hand side looks like a second-order remainder term coming from Taylor’s theorem applied to f at the point x . If we could assume that f were C^2 , then this term would be given by $\frac{1}{2}f''(c)(\bar{x} - x)^2$, thus arriving at our quadratic convergence result. But we’re not assuming that f is C^2 here!

Nevertheless, from our assumption that f is C^1 around $\bar{x}$, we can employ the mean value theorem, i.e., we can write

$$f(x) - f(\bar{x}) = f'(c)(x - \bar{x}) \quad \text{where } c \text{ lies between } \bar{x} \text{ and } x. \quad (5.118)$$

We may then rewrite (5.117) as

$$f(\bar{x}) - f(x) - f'(x)(\bar{x} - x) = f'(c)(\bar{x} - x) - f'(x)(\bar{x} - x) = (f'(c) - f'(x))(\bar{x} - x). \quad (5.119)$$

Insertion of this result into (5.117) yields

$$|N(x) - \bar{x}| = \frac{1}{|f'(x)|} |f'(c) - f'(x)| |\bar{x} - x|. \quad (5.120)$$

We would not like to provide an upper bound to the right-hand side. First of all, continuity of f' over the neighbourhood $B_\delta(\bar{x})$ along with the fact that $f'(x) \neq 0$ implies that

$$\frac{1}{|f'(x)|} \leq K \quad (5.121)$$

for some $K > 0$. Furthermore, continuity of f' implies that

$$|f'(c) - f'(x)| \leq M \quad (5.122)$$

for some $M \geq 0$. This yields the result

$$|N(x) - \bar{x}| \leq KM |\bar{x} - x|, \quad x \in B_\delta(\bar{x}). \quad (5.123)$$

The next step is to make $\delta > 0$, hence M , small enough so that $BM < 1$, thus making N a contraction mapping. In any case, this result is not very exciting, since the convergence is only linear.

Let us now make an additional assumption on f' , namely, that it is Lipschitz continuous, i.e., that

$$|f'(x) - f'(y)| \leq L|x - y| \quad (5.124)$$

for some $L \geq 0$. (Recall that Lipschitz continuity is stronger than continuity but not as strong as differentiability.) Substitution into (5.120) gives

$$|N(x) - \bar{x}| \leq KL |\bar{x} - x|^2, \quad x \in B_\delta(\bar{x}), \quad (5.125)$$

implying quadratic convergence. And we can also come up with $\delta > 0$, for example $\delta = \frac{1}{2KL}$, so that N is a contraction mapping on $B_\delta(\bar{x})$.

In the more general Banach space setting, it is necessary to ensure the existence of the Fréchet derivative over a neighbourhood of the zero $\bar{x}$. In what follows, we give an idea of this aspect of the proof by examining further the much simpler case considered above.

In the scalar case, the goal is to have some control on the term $\frac{1}{|f'(x)|}$ so that it does not “blow up”. Once again, we start with the assumption that f' is continuous in a neighbourhood of $\bar{x}$. This implies that $f'(x)$ is “close” to $f'(\bar{x})$ for x near $\bar{x}$. But how does this closeness translate to the reciprocals $\frac{1}{f'(x)}$ and $\frac{1}{f'(\bar{x})}$?

For simplicity of notation, we let $A = f'(\bar{x})$ and $B = f'(x)$. What can we say about $|B^{-1} - A^{-1}|$ in terms of $|B - A|$? Well,

$$\frac{1}{B} - \frac{1}{A} = \frac{1}{AB}(A - B) = \frac{1}{A} \frac{A - B}{B} = \frac{A^{-1}(A - B)}{A + B - A} = \frac{A^{-2}(A - B)}{1 - A^{-1}(A - B)}. \quad (5.126)$$

Taking absolute values,

$$\left| \frac{1}{B} - \frac{1}{A} \right| = \frac{|A^{-2}|^2 |A - B|}{|1 - A^{-1}(A - B)|} \leq \frac{|A^{-1}|^2 |A - B|}{1 - |A^{-1}| |A - B|}, \quad (5.127)$$

for $|A-B||A^{-1}| < 1$. As expected, we have $B^{-1} \rightarrow A^{-1}$ as $B \rightarrow A$. The next step in the proof is to bound the term B^{-1} in terms of A . With a little work, we can ensure that

$$\frac{1}{|B|} \leq \frac{2}{|A|} \quad (5.128)$$

over a suitable neighbourhood $B_{\delta_1}(\bar{x})$.

There is an alternative to this theorem that does not assume the existence of the root $\bar{x}$.

Theorem 5.9.6 Kantorovich

Let X be a Banach space, $F : X \rightarrow X$, and suppose F is Fréchet differentiable, with

$$\|F'(x) - F'(y)\| \leq L \|x - y\|$$

for x, y in some open convex set $D \subset X$. Further, assume that $\|F'(x_0)^{-1}\| \leq a$, $\|F'(x_0)^{-1}F(x_0)\| \leq b$, $abL < \frac{1}{2}$, $\bar{t} = \frac{1-\sqrt{1-2abL}}{aL}$, and $\overline{B_{\bar{t}-b}(x_1)} \subset D$. If (x_n) is the sequence given by $x_{n+1} = N(x_n)$ for all $n \geq 0$, then this sequence converges quadratically to the unique root $\bar{x}$ of F in D .

Example 5.9.9 Using the Newton-Kantorovich Method to Locate Fixed Points

Recall that in a previous example above we looked at the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(x, y) = \begin{pmatrix} \frac{1}{2}x + xy + \frac{1}{2}y^2 \\ x^2y \end{pmatrix}.$$

Its fixed points were

$$(\bar{x}_1, \bar{y}_1) = (0, 0), \quad (\bar{x}_2, \bar{y}_2) = (-1, 1), \quad (\bar{x}_3, \bar{y}_3) = (1, -1 + \sqrt{2}), \quad (\bar{x}_4, \bar{y}_4) = (1, -1 - \sqrt{2}),$$

and we found, using the Fréchet derivative, that only $(0, 0)$ was locally attractive. Thus, only this fixed point could be detected numerically by the iteration procedure $\mathbf{x}_{n+1} = F(\mathbf{x}_n)$.

We now devise a scheme to detect *all fixed points* of this function using the Newton-Kantorovich method. First, define the function

$$G(x, y) = F(x, y) - (x, y) = \begin{pmatrix} -\frac{1}{2}x + xy + \frac{1}{2}y^2 \\ x^2y - y \end{pmatrix}.$$

Clearly, zeros of G are fixed points of F . We now apply the NK scheme to G .

The Fréchet derivative of G is

$$DG(x, y) = \begin{pmatrix} -\frac{1}{2} + y & x + y \\ 2xy & x^2 - 1 \end{pmatrix}.$$

The Newton-Kantorovich function associated with G is then

$$NK(x, y) = (x, y) - [DG(x, y)]^{-1}G(x, y),$$

which we shall not write out explicitly.

The Newton-Kantorovich method is guaranteed to converge locally to zeros of G , i.e., to fixed points of F . This is observed numerically.

5.9.3 Application: Stability of Dynamical Systems

In this section, we indicate how the stability of equilibria can be determined in terms of a Fréchet derivative.

Non-Linear System of ODEs

Given a non-linear autonomous dynamical system

$$\dot{x} = F(x), \quad x : \mathbb{R} \rightarrow \mathbb{R}^n, \quad F : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad (5.129)$$

(autonomous means that F is not explicitly a function of t) where $\dot{x} \equiv \frac{dx}{dt}$, let $\bar{x} \in \mathbb{R}^n$ be an **equilibrium point** of the the system, i.e., suppose

$$F(\bar{x}) = 0. \quad (5.130)$$

Then, $x(t) = \bar{x}$ is a solution to (5.129).

Definition 5.9.2 Stable Equilibrium Point

An equilibrium point $\bar{x}$ of a non-linear autonomous system of ODEs is called **(locally) stable** if for all $\epsilon > 0$ there exists $\delta > 0$ such that $\|x(0) - \bar{x}\| < \delta$ implies $\|x(t) - \bar{x}\| < \epsilon$ for all $t \geq 0$.

If $\bar{x}$ is not (locally) stable, then it is called **(locally) unstable**.

To investigate the stability of the equilibrium point $\bar{x}$, we will *linearise* the system of ODEs in (5.129): we let

$$x(t) = \bar{x} + y(t),$$

where $\|y\|$ is “small”. Furthermore, consider the approximation of F near $\bar{x}$:

$$F(x) \approx F(\bar{x}) + DF(\bar{x})(x - \bar{x}) = DF(\bar{x})(x - \bar{x}),$$

where we used the fact that $F(\bar{x}) = 0$. Substituting this into (??) gives

$$\dot{\bar{x}} + \dot{y} = DF(\bar{x})(x - \bar{x}) = Ay,$$

where we let $A := DF(\bar{x})$. Since the Fréchet derivative $DF(\bar{x})$ is just the Jacobian matrix of F , the resulting system of ODEs in y is a *linear system* which has equilibrium solution $y(t) = 0$ for all t .

From the theory of linear ODEs, we then have that:

1. If all eigenvalues of A have negative real part, then all solutions $y(t) \rightarrow 0$ as $t \rightarrow \infty$, so that $y = 0$ is a locally stable equilibrium.

2. If all eigenvalues of A have positive real part, then $y = 0$ is a locally unstable equilibrium.
3. If some eigenvalues of A have positive real part and some negative, then $y = 0$ is neither locally stable nor locally unstable ("hyperbolic").

Non-Linear Discrete Systems

Now consider the non-linear iteration process

$$x_{n+1} = F(x_n), \quad F : \mathbb{R}^n \rightarrow \mathbb{R}^n. \quad (5.131)$$

Let $p \in \mathbb{R}^n$ be a fixed point of F , i.e., let p satisfy $F(p) = p$. Then, if $x_0 = p$, we have $x_n = p$ for all $n \geq 1$.

What happens to iterates near p ? Are they locally attractive/stable, or are they locally repulsive/unstable?

Once again, to investigate the stability, let

$$x_n = p + y_n, \quad (5.132)$$

and consider the linear approximation of F near p :

$$F(x) \approx F(p) + DF(p)(x - p). \quad (5.133)$$

Substitution into (5.131) gives

$$p + y_{n+1} \approx \underbrace{F(p)}_p + DF(p)(x_n - p) \Rightarrow y_{n+1} = Ay_n,$$

which is a linear discrete system in $\mathbb{R}^n$ with fixed point $y = 0$.

As before, we look at the eigenvalues λ_i of A :

1. If all eigenvalues of A satisfy $|\lambda_i| < 1$, then $|y_n| \rightarrow 0$ as $n \rightarrow \infty$, so that 0 is locally attractive/stable.
2. If all eigenvalues of A satisfy $|\lambda_i| < 1$, then $|y_n| \rightarrow \infty$ as $n \rightarrow \infty$, and so 0 is locally repulsive/unstable.
3. If some eigenvalues of A satisfy $|\lambda_i| < 1$ and some $|\lambda_i| > 1$, then 0 is neither locally attractive/stable nor locally repulsive/unstable. ("hyperbolic")

6 Inner Product Spaces and Hilbert Spaces

In a normed space, we can add vectors and multiply them by scalars, just as in elementary vector algebra. Furthermore, the norm on such a space generalises the elementary concept of the length of a vector. However, what is still missing in a general normed space, and what we would like to have if possible, is an analogue of the familiar dot product

$$a \cdot b = \alpha_1\beta_1 + \alpha_2\beta_2 + \alpha_3\beta_3,$$

and resulting formulas, notably

$$|a| = \sqrt{a \cdot a}$$

and the condition for orthogonality (perpendicularity)

$$a \cdot b = 0,$$

which are important tools in many applications. Hence, the question arises whether the dot product and orthogonality can be generalised to arbitrary vector spaces. In fact, this can be done and leads to *inner product spaces* and complete inner product spaces, which we call *Hilbert spaces*.

6.1 Definition and Examples

Definition 6.1.1 Inner Product, Inner Product Space

Let X be a (real or complex) vector space. An **inner product** on X is a mapping

$$\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{F}$$

(where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$) satisfying the following properties for all $x, y, z \in X$ and all $\alpha \in \mathbb{F}$:

1. (Linearity in First Argument) $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$;
2. (Homogeneity in First Argument) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$;
3. (Conjugate Symmetry) $\langle x, y \rangle = \overline{\langle y, x \rangle}$;
4. (Positivity) $\langle x, x \rangle \geq 0$;
5. (Strict Positivity) $\langle x, x \rangle = 0 \Leftrightarrow x = 0$.

The pair $(X, \langle \cdot, \cdot \rangle)$ is called an **inner product space**. We write simply X to refer to an inner product space if the inner product is understood.

Proposition 6.1.1

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Then we have for all $x, y, z \in X$ and all scalars α, β

1. $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$;
2. (Conjugate Homogeneity in Second Argument) $\langle x, \alpha y \rangle = \bar{\alpha} \langle x, y \rangle$;
3. (Linearity in Second Argument) $\langle x, \alpha y + \beta z \rangle = \bar{\alpha} \langle x, y \rangle + \bar{\beta} \langle x, z \rangle$;
4. (Bilinearity) $\langle \alpha x + \beta y, \gamma z + \delta w \rangle = \alpha \bar{\gamma} \langle x, z \rangle + \alpha \bar{\delta} \langle x, w \rangle + \beta \bar{\gamma} \langle y, z \rangle + \beta \bar{\delta} \langle y, w \rangle$.
5. $\langle x, 0 \rangle = \langle 0, x \rangle = 0$.

PROOF: To be completed. ■

Theorem 6.1.1

Given an inner product $\langle \cdot, \cdot \rangle$ on a vector space X , define a norm on X , called the *induced norm*, by

$$\|x\| = \sqrt{\langle x, x \rangle} \quad (6.1)$$

and a metric, called the *induced metric*, by

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}. \quad (6.2)$$

Therefore, every inner product space is a normed space and every inner product space is a metric space.

PROOF: To be completed. ■

Definition 6.1.2 Hilbert Space

A complete inner product space is called a **Hilbert space**.

Note that completeness, as with normed spaces, is defined on inner product spaces using the induced metric. This definition, combined with the theorem above and the fact that Banach spaces are (by definition) complete normed spaces, we have the following fact:

All Hilbert spaces are Banach spaces.

Note that, similar to how not all metric spaces are normed spaces, we have that *not all normed spaces are inner product spaces*, and hence, *not all Banach spaces are Hilbert spaces*.

Lemma 6.1.1

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. Then, the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$ for all $x \in X$ satisfies the **parallelogram identity**

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2) \text{ for all } x, y \in X, \quad (6.3)$$

and

$$\langle x, y \rangle = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2 - i\|x - iy\|^2 + i\|x + iy\|^2) \text{ for all } x, y \in X, \quad (6.4)$$

called the **polarisation identity**.

PROOF: We have

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle + \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 - \langle x, y \rangle - \langle y, x \rangle + \|x\|^2 + \|y\|^2 \\ &= 2(\|x\|^2 + \|y\|^2), \end{aligned}$$

as required. Also,

$$\begin{aligned} &\frac{1}{4} (\|x + y\|^2 - \|x - y\|^2 - i\|x - iy\|^2 + i\|x + iy\|^2) \\ &= \frac{1}{4} (\langle x + y, x + y \rangle - \langle x - y, x - y \rangle - i\langle x - iy, x - iy \rangle + i\langle x + iy, x + iy \rangle) \\ &= \frac{1}{4} (2\langle x, y \rangle + 2\langle y, x \rangle - i(i\langle x, y \rangle - i\langle y, x \rangle) + i(-i\langle x, y \rangle + i\langle y, x \rangle)) \\ &= \frac{1}{4} (2\langle x, y \rangle + 2\langle y, x \rangle + \langle x, y \rangle - \langle y, x \rangle + \langle x, y \rangle - \langle y, x \rangle) \\ &= \langle x, y \rangle, \end{aligned}$$

as required. ■

Theorem 6.1.2

Let $(X, \|\cdot\|)$ be a normed linear space. If $\|\cdot\|$ satisfies the parallelogram identity, then X is an inner product space with inner product

$$\langle x, y \rangle = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2 - i\|x - iy\|^2 + i\|x + iy\|^2) \text{ for all } x, y \in X \quad (6.5)$$

PROOF: To be completed. ■

Example 6.1.1 Here are some standard examples of inner product spaces that are also Hilbert spaces.

1. *Euclidean Space $\mathbb{R}^n$* : The space $\mathbb{R}^n$ is a Hilbert space with (standard) inner product defined by

$$\langle x, y \rangle = \xi_1 \eta_1 + \cdots + \xi_n \eta_n, \quad (6.6)$$

where $x = (\xi_1, \dots, \xi_n)$ and $y = (\eta_1, \dots, \eta_n)$. (Is it possible to define other inner products?) From this, we obtain

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\xi_1^2 + \cdots + \xi_n^2},$$

so that the norm induced by the standard inner product is the Euclidean norm, and of course the induced metric is then the Euclidean metric:

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle} = \sqrt{(\xi_1 - \eta_1)^2 + \cdots + (\xi_n - \eta_n)^2}.$$

As we have seen already, $\mathbb{R}^n$ is complete with respect to this metric.

2. *Complex Space $\mathbb{C}^n$* : The space $\mathbb{C}^n$ is a Hilbert space with (standard) inner product given by

$$\langle x, y \rangle = \xi_1 \overline{\eta_1} + \cdots + \xi_n \overline{\eta_n}, \quad (6.7)$$

from which we get the induced norm

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\xi_1 \overline{\xi_1} + \cdots + \xi_n \overline{\xi_n}} = \sqrt{|\xi_1|^2 + \cdots + |\xi_n|^2},$$

and induced metric

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle} = \sqrt{|\xi_1 - \eta_1|^2 + \cdots + |\xi_n - \eta_n|^2}.$$

As we already know, $\mathbb{C}^n$ is complete with respect to this metric.

3. *The Sequence Space ℓ_2* : The space ℓ_2 , which remember is the set of all square-summable (real or complex) sequences, is an inner product space with inner product defined by

$$\langle x, y \rangle = \sum_{j=1}^{\infty} \xi_j \overline{\eta_j}. \quad (6.8)$$

Convergence of this series follows from the Cauchy-Schwarz inequality and the fact that $x, y \in \ell_2$. The induced norm is the two-norm

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{j=1}^{\infty} |\xi_j|^2},$$

and with respect to the corresponding induced norm, we know that this space is complete.

ℓ_2 is the prototype of a Hilbert space. It was introduced and investigated by D. Hilbert (1912) in his work on integral equations. An axiomatic definition of Hilbert space was not given until much later by J. von Neumann (1927) in a paper on the mathematical foundation of quantum mechanics.

Example 6.1.2 Here are some examples of spaces that are *not* inner product spaces.

1. *The Space ℓ_p* : The space $(\ell_p, \|\cdot\|_p)$ with $p \neq 2$ is not an inner product space, hence not a Hilbert space.

PROOF: This statement means that the norm of ℓ_p with $p \neq 2$ cannot be obtained from an inner product. (Another way of stating this is that if we attempted to define an inner product by means of the polarisation identity, it would not satisfy the definition of an inner product because doing so would require the norm to satisfy the parallelogram identity.) We prove this by showing that the norm does not satisfy the parallelogram identity. In fact, let us take $x = (1, 0, 0, \dots) \in \ell_p$ and $y = (1, -1, 0, 0, \dots) \in \ell_p$ and calculate

$$\|x\| = \|y\| = 2^{1/p} \quad \text{and} \quad \|x + y\| = \|x - y\| = 2.$$

It is then clear that the parallelogram identity is not satisfied for $p \neq 2$. ■

But we know that ℓ_p is complete with respect to the norm $\|\cdot\|_p$ defined on it. Hence, ℓ_p with $p \neq 2$ is an example of a Banach space that is not a Hilbert space.

2. *The Space $C[a, b]$* : The space $(C[a, b], \|\cdot\|_\infty)$, which remember is the space of all continuous real-valued functions defined on the interval $[a, b] \subset \mathbb{R}$, is *not* an inner product space, hence not a Hilbert space. (But $\|\cdot\|_\infty$ is not the only norm one can define on $C[a, b]$ (though it is the only norm one can define to make $C[a, b]$ *complete*)—could we generate an inner product using another norm, for example, $\|f\|_p = \left(\int_a^b |f(t)|^p\right)^{1/p}$? Because $(C[a, b], \|\cdot\|_p)$ is not complete, even if we can define an inner product the space won't be Hilbert.)

PROOF: We show that the infinity-norm $\|\cdot\|_\infty$ cannot be obtained from an inner product since this norm does not satisfy the parallelogram identity. Indeed, if we take $x(t) = 1 \ \forall t \in [a, b]$ and $y(t) = \frac{t-a}{b-a}$, then we have $\|x\|_\infty = 1$ and $\|y\|_\infty = 1$, and

$$x(t) + y(t) = 1 + \frac{t-a}{b-a}, \quad x(t) - y(t) = 1 - \frac{t-a}{b-a}.$$

Hence, $\|x + y\|_\infty = 2$, $\|x - y\|_\infty = 1$ and

$$\|x + y\|_\infty^2 + \|x - y\|_\infty^2 = 5 \quad \text{but} \quad 2(\|x\|_\infty^2 + \|y\|_\infty^2) = 4. \quad \blacksquare$$

Lemma 6.1.2 Schwarz Inequality, Triangle Inequality

An inner product and the corresponding norm satisfy the Schwarz inequality and the triangle inequality as follows.

1. (Schwarz Inequality)

$$|\langle x, y \rangle| \leq \|x\| \|y\|, \quad (6.9)$$

where the equality sign holds if and only if $\{x, y\}$ is a linearly dependent set.

2. (Triangle Inequality)

$$\|x + y\| \leq \|x\| + \|y\|, \quad (6.10)$$

where the equality sign holds if and only if $y = 0$ or $x = cy$ for $c \in \mathbb{R}$ and $c \geq 0$.

PROOF:

1. If $y = 0$, then (6.9) holds since $\langle x, 0 \rangle = 0$ for all x . Let $y \neq 0$. For every scalar α , we have

$$0 \leq \|x - \alpha y\|^2 = \langle x - \alpha y, x - \alpha y \rangle = \langle x, x \rangle - \bar{\alpha} \langle x, y \rangle - \alpha [\langle y, x \rangle - \bar{\alpha} \langle y, y \rangle].$$

We see that the expression in the brackets $[\dots]$ is zero if we choose $\bar{\alpha} = \frac{\langle y, x \rangle}{\langle y, y \rangle}$. The remaining inequality is

$$0 \leq \langle x, x \rangle - \frac{\langle y, x \rangle}{\langle y, y \rangle} \langle x, y \rangle = \|x\|^2 - \frac{|\langle x, y \rangle|^2}{\|y\|^2},$$

in which we have used $\langle y, x \rangle = \overline{\langle x, y \rangle}$. Multiplying by $\|y\|^2$, transferring the last term to the left and taking square roots, we obtain the Schwarz inequality (6.9).

Now, equality holds in this derivation if and only if $y = 0$ or $0 = \|x - \alpha y\|^2$, hence $x - \alpha y = 0$, so that $x = \alpha y$, which shows linear dependence.

2. We have

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2.$$

By the Schwarz inequality,

$$|\langle x, y \rangle| = |\langle y, x \rangle| \leq \|x\| \|y\|.$$

By the triangle inequality for real numbers, we thus obtain

$$\|x + y\|^2 \leq \|x\|^2 + 2|\langle x, y \rangle| + \|y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Taking square roots on both sides gives us the required triangle inequality.

Now, equality holds on this derivation if and only if

$$\|x\| \|y\| + \|y\| \|x\| = 2\|x\| \|y\|.$$

The left-hand side of this equation is $2\operatorname{Re}(\langle x, y \rangle)$. From this and the Schwarz inequality,

$$\operatorname{Re}(\langle x, y \rangle) = \|x\| \|y\| \geq |\langle x, y \rangle|. \quad (6.11)$$

Since the real part of a complex number cannot exceed the absolute value, we must have equality, which implies linear dependence by the first part of this proof, say, $y = 0$ or $x = cy$.

We now show that c is real and that it is non-negative. From (6.11) with the equality sign, we have $\operatorname{Re}(\langle x, y \rangle) = |\langle x, y \rangle|$. But if the real part of a complex number equals the absolute value, then the imaginary part must be zero. Hence, $\langle x, y \rangle = \operatorname{Re}(\langle x, y \rangle) \geq 0$ by (6.11), and $c \geq 0$ follows from

$$0 \leq \langle x, y \rangle = \langle cy, y \rangle = c \|y\|^2. \quad \blacksquare$$

Now, we can define sequences and series (and their convergence) in inner product spaces exactly as we did in normed (and metric) spaces because all inner product spaces are normed spaces. We can use sequences to prove that the inner product is a *continuous* function.

Proposition 6.1.2 Continuity of the Inner Product

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. If the sequences (x_n) and (y_n) in X converge to x and y , respectively, then the sequence $(\langle x_n, y_n \rangle)$ of real numbers converges to $\langle x, y \rangle$. Succinctly,

$$\lim_{n \rightarrow \infty} \langle x_n, y_n \rangle = \left\langle \lim_{n \rightarrow \infty} x_n, \lim_{n \rightarrow \infty} y_n \right\rangle \quad (6.12)$$

(provided the limits on the right-hand side exist).

PROOF: Subtracting and adding a term, using the triangle inequality for numbers and, finally, the Schwarz inequality, we obtain

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &= |\langle x_n, y_n \rangle - \langle x_n, y \rangle + \langle x_n, y \rangle - \langle x, y \rangle| \\ &\leq |\langle x_n, y_n - y \rangle| + |\langle x_n - x, y \rangle| \\ &\leq \|x_n\| \|y_n - y\| + \|x_n - x\| \|y\|, \end{aligned}$$

which goes to zero as n goes to infinity since $y_n - y \rightarrow 0$ and $x_n - x \rightarrow 0$ as $n \rightarrow \infty$. $\blacksquare$

Definition 6.1.3 Subspace

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space and a *vector subspace* $Y \subset X$. Then $(Y, \langle \cdot, \cdot \rangle|_{Y \times Y})$ is an inner product space, called the **subspace** of $(X, \langle \cdot, \cdot \rangle)$.

REMARK: As a reminder, a *vector subspace* is a non-empty subset Y of X such that for all $y_1, y_2 \in Y$ and all scalars α, β we have $\alpha y_1 + \beta y_2 \in Y$. Hence, the subspace Y is itself a vector space with the operations of addition and scalar multiplication induced from those on X .

Observe that a subspace Y of a vector space (and hence a subspace of an inner product space) is *convex*. Indeed, for every $x, y \in Y$, the segment joining x and y , that is, the set of points

$$z = \alpha x + (1 - \alpha)y \quad \text{for all } \alpha \in [0, 1]$$

is contained in Y by definition.

Note that a subspace may not be complete even if the containing space is complete. Hence, a subspace of a Hilbert space may not be Hilbert.

Also, note that

Theorem 6.1.3 Subspace

Let Y be a subspace of a Hilbert space H .

1. Y is complete if and only if Y is closed in H .
2. If Y is finite-dimensional, then Y is complete.
3. If H is separable, so is Y . More generally, every subset of a separable inner product space is separable.

PROOF:

1. Follows immediately from Theorem 3.5.1.
2. Follows immediately from Theorem 5.2.8.
3. To be completed. ■

Definition 6.1.4 Isomorphism

An **isomorphism** T of an inner product space $(X, \langle \cdot, \cdot \rangle_X)$ into an inner product space $(\tilde{Y}, \langle \cdot, \cdot \rangle_Y)$ over the same field is a bijective linear operator $T : X \rightarrow Y$ that preserves the inner product, that is, for all $x, y \in X$,

$$\langle T(x), T(y) \rangle_Y = \langle x, y \rangle_X.$$

X is then called **isomorphic** to Y and we sometimes write $X \cong Y$.

REMARK: Note that the bijectivity and linearity guarantees that T is a vector space isomorphism of X onto Y , so that T preserves the whole structure of inner product space. T is also an isometry of X onto Y because distances in X and Y are determined by the induced norms.

Also, inner product isomorphism $\cong$ is an equivalence relation.

Theorem 6.1.4 Isomorphism and Hilbert Dimension

Two Hilbert space H and $\tilde{H}$, both real or both complex, are isomorphic if and only if they have the same dimension.

Example 6.1.3 Let $T : X \rightarrow X$ be a bounded linear operator on a complex inner product space $(X, \langle \cdot, \cdot \rangle)$. If $\langle T(x), x \rangle = 0$ for all $x \in X$, show that $T = 0$. Show that this does not hold in the case of a real inner product space. (Hint: Consider a rotation of the Euclidean plane.)

SOLUTION:

6.2 Properties of Inner Product and Hilbert Spaces

6.2.1 Completion

We now show that every inner product space can be completed.

Theorem 6.2.1 Completion

For any inner product space X there exists a Hilbert space H and an isomorphism A from X onto a dense subspace $W \subset H$. The space H is unique up to isomorphism.

Example 6.2.1 We briefly discussed the L_p spaces in §3.7 and we mentioned the following important fact: for all $[a, b] \subset \mathbb{R}$, the space $(L_p[a, b], d_p)$ is the completion of $(C[a, b], d_p)$. This fact holds even if we decide to make the functions in these spaces *complex-valued* (keeping $t \in [a, b] \subset \mathbb{R}$, as before).

We now consider the space $L_2[a, b]$. This is an inner product space with inner product

$$\langle x, y \rangle = \int_a^b x(t) \overline{y(t)} dt. \quad (6.13)$$

The induced norm is then

$$\|x\| = \sqrt{\int_a^b |x(t)|^2 dt}, \quad (6.14)$$

where here $|\cdot|$ denotes the *modulus* of the (generally complex) number $x(t)$, so that $|x(t)| = x(t)\overline{x(t)}$.

Because $L_2[a, b]$ (with the inner product as in (6.13)) is the completion of $C[a, b]$, by the completion theorem above we have that $L_2[a, b]$ is complete for all $[a, b] \subset \mathbb{R}$, so that in fact $L_2[a, b]$ is a Hilbert space.

6.2.2 Orthogonality

Definition 6.2.1 Orthogonality

Let $(X, \langle \cdot, \cdot \rangle)$ be an inner product space. An element $x \in X$ is said to be **orthogonal** to an element $y \in X$ if

$$\langle x, y \rangle = 0,$$

and we sometimes write $x \perp y$.

If $A, B \subset X$, then we say that x is *orthogonal to A* if $\langle x, a \rangle = 0$ for all $a \in A$, and we say that A is *orthogonal to B* if $\langle a, b \rangle = 0$ for all $a \in A$ and all $b \in B$.

Now, in a metric space (X, d) , the *distance*, denoted δ , from an element $x \in X$ to a non-empty subset

$M \subset X$ is defined to be

$$\delta := \inf_{\tilde{y} \in M} d(x, \tilde{y}).$$

In a normed space, this becomes

$$\delta = \inf_{\tilde{y} \in M} \|x - \tilde{y}\|, \quad (M \neq \emptyset). \quad (6.15)$$

It is important to know whether there is a $y \in M$ such that

$$\delta = \|x - y\|, \quad (6.16)$$

that is, intuitively speaking, a point $y \in M$ that is closest to the given x , and if such an element exists, whether it is unique. For general normed spaces, this can be a difficult question to answer, but for Hilbert spaces the situation becomes relatively simpler.

Theorem 6.2.2 Minimising Vector

Let X be an inner product space and $M \neq \emptyset$ a convex subset that is complete (in the metric induced by the inner product). Then, for every given $x \in X$ there exists a unique $y \in M$ such that

$$\delta = \inf_{\tilde{y} \in M} \|x - \tilde{y}\| = \|x - y\|. \quad (6.17)$$

PROOF:

1. *Existence:* By the definition of an infimum, there is a sequence (y_n) such that

$$\delta_n \rightarrow \delta, \quad \text{where} \quad \delta_n := \|x - y_n\|. \quad (6.18)$$

We show that (y_n) is Cauchy. Writing $y_n = x - v_n$, we have $\|v_n\| = \delta_n$ and

$$\|v_n + v_m\| = \|y_n + y_m - 2x\| = 2 \left\| \frac{1}{2}(y_n + y_m) - x \right\| \geq 2\delta,$$

because M is convex, so that $\frac{1}{2}(y_n + y_m) \in M$. Furthermore, we have $y_n - y_m = v_n - v_m$. Hence, by the parallelogram identity,

$$\|y_n - y_m\|^2 = \|v_n - v_m\|^2 = -\|v_n + v_m\|^2 + 2(\|v_n\|^2 + \|v_m\|^2) \leq -(2\delta)^2 + 2(\delta_n^2 + \delta_m^2),$$

and (6.18) implies that (y_n) is Cauchy. Since M is complete, (y_n) converges, say to $y \in M$. Because of this, we have $\|x - y\| \geq \delta$. Also, by (6.18)

$$\|x - y\| \leq \|x - y_n\| + \|y_n - y\| = \delta_n + \|y_n - y\| \rightarrow \delta.$$

This shows that $\|x - y\| = \delta$.

2. *Uniqueness*: We assume that $y \in M$ and $y_0 \in M$ both satisfy

$$\|x - y\| = \delta \quad \text{and} \quad \|x - y_0\| = \delta$$

and show that then $y_0 = y$. By the parallelogram equality,

$$\begin{aligned} \|y - y_0\| &= \|(y - x) - (y_0 - x)\|^2 \\ &= 2\|y - x\|^2 + 2\|y_0 - x\|^2 - \|(y - x) + (y_0 - x)\|^2 \\ &= 2\delta^2 + 2\delta^2 - 2^2 \left\| \frac{1}{2}(y + y_0) - x \right\|^2. \end{aligned}$$

On the right, $\frac{1}{2}(y + y_0) \in M$, so that

$$\left\| \frac{1}{2}(y + y_0) - x \right\| \geq \delta. \quad (6.19)$$

This implies that the right-hand side is less than or equal to $2\delta^2 + 2\delta^2 - 4\delta^2 = 0$. Hence, we have the inequality $\|y - y_0\| \leq 0$. Clearly, $\|y - y_0\| \geq 0$, so that we must have equality, meaning $y_0 = y$. ■

Turning from arbitrary convex sets to subspaces, we obtain a lemma that generalises the familiar idea of elementary geometry that the unique point y in a given subspace Y closes to a given x is found by “dropping a perpendicular from x to Y ”.

Lemma 6.2.1 Orthogonality

Let X be an inner product space and $Y \neq \emptyset$ a complete subspace. Assume $x \in X$ is fixed. Then $z = x - y$ is orthogonal to Y .

PROOF: If $z \perp Y$ were false, then there would be a $y_1 \in Y$ such that

$$\langle z, y_1 \rangle = \beta \neq 0. \quad (6.20)$$

Clearly, $y_1 \neq 0$ since otherwise $\langle z, y_1 \rangle = 0$. Furthermore, for any scalar α ,

$$\begin{aligned} \|z - \alpha y_1\|^2 &= \langle z - \alpha y_1, z - \alpha y_1 \rangle \\ &= \langle z, z \rangle - \bar{\alpha} \langle z, y_1 \rangle - \alpha [\langle y_1, z \rangle - \bar{\alpha} \langle y_1, y_1 \rangle] \\ &= \langle z, z \rangle - \bar{\alpha} \beta - \alpha [\bar{\beta} - \bar{\alpha} \langle y_1, y_1 \rangle]. \end{aligned}$$

The expression in the brackets $[\dots]$ is zero if we choose

$$\bar{\alpha} \frac{\bar{\beta}}{\langle y_1, y_1 \rangle}.$$

From (6.17), we have $\|z\| = \|x - y\| = \delta$, so that our equation now yields

$$\|z - \alpha y_1\|^2 = \|z\|^2 - \frac{|\beta|^2}{\langle y_1, y_1 \rangle} < \delta^2.$$

But this is impossible because we have

$$z - \alpha y_1 = x - y_2 \quad \text{where} \quad y_2 = y + \alpha y_1 \in Y,$$

so that $\|z - \alpha y_1\| \geq \delta$ by the definition of δ . Hence (6.20) cannot hold, and the lemma is proved. ■

Definition 6.2.2 Direct Sum

A vector space X is said to be the **direct sum** of two subspaces Y and Z of X , written

$$X = Y \oplus Z,$$

if each $x \in X$ has a unique representation

$$x = y + z \text{ for some } y \in Y \text{ and some } z \in Z.$$

Then Z is called an *algebraic complement* of Y in X and vice versa, and the pair Y, Z is called a *complementary pair* of subspaces in X .

For example, $Y = \mathbb{R}$ is a subspace of the Euclidean plane $\mathbb{R}^2$. Y has infinitely many algebraic complements in $\mathbb{R}^2$, each of which is a real line. But most convenient is a complement that is perpendicular. We make use of this fact when we choose a Cartesian coordinate system. In $\mathbb{R}^3$, the situation is the same in principle.

Definition 6.2.3 Orthogonal Complement

Let X be an inner product space and $Y \subset X$ a subspace. The **orthogonal complement** of Y , denoted $Y^\perp$, is defined as

$$Y^\perp = \{z \in X \mid z \perp Y\},$$

i.e., it is set of all vectors in X that are orthogonal to all vectors in Y .

Proposition 6.2.1

Let Y be a finite-dimensional subspace of an inner product space X . Then,

1. $Y^\perp$ is a subspace of X ; and
2. $Y \cap Y^\perp = \{0\}$.

REMARK: See if the finite-dimensional requirement can be removed.

PROOF:

1. Taking $x, y \in Y^\perp$ implies that for all $v \in Y$ and all scalars α, β ,

$$\langle \alpha x + \beta y, v \rangle = \alpha \langle x, v \rangle + \beta \langle y, v \rangle = 0,$$

hence $\alpha x + \beta y \in Y^\perp$.

2. To be completed. ■

Note that the complement of the complement, i.e., $(Y^\perp)^\perp$, is written $Y^{\perp\perp}$. Then, in general we have

$$Y \subset Y^{\perp\perp} \quad (6.21)$$

because

$$x \in Y \implies x \perp Y^\perp \implies x \in (Y^\perp)^\perp.$$

The reverse containment $M^{\perp\perp} \subset M$ is not always true, as we'll see below.

Also, observe that if we take the direct sum of a subspace Y of an inner product space and its orthogonal complement, i.e., we consider the subspace $S := Y \oplus Y^\perp$ (is the direct sum of two subspaces a subspace?), then we have that for all $s \in S$ there exists $y \in Y$ and $y^\perp \in Y^\perp$ such that $s = y + y^\perp$. Additionally,

$$\|s\| = \|y\| + \|y^\perp\| \text{ for all } s \in S.$$

Proposition 6.2.2

Let S be a subset of a Hilbert space. Then $S^\perp$ is a closed linear subspace.

REMARK: Note that S is a *subset*, not a subspace (as stated in the course notes).

PROOF: To be completed. ■

Theorem 6.2.3 Direct Sum/Projection Theorem

Let Y be any closed subspace of a Hilbert space H . Then

$$H = Y \oplus Y^\perp. \quad (6.22)$$

This representation is unique.

PROOF: Since H is complete and Y is closed, by Theorem 3.5.1 Y is complete. Since Y is convex, Theorem 6.2.2 and Lemma 6.2.1 imply that for every $x \in H$ there is a $y \in Y$ such that

$$x = y + z \text{ for } z \in Y^\perp. \quad (6.23)$$

To prove uniqueness, we assume that

$$x = y + z = y_1 + z_1,$$

where $y, y_1 \in Y$ and $z, z_1 \in Y^\perp$. Then, $y - y_1 = z_1 - z$. Since $y - y_1 \in Y$, whereas $z_1 - z \in Y^\perp$, we see that $y - y_1 \in Y \cap Y^\perp = \{0\}$. This implies that $y = y_1$. Hence also $z = z_1$. ■

REMARK: There is no reason to stop the discussion of direct sums with two subspaces. For example, in the statement of the theorem, $H = Y \oplus Z$, it may be possible to divide the subspace $Z \subset H$ into a pair of orthogonal complements, i.e.,

$$Z = Z_1 \oplus Z_2.$$

Then we may write

$$H = Y \oplus Z_1 \oplus Z_2.$$

This means that for every $x \in H$ there correspond unique $y \in Y$, $z_1 \in Z_1$ and $z_2 \in Z_2$ such that

$$x = y + z_1 + z_2.$$

And it may be possible to perform further “splitting” of the spaces. The above notation is awkward for such a generalised treatment. It is often the practice to write the decomposition more compactly as

$$H = E_1 \oplus E_2 \oplus \cdots \oplus E_n,$$

where the subspaces E_k are orthogonal to each other, i.e.,

$$E_k \perp E_\ell, \quad k \neq \ell.$$

This means that for any $x \in E_k$ and $y \in E_\ell$, we have $\langle x, y \rangle = 0$.

Definition 6.2.4 Orthogonal Projection

Let X be an inner product space and $Y \subset X$ a subspace. Then, any $s \in S := Y \oplus Y^\perp$ can be written as

$$s = y + z,$$

where $y \in Y$ and $z \in Y^\perp$. y is called the **orthogonal projection** (or often just **projection**) of s onto Y , often denoted $y \equiv \text{proj}_Y(s)$, and z is called the *perpendicular to the projection*, often denoted $z \equiv \text{perp}_Y(s)$.

Definition 6.2.5 Orthogonal Projection Operator

Let X be an inner product space and $Y \subset X$ a subspace. Then, any $s \in S := Y \oplus Y^\perp$ can be written as

$$s = y + z,$$

The mapping

$$P : S \rightarrow Y, \quad s \mapsto y = P(s)$$

is called the **(orthogonal) projection operator** of S onto Y .

Proposition 6.2.3 Properties of the Projection Operator

Let $P : S \rightarrow Y$ be the orthogonal projection operator on a subspace S of an inner product space into Y . Then, P is

1. Bounded and linear;
2. Maps S onto Y ;
3. Y onto itself;
4. $Y^\perp$ onto $\{0\}$;
5. is *idempotent*, i.e., $P^2 = P$; and
6. $P|_Y = I_Y$.

PROOF: To be completed. ■

Lemma 6.2.2 Closed Subspace

If Y is a closed subspace of a Hilbert space H , then

$$Y = Y^{\perp\perp} \quad (6.24)$$

PROOF: We already have that $Y \subset Y^{\perp\perp}$ from (6.21). We therefore only show that $Y \supset Y^{\perp\perp}$. Let $x \in Y^{\perp\perp}$. Then, $x = y + z$ by Theorem 6.2.3, where $y \in Y \subset Y^{\perp\perp}$ by (6.21). Since $Y^{\perp\perp}$ is a vector space and $x \in Y^{\perp\perp}$ by assumption, we also have $z = x - y \in Y^{\perp\perp}$, hence $z \perp Y^\perp$. But $z \in Y^\perp$ by Theorem 6.2.3. Together, $z \perp z$, hence $z = 0$, so that $x = y$, that is, $x \in Y$. Since $x \in Y^{\perp\perp}$ was arbitrary, this proves that $Y \supset Y^{\perp\perp}$. ■

Lemma 6.2.3 Dense Set

For any subset $M \neq \emptyset$ of a Hilbert space H , the span of M is dense in H if and only if $M^\perp = \{0\}$.

PROOF: ($\Rightarrow$) Let $x \in M^\perp$ and assume $V = \text{span}(M)$ to be dense in H . Then, $x \in \overline{V} = H$. By Theorem 3.3.1, there is a sequence (x_n) in V such that $\lim_{n \rightarrow \infty} x_n = x$. Since $x \in M^\perp$ and $M^\perp \perp V$, we have that $\langle x_n, x \rangle = 0$. The continuity of the inner product implies that $\lim_{n \rightarrow \infty} \langle x_n, x \rangle \rightarrow \langle x, x \rangle$. Together, $\langle x, x \rangle = \|x\|^2 = 0$, so that $x = 0$. Since $x \in M^\perp$ was arbitrary, this shows that $M^\perp = \{0\}$.

($\Leftarrow$) Conversely, suppose that $M^\perp = \{0\}$. If $x \perp V$, then $x \perp M$, so that $x \in M^\perp$ and $x = 0$. Hence $V^\perp = \{0\}$. Noting that V is a subspace of H , we thus obtain $\overline{V} = H$ from Theorem 6.2.3 with $Y = \overline{V}$. ■

6.2.3 Orthonormal Sets and Sequences

Orthogonality of elements plays a basic role in inner product spaces and Hilbert spaces. A first impression of this fact was given in the preceding section. Of particular interest are sets whose elements are orthogonal in pairs. To understand this, let us remember a familiar situation in Euclidean space, $\mathbb{R}^3$. In the space $\mathbb{R}^3$, a set of that kind is the set of the three unit vectors in the positive directions of the axes of a rectangular coordinate system; call these vectors e_1, e_2, e_3 . These vectors form a basis for $\mathbb{R}^3$, so that every $x \in \mathbb{R}^3$ has a unique *representation*

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3.$$

Now we see a great advantage of the orthogonality. Given x , we can readily determine the unknown coefficients $\alpha_1, \alpha_2, \alpha_3$ by taking inner product (i.e., dot products in this case). In fact, to obtain α_1 , for example, we must multiply that representation of x by e_1 , that is,

$$\langle x, e_1 \rangle = \alpha_1 \langle e_1, e_1 \rangle + \alpha_2 \langle e_2, e_1 \rangle + \alpha_3 \langle e_3, e_1 \rangle = \alpha_1.$$

In more general inner product spaces, there are similar and other possibilities for the use of orthogonal and orthonormal sets and sequences.

Definition 6.2.6 Orthonormal Sets and Sequences

An **orthogonal set** M in an inner product space X is a subset $M \subset X$ whose elements are pairwise orthogonal. An **orthonormal set** $M \subset X$ is an orthogonal set in X whose elements have norm one, that is, for all $x, y \in M$,

$$\langle x, y \rangle = \delta_{xy} = \begin{cases} 0, & \text{if } x \neq y \\ 1, & \text{if } x = y. \end{cases} \quad (6.25)$$

If an orthogonal or orthonormal set M is countable, we can arrange it in a sequence (x_n) and call it an **orthogonal** or **orthonormal sequence**, respectively.

Theorem 6.2.4

Pythagorean Identity Let $\{z_1, z_2, \dots, z_n\}$ be an orthogonal set in an inner product space. Then,

$$\|z_1 + z_2 + \dots + z_n\|^2 = \|z_1\|^2 + \|z_2\|^2 + \dots + \|z_n\|^2. \quad (6.26)$$

PROOF: Using the fact that $\langle z_j, z_k \rangle = \delta_{jk}$, we get

$$\left\| \sum_{j=1}^n z_j \right\|^2 = \left\langle \sum_{j=1}^n z_j, \sum_{k=1}^n z_k \right\rangle = \sum_{j,k=1}^n \langle z_j, z_k \rangle = \sum_{k=1}^n \langle z_k, z_k \rangle = \sum_{k=1}^n \|z_k\|^2. \quad \blacksquare$$

Now, suppose that $\{x_1, x_2, \dots, x_n\}$ is an orthogonal set in an inner product space X . Defining the sets

$$E_k := \text{span}(x_k) = \{cx_k \mid c \in \mathbb{R}\}, \quad k = 1, 2, \dots, n,$$

we have that each subset E_k is a one-dimensional closed subspace of X . From the orthogonality of the $\{x_k\}$, it follows that

$$E_k \perp E_\ell \text{ for all } k \neq \ell.$$

In other words, the sets E_k are orthogonal to each other. If $k \neq \ell$, then for all $x \in E_k$ and all $y \in E_\ell$, $\langle x, y \rangle = 0$.

Lemma 6.2.4 Linear Independence

An orthonormal set in an inner product space is linearly independent.

PROOF: Let the set $\{e_1, \dots, e_n\}$ in an inner product space $(X, \langle \cdot, \cdot \rangle)$ be orthonormal and consider the equation

$$\alpha_1 e_1 + \dots + \alpha_n e_n = 0.$$

Multiplication by a fixed e_j gives

$$\left\langle \sum_{k=1}^n \alpha_k e_k, e_j \right\rangle = \sum_{k=1}^n \alpha_k \langle e_k, e_j \rangle = \alpha_j \langle e_j, e_j \rangle = \alpha_j = 0,$$

proving linear independence for any *finite* orthonormal set. This also implies linear independence if the given orthonormal set is infinite. (How?) ■

Recall from linear algebra the definition of a *basis* of a vector space: it is linearly independent set that spans the vector space. From this lemma, we see that every orthonormal set in an inner product space is linearly independent. Therefore, every orthonormal set in an inner product space is a basis for the subspace that it spans. In particular, if the set spans the entire space, then we have a basis of the inner product space.

Example 6.2.2 Here are some examples of orthonormal sets in standard inner product spaces.

1. *Euclidean Space $\mathbb{R}^3$* : In the space $\mathbb{R}^3$, the three unit vectors $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ in the direction of the three axes of a rectangular coordinate system form an orthonormal set.
2. *The Space ℓ_2* : In the space ℓ_2 , an orthonormal sequence is (e_n) , where $e_n = \delta_{nj}$ has the n th element one and all others zero.
3. *Continuous Functions*: Let X be the inner product space of all real-valued continuous functions on $[0, 2\pi]$ with inner product defined by

$$\langle x, y \rangle = \int_0^{2\pi} x(t)y(t) dt.$$

An orthogonal sequence in X is (u_n) , where

$$u_n(t) = \cos(nt), \quad n = 0, 1, \dots$$

Another orthonormal sequence in X is (v_n) , where

$$v_n(t) = \sin(nt), \quad n = 1, 2, \dots$$

In fact, by integration we obtain

$$\langle u_m, u_n \rangle = \int_0^{2\pi} \cos(mt) \cos(nt) dt = \begin{cases} 0, & \text{if } m \neq n \\ \pi, & \text{if } m = n = 1, 2, \dots \\ 2\pi, & \text{if } m = n = 0, \end{cases} \quad (6.27)$$

and similarly for (v_n) . Hence, an orthonormal sequence is (e_n) , where

$$e_0(t) = \frac{1}{\sqrt{2\pi}}, \quad e_n(t) = \frac{u_n(t)}{\|u_n\|} = \frac{\cos(nt)}{\sqrt{\pi}}, \quad n = 1, 2, \dots$$

From (v_n) we obtain the orthonormal sequence $(\tilde{e}_n)$, where

$$\tilde{e}_n(t) = \frac{v_n(t)}{\|v_n\|} = \frac{\sin(nt)}{\sqrt{\pi}}, \quad n = 1, 2, \dots$$

Note that we even have $u_m \perp v_n$ for all m and n (prove this!) These sequences appear of course in *Fourier series*.

A great advantage of orthonormal sequences over arbitrary linearly independent sequences is the following. If we know that a given x can be represented as a linear combination of some elements of an orthonormal sequences, then the orthonormality makes the actual determination of the coefficients very easy.

Theorem 6.2.5 Expansion Coefficients

If $E = (e_1, e_2, \dots)$ is an orthonormal sequence in an inner product space X and we have $x \in \text{span}\{e_1, \dots, e_n\}$, where n is fixed, then by the definition of the span

$$x = \sum_{k=1}^n \alpha_k e_k, \quad (6.28)$$

Then, the coefficients α_k are given by

$$\alpha_k = \langle x, e_k \rangle \quad \text{for all } 1 \leq k \leq n.$$

These are sometimes called the **Fourier coefficients** of x with respect to the set E .

REMARK: Note that this formula for the Fourier coefficients applies only when E is an *orthonormal set*. If E is merely *orthogonal*, then we have

$$\alpha_k = \frac{\langle x, e_k \rangle}{\|e_k\|} \quad \text{for all } 1 \leq k \leq n.$$

PROOF: Taking the inner product of (6.28) by a fixed e_j , we obtain

$$\langle x, e_j \rangle = \left\langle \sum_{k=1}^n \alpha_k e_k, e_j \right\rangle = \sum_{k=1}^n \alpha_k \langle e_k, e_j \rangle = \alpha_j. \quad \blacksquare$$

So the *expansion* of x with respect to an orthonormal set $(e_1, e_2, \dots, e_n)$, i.e., the expansion of an $x \in \text{span}\{e_1, \dots, e_n\}$ is given by

$$x = \sum_{k=1}^n \langle x, e_k \rangle e_k. \quad (6.29)$$

More generally, if we consider any x in an inner product space X , not necessarily in $Y_n := \text{span}\{e_1, \dots, e_n\}$, we can define $y \in Y_n$ by setting

$$y = \sum_{k=1}^n \langle x, e_k \rangle e_k, \quad (6.30)$$

where n is fixed, as before, and then define z by setting

$$x = y + z, \quad (6.31)$$

i.e., $z = x - y$. We want to show that z is orthogonal to y . To really understand what is going on here, note the following. *Every* $y \in Y_n$ is a linear combination

$$y = \sum_{k=1}^n \alpha_k e_k.$$

Here, $\alpha_k = \langle y, e_k \rangle$, as we have shown already. Our claim is that for the *particular choice* $\alpha_k = \langle x, e_k \rangle$ for $k = 1, 2, \dots, n$, we shall obtain a y such that $z = x - y \perp y$. (Think about this in the context of Theorem 6.2.2 and Theorem 6.2.3.) More specifically, we have the following:

Theorem 6.2.6

Let X be an inner product space and let $E = \{e_1, e_2, \dots, e_n\}$ be an orthonormal set in X . Let $x \in X$ be arbitrary. Then, the function $f : \mathbb{C}^n \rightarrow \mathbb{R}$ defined by

$$f(\alpha_1, \alpha_2, \dots, \alpha_n) = \left\| x - \sum_{k=1}^n \alpha_k e_k \right\|^2$$

attains an absolute minimum value at one and only one point $(\alpha_1, \dots, \alpha_n) \in \mathbb{C}^n$, namely

$$\alpha_k = \langle x, e_k \rangle, \quad k = 1, 2, \dots, n.$$

Furthermore,

$$\sum_{k=1}^n |\langle x, e_k \rangle|^2 \leq \|x\|^2.$$

REMARK: Minimising f here may be viewed as minimising the distance between x and the convex set $Y = \text{span}\{e_1, \dots, e_n\}$. The element $y = \sum_{k=1}^n \langle x, e_k \rangle e_k$ is the *unique* point in Y that *lies closest* to $x \in X$. The element $y \in Y$ may also be viewed as the *best approximation* to x in the set Y .

Also, note that if $x \in Y$, then $f_{\min} = 0$, as expected.

PROOF: We first note that, by the orthonormality,

$$\|y\|^2 = \left\langle \sum_{k=1}^n \langle x, e_k \rangle e_k, \sum_{m=1}^n \langle x, e_m \rangle e_m \right\rangle = \sum_{k=1}^n |\langle x, e_k \rangle|^2. \quad (6.32)$$

Using this, we can now show that $z \perp y$:

$$\begin{aligned} \langle z, y \rangle &= \langle x - y, y \rangle = \langle x, y \rangle - \langle y, y \rangle = \left\langle x, \sum_{k=1}^n \langle x, e_k \rangle e_k \right\rangle - \|y\|^2 \\ &= \sum_{k=1}^n \langle x, e_k \rangle \overline{\langle x, e_k \rangle} - \sum_{k=1}^n |\langle x, e_k \rangle|^2 \\ &= 0. \end{aligned}$$

Hence, by the Pythagorean identity, we get

$$\|x\|^2 = \|y\|^2 + \|z\|^2. \quad \blacksquare \quad (6.33)$$

Furthermore, by (6.32) it follows that

$$\|z\|^2 = \|x\|^2 - \|y\|^2 = \|x\|^2 - \sum_{k=1}^n |\langle x, e_k \rangle|^2. \quad (6.34)$$

Since $\|z\| \geq 0$, we have for every $n = 1, 2, \dots$

$$\sum_{k=1}^n |\langle x, e_k \rangle|^2 \leq \|x\|^2. \quad (6.35)$$

PROOF: (Alternate) We consider the square of f :

$$\begin{aligned} \left\| x - \sum_{k=1}^n \alpha_k e_k \right\|^2 &= \left\langle x - \sum_{k=1}^n \alpha_k e_k, x - \sum_{\ell=1}^n \alpha_\ell e_\ell \right\rangle \\ &= \|x\|^2 - \left\langle \sum_{k=1}^n \alpha_k e_k, x \right\rangle - \left\langle x, \sum_{\ell=1}^n \alpha_\ell e_\ell \right\rangle + \sum_{k=1}^n \sum_{\ell=1}^n \langle \alpha_k e_k, \alpha_\ell e_\ell \rangle \\ &= \|x\|^2 - \sum_{k=1}^n \left[\bar{\alpha}_k \langle x, e_k \rangle + \alpha_k \overline{\langle x, e_k \rangle} + |\alpha_k|^2 \right] \\ &= \|x\|^2 + \sum_{k=1}^n \left[|\langle x, e_k \rangle|^2 - \bar{\alpha}_k \langle x, e_k \rangle - \alpha_k \overline{\langle x, e_k \rangle} + |\alpha_k|^2 \right] - \sum_{k=1}^n |\langle x, e_k \rangle|^2 \\ &= \|x\|^2 + \sum_{k=1}^n |\langle x, e_k \rangle - \alpha_k|^2 - \sum_{k=1}^n |\langle x, e_k \rangle|^2. \end{aligned}$$

Now, the first and last terms are fixed. The middle term is a sum on non-negative numbers. The minimum value is attained when all of these terms are zero. Consequently, $f(\alpha_1, \dots, \alpha_n)$ is a minimum if and only if $\alpha_k = \langle x, e_k \rangle$ for all $k = 1, 2, \dots, n$. In this case, we see that

$$\|x\|^2 - \sum_{k=1}^n |\langle x, e_k \rangle|^2 \geq 0. \quad \blacksquare$$

Now, the sums in (6.35) have non-negative terms, so that they form a monotonically increasing sequence. This sequence converges because it is bounded by $\|x\|^2$. This sequence can be viewed as as sequence of partial sums corresponding to an infinite series. Because the sequence converges, so does the series. Therefore, we have the following result.

Theorem 6.2.7 Bessel Inequality

Let (e_k) be an orthonormal sequence in an inner product space X . Then, for every $x \in X$,

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2, \quad (6.36)$$

which is called the **Bessel inequality**.

Note that if X is finite dimensional, then every orthonormal set in X must be finite because every orthonormal set is linearly independent, as we have seen. Hence we must have a finite sum in (6.36).

Definition 6.2.7 Orthogonal Projection and Perpendicular Onto a Subspace

Suppose S is a k -dimensional subspace of an inner product space X and that $\{w_j\}_{j=1}^k$ is an orthogonal basis for S . For any $v \in X$, we define the **orthogonal projection** of v onto S by

$$\text{proj}_S(v) = \frac{\langle w_1, v \rangle}{\|w_1\|^2} w_1 + \cdots + \frac{\langle w_k, v \rangle}{\|w_k\|^2} w_k.$$

The **perpendicular of the projection**, denoted $\text{perp}_S v$, is defined as

$$\text{perp}_S(v) = v - \text{proj}_S v = \frac{\langle w_{k+1}, v \rangle}{\|w_{k+1}\|^2} w_{k+1} + \cdots + \frac{\langle w_n, v \rangle}{\|w_n\|^2} w_n.$$

Theorem 6.2.8

Suppose S is a k -dimensional subspace of an inner product space X . Then, for any $v \in X$,

$$\text{perp}_S(v) \in S^\perp.$$

PROOF: Let $\{w_j\}_{j=1}^k$ be an orthogonal basis for S . Then, we can write any $u \in X$ as

$$u = \sum_{j=1}^k c_j w_j.$$

Now, let $\tilde{w} := \text{perp}_S(v)$ and $w = \text{proj}_S(v)$. Then,

$$\langle \tilde{w}, u \rangle = \langle v - w, u \rangle = \langle v, u \rangle - \langle w, u \rangle.$$

Now, observe that

$$\langle v, u \rangle = \left\langle v, \sum_{j=1}^k c_j w_j \right\rangle = \sum_{j=1}^k c_j \langle v, w_j \rangle,$$

and using the fact that $\{w_j\}_{j=1}^k$ is an orthogonal basis, we get by definition of the projection

$$\langle w, u \rangle = \left\langle \sum_{j=1}^k \frac{\langle v, w_j \rangle}{\|w_j\|^2} w_j, \sum_{j=1}^k c_j w_j \right\rangle = \sum_{j=1}^k c_j \frac{\langle v, w_j \rangle}{\|w_j\|^2} \langle w_j, w_j \rangle = \sum_{j=1}^k c_j \langle v, w_j \rangle.$$

Thus,

$$\langle v, u \rangle - \langle w, u \rangle = 0,$$

and hence $\tilde{w}$ is orthogonal to every $u \in S$, and so $\tilde{w} \in S^\perp$ by definition of the orthogonal complement. ■

The Gram-Schmidt Process

We have seen that orthonormal sequences are very convenient to work with. We now want to ask how to obtain an orthonormal sequence if an arbitrary linearly independent sequence is given. This is accomplished by a constructive procedure called the *Gram-Schmidt process* for orthonormalising a linearly independent sequence (x_i) in an inner product space. The resulting orthonormal sequence (e_j) has the property that for every n ,

$$\text{span}\{e_1, \dots, e_n\} = \text{span}\{x_1, \dots, x_n\}.$$

The process is as follows.

1st Step

The first element of (e_k) is

$$e_1 = \frac{1}{\|x_1\|} x_1.$$

2nd Step

x_2 can be written as

$$x_2 = \langle x_2, e_1 \rangle e_1 + v_2$$

Then $v_2 = x_2 - \langle x_2, e_1 \rangle e_1$ is not the zero vector since (x_j) is linearly independent; also, we have that $v_2 \perp e_1$ since $\langle v_2, e_1 \rangle = 0$, so that we can take

$$e_2 = \frac{1}{\|v_2\|} v_2.$$

3rd Step

The vector

$$v_3 = x_3 - \langle x_3, e_1 \rangle e_1 - \langle x_3, e_2 \rangle e_2$$

is not the zero vector, and $v_3 \perp e_1$ as well as $v_3 \perp e_2$. So we take

$$e_3 = \frac{1}{\|v_3\|} v_3.$$

***n*th Step**

The vector

$$v_n = x_n - \sum_{k=1}^{n-1} \langle x_n, e_k \rangle e_k \quad (6.37)$$

is not the zero vector and is orthogonal to all $e_1, \dots, e_{n-1}$. From this, we obtain

$$e_n = \frac{1}{\|v_n\|} v_n \quad (6.38)$$

Note that the sum that is subtracted on the right-hand side of (6.37) is the projection of x_n on $\text{span}\{e_1, \dots, e_{n-1}\}$. In other words, in each step we subtract from x_n its “components” in the directions of the previously orthogonalised vectors. This gives v_n , which is then multiplied by $\frac{1}{\|v_n\|}$ so that we get a vector that is normalised. Note that v_n cannot be the zero vector for any n . In fact, if n were the smallest subscript for which $v_n = 0$, then (6.37) shows that x_n would be a linear combination of $e_1, \dots, e_{n-1}$, hence a linear combination of $x_1, \dots, x_{n-1}$, contradicting the assumption that $\{x_1, \dots, x_n\}$ is linearly independent.

Additional notes on projections and approximations

(to supplement, Section 4.3, “Projection Theorem,” of the Course Notes)

In a metric space X , the *distance* from an element $x \in X$ to a nonempty subset $Y \subset X$ is defined to be

$$\delta = \inf_{z \in Y} d(x, z). \quad (1)$$

In a normed space, this becomes

$$\delta = \inf_{z \in Y} \|x - z\|. \quad (2)$$

It is most often important to know whether

1. there exists a point $y \in Y$ such that

$$d(x, y) \text{ or } \|x - y\| = \delta, \quad (3)$$

and

2. if such a point exists, whether it is unique.

In other words, we are concerned with *existence* and *uniqueness* of such a closest point y . This is important in the context of *approximation theory*. If there exists a unique element $y \in Y$ for which Eq. (3) is satisfied, then we y could be viewed as the **best approximation** in Y to $x \in X$. The value δ may be viewed as the **error** in the approximation $x \simeq y$. We shall return to this idea later in this section.

Here we simply state that there is a quite significant difference between Banach spaces and Hilbert spaces with regard to Eq. (3). In the case of Banach spaces, is not always guaranteed that a $y \in Y$ satisfying (3) exists. The following result is to be found in *Linear Operator Theory in Science and Engineering*, by A.W. Naylor and G.R. Sell (Theorem 5.14.3, p. 285):

Theorem 1 *Let X be a Banach space and let Y be a closed linear subspace of X . Let $x \in X$ and define δ as in (2). Then for each $\eta > 0$, there is a $y \in Y$ such that*

$$\delta \leq \|x - y\| < \delta + \eta. \quad (4)$$

In other words, there are approximations to x in Y such that the error $\|x - y\|$ is arbitrarily close to δ . But the theorem does not say that the infimum value δ can actually be achieved. Indeed, even if X is complete and Y closed, the existence of a point y for which the infimum value δ is not always guaranteed. The example discussed by Naylor and Sell following Theorem 5.14.3 on page 285 illustrates this point.

That being said, the situation for Banach spaces is not as grim as it may appear from the above. Many Banach spaces, including those employed in applications, possess an additional property that guarantees the existence of a unique minimizes $y \in Y$. We'll return to this idea later in this section.

On the other hand, the problem of nonexistence and nonuniqueness cannot occur in Hilbert space, as we sketch below.

Theorem 2 (*Minimizing vector*) *Let X be an inner product space and $Y \subset X$ a nonempty convex subset which is complete in the metric induced by the inner product on X . Let $x \in X$ and define δ by (2). Then there exists a unique $y \in Y$ such that*

$$\|x - y\| = \delta. \quad (5)$$

Proof: See Kreyszig, pp. 144-145. The proof is quite similar to that proof of existence/uniqueness in the “Projection Theorem” of the AMATH 731 Course Notes, Section 4.3, p. 64.

If $Y \subset X$ is now assumed to be a *complete linear subspace*, then we have the following result.

Theorem 3 (Orthogonality) *Let Y in the previous theorem be a complete linear subspace and $x \in X$ fixed. Then $z = x - y$ is orthogonal to Y .*

Proof: See Kreyszig, p. 145.

A consequence of the latter result is that the inner product space X decomposes into the direct sum $X = Y \oplus Y^\perp$.

The above two results, *Minimizing vector* and *Orthogonality*, in the case that $X = H$ is a Hilbert space, comprise the “Projection Theorem” of Section 4.3 in the AMATH 731 Course Notes. As discussed in the Course Notes, and in more detail in the Supplementary Notes, an important case of best approximation is when the subspace Y is the closed linear subspace,

$$Y_n = \text{span}\{e_1, e_2, \dots, e_n\}, \quad (6)$$

for some $n \geq 1$, where $\{e_1, e_2, \dots, e_n\}$ is an orthonormal set. In this case, the best approximation $y \in Y_n$ to an element $x \in H$ is unique and given by

$$y = \sum_{k=1}^n \langle x, e_k \rangle e_k. \quad (7)$$

The $\langle x, e_k \rangle$ are the *Fourier coefficients* of x .

Best approximation in Banach spaces

The discussion at the beginning of this section was meant to highlight some basic differences between Banach and Hilbert spaces. As mentioned earlier, if a Banach space X satisfies an additional condition, to be discussed below, then the existence of a unique minimizer/best approximation is guaranteed.

Much of the following material is based on the contents of Section 4.2, “Theory of approximation in a normed linear space,” in the book, *Functional Analysis: Applications in Mechanics and Inverse Problems*, Second Edition, by L.P. Lebedev, I.I. Vorovich and G.M.L. Gladwell. Proofs, which are to be found in this book, are omitted here.

In what follows, we shall consider a quite simple, yet very common set of approximation problems, along the lines of Eq. (6). Given a Banach space X , we consider the approximation space Y_n to be the closed linear subspace,

$$Y_n = \text{span}\{v_1, v_2, \dots, v_n\}, \quad (8)$$

where the $v_i \in X$ are nonzero and linearly independent. In other words, the set of elements $\{v_i\}_{i=1}^n \subset X$ forms a basis in Y_n . (Note that we are not assuming that the v_i are normalized, i.e., $\|v_i\| = 1$.) The Hilbert space approximation problem, Eq. (6), is a special case of this class of problems.

Let X be a Banach space with norm $\|\cdot\|$. Given an element $x \in X$, the **best approximation** $y_n^* \in Y_n$ to x is defined as

$$y_n^* = \sum_{k=1}^n a_k v_k, \quad (9)$$

such that

$$\|x - y_n^*\| = \min_{c_1, c_2, \dots, c_n} \left\| x - \sum_{k=1}^n c_k v_k \right\|, \quad (10)$$

provided that such a minimizer exists (with no constraint on uniqueness). Alternatively, we may write that the expansion coefficients $\mathbf{a} = (a_1, a_2, \dots, a_n)$ of the best approximation $y_n^* \in Y_n$ in (9) are given by

$$\mathbf{a} = (a_1, a_2, \dots, a_n) = \arg \min_{\mathbf{c} \in \mathbf{R}^n} \left\| x - \sum_{k=1}^n c_k v_k \right\|, \quad (11)$$

With reference to Eq. (10), the quantity

$$\Delta_n = \|x - y_n^*\| \quad (12)$$

may be viewed as the *approximation error* associated with the approximation

$$x \approx y_n^* \in Y_n. \quad (13)$$

It should be clear that

$$Y_{n_1} \subset Y_{n_2} \text{ for } n_1 < n_2. \quad (14)$$

As such, we would expect/hope that the best approximations y_n^* “get better,” or at least “do not get worse,” with increasing n , i.e.,

$$\Delta_{n+1} \leq \Delta_n, \quad n = 1, 2, \dots. \quad (15)$$

Of course, we would like to go one step further and be assured that

$$\Delta_n \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (16)$$

This, however, will depend upon whether or not we can find a complete or *maximal* set of basis elements $\{v_k\}_{k=1}^\infty \in X$. That is beyond the scope of this discussion.

Theorem 1: A solution (not necessarily unique) to the above minimization problem exists. (In other words, we have *existence*.)

Theorem 2: If the Banach space X is **strictly normed**, then a **unique** solution to the minimization problem in (9) **exists**.

Definition 1: A normed linear space X is said to be **strictly normed** if the equality

$$\|x + y\| = \|x\| + \|y\| \quad x \neq 0, \quad (17)$$

implies that $y = \lambda x$ and $\lambda \geq 0$.

Remarks:

1. The Banach spaces L^p and l^p for $1 < p < \infty$ are strictly normed. Note that L^1 is not strictly normed, which means that uniqueness of the minimizer is not guaranteed. In fact, simple examples can be constructed to show this.
2. The Sobolev spaces $W^{m,p}$, to be discussed later in this course, are strictly normed for $1 < p < \infty$.
3. A Hilbert space H is strictly normed. (This is consistent with the existence/uniqueness result of Projection Theorem.) In this case, the norm is given by $\|x\| = \sqrt{\langle x, x \rangle}$. If we further assume that the elements $v_k \in H$ comprise an orthonormal set then the coefficients a_k are the Fourier coefficients of x , i.e., $a_k = \langle x, v_k \rangle$.

4. Indeed, for Banach spaces X that are not Hilbert spaces, the coefficients a_k of best approximations are not, in general (dare we say almost never), expressible in terms of simple formulas as in the Hilbert space case. This is a major reason why working in appropriate Hilbert spaces is desirable.

Furthermore, working with an orthonormal basis $\{e_k\}$ in a Hilbert space H provides an additional bonus: The coefficients $(a_1, a_2, \dots, a_n)$ employed in the best approximation $y_n^* \in Y_n$ are also used in all “higher order” approximations $y_m^* \in Y_m$ for $m > n$. *They do not have to be recomputed.* As such, one may obtain y_{n+1}^* from y_n^* by simply computing the additional coefficient $a_{n+1} = \langle x, e_{n+1} \rangle$.

Some numerical examples

Here we examine the best L^1 and L^2 approximations of two simple functions $f : [0, 1] \rightarrow \mathbf{R}$. As mentioned earlier, the computation of best approximation coefficients a_k in L^1 cannot be done in closed form, so we resort to numerical methods. In order that the approximations can be fairly compared, the best approximations in L^2 will also be computed numerically, as opposed to analytically.

The numerical approaches will involve a discretization of the functions over a set of N equally-spaced mesh points $x_i \in [0, 1]$, i.e.,

$$x_i = i \Delta x, \quad 1 \leq i \leq N, \quad \text{where} \quad \Delta x = \frac{1}{N}. \quad (18)$$

For a given approximation space Y_n spanned by the basis functions v_k , $1 \leq k \leq n$, the function value $f(x_i)$ will be approximated at each mesh point x_i as follows,

$$f(x_i) \approx \sum_{k=1}^n c_k v_k(x_i), \quad 1 \leq i \leq N. \quad (19)$$

This may be expressed in vector/matrix form as follows,

$$\mathbf{f} \approx \mathbf{B}\mathbf{c}, \quad (20)$$

where

$$\mathbf{f} = (f_1, f_2, \dots, f_N)^T \in \mathbf{R}^N, \quad (21)$$

with components

$$f_i = f(x_i), \quad (22)$$

$\mathbf{c}$ is the vector of expansion coefficients, i.e.,

$$\mathbf{c} = (c_1, c_2, \dots, c_n)^T \in \mathbf{R}^n \quad (23)$$

and $\mathbf{B}$ is an $N \times n$ matrix with elements,

$$b_{ij} = v_j(x_i), \quad 1 \leq i \leq N, \quad 1 \leq j \leq n. \quad (24)$$

In a given Banach space (here, L^1 or L^2), the coefficients a_k of the best approximation y_n^* in (9) will be given by

$$\mathbf{a} = \arg \min_{\mathbf{c} \in \mathbf{R}^n} \|\mathbf{B}\mathbf{c} - \mathbf{f}\|. \quad (25)$$

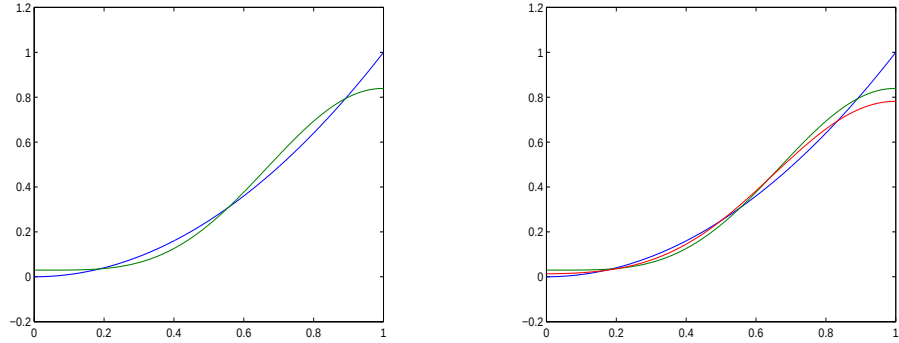
The following set of linearly independent functions on $[0, 1]$ were employed in all computations:

$$v_1(x) = 1, \quad v_k(x) = \cos[(k-1)\pi x], \quad k = 2, 3, \dots. \quad (26)$$

These functions form an orthogonal (but not orthonormal) basis in the Hilbert space $L^2([-1, 1])$. (The cosine functions would have to be multiplied by the factor $\sqrt{2}$ to produce an orthonormal basis.)

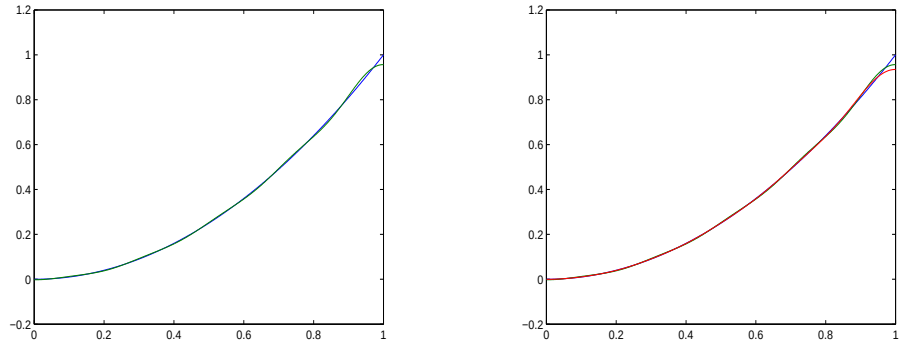
Example 1: We first consider the function $f(x) = x^2$ on $[0, 1]$.

In the figure below on the left are plotted the graphs of $f(x)$ and its best L^2 approximation in Y_3 , i.e., using the three functions $\{v_1, v_2, v_3\}$. On the right, for purposes of comparison are plotted the best L^1 and L^2 approximations in Y_3 to $f(x)$. The L^1 approximation is observed to lie slightly farther away from the graph of $f(x)$ in the region near $x = 1$.



Approximation function $f(x) = x^2$ on $[0, 1]$. **Left:** Best L^2 approximation using 3 basis functions. **Right:** Best L^1 and L^2 approximations using 3 basis functions.

Below are shown the the best L^1 and L^2 approximations to $f(x)$ in Y_{10} , i.e., using the 10 functions $\{v_1, v_2, \dots, v_{10}\}$. As expected, both approximations are much better than their Y_3 counterparts.

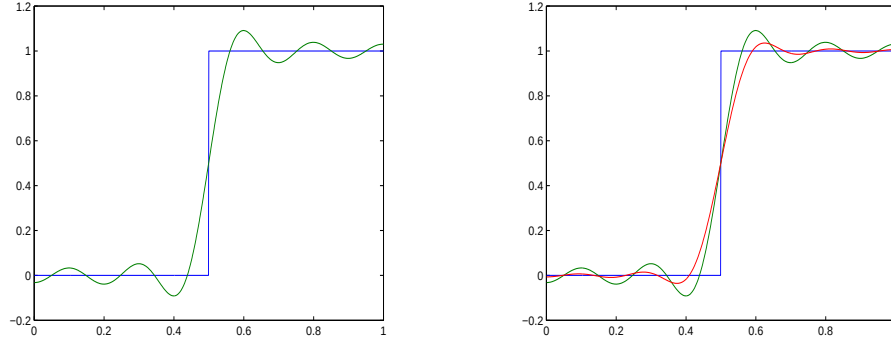


Approximation function $f(x) = x^2$ on $[0, 1]$. **Left:** Best L^2 approximation using 10 basis functions. **Right:** Best L^1 and L^2 approximations using 10 basis functions.

Example 2: We now consider the following step function on $[0, 1]$,

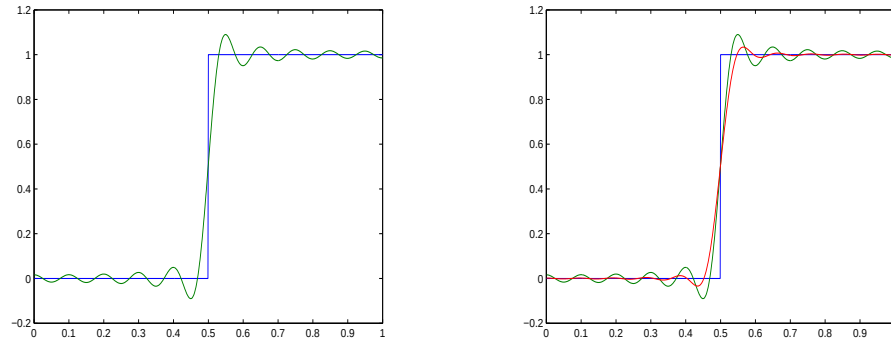
$$f(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{2}, \\ 1, & \frac{1}{2} < x \leq 1, \end{cases} \quad (27)$$

In the figure below are shown the best L^1 and L^2 approximations to $f(x)$ in Y_{10} , i.e., using the 10 functions $\{v_1, v_2, \dots, v_{10}\}$.



Step function of Example 2. **Left:** Best L^2 approximation using 10 basis functions. **Right:** Best L^1 and L^2 approximations using 10 basis functions.

The best L^1 and L^2 approximations in Y_{20} , i.e., using the 20 functions $\{v_1, v_2, \dots, v_{20}\}$, are shown below.



Step function of Example 2. **Left:** Best L^2 approximation using 20 basis functions. **Right:** Best L^1 and L^2 approximations using 20 basis functions.

In both sets of results, a rather interesting observation of the L^1 approximations can be made. They seem to oscillate with a lesser amplitude than their L^2 counterparts, staying closer to the horizontal pieces that comprise the graph of $f(x)$. In terms of the L^2 metric, they are not optimal – the L^2 best approximations are optimal. Perhaps the seemingly better performance of the L^1 approximations about the horizontal components of $f(x)$ is overridden by their poorer approximations to $f(x)$ near the discontinuous jump at $x = 1/2$.

6.2.4 Series Related to Orthonormal Sequences and Sets

Recall the idea of the convergence of a series in a Banach space X : given an infinite sequence $(x_n) \subset X$, we said that the series $\sum_{n=1}^{\infty} x_n$ converges to $x \in X$, i.e., that $x = \sum_{n=1}^{\infty} x_n$, if

$$\lim_{n \rightarrow \infty} \|x - s_n\| = 0, \quad \text{where} \quad s_n = \sum_{k=1}^n x_k.$$

Theorem 6.2.9 Convergence of Series in Hilbert Spaces

Let $\{z_n\}_{n=1}^{\infty} \subset H$ be an orthogonal set in a Hilbert space H . Then,

1. $\sum_{n=1}^{\infty} z_n$ converges if and only if $\sum_{n=1}^{\infty} \|z_n\|^2 < \infty$;
2. If $\sum_{n=1}^{\infty} z_n = z$, then $\sum_{n=1}^{\infty} \|z_n\|^2 = \|z\|^2$.

PROOF:

1. Let $s_n = \sum_{k=1}^n z_k$. For $n > m$, it follows that

$$\|s_n - s_m\|^2 = \left\| \sum_{k=m+1}^n z_k \right\|^2 = \sum_{k=m+1}^n \|z_k\|^2 = |t_n - t_m|,$$

where $t_n := \sum_{k=1}^n \|z_k\|^2$. Thus, the sequences of partial sums (s_n) is a Cauchy sequence in H if and only if the sequence of partial sums (t_n) is a Cauchy sequence in $\mathbb{R}$.

2. Now, suppose that $\sum_{k=1}^{\infty} z_k = z$. Defining s_n and t_n as above, we have $\|z - s_n\| \rightarrow 0$ and $\|s_n\|^2 = t_n$. Thus, $(\|z\| + \|s_n\|)$ is a bounded sequence of numbers. Also, we have (from the triangle inequality)

$$|\|z\| - \|s_n\|| \leq \|z - s_n\| \rightarrow 0,$$

so that

$$|\|z\|^2 - t_n| = |\|z\|^2 - \|s_n\|^2| = (\|z\| + \|s_n\|)|\|z\| - \|s_n\|| \rightarrow 0,$$

i.e.,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \|z_k\|^2 = \|z\|^2. \quad \blacksquare$$

The following result follows directly from this theorem (but we prove it in full anyway).

Theorem 6.2.10

Let (e_k) be an orthonormal sequence in a Hilbert space H . Then

1. The series

$$\sum_{k=1}^{\infty} \alpha_k e_k, \quad (6.39)$$

where α_k are scalars, converges (in the norm on H) if and only if the series

$$\sum_{k=1}^{\infty} |\alpha_k|^2 \quad (6.40)$$

converges.

2. If (6.39) converges, then the coefficients α_k are the Fourier coefficients $\langle x, e_k \rangle$, where x denotes the sum of (6.39); hence, in this case, (6.39) can be written

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k. \quad (6.41)$$

3. For any $x \in H$, the series (6.39) with $\alpha_k = \langle x, e_k \rangle$ converges (in the norm of H).

PROOF:

1. Let

$$s_n = \alpha_1 e_1 + \cdots + \alpha_n e_n \quad \text{and} \quad \sigma_n = |\alpha_1|^2 + \cdots + |\alpha_n|^2.$$

Then, because of the orthonormality, for any m and $n > m$,

$$\begin{aligned} \|s_n - s_m\|^2 &= \|\alpha_{m+1} e_{m+1} + \cdots + \alpha_n e_n\|^2 \\ &= |\alpha_{m+1}|^2 + \cdots + |\alpha_n|^2 = \sigma_n - \sigma_m. \end{aligned}$$

Hence, (s_n) is Cauchy in H if and only if (σ_n) is Cauchy in $\mathbb{R}$. Since H and $\mathbb{R}$ are complete, the first statement of the theorem follows.

2. Taking the inner product of s_n and e_j , and using the orthonormality, we have

$$\langle s_n, e_j \rangle = \alpha_j \quad \text{for } j = 1, 2, \dots, k \quad (k \leq n \text{ fixed}).$$

By assumption, $s_n \rightarrow x$. Since the inner product is continuous, we have

$$\alpha_j = \langle s_n, e_j \rangle \rightarrow \langle x, e_j \rangle \quad (j \leq k).$$

Here, we can take k ($\leq n$) as large as we please because $n \rightarrow \infty$, so that we have $\alpha_j = \langle x, e_j \rangle$ for every $j = 1, 2, \dots$

3. From the Bessel inequality, we see that the series

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2$$

converges. From this and Part 1., we conclude that Part 3. must hold. ■

Lemma 6.2.5 Fourier Coefficients

Any x in an inner product space X can have at most countably many non-zero Fourier coefficients $\langle x, e_\kappa \rangle$ with respect to an orthonormal family (e_κ) , $\kappa \in I$, in X .

REMARK: Hence, with any fixed $x \in H$, a Hilbert space, we can associate a series similar to (6.41),

$$\sum_{\kappa \in I} \langle x, e_\kappa \rangle e_\kappa, \quad (6.42)$$

and we can arrange the e_κ with $\langle x, e_\kappa \rangle \neq 0$ in a sequence $(e_1, e_2, \dots)$, so that (6.42) takes the form (6.41). Convergence follows from the previous theorem. We show in the proof below that the sum does not depend on the order in which those e_κ are arranged in a sequence.

PROOF: Let (w_m) be a rearrangement of (e_n) . By definition, this means that there is a bijective mapping $n \mapsto m(n)$ of $\mathbb{N}$ onto itself such that corresponding terms of the two sequences are equal, that is, $w_{m(n)} = e_n$. We set

$$\alpha_n := \langle x, e_n \rangle \quad \text{and} \quad \beta_m := \langle x, w_m \rangle,$$

and

$$x_1 := \sum_{n=1}^{\infty} \alpha_n e_n \quad \text{and} \quad x_2 := \sum_{m=1}^{\infty} \beta_m w_m.$$

Then, by Part 2. of the theorem above,

$$\alpha_n = \langle x, e_n \rangle = \langle x_1, e_n \rangle, \quad \text{and} \quad \beta_m = \langle x, w_m \rangle = \langle x_2, w_m \rangle.$$

Since $e_n = w_{m(n)}$, we thus obtain

$$\langle x_1 - x_2, e_n \rangle = \langle x_1, e_n \rangle - \langle x_2, w_{m(n)} \rangle = \langle x, e_n \rangle - \langle x, w_{m(n)} \rangle = 0,$$

and similarly, $\langle x_1 - x_2, w_m \rangle = 0$. This implies

$$\begin{aligned} \|x_1 - x_2\|^2 &= \left\langle x_1 - x_2, \sum_{n=1}^{\infty} \alpha_n e_n - \sum_{m=1}^{\infty} \beta_m w_m \right\rangle \\ &= \sum_{n=1}^{\infty} \bar{\alpha}_n \langle x_1 - x_2, e_n \rangle - \sum_{m=1}^{\infty} \bar{\beta}_m \langle x_1 - x_2, w_m \rangle = 0. \end{aligned}$$

Consequently, $x_1 - x_2 = 0$, so that $x_1 = x_2$. Since the rearrangement (w_m) of (e_n) was arbitrary, the proof is complete. ■

Theorem 6.2.11

Let $\{z_k\}_{k=1}^{\infty}$ be an orthonormal set in a Hilbert space H . For every $x \in H$, the vector $y = \sum_{k=1}^{\infty} \langle x, z_k \rangle z_k$ exists in H , and $x - y$ is orthogonal to every z_k .

PROOF: The existence of y follows from Theorem 6.2.9 and Bessel's inequality. Let $m \in \mathbb{N}$. We must show that $\langle x - y, z_m \rangle = 0$ for all m . For each $n \in \mathbb{N}$, define $y_n := \sum_{k=1}^n \langle x, z_k \rangle z_k$. From the identity

$$\langle x - y, z_m \rangle = \langle x - y_n, z_m \rangle + \langle y_n - y, z_m \rangle,$$

it follows that

$$\begin{aligned}
 |\langle x - y, z_m \rangle| &\leq |\langle x - y_n, z_m \rangle| + |\langle y_n - y, z_m \rangle| \\
 &\leq \left| \langle x, z_m \rangle - \sum_{k=1}^n \langle x, z_k \rangle \langle z_k, z_m \rangle \right| + \|y_n - y\| \|z_m\| \\
 &= 0 + \|y_n - y\|.
 \end{aligned}$$

(Technically, n must be greater than n for the first term on the left-hand side to vanish.) Since $\|y_n - y\| \rightarrow 0$, it follows that $\langle x - y, z_m \rangle = 0$ for all m . ■

6.3 Total Orthonormal Sets and Sequences

The truly interesting orthonormal sets in inner product spaces and Hilbert spaces are those that consist of “sufficiently many” elements so that every element in the space can be represented or sufficiently accurately approximated by the use of those orthonormal sets. In finite-dimensional (n -dimensional) spaces, the situation is simple: all we need is an orthonormal set of n elements. The question is what can be done to take care of infinite-dimensional spaces, too.

Definition 6.3.1 Total/Maximal Orthonormal Set

A *total set* (or *maximal set*) in a normed space X is a subset $M \subset X$ whose span is dense in X . Accordingly, an orthonormal set (or sequence or family) in an inner product space X that is total in X is called a **total/maximal orthonormal set** (or sequence or family, respectively) in X .

M is total in X if and only if $\overline{\text{span}(M)} = X$. This is obvious from the definition.

A total orthonormal family in X is sometimes called an **orthonormal basis** for X . However, it is important to note that this is not a basis, in the sense of linear algebra, for X as a vector space, unless X is finite dimensional.

Theorem 6.3.1

In every Hilbert space $H \neq \{0\}$, there exists a total orthonormal set.

For a finite-dimensional Hilbert space, this is clear. For an infinite-dimensional separable H , it follows from the Gram-Schmidt process by (ordinary) induction. For a non-separable H , a (non-constructive) proof results from Zorn’s lemma.

Theorem 6.3.2

All total orthonormal sets in a given Hilbert space $H \neq \{0\}$ have the same cardinality. The latter is called the **Hilbert dimension** or **orthogonal dimension** of H . (If $H = \{0\}$, then this dimension is defined to be 0.)

For a finite-dimensional Hilbert space, the statement is clear since then the Hilbert dimension is the dimension in the sense of linear algebra. For an infinite-dimensional separable H , the statement will

readily follow from Theorem 6.3.6 below, and for a general H the proof would require somewhat more advanced tools from set theory.

Theorem 6.3.3

Every inner product space that is not $\{0\}$ contains a complete orthonormal set. In fact, every orthonormal subset of H is contained in a complete orthonormal set.

The following theorem shows that a total orthonormal set cannot be augmented to a more extensive orthonormal set by the adjunction of new elements.

Theorem 6.3.4 Totality

Let M be a subset of an inner product space X . Then,

1. If M is total in X , then there does not exist a non-zero $x \in X$ that is orthogonal to every element of M , i.e.,

$$x \perp M \implies x = 0. \quad (6.43)$$

2. If X is complete, then (6.43) is also sufficient for the totality of M in X , i.e., if X is complete then $x \perp M \iff x = 0$.

Another important criterion for totality can be obtained from the Bessel inequality, which, recall, is

$$\sum_k |\langle x, e_k \rangle|^2 \leq \|x\|^2, \quad (6.44)$$

where the left-hand side is either an infinite series or a finite sum, and (e_k) is an orthonormal set. With the equation sign, this becomes the **Parseval relation**

$$\sum_k |\langle x, e_k \rangle|^2 = \|x\|^2. \quad (6.45)$$

Theorem 6.3.5 Totality

An orthonormal set M in a Hilbert space H is total in H if and only if for all $x \in H$ the Parseval relation (6.45) holds (summation is over all non-zero Fourier coefficients of x with respect to M).

Let us turn to Hilbert spaces that are separable. Recall that such a space has a countable dense subset that is dense in the space. Separable Hilbert spaces are simpler than non-separable ones since they cannot contain uncountable orthonormal sets.

Theorem 6.3.6

Let H be a Hilbert space. Then,

1. If H is separable, then every orthonormal set in H is countable.
2. If H contains an orthonormal sequence that is total in H , then H is separable.

Lemma 6.3.1

Let $L_y(x) = \langle x, y \rangle$ be continuous on a Hilbert space H . If $\sum a_n x_n$ converges in H , then $\langle \sum a_n x_n, y \rangle = \sum a_n \langle x_n, y \rangle$.

PROOF: We have

$$|L_y(x_1) - L_y(x_2)| = |\langle x_1 - x_2, y \rangle| \leq \|x_1 - x_2\| \|y\|.$$

This shows that L_y is continuous on H . Then,

$$L_y\left(\sum a_n x_n\right) = \sum L_y(a_n x_n) = \sum \langle a_n x_n, y \rangle = \sum a_n \langle x_n, y \rangle. \quad \blacksquare$$

Theorem 6.3.7 Generalised Fourier Series

Let (e_n) be an orthonormal sequence in a separable Hilbert space H . Then the following are equivalent.

1. (e_n) is maximal.
2. For any $x \in H$, $x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$ (with convergence in H).
3. For any $x \in H$, $\|x\|^2 = \sum_{n=1}^{\infty} \langle x, e_n \rangle^2$ (with convergence in $\mathbb{R}$).

Such an (e_n) is an orthonormal basis for H .

PROOF:

- 1. $\Rightarrow$ 2.: For $x \in H$, $\sum \langle x, e_n \rangle^2$ converges by the Bessel inequality, which implies that $\sum_{k=1}^{\infty} \langle x, e_n \rangle e_n$ converges by Theorem 6.2.10 Part 1. Now, let $y = \sum_{k=1}^{\infty} \langle x, e_n \rangle e_n$. We show that $y = x$. By the Lemma above,

$$\langle y, e_m \rangle = \sum \langle x, e_n \rangle \langle e_n, e_m \rangle = \langle x, e_m \rangle \Rightarrow \langle y - x, e_m \rangle = 0 \quad \forall m \Rightarrow y - x = 0$$

with the last step following from the fact that $\{e_n\}$ is maximal. Therefore, $x = \sum \langle x, e_n \rangle e_n$.

- 2. $\Rightarrow$ 3.: Let $s_n = \sum_{k=1}^n \langle x, e_k \rangle e_k$. We have that $s_n \rightarrow x$ as $n \rightarrow \infty$. Then,

$$\|x - s_n\|^2 = \|x\|^2 - \sum_{k=1}^n \langle x, e_k \rangle^2.$$

Letting $n \rightarrow \infty$, we get

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle^2 = \|x\|^2. \quad (6.46)$$

- 3. $\Rightarrow$ 1.: Suppose $\langle x, e_n \rangle = 0$ for all n . Then, by (6.46), we have

$$\|x\|^2 = \sum_{n=1}^{\infty} \langle x, e_n \rangle^2 = 0 \Rightarrow x = 0.$$

Thus, $\{e_n\}$ is maximal. $\blacksquare$

Example 6.3.1 For $L_2[0, 1]$, let $\{e_k\} = \{1, \sqrt{2}\cos(2\pi t), \sqrt{2}\sin(2\pi t), \dots, \sqrt{2}\cos(2n\pi t), \sqrt{2}\sin(2n\pi t), \dots\}$. Then, $\{e_k\}$ is an orthonormal basis. Indeed, we have

$$\begin{aligned}\langle 1, 1 \rangle &= 1, \\ \langle \sqrt{2}\cos(2\pi nt), \sqrt{2}\cos(2\pi mt) \rangle &= \delta_{n,m}, \\ \langle \sqrt{2}\cos(2\pi nt), \sqrt{2}\sin(2\pi mt) \rangle &= 0, \\ \langle \sqrt{2}\sin(2\pi nt), \sqrt{2}\sin(2\pi mt) \rangle &= \delta_{n,m}.\end{aligned}$$

Then, for $f \in L_2[0, 1]$,

$$\begin{aligned}f(t) &= \langle f, 1 \rangle + \sum_{n=1}^{\infty} [\langle f, \sqrt{2}\cos(2\pi nt) \rangle \sqrt{2}\cos(2\pi nt) + \langle f, \sqrt{2}\sin(2\pi nt) \rangle \sqrt{2}\sin(2\pi nt)] \\ &= a_0 + \sum_{n=1}^{\infty} [a_n \sqrt{2}\cos(2\pi nt) + b_n \sqrt{2}\sin(2\pi nt)],\end{aligned}$$

where the convergence is in L_2 :

$$S_N = a_0 + \sum_{n=1}^N [a_n \sqrt{2}\cos(2\pi nt) + b_n \sqrt{2}\sin(2\pi nt)],$$

and $\|f - S_N\|_2 \rightarrow 0$ as $n \rightarrow \infty$.

Example 6.3.2 $L_2[a, b]$ is separable since $C[a, b]$ is dense in $L_2[a, b]$ with respect to the $\|\cdot\|_2$ norm. Also, P , the set of polynomials with rational coefficients, is dense in $C[a, b]$ with respect to the $\|\cdot\|_{\infty}$ norm. To show the latter, let $f \in L_2[a, b]$ and $\epsilon > 0$. Then there exists $g \in C[a, b]$ such that

$$\|f - g\|_2 < \frac{\epsilon}{2}$$

(because $C[a, b]$ is dense in L_2 with respect to the $\|\cdot\|_2$ norm, as expressed above). Then, there exists $p \in P$ such that

$$\|g - p\|_{\infty} < \frac{\epsilon}{2\sqrt{b-a}}$$

(because P is dense in $C[a, b]$ with respect to the $\|\cdot\|_{\infty}$ norm, as expressed above). Therefore,

$$\|g - p\|_2 = \sqrt{\int_a^b |g - p|^2 dt} \leq \|g - p\|_{\infty} \sqrt{b-a} < \frac{\epsilon}{2},$$

which means that $\|f - p\|_2 \leq \|f - g\|_2 + \|g - p\|_2 < \epsilon$.

Theorem 6.3.8

The set $\{\phi_n\} = \{1, \sqrt{2} \cos(2\pi t), \sqrt{2} \sin(2\pi t), \dots, \sqrt{2} \cos(2\pi nt), \sqrt{2} \sin(2\pi nt), \dots\}$ is an orthonormal basis for $L_2[0, 1]$.

PROOF: We have already seen that this set is orthonormal, and hence we know that it is linearly independent. By the Generalised Fourier Series Theorem, it suffices to show that the set is maximal.

Suppose that there exists $f \neq 0$ such that $\langle f, \phi_n \rangle = 0$ for all n . We have two cases: $f \in C[0, 1]$ and $f \notin C[0, 1]$.

 f is continuous

Now, there exists $t_0 \in (0, 1)$ such that $f(t_0) \neq 0$. Take $f(t_0) > 0$. Then, for all $\delta > 0$ such that $f(t) \geq b > 0$ or $|t - t_0| \leq \delta$, we have $[t_0 - \delta, t_0 + \delta] \subset [0, 1]$. Then, let

$$\psi(t) = 1 + \cos(2\pi(t - t_0)) - \cos(2\pi\delta), \quad p(t) = \psi(t)^N.$$

Note that p is a linear combination of the functions in $\{\phi_n\}$.

Let $k = \psi(t_0 + \frac{\delta}{2}) = 1 + \cos(\pi\delta) - \cos(2\pi\delta) > 1$ (since $\delta \leq 1/2$). By the continuity of f , $|f(t)| \leq M$ for all $t \in [0, 1]$.

For $|t - t_0| \leq \frac{\delta}{2}$, $\psi(t) \geq k \Rightarrow p(t) \geq k^N$ and $f(t) \geq b$.

For $|t - t_0| > \delta$, $t \in [0, 1]$, we have $|\psi(t)| < 1 \Rightarrow |p(t)| < 1$ and $|f(t)| \leq M$.

For $\frac{\delta}{2} \leq |t - t_0| \leq \delta$, we have $\psi(t) \geq 1$ and $f(t) \geq b$, which implies that $p(t)f(t) \geq b > -M$. Thus,

$$0 = \langle p, f \rangle = \int_0^1 p(t)f(t) dt = \left(\int_0^{t_0 - \frac{\delta}{2}} + \int_{t_0 - \frac{\delta}{2}}^{t_0 + \frac{\delta}{2}} + \int_{t_0 + \frac{\delta}{2}}^1 \right) p(t)f(t) dt \geq -M(1 - \delta) + k^N b.$$

Then, letting $N \rightarrow \infty$ the right-hand side of the above equation approaches infinity, a contradiction.

 f not continuous

In this case, let $F(t) = \int_0^t f(s) ds$. Then, since $f \in L_2[0, 1]$, we have that $f \in L_1[0, 1]$, which means that $F'(t) = f(t)$ almost everywhere and $F \in C[0, 1]$. Now,

$$\langle f, \phi_n \rangle = \int_0^1 f(t)\phi_n(t) dt = \int_0^1 F'(t)\phi_n(t) dt = F(t)\phi_n(t)|_0^1 - \int_0^1 F(t)\phi_n'(t) dt.$$

Using $F(0) = 0$ and $F(1) = \int_0^1 f(s) ds = \langle f, 1 \rangle = 0$, we have that $\langle f, \phi_n \rangle = -\langle F, \phi_n' \rangle$.

Let $G(t) = F(t) - \int_0^1 F(t) dt$. Then, $\int_0^1 G(t) dt = 0$ and

$$\langle G, \phi_n' \rangle = \langle F, \phi_n' \rangle = 0 \Rightarrow \langle G, \phi_n \rangle = 0 \quad \forall n \neq 1.$$

Thus, $\langle G, \phi_n \rangle = 0$ for all n (since $\langle G, 1 \rangle = 0$). By the first part, $G = 0$. Therefore, $f = F' = G' = 0$ almost everywhere. ■

Theorem 6.3.9

Any separable Hilbert space is isomorphic to ℓ_2 .

PROOF: Let $\{e_n\}$ be an orthonormal basis in a separable Hilbert space H and let $\hat{e}_n = (0, \dots, 0, 1, 0, \dots)$, with a one only in the n th slot. Then $\{\hat{e}_n\}$ is an orthonormal basis for ℓ_2 since $\langle x, \hat{e}_n \rangle = 0$ for all n , which means that $x = 0$.

Now, for $x \in H$, we can write $x = \sum \langle x, e_n \rangle e_n$. Define $T : H \rightarrow \ell_2$ by $T(x) = \sum \langle x, e_n \rangle \hat{e}_n$. Then, $\sum \langle x, e_n \rangle \hat{e}_n$ converges since $\sum \langle x, e_n \rangle^2$ converges. We now show that T preserves the inner product:

$$\begin{aligned} \langle T(x), T(y) \rangle &= \left\langle \sum \langle x, e_n \rangle \hat{e}_n, \sum \langle y, e_m \rangle \hat{e}_m \right\rangle \\ &= \sum \langle x, e_n \rangle \langle y, e_n \rangle = \lim_{N \rightarrow \infty} \sum_{n=1}^N \langle x, e_n \rangle \langle y, e_n \rangle \\ &= \lim_{N \rightarrow \infty} \left\langle \sum_{n=1}^N \langle x, e_n \rangle e_n, \sum_{m=1}^N \langle y, e_m \rangle e_m \right\rangle \\ &= \langle x, y \rangle. \quad \blacksquare \end{aligned}$$

6.3.1 Legendre, Laguerre, and Hermite Polynomials**6.4 Representation of Functionals**

It is of practical importance to know the general form of bounded linear functionals on various spaces. For general Banach spaces, such formulas and their derivation can sometimes be complicated. However, for a Hilbert space, the situation is surprisingly simple.

Lemma 6.4.1

If T is a bounded linear operator $T : X \rightarrow Y$ between normed linear spaces X and Y , then the null space $\mathcal{N}(T) = \{x \in X \mid T(x) = 0\}$ is a closed linear subspace.

PROOF: To be completed. $\blacksquare$

Theorem 6.4.1 Riesz (Functionals on Hilbert Space)

Every bounded linear functional f on a Hilbert space H can be represented in terms of the inner product, namely,

$$f(x) = \langle x, z \rangle, \quad (6.47)$$

where z depends on f , is uniquely determined by f , and has norm

$$\|z\| = \|f\|. \quad (6.48)$$

PROOF: The proof has the following steps:

1. Showing f has a representation (6.47),
2. Showing that z in (6.47) is unique,
3. Showing that (6.48) holds.

The details are as follows.

1. If $f = 0$, then (6.47) and (6.48) hold if we take $z = 0$. Let $f \neq 0$. To motivate the idea of the proof, let us ask what properties z must have if a representation (6.47) exists. First of all, $z \neq 0$ since otherwise $f = 0$. Second, $\langle x, z \rangle = 0$ for all x for which $f(x) = 0$, that is, for all x in the null space $\mathcal{N}(f)$ of f . Hence, $z \perp \mathcal{N}(f)$. This suggests that we consider $\mathcal{N}(f)$ and its orthogonal complement $\mathcal{N}(f)^\perp$.

We know that $\mathcal{N}(f)$ is a vector space and is closed by the above Lemma. Furthermore, $f \neq 0$ implies that $\mathcal{N}(f) \neq H$, so that $\mathcal{N}(f)^\perp \neq \{0\}$ by the projection theorem. Hence, $\mathcal{N}(f)^\perp$ contains a $z_0 \neq 0$. We set

$$v = f(x)z_0 - f(z_0)x$$

where $x \in H$ is arbitrary. Applying f , we obtain

$$f(v) = f(x)f(z_0) - f(z_0)f(x) = 0.$$

This shows that $v \in \mathcal{N}(f)$. Since $z_0 \perp \mathcal{N}(f)$, we have

$$0 = \langle v, z_0 \rangle = \langle f(x)z_0 - f(z_0)x, z_0 \rangle = f(x)\langle z_0, z_0 \rangle - f(z_0)\langle x, z_0 \rangle.$$

Noting that $\langle z_0, z_0 \rangle = \|z_0\|^2 \neq 0$, we can solve for $f(x)$. The result is

$$f(x) = \frac{f(z_0)}{\langle z_0, z_0 \rangle} \langle x, z_0 \rangle.$$

This can be written in the form (6.47), where

$$z = \frac{\overline{f(z_0)}}{\langle z_0, z_0 \rangle} z_0.$$

Since $x \in H$ was arbitrary, (6.47) is proved.

2. We now prove that z in (6.47) is unique. Suppose that for all $x \in H$ $f(x) = \langle x, z_1 \rangle = \langle x, z_2 \rangle$. Then, $\langle x, z_1 - z_2 \rangle = 0$ for all x . Choosing the particular $x = z_1 - z_2$, we have

$$\langle x, z_1 - z_2 \rangle = \langle z_1 - z_2, z_1 - z_2 \rangle = \|z_1 - z_2\|^2 = 0.$$

Hence, $z_1 - z_2 = 0$, so that $z_1 = z_2$, as required.

3. We finally prove (6.48). If $f = 0$, then $z = 0$ and (6.48) holds. Let $f \neq 0$, therefore. Then, $z \neq 0$. From (6.47) with $x = z$, we have

$$\|z\|^2 = \langle z, z \rangle = |\langle z, z \rangle| = |f(z)| \leq \|f\| \|z\|.$$

Division by $\|z\| \neq 0$ gives $\|z\| \leq \|f\|$. It remains to show that $\|f\| \leq \|z\|$. From (6.47) and the Cauchy-Schwarz inequality, we see that

$$|f(x)| = |\langle x, z \rangle| \leq \|x\| \|z\|.$$

This implies that

$$\|f\| = \sum_{\|x\|=1} |\langle x, z \rangle| \leq \|z\|. \quad \blacksquare$$

REMARK: For convenience in the argument to be presented here, let $z_1 := \frac{z_0}{\|z_0\|}$. Then, we can write the v used in the proof as (up to a minus sign)

$$v = f(z_1)x - f(x)z_1.$$

Note that

$$f(v) = f(z_1)f(x) - f(x)f(z_1) = 0,$$

implying that $v \in \mathcal{N}(f)$.

Now, rewrite the expression for v as follows:

$$x = \frac{1}{f(z_1)}v + \frac{f(x)}{f(z_1)}z_1, \quad x \in H.$$

Recalling that $v \in \mathcal{N}(f)$, and $z_1 \in \mathcal{N}(f)^\perp$, this is the *unique* orthogonal decomposition of $x \in H$ as a sum of its components in $\mathcal{N}(f)$ and $\mathcal{N}(f)^\perp$. But recall that $z_0 \in \mathcal{N}(f)^\perp$, hence $z_1 = \frac{z_0}{\|z_0\|}$ is independent of x . No matter what $x \in H$ we choose, its component, or projection, in $\mathcal{N}(f)^\perp$ is a multiple of z_1 . This implies that the space $\mathcal{N}(f)^\perp$ is *one-dimensional*, i.e.,

$$\mathcal{N}(f)^\perp = \text{span}\{z_1\}.$$

These results may be summarised as follows:

If f is a non-zero linear continuous functional on a Hilbert space, then the null space $\mathcal{N}(f)$ of f is a closed subspace and its orthogonal complement $\mathcal{N}(f)^\perp$ has dimension one, i.e., $\dim(\mathcal{N}(f)^\perp) = 1$.

The idea of the uniqueness proof in the second part of the proof above is worth noting for later use.

Lemma 6.4.2 Equality

If $\langle v_1, w \rangle = \langle v_2, w \rangle$ for all w in an inner product space X , then $v_1 = v_2$. In particular, $\langle v_1, w \rangle = 0$ for all $w \in X$ implies $v_1 = 0$.

PROOF: By assumption, for all w ,

$$\langle v_1 - v_2, w \rangle = \langle v_1, w \rangle - \langle v_2, w \rangle = 0.$$

For $w = v_1 - v_2$, this gives $\|v_1 - v_2\|^2 = 0$. Hence, $v_1 - v_2 = 0$, so that $v_1 = v_2$. In particular, $\langle v_1, w \rangle = 0$ with $w = v_1$ gives $\|v_1\|^2 = 0$, so that $v_1 = 0$. $\blacksquare$

The practical usefulness of bounded linear functionals on Hilbert spaces results to a large extend from the simplicity of the Riesz representation (6.47).

Furthermore, (6.47) is quite important in the theory of operators on Hilbert spaces. In particular, this refers to the *Hilbert-adjoint operator* T^* of a bounded linear operator T , which we'll see in the next section.

Definition 6.4.1 Sesquilinear Form

Let X and Y be vector spaces over the same field $\mathbb{F}$ (either $\mathbb{R}$ or $\mathbb{C}$). Then, a **sesquilinear form** (or **sesquilinear functional**) h on $X \times Y$ is a mapping

$$h : X \times Y \rightarrow \mathbb{F}$$

such that for all $x, x_1, x_2 \in X$ and all $y, y_1, y_2 \in Y$ and all scalars α, β , we have

1. $h(x_1 + x_2, y) = h(x_1, y) + h(x_2, y)$;
2. $h(x, y_1 + y_2) = h(x, y_1) + h(x, y_2)$;
3. $h(\alpha x, y) = \alpha h(x, y)$;
4. $h(x, \beta y) = \overline{\beta} h(x, y)$.

Hence, a sesquilinear form is linear in the first argument and conjugate linear in the second one. If X and Y are real, then the fourth condition above is simply

$$h(x, \beta y) = \beta h(x, y),$$

in which case h is called a **bilinear form** because it is linear in both arguments.

If X and Y are normed spaces and if there is a real number c such that

$$|h(x, y)| \leq c \|x\| \|y\| \text{ for all } x, y, \quad (6.49)$$

then h is called *bounded*, and the number

$$\|h\| := \sup_{0 \neq x \in X, 0 \neq y \in Y} \frac{|h(x, y)|}{\|x\| \|y\|} = \sup_{\|x\|=1, \|y\|=1} |h(x, y)| \quad (6.50)$$

is called the *norm* of h .

For example, the inner product is sesquilinear and bounded. Note that from (6.49) and (6.50), we have

$$|h(x, y)| \leq \|h\| \|x\| \|y\|. \quad (6.51)$$

Theorem 6.4.2 Riesz (General)

Let H_1, H_2 be Hilbert spaces and

$$h : H_1 \times H_2 \rightarrow \mathbb{F}$$

a bounded sesquilinear form ($\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$). Then, h has a representation

$$h(x, y) = \langle Sx, y \rangle, \quad (6.52)$$

where $S : H_1 \rightarrow H_2$ is a bounded linear operator. S is uniquely determined by h and has norm

$$\|S\| = \|h\|. \quad (6.53)$$

PROOF: We consider $\overline{h(x, y)}$. This is linear in y because of the conjugation. To make the first Riesz theorem applicable, we keep x fixed. Then, that theory yields a representation in which y is variable, say

$$\overline{h(x, y)} = \langle y, z \rangle.$$

Hence,

$$h(x, y) = \langle z, y \rangle. \quad (6.54)$$

Here, $z \in H_2$ is unique but, of course, depends on our fixed $x \in H_1$. It follows that (6.54) with variable x defines an operator

$$S : H_1 \rightarrow H_2 \quad \text{given by} \quad z = S(x). \quad (6.55)$$

Substituting $z = S(x)$ in (6.54), we have (6.52).

S is linear. In fact, its domain is the vector space H_1 and from (6.52) and the sesquilinearity, we obtain

$$\begin{aligned} \langle S(\alpha x_1 + \beta x_2), y \rangle &= h(\alpha x_1 + \beta x_2, y) = \alpha h(x_1, y) + \beta h(x_2, y) \\ &= \alpha \langle S(x_1), y \rangle + \beta \langle S(x_2), y \rangle \\ &= \langle \alpha S(x_1) + \beta S(x_2), y \rangle \end{aligned}$$

for all $y \in H_2$, so that by the Equality lemma above,

$$S(\alpha x_1 + \beta x_2) = \alpha S(x_1) + \beta S(x_2).$$

S is bounded. Indeed, leaving aside the trivial case $S = 0$, we have from (6.50) and (6.52),

$$\|h\| = \sup_{x \neq 0, y \neq 0} \frac{|\langle S(x), y \rangle|}{\|x\| \|y\|} \geq \sup_{x \neq 0, S(x) \neq 0} \frac{|\langle S(x), S(x) \rangle|}{\|x\| \|S(x)\|} = \sup_{x \neq 0} \frac{\|S(x)\|^2}{\|x\|^2} = \|S\|^2.$$

This proves boundedness (why?). Moreover, $\|h\| \geq \|S\|^2$.

We now obtain (6.53) by noting that $\|h\| \leq \|S\|^2$ follows by an application of the Schwarz inequality:

$$\|h\| = \sup_{x \neq 0, y \neq 0} \frac{|\langle S(x), y \rangle|}{\|x\| \|y\|} \leq \sup_{x \neq 0} \frac{\|S(x)\| \|y\|}{\|x\| \|y\|} = \|S\|.$$

S is unique. In fact, assuming that there is a linear operator $T : H_1 \rightarrow H_2$ such that for all $x \in H_1$ and all $y \in H_2$ we have

$$h(x, y) = \langle S(x), y \rangle = \langle T(x), y \rangle,$$

we see that $S(x) = T(x)$ by the Equality lemma for all $x \in H_1$. Hence, $S = T$ by definition. This completes the proof. ■

6.5 The Hilbert Adjoint Operator

The results of the previous section will now enable us to introduce the Hilbert-adjoint operator of a bounded linear operator on a Hilbert space. This operator was suggested by problems in matrices and linear differential and integral equations. We shall see that it also helps to define three important classes of operators, called *self-adjoint*, *unitary*, and *normal operators*, which have been studied extensively because they play a key role in various applications.

Definition 6.5.1 Hilbert-Adjoint Operator

Let $T : H_1 \rightarrow H_2$ be a bounded linear operator, where H_1 and H_2 are Hilbert spaces. Then, the **Hilbert-adjoint operator**, denoted T^* , of T is the operator

$$T^* : H_2 \rightarrow H_1$$

such that for all $x \in H_1$ and all $y \in H_2$,

$$\langle T(x), y \rangle_{H_2} = \langle x, T^*(y) \rangle_{H_1}. \quad (6.56)$$

REMARK: As shown in (6.56), remember to keep in mind the space on which the inner product is being computed. For convenience, for the rest of this section, we will omit the explicit reference to the Hilbert space on which the inner product is being taken.

Of course, we should first show that this definition is worth making, i.e., we should prove that for a given T such a T^* does indeed exist.

Theorem 6.5.1 Existence

The Hilbert-adjoint operator T^* of T exists, is unique, and is a bounded linear operator with norm

$$\|T^*\| = \|T\|. \quad (6.57)$$

PROOF: The formula

$$h(y, x) = \langle y, T(x) \rangle \quad (6.58)$$

defines a sesquilinear form on $H_2 \times H_1$ because the inner product is sesquilinear and T is linear. In fact, conjugate linearity of the form is seen from

$$\begin{aligned} h(y, \alpha x_1 + \beta x_2) &= \langle y, T(\alpha x_1 + \beta x_2) \rangle = \langle y, \alpha T(x_1) + \beta T(x_2) \rangle = \bar{\alpha} \langle y, T(x_1) \rangle + \bar{\beta} \langle y, T(x_2) \rangle \\ &= \bar{\alpha} h(y, x_1) + \bar{\beta} h(y, x_2). \end{aligned}$$

h is bounded. Indeed, by the Schwarz inequality,

$$|h(x, y)| = |\langle y, T(x) \rangle| \leq \|y\| \|T(x)\| \leq \|T\| \|x\| \|y\|.$$

This also implies that $\|h\| \leq \|T\|$. Moreover, we have $\|h\| \geq \|T\|$ from

$$\|h\| = \sup_{x \neq 0, y \neq 0} \frac{|\langle y, T(x) \rangle|}{\|y\| \|x\|} \geq \sup_{x \neq 0, T(x) \neq 0} \frac{|\langle T(x), T(x) \rangle|}{\|T(x)\| \|x\|} = \|T\|.$$

Together,

$$\|h\| = \|T\|. \quad (6.59)$$

The general Riesz representation theorem gives a Riesz representation for h . Writing T^* for S , we have

$$h(y, x) = \langle T^*(y), x \rangle, \quad (6.60)$$

and we know from that theorem that $T^* : H_2 \rightarrow H_1$ is a uniquely determined bounded linear operator with norm

$$\|T^*\| = \|h\| = \|T\|.$$

This proves (6.57). Also, $\langle y, T(x) \rangle = \langle T^*(x), y \rangle$ by comparing (6.58) and (6.60), so that we have (6.56) by taking conjugates, and we now see that T^* is in fact the operator we are looking for. ■

Lemma 6.5.1 Zero Operator

Let X and Y be inner product spaces and $Q : X \rightarrow Y$ a bounded linear operator. Then,

1. $Q = 0$ if and only if $\langle Q(x), y \rangle = 0$ for all $x \in X$ and all $y \in Y$;
2. If $Q : X \rightarrow X$, where X is complex, and $\langle Q(x), x \rangle = 0$ for all $x \in X$, then $Q = 0$.

PROOF:

1. By definition, $Q = 0$ means that $Q(x) = 0$ for all x , and implies

$$\langle Q(x), y \rangle = \langle 0, y \rangle = 0 \quad \langle w, y \rangle = 0.$$

Conversely, $\langle Q(x), y \rangle = 0$ for all x and y implies $Q(x) = 0$ for all x by the Equality lemma, so that $Q = 0$ by definition.

2. By assumption, $\langle Q(v), v \rangle = 0$ for every $v = \alpha x + y \in X$, i.e.,

$$0 = \langle Q(\alpha x + y), \alpha x + y \rangle = |\alpha|^2 \langle Q(x), x \rangle + \langle Q(y), y \rangle + \alpha \langle Q(x), y \rangle + \bar{\alpha} \langle Q(y), x \rangle.$$

The first two terms on the right are zero by assumption. $\alpha = 1$ gives

$$\langle Q(x), y \rangle + \langle Q(y), x \rangle = 0.$$

$\alpha = i$ gives $\bar{\alpha} = -i$, and

$$\langle Q(x), y \rangle - \langle Q(y), x \rangle = 0.$$

By addition, $\langle Q(x), y \rangle = 0$, and so $Q = 0$ follows from the first part. ■

In the second part of this lemma, it is essential that X be complex. Indeed, the conclusion may not hold if X is real. A counterexample is a rotation Q of the plane $\mathbb{R}^2$ through a right angle. Q is linear, and $Q(x) \perp x$, hence $\langle Q(x), x \rangle = 0$ for all $x \in \mathbb{R}^2$, but $Q \neq 0$. (What about such a rotation in the complex plane?)

Theorem 6.5.2 Properties of Hilbert-Adjoint Operators

Let H_1, H_2 be Hilbert spaces, $S : H_1 \rightarrow H_2$ and $T : H_1 \rightarrow H_2$ bounded linear operators, and α any scalar. Then we have

1. $\langle T^*(y), x \rangle = \langle y, T(x) \rangle$ for all $x \in H_1$ and all $y \in H_2$.
2. $(S + T)^* = S^* + T^*$.
3. $(\alpha T)^* = \bar{\alpha} T^*$.
4. $(T^*)^* = T$.
5. $\|T^* T\| = \|T T^*\| = \|T\|^2$.
6. $T^* T = 0 \iff T = 0$.
7. $(ST)^* = T^* S^*$ (assuming $H_1 = H_2$).

PROOF:

1. From (6.56), we have

$$\langle T^*(y), x \rangle = \overline{\langle x, T^*(y) \rangle} = \overline{\langle T(x), y \rangle} = \langle y, T(x) \rangle.$$

2. By (6.56), for all x and y ,

$$\begin{aligned} \langle x, (S + T)^*(y) \rangle &= \langle (S + T)(x), y \rangle = \langle S(x), y \rangle + \langle T(x), y \rangle \\ &= \langle x, S^*(y) \rangle + \langle x, T^*(y) \rangle = \langle x, (S^* + T^*)(y) \rangle. \end{aligned}$$

Hence, $(S + T)^*(y) = (S^* + T^*)(y)$ for all y by the Equality lemma.

3. Do not confuse this formula with the formula $T^*(\alpha x) = \alpha T^*(x)$. It is obtained from the following calculation and subsequent application of the first part of the lemma above to $Q = (\alpha T)^* - \bar{\alpha} T^*$:

$$\begin{aligned} \langle (\alpha T)^*(y), x \rangle &= \langle y, (\alpha T)(x) \rangle = \langle y, \alpha(T(x)) \rangle = \bar{\alpha} \langle y, T(x) \rangle \\ &= \bar{\alpha} \langle T^*(y), x \rangle \\ &= \langle \bar{\alpha} T^*(y), x \rangle. \end{aligned}$$

4. Let $(T^*)^* \equiv T^{**}$. For all $x \in H_1$ and all $y \in H_2$, we have from the first part of this theorem and (6.56),

$$\langle T^{**}(x), y \rangle = \langle x, T^*(y) \rangle = \langle T(x), y \rangle,$$

and the result follows from the first part of the lemma above applied to $Q = T^{**} - T$.

5. We see that $T^*T : H_1 \rightarrow H_1$, but $TT^* : H_2 \rightarrow H_2$. By the Schwarz inequality,

$$\|T(x)\|^2 = \langle T(x), T(x) \rangle = \langle T^*T(x), x \rangle \leq \|T^*T(x)\| \|x\| \leq \|T^*T\| \|x\|^2.$$

Taking the supremum over all x of norm one, we obtain $\|T\|^2 \leq \|T^*T\|$. We thus have

$$\|T\|^2 \leq \|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2.$$

Hence $\|T^*T\| = \|T\|^2$. Replacing T with T^* , we have

$$\|T^{**}T^*\| = \|T^*\|^2 = \|T\|^2.$$

But $T^{**} = T$ by the previous part, so this completes the proof.

6. Immediate from the previous part.

7. Repeated application of (6.56) gives

$$\langle x, (ST)^*(y) \rangle = \langle (ST)(x), y \rangle = \langle T(x), S^*(y) \rangle = \langle x, T^*S^*(y) \rangle.$$

Hence, $(ST)^*(y) = T^*S^*(y)$ by the Equality lemma, completing the proof. ■

Theorem 6.5.3

Let T be a bounded linear (therefore continuous) linear transformation of a Hilbert space H into itself. A closed linear subspace M of H is invariant under L if and only if $M^\perp$ is invariant under L^* .

REMARK: Recall the definition of *invariant subspace*.

Definition 6.5.2 Invariant Subspace

Let X be a vector space. If $T : X \rightarrow X$ is a linear transformation and M is a linear subspace of X such that $T(M) \subset M$, then M is called **invariant** under T .

PROOF: If M is invariant under L , then $L(M) \subset M$. This means that $\langle x, L(y) \rangle = 0$ for all $y \in M$ and all $x \in M^\perp$. But this means that $\langle L^*(x), y \rangle = 0$ for all $y \in M$ and all $x \in M^\perp$. Thus, $L^*(x) \in M^\perp$ for all $x \in M^\perp$, implying that $L^*(M^\perp) \subset M^\perp$. If $M^\perp$ is invariant under L^* , the same type of argument shows that M is invariant under L . ■

Theorem 6.5.4

Let T be a bounded linear operator of a Hilbert space H into itself. Then,

$$\overline{\mathcal{R}(T)} = \mathcal{N}(T^*)^\perp \quad \text{and} \quad \mathcal{N}(T)^\perp = \overline{\mathcal{R}(T^*)},$$

where $\mathcal{N}(T)$ and $\mathcal{N}(T^*)$ are the null spaces of T and T^* , respectively, and $\overline{\mathcal{R}(T)}$ and $\overline{\mathcal{R}(T^*)}$ are the closures of the ranges of T and T^* , respectively.

PROOF: Since $T^{**} = T$, it will suffice to prove just the first one. A point $z \in H$ is in $\mathcal{R}(T)^\perp$ if and only if $\langle z, T(x) \rangle = 0$ for all $x \in H$. But by definition of the adjoint, it follows that

$$\langle T^*(z), x \rangle = 0$$

for all $x \in H$. So $z \in \mathcal{R}(T)^\perp$ if and only if $T^*(z) = 0$. Hence, we have shown that $\mathcal{R}(T)^\perp = \mathcal{N}(T^*)$.

However, now $\mathcal{R}(T)$ may not be closed, so it is not necessarily the case that $\mathcal{R}(T)^{\perp\perp} = \mathcal{R}(T)$ (remember this condition for a subspace to be closed!); but, we do always have that $\overline{\mathcal{R}(T)}^{\perp\perp} = \overline{\mathcal{R}(T)}$, completing the proof. ■

Example 6.5.1 Let H be a Hilbert space and $T : H \rightarrow H$ a bijective bounded linear operator whose inverse is bounded. Show that $(T^*)^{-1}$ exists and that $(T^*)^{-1} = (T^{-1})^*$.

SOLUTION: To be completed.

Example 6.5.2 If (T_n) is a sequence of bounded linear operators on a Hilbert space and $T_n \rightarrow T$, show that $T_n^* \rightarrow T^*$.

SOLUTION: To be completed.

Example 6.5.3 Let $I = [a, b] \subset \mathbb{R}$ and let $k : I \times I \rightarrow \mathbb{C}$ be such that

$$\iint_I |k(s, t)|^2 ds dt < \infty.$$

Define the integral operator $K : L_2[a, b] \rightarrow L_2[a, b]$ by

$$K(x)(s) = \int_a^b k(s, t)x(t) dt,$$

and take as the inner product on $L_2[a, b]$ the usual inner product

$$\langle x, y \rangle = \int_a^b x(t)\overline{y(t)} dt.$$

We will show that K^* is also an integral operator. In this case, we get

$$\begin{aligned} \langle K(x), y \rangle &= \int_a^b \left[\int_a^b k(s, t)x(t) dt \right] \overline{y(s)} ds = \int_a^b \int_a^b k(s, t)x(t)\overline{y(s)} ds dt \\ &= \int_a^b x(t) \left[\int_a^b \overline{k(s, t)}y(s) ds \right] dt = \langle x, K^*(y) \rangle. \end{aligned}$$

Hence, after interchanging the s and t variables, we get

$$K^*(y)(s) = \int_a^b \overline{k(t, s)}y(t) dt.$$

Example 6.5.4 Volterra Integral Operator

Let $I = [0, T]$ and consider the linear operator $K : C(I) \rightarrow C(I)$ defined as

$$y(t) = K(x)(t) = \int_0^t k(t, s)x(s) \, ds.$$

In the usual case of the Volterra integral operator, k needs only to be defined in the region $s \leq t$, $t \in [0, T]$. However, if we wish to define an adjoint operator, essentially replacing $k(t, s)$ with $k(s, t)$, we shall have to extend the definition of k .

If we set $k(t, s) = 0$ for $s > t$, then we get

$$y(t) = \int_I k(t, s)x(s) \, ds.$$

Then, from the above example, we have

$$K^*(y)(t) = \int_I \overline{k(s, t)}y(s) \, ds = \int_t^T \overline{k(s, t)}y(s) \, ds.$$

Thus, the adjoint of a Volterra integral operator is also a Volterra integral operator. But if K depends on the “past” (i.e., $y(t) = K(x)(t)$ is determined by $x(s)$ for $0 \leq s \leq t$), then K^* depends on the “future” (i.e., $K^*(y)(t)$ is determined by $y(s)$ for $t \leq s \leq T$).

The Volterra integral operator is an example of a *causal operator* and its adjoint is an example of an *anti-causal operator*.

Example 6.5.5 Consider the operator

$$F : x(t) \mapsto f(t)x(t)$$

on $L^2(-\infty, \infty)$, where $|f(t)| \leq B < \infty$ for all t . (Note that we do not require that $f \in L^2$. After all, multiplication of x by a constant $k \in \mathbb{R}$ is a special case.) Then, F is a bounded linear operator with

$$\|F\| = \sup_{t \in \mathbb{R}} |f(t)| = \|f\|_\infty.$$

Now, consider the following:

$$\langle F(x), y \rangle = \int_{-\infty}^{\infty} f(t)x(t)\overline{y(t)} \, dt = \int_{-\infty}^{\infty} x(t)\overline{f(t)y(t)} \, dt = \langle x, F^*(y) \rangle.$$

This implies that the associated adjoint operator is given by

$$F^* : y(t) \mapsto \overline{f(t)}y(t).$$

The condition that f be strictly bounded may be replaced by the condition $|f(t)| \leq B < \infty$ for almost all $t \in \mathbb{R}$. Then,

$$\|F\| = \text{ess. sup}_{t \in \mathbb{R}} |f(t)| = \|F\|_\infty,$$

where $\|\cdot\|_\infty$ denotes the L_∞ norm, i.e., $|f(t)| \leq \|f\|_\infty$ “almost everywhere”.

6.6 Self-Adjoint, Unitary and Normal Operators

Classes of bounded linear operators of great practical importance can be defined by the use of the Hilbert adjoint operator as follows.

Definition 6.6.1 Self-Adjoint, Unitary, Normal Operator

A bounded linear operator $T : H \rightarrow H$ on a Hilbert space H is said to be

- **Self-Adjoint**, or **Hermitian**, if $T^* = T$;
- **Unitary** if T is bijective and $T^* = T^{-1}$;
- **Normal** if $TT^* = T^*T$.

Recall that the Hilbert adjoint operator T^* of T was defined in (6.56) as

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle.$$

If T is self-adjoint, we see that the formula becomes

$$\langle T(x), y \rangle = \langle x, T(y) \rangle. \quad (6.61)$$

Proposition 6.6.1

If a bounded linear operator T on a Hilbert space is self-adjoint or unitary, then it is normal.

PROOF: Immediate from the definition. ■

Of course, a normal operator need not be self-adjoint or unitary. For example, if $I : H \rightarrow H$ is the identity operator, then $T = 2iI$ is normal since $T^* = -2iI$, so that $TT^* = T^*T = 4I$, but $T^* \neq T$ as well as $T^* \neq T^{-1} = -\frac{1}{2}iI$.

Example 6.6.1 Matrices

We consider the Hilbert space $\mathbb{C}^n$ with the standard inner product

$$\langle x, y \rangle = x^T \bar{y}, \quad (6.62)$$

where x and y are written as column vectors, so that the multiplication is matrix multiplication.

Let $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a linear operator (which, remember, is bounded—why?). A basis for $\mathbb{C}^n$ being given, we can represent T and its Hilbert adjoint operator T^* by two n -rowed square matrices, say, A and B , respectively.

Using (6.62) and the familiar rule $(B(x))^T = x^T B^T$ for the transposition of a product, we obtain

$$\langle T(x), y \rangle = (Ax)^T \bar{y} = x^T A^T \bar{y},$$

and

$$\langle x, T^*(y) \rangle = x^T \bar{B} \bar{y}.$$

By (6.56), the left-hand sides of the above equations are equal for all $x, y \in \mathbb{C}^n$. Hence, we must have $A^T = \bar{B}$. Consequently,

$$B = \bar{A}^T.$$

Therefore,

If a basis for $\mathbb{C}^n$ is given and a linear operator on $\mathbb{C}^n$ is represented by a certain matrix, then its Hilbert adjoint operator is represented by the complex conjugate transpose of that matrix.

Consequently, representing matrices are

- *Hermitian* if T is self-adjoint (Hermitian);
- *Unitary* if T is unitary;
- *Normal* if T is normal.

Similarly, for a linear operator $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, representing matrices are

- *Real symmetric* if T is self-adjoint;
- *Orthogonal* if T is unitary.

In this connection, remember the following definitions. A square matrix $A = (\alpha_{jk})$ is said to be

- *Hermitian* if $\bar{A}^T = A$ (hence $\bar{\alpha}_{kj} = \alpha_{jk}$);
- *Skew-Hermitian* if $\bar{A}^T = -A$ (hence $\bar{\alpha}_{kj} = -\alpha_{jk}$);
- *Unitary* if $\bar{A}^T = A^{-1}$;
- *Normal* if $A\bar{A}^T = \bar{A}^T A$.

A *real* square matrix $A = (\alpha_{jk})$ is said to be

- *(Real) symmetric* if $A^T = A$ (hence $\alpha_{kj} = \alpha_{jk}$);
- *(Real) skew-symmetric* if $A^T = -A$ (hence $\alpha_{jk} = -\alpha_{jk}$);
- *Orthogonal* if $A^T = A^{-1}$.

Hence, a real Hermitian matrix is a (real) symmetric matrix. A real skew-Hermitian matrix is a (real) skew-symmetric matrix. A real unitary matrix is an orthogonal matrix.

Theorem 6.6.1 Self-Adjointness

Let $T : H \rightarrow H$ be a bounded linear operator on a Hilbert space H . Then

1. If T is self-adjoint, then $\langle T(x), x \rangle$ is real for all $x \in H$.
2. If H is complex and $\langle T(x), x \rangle$ is real for all $x \in H$, then the operator T is self-adjoint.

PROOF:

1. If T is self-adjoint, then for all x ,

$$\overline{\langle T(x), x \rangle} = \langle x, T(x) \rangle = \langle T(x), x \rangle.$$

Hence, $\langle T(x), x \rangle$ is equal to its complex conjugate, so that it is real.

2. If $\langle T(x), x \rangle$ is real for all x , then

$$\langle T(x), x \rangle = \overline{\langle T(x), x \rangle} = \overline{\langle x, T^*(x) \rangle} = \langle T^*(x), x \rangle.$$

Hence,

$$0 = \langle T(x), x \rangle - \langle T^*(x), x \rangle = \langle (T - T^*)(x), x \rangle,$$

and $T - T^* = 0$ by the Equality lemma since H is complex. ■

REMARK: In the second part of the above theorem it is essential that H be complex. This is clear since for a real H the inner product is real-valued, which makes $\langle T(x), x \rangle$ real without any further assumptions about the linear operator T .

Products (i.e., composites) of self-adjoint operators appear quite often in applications, so that the following theorem will be useful.

Theorem 6.6.2 Self-Adjointness of Product

The product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, i.e.,

$$ST = TS.$$

PROOF: We have

$$(ST)^* = T^*S^* = TS,$$

with the last equality coming from the assumption that ST is self-adjoint. Then, it is clear that

$$ST = (ST)^* \Leftrightarrow ST = TS. \quad \blacksquare$$

Theorem 6.6.3 Sequences of Self-Adjoint Operators

Let (T_n) be a sequence of bounded self-adjoint linear operators $T_n : H \rightarrow H$ on a Hilbert space H . Suppose that (T_n) converges, say, to an operator T , i.e., $\lim_{n \rightarrow \infty} T_n = T$ (in the operator norm). Then, the limit operator T is a bounded self-adjoint linear operator on H .

PROOF: We must show that $T^* = T$. This follows from $\|T - T^*\| = 0$, where $\|\cdot\|$ is the operator norm. To prove the latter, we recall that the norm of the adjoint is the same as the norm of the original operator, so that

$$\|T_n^* - T^*\| = \|(T_n - T)^*\| = \|T_n - T\|,$$

and we obtain by the triangle inequality in $B(H)$,

$$\begin{aligned} \|T - T^*\| &\leq \|T - T_n\| + \|T_n + T_n^*\| + \|T_n^* - T^*\| \\ &= \|T - T_n\| + 0 + \|T_n - T\| \\ &= 2\|T_n - T\|, \end{aligned}$$

and the last line goes to zero as n goes to ∞ by definition of the convergence of the sequence (T_n) . Hence, $\|T - T^*\| = 0$, meaning that $T^* = T$. ■

We now turn to unitary operators and consider some of their basic properties.

Theorem 6.6.4 Unitary Operators

Let the operators $U : H \rightarrow H$ and $V : H \rightarrow H$ be unitary, where H is a Hilbert space. Then,

1. U is isometric, i.e., $\|U(x)\| = \|x\|$ for all $x \in H$.
2. $\|U\| = 1$ provided $H \neq \{0\}$.
3. U^{-1} is unitary (i.e., U^* is unitary).
4. UV is unitary.
5. U is normal.

PROOF:

1. This can be seen from

$$\|U(x)\|^2 = \langle U(x), U(x) \rangle = \langle x, U^*(U(x)) \rangle = \langle x, I(x) \rangle = \|x\|^2.$$

2. This follows immediately from the above equation.
3. Since U is bijective, so is U^{-1} , and since $U^{**} = U$, we have

$$(U^{-1})^* = U^{**} = U = (U^{-1})^{-1}.$$

4. UV is bijective, and so we get

$$(UV)^* = V^*U^* = V^{-1}U^{-1} = (UV)^{-1}.$$

5. This follows from the fact that $U^{-1} = U^*$ and $UU^{-1} = U^{-1}U = I$. ■

Proposition 6.6.2 Unitary Operators

A bounded linear operator T on a complex Hilbert space H is unitary if and only if T is isometric and surjective.

PROOF: Suppose that T is isometric and surjective. Isometry implies injectivity (check this!), so that T is bijective. We show that $T^* = T^{-1}$. By this isometry,

$$\langle T^*(T(x)), x \rangle = \langle T(x), T(x) \rangle = \langle x, x \rangle = \langle Ix, x \rangle.$$

Hence,

$$\langle (T^*T - I)(x), x \rangle = 0,$$

so that $T^*T - I = 0$ by the Equality lemma, meaning $T^*T = I$. From this,

$$TT^* = TT^*(TT^{-1}) = T(T^*T)T^{-1} = TIT^{-1} = I.$$

Together, $T^*T = TT^* = I$. Hence, $T^* = T^{-1}$, so that T is unitary. The converse is clear since T is isometric by the first part of the above theorem and surjective by definition. ■

Note that an isometric operator need not be unitary since it may fail to be surjective. An example is the *right-shift operator* $T : \ell_2 \rightarrow \ell_2$ given by

$$(\xi_1, \xi_2, \xi_3 \rightarrow) \mapsto (0, \xi_1, \xi_2, \xi_3, \dots),$$

where $x = (\xi_j) \in \ell_2$.

Example 6.6.2 If S and T are bounded self-adjoint linear operators on a Hilbert space H and α and β are real, show that $\tilde{T} := \alpha S + \beta T$ are self-adjoint.

SOLUTION: To be completed.

Example 6.6.3 Show that for any bounded linear operator T on a Hilbert space H that the operators

$$T_1 := \frac{1}{2}(T + T^*) \quad \text{and} \quad T_2 = \frac{1}{2}(T - T^*)$$

are self-adjoint. Show that

$$T = T_1 + iT_2, \quad T^* = T_1 - iT_2.$$

Show uniqueness, that is, that $T_1 + iT_2 = S_1 + iS_2$ implies $S_1 = T_1$ and $S_2 = T_2$. Here, S_1 and S_2 are self-adjoint operators by assumption.

SOLUTION: To be completed.

Example 6.6.4 Show that an isometric linear operator $T : H \rightarrow H$ satisfies $T^*T = I$, where I is the identity operator on the Hilbert space H .

SOLUTION: To be completed.

Example 6.6.5 Delay/Shift Operator

Let $H = L_2(-\infty, \infty)$ and consider the *delay/shift operator* $S_\tau : H \rightarrow H$, $\tau \in \mathbb{R}$, defined by

$$S_\tau(x)(t) = x(t - \tau) \text{ for all } t \in (-\infty, \infty).$$

Note that S_τ has an inverse $S_\tau^{-1} = S_{-\tau}$. S_τ is also a unitary operator since

$$\langle S_\tau(x), S_\tau(y) \rangle = \int_{-\infty}^{\infty} \overline{y(t - \tau)} x(t - \tau) dt = \int_{-\infty}^{\infty} \overline{y(t)} x(t) dt$$

for all $x, y \in H$. It can be shown that $\|S_\tau\| = 1$ (do it!).

Note also that

$$\langle S_\tau(x), y \rangle = \int_{-\infty}^{\infty} \overline{y(t)} x(t - \tau) dt = \int_{-\infty}^{\infty} \overline{y(t + \tau)} x(t) dt$$

for all $x, y \in H$. This implies that the adjoint of S_τ is defined by

$$S_\tau^*(y)(t) = y(t + \tau) \text{ for all } t \in (-\infty, \infty).$$

Therefore, as expected (since S_τ was found to be unitary),

$$S_\tau^* = S_{-\tau} = S_\tau^{-1}.$$

If we interpret S_τ for $\tau > 0$ as a causal operator, then S_τ^* is anti-causal.

6.6.1 Application: The Fourier Transform

The Fourier transform of a function f , denoted $\hat{f}$, is defined as

$$\hat{f}(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iyx} f(x) dx, \quad (6.63)$$

with inverse given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iyx} \hat{f}(y) dy. \quad (6.64)$$

The representation for F and F^{-1} given by (??) and (6.64) is valid for functions f and $\hat{f}$ in $L_1(-\infty, \infty) \cap L_2(-\infty, \infty)$.

In order to discuss arbitrary functions in $L_2(-\infty, \infty)$, we need a different representation, namely,

$$\hat{f}(y) = F(f)(y) = \frac{1}{\sqrt{2\pi}} \frac{d}{dy} \int_{-\infty}^{\infty} \frac{e^{-ixy} - 1}{-ix} f(x) dx \quad (6.65)$$

and

$$f(x) = F^{-1}(\hat{f})(x) = \frac{1}{\sqrt{2\pi}} \frac{d}{dx} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} \hat{f}(y) dy. \quad (6.66)$$

If f and $\hat{f}$ belong to $L_1(-\infty, \infty)$, then we can bring the derivative inside the integral and then (6.65) and (6.66) reduce to (6.63) and (6.64). The transformation F defined by (6.65) is sometimes called the *Fourier-Plancherel transform*.

Let us now show that F and F^{-1} are unitary operators on $L_2(-\infty, \infty)$. For this purpose, we shall denote the operator defined by (6.66) as G . We will then show that F and G are unitary and that $F^* = G$, which implies that $F^{-1} = G$.

Now, define

$$H(y, x) = \frac{1}{\sqrt{2\pi}} \frac{e^{-ixy} - 1}{-ix}, \quad K(x, y) = \frac{1}{\sqrt{2\pi}} \frac{e^{ixy} - 1}{iy},$$

and let

$$\phi_r(x) = \begin{cases} +1, & 0 \leq x \leq r, & 0 \leq r \\ -1, & r \leq x \leq 0, & r \leq 0, \\ 0, & \text{otherwise} \end{cases}$$

Now, for $r \geq 0$ one has

$$F(\phi_r)(y) = \frac{1}{\sqrt{2\pi}} \frac{d}{dy} \int_0^r \frac{e^{-ixy} - 1}{-ix} dx = \frac{1}{\sqrt{2\pi}} \int_0^r e^{-ixy} dx = H(r, y).$$

Similarly, for $r \leq 0$, one has $F(\phi_r)(y) = H(r, y)$. Likewise, we get $G(\phi_r)(x) = K(r, x)$.

Since $\phi(y) := \text{Im}(H(r, y)\overline{H(s, y)})$ is an odd function in $L_1(-\infty, \infty)$, one has

$$\int_{-\infty}^{\infty} \phi(y) dy = 0.$$

Hence,

$$\langle F(\phi_r), F(\phi_r) \rangle = \int_{-\infty}^{\infty} H(r, y) \overline{H(s, y)} dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\cos(s-r)y - \cos(sy) - \cos(ry) + 1}{y^2} dy.$$

By using the trigonometric identity $\cos \theta = 1 - 2 \sin^2(\theta/2)$, and by chaing variables, we get

$$\langle F(\phi_r), F(\phi_r) \rangle = \frac{1}{2\pi} (|r| + |s| - |r-s|) \int_{-\infty}^{\infty} \frac{\sin^2 u}{u^2} du.$$

Using the fact that

$$\int_{-\infty}^{\infty} \frac{\sin^2 u}{u^2} du = \pi,$$

we get

$$\langle F(\phi_r), F(\phi_r) \rangle = \begin{cases} \min\{|r|, |s|\}, & \text{if } rs \geq 0, \\ 0, & \text{if } rs \leq 0 \end{cases} = \langle \phi_r, \phi_r \rangle.$$

Similarly, one has

$$\langle G(\phi_r), G(\phi_r) \rangle = \langle \phi_r, \phi_r \rangle.$$

Furthermore, by a simple change of variables, we get

$$\langle F(\phi_r), \phi_s \rangle = \langle \phi_r, G(\phi_s) \rangle.$$

If f and g are now finite linear combinations of the functions ϕ_r , that is, if they are step-functions, then one has

$$\begin{aligned} \langle F(f), F(g) \rangle &= \langle f, g \rangle, \\ \langle G(f), G(g) \rangle &= \langle f, g \rangle, \\ \langle F(f), g \rangle &= \langle f, G(g) \rangle. \end{aligned} \tag{6.67}$$

However, the step functions are dense in $L_2(-\infty, \infty)$, therefore (6.67) is valid for all f and g in $L_2(-\infty, \infty)$. This shows that F and G are unitary and that $F^* = G$.

Next, one can prove that the Fourier transform F is given by (6.63) for all $f \in L_2(-\infty, \infty)$ if one interprets the integral in (6.63) in the following way:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iyx} f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-N}^N e^{-iyx} f(x) dx,$$

where $\hat{\lim}$ means “limit in the mean”, that is,

$$\int \left| \hat{f}(y) - \frac{1}{\sqrt{2\pi}} \int_{-N}^N e^{-iyx} f(x) dx \right|^2 dy \rightarrow 0 \text{ as } N \rightarrow \infty.$$

The Fourier transform is a fundamental tools in the operational calculus of differential operators. The following theorem is the cornerstone of this theory.

Theorem 6.6.5

Let P and Q be the linear operators defined by

$$\begin{aligned} P : u(x) &\mapsto i \frac{du}{dx} \\ Q : u(x) &\mapsto xu(x), \end{aligned}$$

where the domains are

$$\begin{aligned} \mathcal{D}_P &= \{u \in L_2(-\infty, \infty) \mid u \text{ is absolutely continuous and } u' \in L_2(-\infty, \infty)\} \\ \mathcal{D}_Q &= \{u \in L_2(-\infty, \infty) \mid xu(x) \in L_2(-\infty, \infty)\}. \end{aligned}$$

Then, the Fourier transform F sets up a one-to-one correspondence between $\mathcal{D}_P$ and $\mathcal{D}_Q$ in such a way that

$$P = FQF^{-1} \quad \text{and} \quad Q = F^{-1}PF.$$

PROOF: The first step is to show that if $u \in \mathcal{D}_Q$, then $F(u) \in \mathcal{D}_P$ and $P(F(u)) = F(Q(u))$. Let $u \in \mathcal{D}_Q$. Then, one can show that $\int_{-\infty}^{\infty} |u(x)| dx < \infty$. Thus, $v = F(u)$ is given by

$$v(y) = F(u)(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iyx} u(x) dx.$$

However, $F(Q(u)) \in L_2(-\infty, \infty)$, and

$$\begin{aligned}(FQ)(u)(y) &= \frac{1}{\sqrt{2\pi}} \frac{d}{dy} \int_{-\infty}^{\infty} \frac{e^{-iyx} - 1}{-ix} xu(x) dx \\ &= i \frac{d}{dy} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (e^{-iyx} - 1)u(x) dx \\ &= i \frac{d}{dy} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iyx} u(x) dx \\ &= (PF)(u)(x).\end{aligned}$$

Hence, $F(u) \in \mathcal{D}_p$ and $(FQ)(u) = (PF)(u)$.

The second step is to show that if $v \in \mathcal{D}_p$, then $F^{-1}(v) \in \mathcal{D}_Q$ and $(F^{-1}P)(v) = (QF^{-1})(v)$. Let $v \in \mathcal{D}_p$. Then, one can show that

$$\lim_{x \rightarrow \pm\infty} v(x) = 0.$$

Furthermore, $(F^{-1}P)(v)$ is in $L_2(-\infty, \infty)$, and if we integrate by parts, we get

$$\begin{aligned}(F^{-1}P)(v)(x) &= \frac{d}{dx} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} i \frac{dv(y)}{dy} dy \\ &= \frac{d}{dx} \frac{-1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy}(ixy) - (e^{ixy} - 1)}{y^2} v(y) dy \\ &= \frac{d}{dx} \frac{x}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} v(y) dy + \frac{d}{dx} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1 - ixy}{y^2} v(y) dy.\end{aligned}$$

Since

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} v(y) dy = -\frac{d}{dx} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1 - ixy}{y^2} v(y) dy,$$

we get

$$(F^{-1}P)(v)(x) = x \frac{d}{dx} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{ixy} - 1}{iy} v(y) dy = xF^{-1}(v)(x).$$

Hence, $F^{-1}(v) \in \mathcal{D}_Q$ and $(F^{-1}P)(v) = (QF^{-1})(v)$. ■

On $\mathbb{R}^n$, the Fourier transform takes on the form

$$\hat{f}(y) = \mathcal{F}(f)(y) = \int_{\mathbb{R}^n} e^{-ix \cdot y} f(x) dx, \quad (6.68)$$

and

$$f(x) = \mathcal{F}^{-1}(\hat{f})(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{ix \cdot y} \hat{f}(y) dy, \quad (6.69)$$

where $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are points in $\mathbb{R}^n$ and

$$x \cdot y = x_1 y_1 + \dots + x_n y_n.$$

(6.68) and (6.69) are valid for f and $\hat{f}$ in $L_1(\mathbb{R}^n) \cap L_2(\mathbb{R}^n)$, and they are also valid for all f and $\hat{f}$ in $L_2(\mathbb{R}^n)$ provided that we compute the integral as a limit in the mean of integrals over bounded regions.

One can prove that

$$\langle \mathcal{F}(f), \mathcal{F}(g) \rangle = (2\pi)^n \langle f, g \rangle$$

for all $f, g \in L_2(\mathbb{R}^n)$. This is done by applying the Fourier transform on $\mathbb{R}$ to each of the variables $x_1, x_2, \dots, x_n$ successively. Similarly, one gets

$$\langle \mathcal{F}^{-1}(f), \mathcal{F}^{-1}(g) \rangle = (2\pi)^{-n} \langle f, g \rangle.$$

By the same method as before, one can prove the following theorem.

Theorem 6.6.6

Let P_k and Q_k be the linear operators defined by

$$\begin{aligned} P_k : u(x_1, \dots, x_n) &\mapsto i \frac{\partial u}{\partial x_k} \\ Q_k : u(x_1, \dots, x_n) &\mapsto x_k u(x_1, \dots, x_n), \end{aligned}$$

where the domains are

$$\begin{aligned} \mathcal{D}_{P_k} &= \{u \in L_2(\mathbb{R}^n) \mid P_k(u) \in L_2(\mathbb{R}^n)\} \\ \mathcal{D}_{Q_k} &= \{u \in L_2(\mathbb{R}^n) \mid Q_k(u) \in L_2(\mathbb{R}^n)\}. \end{aligned}$$

Then, the Fourier transform $\mathcal{F}$ sets up a one-to-one correspondence between $\mathcal{D}_{P_k}$ and $\mathcal{D}_{Q_k}$, for all k , in such a way that

$$P_k = \mathcal{F} Q_k \mathcal{F}^{-1} \quad \text{and} \quad Q_k \mathcal{F}^{-1} P_k \mathcal{F} \text{ for all } k.$$

6.6.2 Application: Quantum Mechanics

6.7 Compact Operators

Compact linear operators are very important in applications. For instance, they play a central role in the theory of integral equations and in various problems of mathematical physics.

Most of the discussion in this section applies generally to merely normed linear spaces (i.e., we don't need inner product spaces or Hilbert spaces).

Definition 6.7.1 Compact Linear Operator

Let X and Y be normed spaces. An operator $T : X \rightarrow Y$ is called a **compact linear operator** if T is linear if it maps every bounded sequence in X into a sequence that has a convergent subsequence.

REMARK: An equivalent way of defining a compact linear operator is as follows: T is a compact linear operator if for every bounded subset M of X the closure of the image, $\overline{T(M)}$, is compact.

Proof of the equivalence is as follows: if T is compact and (x_n) is bounded, then the closure of $(T(x_n))$ in Y is compact and hence $(T(x_n))$ contains a convergence subsequence. Conversely, assume that every bounded sequence (x_n) contains a subsequence (x_{n_k}) such that $(T(x_{n_k}))$ converges in Y . Consider any bounded subset $B \subset X$, and let (y_n) be any sequence in $T(B)$. Then, $y_n = T(x_n)$ for some $x_n \in B$, and (x_n) is bounded since B is bounded. By assumption, $(T(x_n))$ contains a convergent subsequence. Hence, $\overline{T(B)}$ is compact because (y_n) was arbitrary. By definition, this shows that T is compact.

Example 6.7.1 Every linear operator defined on a finite-dimensional normed linear space is compact.

Lemma 6.7.1 Continuity

Let X and Y be normed spaces. Then,

1. Every compact linear operator $T : X \rightarrow Y$ is bounded, hence continuous.
2. If $\dim(X) = \infty$, the identity operator $I_X : X \rightarrow X$ (which is continuous) is not compact.

PROOF:

1. The unit sphere $U = \{x \in X \mid \|x\| = 1\}$ is bounded. Since T is compact, $\overline{T(U)}$ is compact, and is bounded, so that

$$\sup_{\|x\|=1} \|T(x)\| < \infty.$$

Hence, T is bounded, and by Theorem 5.5.2, it is continuous.

2. Of course, the closed unit ball $M = \{x \in X \mid \|x\| \leq 1\}$ is bounded. If $\dim(X) = \infty$, then Theorem 5.5.1 implies that M cannot be compact; thus, $I(M) = M = \overline{M}$ is not relatively compact, i.e., $I(M)$ is not compact. ■

It is important to note that the converse of the first part of the above theorem is not true, i.e., L bounded does not imply that L is compact. To illustrate this, let us return to the example, see earlier, of a bounded sequence in a normed space that is not compact, namely, the sequence of basis vectors (e_n) in ℓ_2 defined as

$$e_n = (e_{n,1}, e_{n,2}, \dots),$$

where $e_{n,n} = 1$ and $e_{n,k} = 0$ for $n \neq k$. Clearly, the identity mapping I on ℓ_2 , which is bounded, is not compact, since it maps the sequence (e_n) to itself.

Theorem 6.7.1 Finite Dimensional Domain or Range

Let X and Y be normed spaces and $T : X \rightarrow Y$ a linear operator. Then,

1. If T is bounded and $\dim(X) < \infty$, then T is compact.
2. If $\dim(X) < \infty$, the operator is compact.

PROOF:

1. Let (x_n) be any bounded sequence in X . Then, the inequality $\|T(x_n)\| \leq \|T\| \|x_n\|$ shows that $(T(x_n))$ is bounded. Hence, $(T(x_n))$ is relatively compact by Theorem 5.2.10 since $\dim(X) < \infty$. It follows that $(T(x_n))$ has a convergent subsequence. Since (x_n) was an arbitrary bounded sequence in X , the operator T is compact by definition.
2. This follows from the first part by noting that $\dim(X) < \infty$ implies boundedness of T by Theorem 5.5.1 and $\dim(T(X)) \leq \dim(X)$ by Theorem 5.4.1. ■

Example 6.7.2 Let $L : C[a, b] \rightarrow C[a, b]$ be defined by

$$L(f)(x) = \int_a^b p(x, y)f(y) dy,$$

where $p(x, y)$ is a polynomial of degree N . Then, we can write

$$L(f)(x) = \sum_{n=0}^N \beta_n x^n,$$

which implies that $\dim(\mathcal{R}(L)) \leq N + 1$. Therefore, by the above theorem, L is compact.

Definition 6.7.2 Operator of Finite Rank

An operator $T \in B(X)$ with $\dim(T(X)) < \infty$ is often called an **operator of finite rank**.

The following theorem states conditions under which the limit of a sequence of compact linear operators is compact. The theorem is also important as a tool for proving compactness of a given operator by exhibiting it as the uniform operator limit of a sequence of compact linear operators.

Lemma 6.7.2

Let A and B be compact linear operators between normed spaces X and Y . Then,

1. $A + B$ is compact.
2. cA is compact for all $c \in \mathbb{R}$ (or $c \in \mathbb{C}$).

PROOF: To be completed. ■

Lemma 6.7.3

Let $T : X \rightarrow X$ be compact linear operator and $S : X \rightarrow X$ be a bounded linear operator. Then TS and ST are compact linear operators.

PROOF: Let $B \subset X$ be any bounded set. Since S is a bounded operators, $S(B)$ is a bounded set, and the set $T(S(B)) = TS(B)$ is relatively compact because T is compact. Hence, TS is a compact linear operator.

Also, letting (x_n) be any bounded sequence in X , we have that $(T(x_n))$ has a convergent subsequence $(T(x_{n_k}))$ by definition, and $(ST(x_{n_k}))$ converges. Hence, by definition, ST is compact. ■

Theorem 6.7.2 Sequence of Compact Linear Operators

Let (T_n) be a sequence of compact linear operators from a normed space X into a Banach space Y . If (T_n) is convergent in the operator norm, say $\lim_{n \rightarrow \infty} T_n = T$, then T is compact.

PROOF: Using a “diagonal method”, we show that for any bounded sequence (x_m) in X the image $(T(x_m))$ has a convergence subsequence.

Since T_1 is compact, (x_m) has a subsequence $(x_{1,m})$ such that $(T_1(x_{1,m}))$ is Cauchy. Similarly, $(x_{1,m})$ has a subsequence $(x_{2,m})$ such that $(T_2(x_{2,m}))$ is Cauchy. Continuing in this way, we see that the “diagonal sequence” $(y_m) := (x_{m,m})$ is a subsequence of (x_m) such that for every fixed positive integer n the sequence $(T_n(y_m))_{m \in \mathbb{N}}$ is Cauchy. (x_m) is bounded, say, $\|x_m\| \leq c$ for all m . Hence $\|y_m\| \leq c$ for all m . Hence, $\|y_m\| \leq c$ for all m . Let $\epsilon > 0$. Since $T_m \rightarrow T$, there exists $n = p$ such that $\|T - T_p\| < \frac{\epsilon}{3c}$. Since $(T_p(y_m))_{m \in \mathbb{N}}$ is Cauchy, there exists N such that

$$\|T_p(y_j) - T_p(y_k)\| < \frac{\epsilon}{3} \text{ for all } j, k > N.$$

Hence we obtain for $j, k > N$,

$$\begin{aligned} \|T(y_j) - T(y_k)\| &\leq \|T(y_j) - T_p(y_j)\| + \|T_p(y_j) - T_p(y_k)\| + \|T_p(y_k) - T(y_k)\| \\ &\leq \|T - T_p\| \|y_j\| + \frac{\epsilon}{3} + \|T_p - T\| \|y_k\| \\ &< \frac{\epsilon}{3c}c + \frac{\epsilon}{3} + \frac{\epsilon}{3c}c = \epsilon. \end{aligned}$$

This shows that $(T(y_m))$ is Cauchy and converges since Y is complete. Remembering that (y_m) is a subsequence of the arbitrary bounded sequence (x_m) , we have by definition that T is compact. ■

Note that the present theorem becomes false if we replace the uniform operator convergence by strong/pointwise operator convergence $\|T_n(x) - T(x)\| \rightarrow 0$. This can be seen from $T_n : \ell_2 \rightarrow \ell_2$ defined by $T_n(x) = (\xi_1, \dots, \xi_n, 0, \dots)$, where $x = (\xi_j) \in \ell_2$. Since T_n is linear and bounded, T_n is compact by the first part of Theorem 6.7.1. Clearly, $T_n(x) \rightarrow x = I(x)$, but I is not compact since $\dim(\ell_2) = \infty$.

The following example illustrates how the theorem can be used to prove compactness of an operator.

Example 6.7.3 Prove the compactness of $T : \ell_2 \rightarrow \ell_2$ defined by $y = (\eta_j) = T(x)$ ($x = (\xi_j)$), where $\eta_j = \frac{\xi_j}{j}$ for $j = 1, 2, \dots$.

SOLUTION: T is linear. If $x = (\xi_j) \in \ell_2$, then $y = (\eta_j) \in \ell_2$. Let $T_n : \ell_2 \rightarrow \ell_2$ be defined by

$$T_n(x) = \left(\xi_1, \frac{\xi_2}{2}, \frac{\xi_3}{3}, \dots, \frac{\xi_n}{n}, 0, 0, \dots \right).$$

T_n is linear and bounded, and is compact by the first part of (6.7.1). Furthermore,

$$\|(T - T_n)(x)\| = \sum_{j=n+1}^{\infty} |\eta_j|^2 = \sum_{j=n+1}^{\infty} \frac{1}{j^2} |\xi_j|^2 \leq \frac{1}{(n+1)^2} \sum_{j=n+1}^{\infty} |\xi_j|^2 \leq \frac{\|x\|^2}{(n+1)^2}.$$

Taking the supremum over all x of norm one, we see that

$$\|T - T_n\| \leq \frac{1}{n+1}.$$

Hence, $T_n \rightarrow T$, and T is compact by Theorem 6.7.2.

Example 6.7.4 Show that the compact linear operators from X into Y constitute a subspace of $B(X, Y)$.

SOLUTION: To be completed.

Theorem 6.7.3 Separability of Range

The range $\mathcal{R}(T)$ of a compact linear operator $T : X \rightarrow Y$ is separable, where X and Y are normed spaces.

Theorem 6.7.4 Compact Extension

A compact linear operator $T : X \rightarrow Y$ from a normed space X into a Banach space Y has a compact linear extension $\tilde{T} : \hat{X} \rightarrow Y$, where $\hat{X}$ is the completion of X .

Theorem 6.7.5 Compact Operators on a Hilbert Space

If $L : H \rightarrow H$ is a bounded linear operator on a separable Hilbert space H , $\{e_n\}$ is an orthonormal basis for H , and $\sum_{n=1}^{\infty} \|L(e_n)\|^2 < \infty$, then L is compact.

PROOF: For any v in H , we know that we can write it in the basis $\{e_n\}$ as

$$v = \sum_{i=1}^{\infty} \langle v, e_i \rangle e_i \Rightarrow L(v) = \sum_{i=1}^{\infty} \langle v, e_i \rangle L(e_i).$$

Now, consider the approximate operator $L_n(v) = \sum_{i=1}^n \langle v, e_i \rangle L(e_i)$. L_n is compact since it has a finite-dimensional range (each $\text{span}\{L(e_i)\}$ is one-dimensional).

Now, if we can prove that $\|L_n - L\| \rightarrow 0$, then we will have that L is compact. Indeed,

$$\begin{aligned} \|L_n(v) - L(v)\| &= \left\| \sum_{i=n+1}^{\infty} \langle v, e_i \rangle L(e_i) \right\|^2 \leq \left(\sum_{i=n+1}^{\infty} |\langle v, e_i \rangle| \|L(e_i)\| \right)^2 \\ &\leq \sum_{i=n+1}^{\infty} \langle v, e_i \rangle^2 \sum_{i=n+1}^{\infty} \|L(e_i)\|^2 \leq \|v\|^2 \sum_{i=n+1}^{\infty} \|L(e_i)\|^2. \end{aligned}$$

Given $\epsilon > 0$, there exists N such that $\sum_{i=n+1}^{\infty} \|L(e_i)\|^2 < \epsilon^2$ for $n > N$. Therefore,

$$\|L_n(v) - L(v)\| < \epsilon \|v\| \Rightarrow \|L_n - L\| < \epsilon$$

for all $n > N$. Therefore, (L_n) converges to L , and therefore L is compact. ■

Example 6.7.5 Let $k \in C([a, b] \times [a, b])$, $L : C[a, b] \rightarrow C[a, b]$, $\|\cdot\| = \|\cdot\|_{\infty}$, and

$$L(f)(x) = \int_a^b k(x, y) f(y) dy.$$

For every n there exists a polynomial $k_n(x, y)$ such that $|k(x, y) - k_n(x, y)| < \frac{1}{n}$ for all $(x, y) \in [a, b]^2$ (this is by the Weierstrass approximation theorem). Letting

$$L_n(f)(x) := \int_a^b k_n(x, y) f(y) dy,$$

we see that $L_n : C[a, b] \rightarrow C[a, b]$ is compact. Then,

$$|L(f)(x) - L_n(f)(x)| = \left| \int_a^b (k(x, y) - k_n(x, y)) f(y) dy \right| \leq \int_a^b |k(x, y) - k_n(x, y)| |f(y)| dy \leq \frac{b-a}{n} \|f\|,$$

which implies that

$$\|L - L_n\| \leq \frac{b-a}{n} \Rightarrow \|L - L_n\| \rightarrow 0 \text{ as } n \rightarrow \infty.$$

This means that the sequence (L_n) converges to L , which means that L is compact.

Proposition 6.7.1

If $\{g_i\}_{i=1}^{\infty}$ is an orthonormal basis for $L_2[a, b]$, then $\{g_i g_j\}_{i,j=1}^{\infty}$, where $(g_i g_j)(x) = g_i(x) g_j(x)$, is an orthonormal basis for $L_2([a, b]^2)$.

PROOF: By the orthonormality of $\{g_i\}$ we have that $\{g_i g_j\}$ is orthonormal. Then, suppose $\langle f, g_i g_j \rangle =$

0 for all i, j . Then,

$$\begin{aligned} \int_a^b \int_a^b f(x, y) g_j(y) g_i(x) dy dx = 0 &\Rightarrow \left\langle \int_a^b f(x, y) g_j(y) dy, g_i \right\rangle_{L_2} = 0 \quad \forall i \\ &\Rightarrow \int_a^b f(x, y) g_j(y) dy = 0 \text{ a.e. } x \\ &\Rightarrow \langle f(x, y), g_j(y) \rangle = 0 \quad \forall j \Rightarrow f(x, y) = 0 \text{ a.e. } y. \end{aligned}$$

Therefore, $f(x, y) = 0$ a.e. $(x, y) \in [a, b]^2$. ■

Example 6.7.6 Define $L : L_2[a, b] \rightarrow L_2[a, b]$. $L(f)(x) = \int_a^b k(x, y) f(y) dy$ for $k \in L_2([a, b]^2)$, and let $\{g_i\}_{i=1}^\infty$ be an orthonormal basis for $L_2[a, b]$. Show that L is compact.

SOLUTION: We can write the action of L on f in the basis $\{g_i\}$ as

$$L(f) = \sum_{i=1}^{\infty} \langle L(f), g_i \rangle g_i \Rightarrow \|L(f)\|^2 = \sum_{i=1}^{\infty} \langle L(f), g_i \rangle^2.$$

Now,

$$\sum_{j=1}^{\infty} \|L(g_j)\|^2 = \sum_{i,j=1}^{\infty} \langle L(g_j), g_i \rangle^2 = \sum_{i,j=1}^{\infty} \left(\int_a^b \int_a^b k(x, y) g_j(y) g_i(x) dy dx \right)^2.$$

Since $\{g_i g_j\}$ is an orthonormal basis for $L_2([a, b]^2)$, we have

$$\sum_{i,j=1}^{\infty} \langle k(x, y), g_i(y) g_j(x) \rangle^2 = \|k\|_{L_2([a, b]^2)}^2 = \int_a^b \int_a^b k(x, y)^2 dy dx.$$

Thus, $\sum_{j=1}^{\infty} \|L(g_j)\|^2 < \infty$. By Theorem 6.7.5, L is compact.

Example 6.7.7 Canonical Example

Define the map $L : \ell_2 \rightarrow \ell_2$ by

$$L(x_1, x_2, \dots) = (\alpha_1 x_1, \alpha_2 x_2, \dots),$$

where (α_n) is a bounded sequence in $\mathbb{R}$, so that $|\alpha_n| \leq M$ for all n . Therefore, we can write

$$\|L(x)\|^2 = \sum_{n=1}^{\infty} \alpha_n^2 x_n^2 \leq M^2 \sum_{n=1}^{\infty} x_n^2 \Rightarrow \|L(x)\| \leq M \|x\|.$$

Thus, L is a bounded linear operator. Let us now know that L is compact if and only if $\lim_{n \rightarrow \infty} \alpha_n = 0$.

First, suppose $\lim_{n \rightarrow \infty} \alpha_n \neq 0$. Then there exists $\epsilon > 0$ and (n_k) such that $|\alpha_{n_k}| \geq \epsilon$ for all k . Now, consider the sequence (e_{n_k}) such that $L(e_{n_k}) = \alpha_{n_k} e_{n_k}$. Then,

$$\|L(e_{n_k}) - L(e_{n_\ell})\|^2 = \alpha_{n_k}^2 + \alpha_{n_\ell}^2 \geq 2\epsilon^2 \text{ for } k \neq \ell \Rightarrow \|L(e_{n_k}) - L(e_{n_\ell})\| \geq \sqrt{2}\epsilon.$$

Thus, $(L(e_{n_k}))$ does not have a convergent subsequence. Therefore, L is not compact.

Now, suppose $\lim_{n \rightarrow \infty} \alpha_n = 0$. Let $L_N(x_1, x_2, \dots) := (\alpha_1 x_1, \alpha_2 x_2, \dots, \alpha_N x_N, 0, \dots)$. Then $L_N : \ell_2 \rightarrow \ell_2$ and L_N is compact. Then, given $\epsilon > 0$, there exists N_1 such that $|\alpha_n| < \epsilon$ for all $n > N_1$. Then,

$$\|(L - L_N)(x)\|^2 = \sum_{n=N+1}^{\infty} (\alpha_n x_n)^2 < \epsilon^2 \sum_{n=N+1}^{\infty} x_n^2 < \epsilon^2 \|x\|^2 \text{ for } N > N_1.$$

This implies that $\|L - L_N\| < \epsilon$ for $N > N_1$. So $\|L - L_N\| \rightarrow 0$ as $N \rightarrow \infty$. Therefore, L is compact.

Theorem 6.7.6 Adjoint Operator

Let $T : X \rightarrow Y$ be a linear operator. If T is compact, so is its adjoint operator $T^\times : Y' \rightarrow X'$, where X and Y are normed spaces and X' and Y' are the dual spaces of X and Y , respectively.

Theorem 6.7.7 Hilbert-Adjoint Operator

Let A be a continuous linear operator on a Hilbert space H_1 into a Hilbert space H_2 . If A^*A is compact, then A is compact.

PROOF: Let S be a bounded set in H_1 . The operator A^*A is a compact operator in H_1 by assumption. It therefore maps S into a precompact set. Thus, there is a sequence $(x_{n_k}) \subset S$ such that $(A^*A(x_{n_k}))$ is a convergence sequence, and

$$\begin{aligned} \|A(x_n) - A(x_m)\|_2^2 &= \langle A(x_n) - A(x_m), A(x_n) - A(x_m) \rangle_2 \\ &= \langle x_n - x_m, A^*A(x_n) - A^*A(x_m) \rangle_1 \\ &\leq \|x_n - x_m\|_1 \cdot \|A^*A(x_n) - A^*A(x_m)\|_1. \end{aligned}$$

But $\|A^*A(x_n) - A^*A(x_m)\|_1 \rightarrow 0$, and $\|x_n - x_m\|_1$ is bounded, so that $\|A(x_n) - A(x_m)\|_2 \rightarrow 0$. Thus, $(A(x_n))$ is a Cauchy sequence in H_2 , but H_2 is complete, so that $(A(x_n))$ is a convergence sequence. Thus, A maps S into a precompact set, meaning that A is compact. ■

Corollary 6.7.1

If $A : H_1 \rightarrow H_2$ between Hilbert spaces is compact, then so is A^* .

PROOF: This follows immediately from the above theorem and the fact that $AA^* = (A^*)^*A^*$ is compact by Lemma 6.7.3. ■

Theorem 6.7.8 Bounded Inverse Theorem (Banach)

If X and Y are Banach spaces and $L : X \rightarrow Y$ is a one-to-one and onto bounded linear operator, then L^{-1} is bounded.

PROOF: To be completed. ■

Theorem 6.7.9 Inverse of a Compact Operator

Let X and Y be infinite-dimensional Banach spaces and $L : X \rightarrow Y$ a one-to-one compact linear operator. Then $\mathcal{R}(L) \neq Y$ and L^{-1} is not bounded.

PROOF: The proof uses three results:

1. If L_1 is compact and L_2 is bounded, then $L_1 L_2$ is compact (this is Lemma 6.7.3);
2. $\overline{B_1(0)}$ is compact in X , a normed linear space, if and only if $\dim(X) < \infty$ (this is Theorem 5.2.11);
3. The Banach inverse theorem above.

Suppose for a contradiction that $\mathcal{R}(L) = Y$. Then L^{-1} is bounded. This means that $LL^{-1} = I$ is compact, so that $B := \overline{B_1(0)}$ in Y is compact, which implies that Y is finite-dimensional. But the boundedness of L^{-1} implies that $LL^{-1} = I$ acts on $\mathcal{R}(L)$, which implies that I is compact, so that $B := \overline{B_1(0)} \cap \mathcal{R}(L)$ is compact, which implies that $\mathcal{R}(L)$ is finite-dimensional, which finally implies that $\mathcal{R}(L^{-1}) = X$ is finite-dimensional, a contradiction to X being infinite-dimensional. ■

Example 6.7.8 Define $L : \ell_2 \rightarrow \ell_2$ by

$$L(x) = \left(x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots \right).$$

L is compact. Also, L is one-to-one since $L(x) = 0 \Rightarrow x = 0$, i.e., its kernel contains just the zero vector.

Now, $L(x) = y \Leftrightarrow x_n = ny_n$, so that

$$L^{-1}(y) = (y_1, 2y_2, 3y_3, \dots).$$

Thus,

$$\mathcal{R}(L) = \left\{ (y_1, y_2, \dots) \mid \sum_{n=1}^{\infty} (ny_n)^2 < \infty \right\}.$$

Observe that $\mathcal{R}(L) \neq \ell_2$ since $y = \left(\frac{1}{n}\right)_{n=1}^{\infty} \in \ell_2$, but $y \notin \mathcal{R}(L)$.

Also, $L^{-1} : \mathcal{R}(L) \rightarrow \ell_2$ is unbounded since for $e_n = (0, \dots, 0, 1, 0, \dots)$ (1 in the n th position, 0 elsewhere) $L^{-1}(e_n) = ne_n \Rightarrow \|L^{-1}(e_n)\| = n \rightarrow \infty$ as $n \rightarrow \infty$; but (e_n) is bounded.

6.8 Closed Linear Operators

Not all linear operators of practical importance are bounded. For instance, in quantum mechanics and other applications, one needs unbounded operators quite frequently. However, practically all of the linear operators that the analyst is likely to use are so-called closed linear operators.

Definition 6.8.1 Closed Linear Operator

Let X and Y be normed spaces and $T : X \rightarrow Y$ a linear operator with domain $\mathcal{D}(T) \subset X$. Then T is called a **closed linear operator** if its *graph*, defined as

$$\mathcal{G}(T) = \{(x, y) \mid x \in \mathcal{D}(T), y = T(x)\},$$

is closed in the normed space $X \times Y$, where the two algebraic operations of a vector space in $X \times Y$ are defined as

$$\begin{aligned} (x_1, y_1) + (x_2, y_2) &= (x_1 + x_2, y_1 + y_2), \\ \alpha(x, y) &= (\alpha x, \alpha y), \end{aligned}$$

(for α a scalar) and the norm on $X \times Y$ is defined as

$$\|(x, y)\| = \|x\| + \|y\| \tag{6.70}$$

REMARK: Note that this is not the only norm one can defined on $X \times Y$. For example, other frequently used norms on the product space $X \times Y$ are defined by

$$\|(x, y)\| = \max\{\|x\|, \|y\|\} \quad \text{and} \quad \|(x, y)\|_0 = \sqrt{\|x\|^2 + \|y\|^2}.$$

It is easy to verify that these are norms.

Theorem 6.8.1 Inverse of Closed Linear Operator

The inverse T^{-1} of a closed linear operator, if it exists, is a closed linear operator.

PROOF: To be completed. ■

Theorem 6.8.2 Closed Graph Theorem

Let X and Y be Banach spaces and $T : X \rightarrow Y$ a closed linear operator with domain $\mathcal{D}(T) \subset X$. Then, if $\mathcal{D}(T)$ is closed in X , the operator T is bounded.

PROOF: We first show that $X \times Y$ with norm defined by (6.70) is complete. Let (z_n) be Cauchy in $X \times Y$, where $z_n := (x_n, y_n)$. Then, for every $\epsilon > 0$ there is an N_ϵ such that

$$\|z_n - z_m\| = \|x_n - x_m\| + \|y_n - y_m\| < \epsilon \quad \forall m, n > N_\epsilon. \tag{6.71}$$

Hence, (x_n) and (y_n) are Cauchy in X and Y , respectively, and converge, say to x and y , because X and Y are complete. This implies that $z_n \rightarrow z = (x, y)$ since from (6.71) with $m \rightarrow \infty$ we have $\|z_n - z\| \leq \epsilon$ for $n > N_\epsilon$. Since the Cauchy sequence (z_n) was arbitrary, $X \times Y$ is complete.

Now, by assumption, $\mathcal{G}(T)$ is closed in $X \times Y$ and $\mathcal{D}(T)$ is closed in X . Hence, $\mathcal{G}(T)$ and $\mathcal{D}(T)$ are complete by Theorem 3.5.1. We now consider the mapping

$$P : \mathcal{G}(T) \rightarrow \mathcal{D}(T), \quad (x, T(x)) \mapsto x.$$

P is certainly linear. It is bounded because

$$\|P(x, T(x))\| = \|x\| \leq \|x\| + \|T(x)\| = \|(x, T(x))\|.$$

P is bijective; in fact, the inverse mapping is

$$P^{-1} : \mathcal{D}(T) \rightarrow \mathcal{G}(T), \quad x \mapsto (x, T(x)).$$

Since $\mathcal{G}(T)$ and $\mathcal{D}(T)$ are complete, we can apply the bounded inverse theorem and see that P^{-1} is bounded, say $\|(x, T(x))\| \leq b \|x\|$ for some b and all $x \in \mathcal{D}(T)$. Hence, T is bounded because

$$\|T(x)\| \leq \|(x, T(x))\| - \|x\| = \|(x, T(x))\| \leq b \|x\|$$

for all $x \in \mathcal{D}(T)$. ■

By definition, $\mathcal{G}(T)$ is closed if and only if $z = (x, y) \in \overline{\mathcal{G}(T)}$ implies $z \in \mathcal{G}(T)$. From Theorem 3.3.1, we see that $z \in \overline{\mathcal{G}(T)}$ if and only if there are $z_n := (x_n, T(x_n)) \in \mathcal{G}(T)$ such that $\lim_{n \rightarrow \infty} z_n = z$, hence

$$x_n \rightarrow x, \quad T(x_n) \rightarrow y, \tag{6.72}$$

and $z = (x, y) \in \mathcal{G}(T)$ if and only if $x \in \mathcal{D}(T)$ and $y = T(x)$. This proves the following useful criterion that expresses the property that is often taken as a definition of closedness of a linear operator.

Theorem 6.8.3 Closed Linear Operator

Let $T : X \rightarrow Y$ be a linear operator with domain $\mathcal{D}(T) \subset X$, where X and Y are normed spaces. Then, T is closed if and only if it has the following property. If $x_n \rightarrow x$, where $(x_n) \subset \mathcal{D}(T)$, and $T(x_n) \rightarrow y$, then $x \in \mathcal{D}(T)$ and $T(x) = y$.

Note well that this property is different from the following property of a bounded linear operator. If a linear operator T is bounded and thus continuous, and if (x_n) is a sequence in $\mathcal{D}(T)$ that converges in $\mathcal{D}(T)$, then $(T(x_n))$ also converges. This need not hold for a closed linear operator. However, if T is closed and two sequences (x_n) and $(\tilde{x}_n)$ in the domain of T converge with the same limit and if the corresponding sequences $(T(x_n))$ and $(T(\tilde{x}_n))$ both converge, then the latter have the same limit.

Example 6.8.1 Differential Operator

Let $X = C^1[0, 1]$ and $Y = C[0, 1]$ and $T : X \rightarrow Y$ such that $T(x) = x'$, where the prime denotes differentiation and the domain $\mathcal{D}(T)$ is the subspace of functions $x \in X$ that have a continuous derivative. Then T is not bounded, but is closed.

PROOF: We already know that T is not bounded because it is the differentiation operator. We prove that T is closed by applying Theorem 6.8.3. Let (x_n) in $\mathcal{D}(T)$ be such that both (x_n) and $(T(x_n))$ converge, say,

$$x_n \rightarrow x \quad \text{and} \quad T(x_n) = x'_n \rightarrow y.$$

Since convergence in the infinity-norm of $C[0, 1]$ is uniform convergence on $[0, 1]$, from $x'_n \rightarrow y$, we have

$$\int_0^t y(\tau) \, d\tau = \int_0^t \lim_{n \rightarrow \infty} x'_n(\tau) \, d\tau = \lim_{n \rightarrow \infty} \int_0^t x'_n(\tau) \, d\tau = x(t) - x(0),$$

that is,

$$x(t) = x(0) + \int_0^t y(\tau) \, d\tau.$$

This shows that $x \in \mathcal{D}(T)$ and $x' = y$. Theorem 6.8.3 now implies that T is closed. ■

It is worth noting that in this example, $\mathcal{D}(T)$ is not closed in X since T would then be bounded by the closed graph theorem. This demonstrates the following fact.

Proposition 6.8.1

Closedness does not imply boundedness of a linear operator. Conversely, boundedness does not imply closedness.

PROOF: The first statement is illustrated by the above example and the second one by the following example. Let $T : X \rightarrow X$ be the identity operator on X , where $\mathcal{D}(T)$ is a proper dense subspace of the normed space X . Then, it is trivial that T is linear and bounded. However, T is not closed. This follows immediately from Theorem 6.8.3 if we take an $x \in X - \mathcal{D}(T)$ and a sequence $(x_n) \subset \mathcal{D}(T)$ that converges to x . ■

Lemma 6.8.1 Closed Operator

Let $T : X \rightarrow Y$ be a bounded linear operator with domain $\mathcal{D}(T) \subset X$, where X and Y are normed spaces. Then,

1. If $\mathcal{D}(T)$ is a closed subset of X , then T is closed.
2. If T is closed and Y is complete, then $\mathcal{D}(T)$ is a closed subset of X .

PROOF:

1. If $(x_n) \subset \mathcal{D}(T)$ and it converges, say to x , and is such that $(T(x_n))$ also converges, then $x \in \overline{\mathcal{D}(T)} = \mathcal{D}(T)$ since $\mathcal{D}(T)$ is closed, and $T(x_n) \rightarrow T(x)$ since T is continuous. Hence, T is closed by Theorem 6.8.3.
2. For $x \in \overline{\mathcal{D}(T)}$, there is a sequence $(x_n) \subset \mathcal{D}(T)$ such that $x_n \rightarrow x$ (this is by Theorem 3.3.1,

remember). Since T is bounded,

$$\|T(x_n) - T(x_m)\| = \|T(x_n - x_m)\| \leq \|T\| \|x_n - x_m\|.$$

This shows that $(T(x_n))$ is Cauchy. $(T(x_n))$ converges, say to $y \in Y$ because Y is complete. Since T is closed, $x \in \mathcal{D}(T)$ by 6.8.3 (and $T(x) = y$). Hence, $\mathcal{D}(T)$ is closed because $x \in \overline{\mathcal{D}(T)}$ was arbitrary. ■

7 Spectral Theory

Spectral theory is one of the main branches of modern functional analysis and its applications. Roughly speaking, it is concerned with certain inverse operators, their general properties and their relations to the original operators. Such inverse operators arise quite naturally in connection with the problem of solving equations (systems of linear algebraic equations, differential equations, integral equations). For instance, the investigations of boundary value problems by Sturm and Liouville, and Fredholm's famous theory of integral equations, were important to the development of the field.

7.1 Finite-Dimensional Normed Spaces

Let X be a finite-dimensional normed space and $T : X \rightarrow X$ a linear operator. Spectral theory of such operators is simpler than that of operators defined on infinite-dimensional spaces. In fact, we know that we can represent T by *matrices* (which depend on the choice of basis for X), and we shall see that spectral theory of T is essentially matrix eigenvalue theory.

Definition 7.1.1 Eigenvalues, Eigenvectors, Eigenspaces, Spectrum, Resolvent Set

An **eigenvalue** of a square matrix $A = (a_{jk})$ is a number λ such that

$$Ax = \lambda x \quad (7.1)$$

has a solution $x \neq 0$. This x is called an **eigenvector** of A corresponding to that eigenvalue λ . The eigenvectors corresponding to that eigenvalue λ and the zero vector form a vector subspace of X , which is called the **eigenspace** of A corresponding to that eigenvalue λ . The set $\sigma(A)$ of all eigenvalues of A is called the **spectrum** of A . The complement of this set, $\rho(A) := \mathbb{C} - \sigma(A)$, in the complex plane, is called the **resolvent set** of A .

For example, by direct calculation, we can verify that

$$x_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{ and } x_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ are eigenvectors of } A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

corresponding to the eigenvalues $\lambda_1 = 6$ and $\lambda_2 = 1$ of A , respectively.

How do we obtain the eigenvalues and eigenvectors of a matrix, and in general what can we say about the existence these objects?

Firstly, note that (7.1) can be written as

$$(A - \lambda I)x = 0, \quad (7.2)$$

where I is the n -rowed unit matrix. This is a homogeneous system of n linear equations in n unknowns, call then $\xi_1, \dots, \xi_n$, the components of x . The determinant of the coefficients is $\det(A - \lambda I)$ and must be zero in order for (7.2) to have a solution $x \neq 0$. This gives us the **characteristic equation** of A :

$$\det(A - \lambda I) = \begin{vmatrix} \alpha_{11} - \lambda & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} - \lambda & \dots & \alpha_{2n} \\ \cdot & \cdot & \dots & \cdot \\ \alpha_{n1} & \alpha_{n2} & \dots & \alpha_{nn} - \lambda \end{vmatrix} = 0. \quad (7.3)$$

$\det(A - \lambda I)$ is called the **characteristic determinant** of A . By developing it, we obtain a polynomial in λ of degree n , called the **characteristic polynomial** of A .

Definition 7.1.2 Multiplicity of an Eigenvalue

The **algebraic multiplicity** of an eigenvalue λ of a matrix A is the multiplicity of λ as a root of the characteristic polynomial, and the dimension of the eigenspace of A corresponding to λ is called the **geometric multiplicity** of λ .

Theorem 7.1.1 Eigenvalues of a Matrix

The eigenvalues of an n -rowed square matrix $A = (\alpha_{jk})$ are given by the solutions to the characteristic equation (7.3) of A . Hence, A has at least one eigenvalue (and at most n numerically different eigenvalues).

The second statement in the above theorem holds since, by the *fundamental theorem of algebra* and the *factorisation theorem*, a polynomial of positive degree n and with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$ (and at most n numerically different roots). Note that the roots may be complex even if A is real.

Example 7.1.1 Let's calculate the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$$

that we saw earlier.

The characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{vmatrix} = \lambda^2 - 7\lambda + 6 = 0,$$

the spectrum is $\{6, 1\}$, and the eigenvectors of A corresponding to 6 and 1, respectively, are obtained from

$$\begin{array}{lcl} -\xi_1 + 4\xi_2 = 0 & \text{and} & 4\xi_1 + 4\xi_2 = 0 \\ \xi_1 - 4\xi_2 = 0 & & \xi_1 + \xi_2 = 0 \end{array}.$$

Observe that in each case we need only one of the two equations because one is a constant multiple of the other.

How can we apply our result to a linear operator $T : X \rightarrow X$ on a normed space X of dimension n ? Let $e = \{e_1, \dots, e_n\}$ be any basis for X and $T_e = (\alpha_{jk})$ the matrix representing T with respect to that basis (whose elements are kept in the given order). Then, the eigenvalues of the matrix T_e are called the **eigenvalues of the operator T** , and similarly for the spectrum and the resolvent set.

Theorem 7.1.2 Eigenvalues of an Operator

All matrices representing a given linear operator $T : X \rightarrow X$ on a finite-dimensional normed space X relative to various bases for X have the same eigenvalues.

PROOF: We must see what happens in the transition from one basis for X to another. Let $e = (e_1, \dots, e_n)$ and $\tilde{e} = (\tilde{e}_1, \dots, \tilde{e}_n)$ be any two bases for X , written as row vectors. By the definition of a basis, each e_j is a linear combination of the $\tilde{e}_k$ s, and conversely. We can write this as

$$\tilde{e} = eC \quad \text{or} \quad \tilde{e}^T = C^T e^T, \quad (7.4)$$

where C is a non-singular n -rowed square matrix. Every $x \in X$ has a unique representation with respect to each of the two bases, say,

$$x = ex_1 = \sum_{j=1}^n \xi_j e_j = \tilde{e}x_2 = \sum_{k=1}^n \tilde{\xi}_k \tilde{e}_k,$$

where $x_1 = (\xi_j)$ and $x_2 = (\tilde{\xi}_k)$ are column vectors. From this and (7.4) we have $ex_1 = \tilde{e}x_2 = eCx_2$. Hence,

$$x_1 = Cx_2. \quad (7.5)$$

Similarly, for $T(x) = y = ey_1 = \tilde{e}y_2$, we have

$$y_1 = Cy_2. \quad (7.6)$$

Consequently, if T_1 and T_2 denote the matrices that represent T with respect to e and $\tilde{e}$, respectively, then

$$y_1 = T_1(x_1) \quad \text{and} \quad y_2 = T_2(x_2),$$

and from this and (7.5) and (7.6),

$$CT_2(x_2) = Cy_2 = y_1 = T_1(x_1) = T_1(Cx_2).$$

Premultiplying by C^{-1} (which exists because C is non-singular), we obtain the transformation law

$$T_2 = C^{-1}T_1C, \quad (7.7)$$

with C determined by the bases according to (7.4) (and independent of T). Using (7.7) and $\det(C^{-1})\det(C) = 1$, we can now show that the characteristic polynomials of T_2 and T_1 are equal:

$$\begin{aligned} \det(T_2 - \lambda I) &= \det(C^{-1}T_1C - \lambda C^{-1}IC) \\ &= \det(C^{-1}(T_1 - \lambda I)C) \\ &= \det(C^{-1})\det(T_1 - \lambda I)\det(C) \\ &= \det(T_1 - \lambda I). \end{aligned} \quad (7.8)$$

Equality of the eigenvalues of T_1 and T_2 now follows from the above theorem. ■

We can also express our result above in terms of the following result, which is of general interest.

Definition 7.1.3 Similar Matrices

An $n \times n$ matrix T_2 is said to be **similar** to an $n \times n$ matrix T_1 if there exists a non-singular matrix C such that (7.7) holds. T_1 and T_2 are then called **similar matrices**, and the transformation given by (7.7) is sometimes called a **similarity transformation**.

In terms of the concept of similar matrices, our proof shows the following:

1. Two matrices representing the same linear operator T on a finite-dimensional normed space X relative to any two bases for X are similar.
2. Similar matrices have the same eigenvalues.

Furthermore, Theorems 7.1.1 and 7.1.2 imply the following.

Theorem 7.1.3 Eigenvalues

A linear operator on a finite-dimensional complex normed space $X \neq \{0\}$ has at least one eigenvalue.

Furthermore, (7.8) with $\lambda = 0$ gives $\det(T_2) = \det(T_1)$. Hence, the value of the determinant represents an intrinsic property of the operator T , so that we can speak unambiguously of the quantity $\det(T)$.

Example 7.1.2

1. (*Hermitian Matrix*) Show that the eigenvalues of a Hermitian matrix are real.
2. (*Skew-Hermitian Matrix*) Show that the eigenvalues of a skew-Hermitian matrix are purely imaginary or zero.
3. (*Unitary Matrix*) Show that the eigenvalues of a unitary matrix have modulus 1.
4. Let X be a finite-dimensional inner product space and $T : X \rightarrow X$ a linear operator. If T is self-adjoint, show that its spectrum is real. If T is unitary, show that its eigenvalues have modulus 1.
5. (*Trace*) Let $\lambda_1, \dots, \lambda_n$ be the n eigenvalues of an n -rowed square matrix $A = (\alpha_{jk})$, where some or all of the λ_j s may be equal. Show that the product of the eigenvalues is equal to $\det(A)$ and that their sum is equal to the *trace* of A , that is, to the sum of the elements of the *principal diagonal*:

$$\text{trace}(A) = \alpha_{11} + \alpha_{22} + \dots + \alpha_{nn}.$$

6. (*Inverse*) Show that the inverse A^{-1} of a square matrix A exists if and only if all the eigenvalues $\lambda_1, \dots, \lambda_n$ of A are different from zero. If A^{-1} exists, show that it has eigenvalues $\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$.
7. (*Multiplicity*) Find the eigenvalues and their multiplicities of the matrix corresponding to the following transformation:

$$\eta_j = \xi_j + \xi_{j+1}, \quad (j = 1, 2, \dots, n-1), \quad \eta_n = \xi_n.$$

Comment on the result. Also, show that the geometric multiplicity of an eigenvalue cannot exceed the algebraic multiplicity.

SOLUTION: To be completed.

7.2 General Normed Spaces

Now we consider normed spaces of any dimension, i.e., infinite dimensions, and we shall see that in infinite-dimensional spaces, spectral theory becomes more complicated.

Definition 7.2.1 Resolvent Operator

Let $X \neq \{0\}$ be a complex normed space and $T : X \rightarrow X$ a linear operator with domain $\mathcal{D}(T) \subset X$. With T we associate the operator

$$T_\lambda = T - \lambda I, \quad (7.9)$$

where λ is a complex number and I is the identity operator on $\mathcal{D}(T)$. If T_λ has an inverse, we denote it $R_\lambda(T)$, i.e.,

$$R_\lambda(T) = T_\lambda^{-1} = (T - \lambda I)^{-1} \quad (7.10)$$

and call it the **resolvent operator** of T , or simply the **resolvent** of T . Instead of $R_\lambda(T)$ we also write R_λ if it is clear to what operator T we refer in a specific discussion.

REMARK: The name “resolvent” is appropriate since $R_\lambda(T)$ helps to solve the equation $T_\lambda(x) = y$. Thus, $x = T_\lambda^{-1}(y) = R_\lambda(T)(y)$, provided $R_\lambda(T)$ exists.

Note that $R_\lambda(T)$ is a linear operator by Theorem 5.4.2.

Definition 7.2.2 Regular Value, Resolvent Set, Spectrum

Let $X \neq \{0\}$ be a complex normed space and $T : X \rightarrow X$ with domain $\mathcal{D}(T) \subset X$. A **regular value** λ of T is a complex number such that

1. $R_\lambda(T)$ exists;
2. $R_\lambda(T)$ is bounded;
3. $R_\lambda(T)$ is defined on a set that is dense in X .

The **resolvent set**, denoted $\rho(T)$, of T is the set of all regular values λ of T . Its complement $\sigma(T) := \mathbb{C} - \rho(T)$ in the complex plane $\mathbb{C}$ is called the **spectrum** of T , and a $\lambda \in \sigma(T)$ is called a **spectral value** of T . Furthermore, the spectrum of $\sigma(T)$ is partitioned into three disjoint sets as follows:

- The **point spectrum**, or **discrete spectrum**, denoted $\sigma_p(T)$, is the set such that $R_\lambda(T)$ does *not* exist. A $\lambda \in \sigma_p(T)$ is called an *eigenvalue* of T .
- The **continuous spectrum**, denoted $\sigma_c(T)$, is the set such that $R_\lambda(T)$ exists and satisfies condition 3 above but not condition 2, i.e., such that $R_\lambda(T)$ is unbounded.
- The **residual spectrum**, denoted $\sigma_r(T)$, is the set such that $R_\lambda(T)$ exists (and may be bounded or not) but does *not* satisfy condition 3, that is, the domain of $R_\lambda(T)$ is not dense in X .

REMARK: To avoid trivial misunderstandings, let us say that some of the sets in this definition may be empty. This is an existence problem that we shall have to discuss. For instance, $\sigma_c(T) = \sigma_r(T) = \emptyset$ in the finite-dimensional case, as we have seen. In other words, the spectrum of a linear operator on a finite-dimensional space is a *purely point spectrum*. This means that every spectral value is an eigenvalue.

We first note that the four sets in the table are disjoint and their union is the whole complex plane:

$$\mathbb{C} = \rho(T) \cup \sigma(T) = \rho(T) \cup \sigma_p(T) \cup \sigma_c(T) \cup \sigma_r(T).$$

Furthermore, as mentioned earlier, if the resolvent $R_\lambda(T)$ exists, it is linear by Theorem 5.4.2. That theorem also shows that $R_\lambda(T) : \mathcal{R}(T_\lambda) \rightarrow \mathcal{D}(T_\lambda)$ exists if and only if $T_\lambda(x) = 0$ implies $x = 0$, that is, the null space of T_λ is $\{0\}$.

Definition 7.2.3 Eigenvector, Eigenspace

If $T_\lambda(x) = (T - \lambda I)(x) = 0$ for some $x \neq 0$, then $\lambda \in \sigma_p(T)$ by definition, that is, λ is an eigenvalue of T . The vector x is called an **eigenvector** of T (or **eigenfunction** of T if X is a function space) corresponding to the eigenvalue λ . The subspace of $\mathcal{D}(T)$ consisting of 0 and all eigenvectors of T corresponding to an eigenvalue λ of T is called the **eigenspace** of T corresponding to that eigenvalue λ .

Example 7.2.1 We consider some basic examples of operators and their spectra.

1. As we have already seen, if L is an $N \times N$ matrix operator acting on $\mathbb{R}^N$, then L has only a point spectrum consisting of no more than N eigenvalues. All other points of the complex plane are regular points—on other words, at these other points, $(\lambda I - L)^{-1}$ exists, and you can solve the equation $x = R_\lambda(f) = (\lambda I - L)^{-1}(f)$.
2. The differentiation operator $L = \frac{d}{dt}$ acting of $C^1(a, b) \subset C[0, 1]$ has only a point spectrum since any point $\lambda \in \mathbb{C}$, the equation

$$L(x) = \lambda x, \quad \text{or} \quad \frac{dx}{dt} = \lambda x,$$

as a solution $x(t) = e^{\lambda t}$.

3. The situation is different for the differentiation operator $L = \frac{d}{dt}$ acting on the linear subspace $X \subset L_2(-\infty, \infty)$ of functions x for which $\frac{dx}{dt}$ is in $L_2(-\infty, \infty)$. Here, the functions $e^{\lambda t}$ do not belong to $L_2(-\infty, \infty)$. As a result, $\operatorname{Re}(\lambda) \neq 0$ on the resolvent set, and $\operatorname{Re}(\lambda) = 0$ is the continuous spectrum of L . (See Naylor and Sell pp. 423-426.)
4. We will see later that if L is a compact operator on a Hilbert space H , then L has only a point spectrum.
5. The so-called “co-ordinate operator” Q on $C[a, b]$ defined by

$$Q(u)(t) = tu(t)$$

has not eigenvalues. ($u(t) = 0$ is not eligible by definition.) To investigate further, consider the equation

$$(\lambda I - Q)(u)(t) = \lambda u(t) - Q(u)(t) = f(t),$$

i.e.,

$$(\lambda - t)u(t) = f(t).$$

If $\lambda \notin [a, b]$, then this equation has the unique solution

$$u(t) = \frac{f(t)}{\lambda - t} = (\lambda - Q)^{-1}(f)(t),$$

which implies that all such λ belong to the resolvent set.

If $\lambda \in [a, b]$, then the inverse function $(\lambda - Q)^{-1}$ is defined for functions f such that $f(\lambda) = 0$. For this set of functions, the domain of $(\lambda - Q)^{-1}$ is not dense in $C[a, b]$, which means that points $\lambda \in [a, b]$ belong to the *residual spectrum* of Q .

6. For the co-ordinate operator Q on $L_2[a, b]$, the domain of $(\lambda I - Q)^{-1}$ is dense in $L_2[a, b]$, but $(\lambda I - Q)^{-1}$ is unbounded (check this!). This implies that $[a, b]$ is the *continuous spectrum* of Q . This is important in quantum mechanics.

Proposition 7.2.1 Invariant Subspace

An eigenspace of a linear operator $T : X \rightarrow X$ on a normed linear space X is invariant under T .

PROOF: To be completed. ■

If X is infinite, then T can have spectral value that are not eigenvalues.

Example 7.2.2 Operator with a Spectral Value that is not an Eigenvalue

On the Hilbert space $X = \ell_2$, we define a linear operator $T : \ell_2 \rightarrow \ell_2$ by

$$T(\xi_1, \xi_2, \dots) = (0, \xi_1, \xi_2, \dots), \quad (7.11)$$

called the *right-shift operator*. T is bounded, and has norm one (i.e., $\|T\| = 1$) because

$$\|T(x)\|^2 = \sum_{j=1}^{\infty} |\xi_j|^2 = \|x\|^2.$$

Now, the operator $R_0(T) = T^{-1} : T(X) \rightarrow X$ exists; in fact, it is the *left-shift operator* given by

$$R_0(T)(\xi_1, \xi_2, \dots) = (\xi_2, \xi_3, \dots).$$

But $R_0(T)$ does not satisfy condition 3 in Definition 7.2.2 because (7.11) shows that $T(X)$ is not dense in X . Indeed, $T(X)$ is the subspace Y consisting of all $y = (\eta_j)$ with $\eta_1 = 0$. Hence, by definition, $\lambda = 0$ is a spectral value of T . Furthermore, $\lambda = 0$ is not an eigenvalue. We can see this directly from (7.11) since $T(x) = 0$ implies $x = 0$ and the zero vector is not an eigenvector (by definition).

Now, the bounded inverse theorem contributes the following. If $T : X \rightarrow X$ is bounded and linear and X is complete, and if for some λ the resolvent $R_\lambda(T)$ exists and is defined on the whole space X , then for that λ the resolvent is bounded.

Lemma 7.2.1 Domain of R_λ

Let X be a complex Banach space, $T : X \rightarrow X$ a linear operator, and $\lambda \in \rho(T)$. Assume that either T is closed or that T is bounded. Then $R_\lambda(T)$ is defined on the whole space X and is bounded.

PROOF:

1. Since T is closed, so is T_λ by Theorem 6.8.3. Hence, R_λ is closed. Also, R_λ is bounded by the second condition in 7.2.2. Hence, its domain $\mathcal{D}(R_\lambda)$ is closed by Part 2 of Lemma 6.8.1 applied to R_λ , so that condition 3 of Definition 7.2.2 implies $\mathcal{D}(R_\lambda) = \overline{\mathcal{D}(R_\lambda)} = X$.

2. Since $\mathcal{D}(T) = X$ is closed, T is closed by Part 2 of Lemma 6.8.1 and the statement follows from the first part of this proof. ■

Example 7.2.3 For the identity operator I on a normed space X , find the eigenvalues and eigenspaces as well as $\sigma(I)$ and $R_\lambda(I)$.

SOLUTION: To be completed.

7.3 Bounded Linear Operators on Normed Spaces

The properties of the spectrum of a given linear operator will depend on the kind of space on which the operator is defined and on the kind of operator we consider.

Theorem 7.3.1 Inverse

Let $T \in B(X)$, where X is a Banach space. If $\|T\| < 1$, then $(I - T)^{-1}$ exists as a bounded linear operator on the whole space X and

$$(I - T)^{-1} = \sum_{j=0}^{\infty} T^j = I + T + T^2 + \cdots, \quad (7.12)$$

where the series on the right-hand side is convergent in the norm on $B(X)$ (i.e., convergent in the operator norm).

PROOF: This is just Theorem 5.7.6. ■

Theorem 7.3.2 Spectrum Closed

The resolvent set $\rho(T)$ of a bounded linear operator T on a complex Banach space X is open; hence, the spectrum $\sigma(T)$ is closed.

PROOF: If $\rho(T) = \emptyset$, it is open. (Actually, $\rho(T) \neq \emptyset$, as we'll see in Theorem 7.3.4.) Let $\rho(T) \neq \emptyset$. For a fixed $\lambda_0 \in \rho(T)$ and any $\lambda \in \mathbb{C}$, we have

$$T - \lambda I = T - \lambda_0 I - (\lambda - \lambda_0)I = (T - \lambda_0 I)[I - (\lambda - \lambda_0)(T - \lambda_0 I)^{-1}].$$

Denoting the operator in the square brackets by V , we can write this in the form

$$T - \lambda I = T_{\lambda_0} V, \quad \text{where } V := I - (\lambda - \lambda_0)R_{\lambda_0}. \quad (7.13)$$

Since $\lambda_0 \in \rho(T)$ and T is bounded, Lemma 7.2.1 implies that $R_{\lambda_0} = T_{\lambda_0}^{-1} \in B(X)$. Furthermore, Theorem 7.3.1 shows that V has an inverse,

$$V^{-1} = \sum_{j=0}^{\infty} [(\lambda - \lambda_0)R_{\lambda_0}]^j = \sum_{j=0}^{\infty} (\lambda - \lambda_0)^j R_{\lambda_0}^j \quad (7.14)$$

in $B(X)$ for all λ such that $\|(\lambda - \lambda_0)R_{\lambda_0}\| < 1$, that is,

$$|\lambda - \lambda_0| < \frac{1}{\|R_{\lambda_0}\|}. \quad (7.15)$$

Since $T_{\lambda_0}^{-1} = R_{\lambda_0} \in B(X)$, we see from this and (7.13) that for every λ satisfying (7.14) the operator T_λ has an inverse,

$$R_\lambda = T_\lambda^{-1} = (T_{\lambda_0} V)^{-1} = V^{-1} R_{\lambda_0}. \quad (7.16)$$

Hence, (7.15) represents a neighbourhood of λ_0 consisting of regular values λ of T . Since $\lambda_0 \in \rho(T)$ was arbitrary, $\rho(T)$ is open, so that its complement $\sigma(T) = \mathbb{C} - \rho(T)$ is closed. ■

It is worth noting that in this proof we have also obtained a basic representation of the resolvent by a power series in powers of λ . In fact, from (7.14) and (7.15) and (7.16), we immediately have the following.

Theorem 7.3.3 Resolvent Representation

Let X be a Banach space and T a bounded linear operator on X . For every $\lambda_0 \in \rho(T)$, the resolvent $R_\lambda(T)$ has the representation

$$R_\lambda = \sum_{j=0}^{\infty} (\lambda - \lambda_0)^j R_{\lambda_0}^{j+1}, \quad (7.17)$$

the series being absolutely convergent for every λ in the open disk given by

$$|\lambda - \lambda_0| < \frac{1}{\|R_{\lambda_0}\|}$$

in the complex plane. This disk is a subset of $\rho(T)$.

Theorem 7.3.4 Spectrum

The spectrum $\sigma(T)$ of a bounded linear operator $T : X \rightarrow X$ on a complex Banach space X is compact and lies in the disk given by

$$|\lambda| \leq \|T\|. \quad (7.18)$$

Hence, the resolvent set $\rho(T)$ of T is not empty.

PROOF: Let $\lambda \neq 0$ and $\kappa = \frac{1}{\lambda}$. From Theorem 7.3.1, we obtain the representation

$$R_\lambda = (T - \lambda I)^{-1} = -\frac{1}{\lambda}(I - \kappa T)^{-1} = -\frac{1}{\lambda} \sum_{j=0}^{\infty} (\kappa T)^j = -\frac{1}{\lambda} \sum_{j=0}^{\infty} \left(\frac{1}{\lambda} T\right)^j, \quad (7.19)$$

where, by Theorem 7.3.1, the series converges for all λ such that

$$\left\| \frac{1}{\lambda} T \right\| = \frac{\|T\|}{|\lambda|} < 1, \quad \text{that is, } |\lambda| \geq \|T\|.$$

The same theorem also shows that any such λ is in $\rho(T)$. Hence, the spectrum $\sigma(T) = \mathbb{C} - \rho(T)$ must lie in the disk (7.18), so that $\sigma(T)$ is bounded. Furthermore, $\sigma(T)$ is closed by Theorem 7.3.2. Hence, $\sigma(T)$ is compact. ■

Since from the theorem just proved we know that for a bounded linear operator T on a complex Banach space the spectrum is bounded, it seems natural to ask for the smallest disk about the origin that contains the whole spectrum.

Definition 7.3.1 Spectral Radius

The **spectral radius**, denoted $r_\sigma(T)$, of an operator $T \in B(X)$ on a complex Banach space X , is the radius

$$r_\sigma(T) = \sup_{\lambda \in \sigma(T)} |\lambda|$$

of the smallest closed disk centred at the origin of the complex λ -plane and containing $\sigma(T)$.

From (??), it is obvious that for the spectral radius of a bounded linear operator T on a complex Banach space, we have

$$r_\sigma(T) \leq \|T\|, \quad (7.20)$$

and we will see later that

$$r_\sigma(T) = \lim_{n \rightarrow \infty} (\|T^n\|)^{1/n}. \quad (7.21)$$

Example 7.3.1 Let $T \in B(X)$, where X is a Banach space. Show that $\|R_\lambda(T)\| \rightarrow 0$ as $\lambda \rightarrow \infty$.

SOLUTION: To be completed.

Our next result will be the important spectral mapping theorem.

If λ is an eigenvalue of a square matrix A , then $Ax = \lambda x$ for some $x \neq 0$. Application of A gives

$$A^2x = A\lambda x = \lambda Ax = \lambda^2x.$$

Continuing in this way, we have for every positive integer m ,

$$A^m x = \lambda^m x,$$

that is, if λ is an eigenvalue of A , then λ^m is an eigenvalue of A^m . More generally, then,

$$p(\lambda) := \alpha_n \lambda^n + \alpha_{n-1} \lambda^{n-1} + \cdots + \alpha_0$$

is an eigenvalue of the matrix

$$p(A) := \alpha_n A^n + \alpha_{n-1} A^{n-1} + \cdots + \alpha_0 I.$$

This property turns out to hold in Banach spaces as well. Before stating the theorem, we define the set

$$p(\sigma(T)) := \{\mu \in \mathbb{C} \mid \mu = p(\lambda), \lambda \in \sigma(T)\}, \quad (7.22)$$

that is, $p(\sigma(T))$ is the set of all complex numbers μ such that $\mu = p(\lambda)$ for some $\lambda \in \sigma(T)$. We shall also use $p(\rho(T))$ in a similar sense.

Theorem 7.3.5 Spectral Mapping Theorem

Let X be a complex Banach space, $T \in B(X)$, and

$$p(\lambda) = \alpha_n \lambda^n + \alpha_{n-1} \lambda^{n-1} + \cdots + \alpha_0, \quad \alpha_n \neq 0.$$

Then,

$$\sigma(p(T)) = p(\sigma(T)), \quad (7.23)$$

that is, the spectrum $\sigma(p(T))$ of the operator

$$p(T) = \alpha_n T^n + \alpha_{n-1} T^{n-1} + \cdots + \alpha_0 I$$

consists precisely of all those values that the polynomial p assumes on the spectrum $\sigma(T)$ of T .

Theorem 7.3.6 Linear Independence

The eigenvectors $x_1, \dots, x_n$ corresponding to different eigenvalues $\lambda_1, \dots, \lambda_n$ of a linear operator T on a vector space X constitute a linearly independent set.

PROOF: We assume for a contradiction that $\{x_1, \dots, x_n\}$ is linearly dependent. Let x_m be the first of the vectors that is a linear combination of its predecessors, say,

$$x_m = \alpha_1 x_1 + \cdots + \alpha_{m-1} x_{m-1}. \quad (7.24)$$

Then, $\{x_1, \dots, x_{m-1}\}$ is linearly independent. Applying $T - \lambda_m I$ on both sides of (7.24), we obtain

$$(T - \lambda_m I)x_m = \sum_{j=1}^{m-1} \alpha_j (T - \lambda_m I)x_j = \sum_{j=1}^{m-1} \alpha_j (\lambda_j - \lambda_m)x_j.$$

Since x_m is an eigenvector corresponding to λ_m , the left-hand side is zero. Since the vectors on the right-hand side form a linearly independent set, we must have

$$\alpha_j (\lambda_j - \lambda_m) = 0, \quad \text{hence} \quad \alpha_j = 0 \quad \forall 1 \leq j \leq m-1$$

since $\lambda_j - \lambda_m \neq 0$. But then $x_m = 0$ by (7.24). This contradicts the fact that $x_m \neq 0$ since x_m is an eigenvector. So the proof is complete. ■

Example 7.3.2 Idempotent Operator

Let T be a bounded linear operator on a Banach space. T is called **idempotent** if $T^2 = T$. We show that if $T \neq 0$ and $T \neq I$, then its spectrum is equal to $\{0, 1\}$.

(To be completed.)

Theorem 7.3.7 Resolvent

If $T \in B(X)$, where X is a complex Banach space, and $\lambda \in \rho(T)$, then

$$\|R_\lambda(T)\| \geq \frac{1}{\delta(\lambda)}, \quad \text{where} \quad \delta(\lambda) = \inf_{s \in \sigma(T)} |\lambda - s| \quad (7.25)$$

is the distance from λ to the spectrum $\sigma(T)$. Hence,

$$\|R_\lambda(T)\| \rightarrow \infty \quad \text{as} \quad \delta(\lambda) \rightarrow 0. \quad (7.26)$$

It is of great theoretical and practical importance that the spectrum of a bounded linear operator T on a complex Banach space can never be the empty set.

Theorem 7.3.8 Spectrum Non-Empty

If $X \neq \{0\}$ is a complex Banach space and $T \in B(X)$, then $\sigma(T) \neq \emptyset$.

Example 7.3.3 Nilpotent Operator

A linear operator T is called **nilpotent** if there is a positive integer m such that $T^m = 0$. Let us determine the spectrum of a nilpotent operator $T : X \rightarrow X$ on a complex Banach space $X \neq \{0\}$.

(To be completed.)

7.4 Compact Linear Operators on Normed Spaces

We now consider the spectral properties of a compact linear operator $T : X \rightarrow X$ on a normed space X . For this, we shall again use the operator

$$T_\lambda = T - \lambda I, \quad \lambda \in \mathbb{C}, \quad (7.27)$$

and the basic concepts of spectral theory that we have seen already.

The spectral theory of compact linear operators is a relatively simple generalisation of the eigenvalue theory of finite matrices and resembles that finite-dimensional case in many ways.

Theorem 7.4.1 Eigenvalues of a Compact Operator

The set of eigenvalues of a compact linear operator $T : X \rightarrow X$ on a normed space X is countable (perhaps finite or even empty), and the only possible point of accumulation is $\lambda = 0$.

REMARK: This theorem shows us that if a compact linear operator on a normed space has infinitely many eigenvalues, we can arrange these eigenvalues in a sequence converging to zero.

PROOF: It suffices to show that for every real $k > 0$ the set of all $\lambda \in \sigma_p(T)$ such that $|\lambda| \geq k$ is finite.

Suppose the contrary for some $k_0 > 0$. Then there is a sequence (λ_n) of infinitely-many distinct eigenvalues such that $|\lambda_n| \geq k_0$. Also, $T(x_n) = \lambda_n x_n$ for some $x_n \neq 0$. The set of all the x_n s is linearly independent by Theorem 7.3.6. Let $M_n = \text{span}\{x_1, \dots, x_n\}$. Then, every $x \in M_n$ has a unique representation

$$x = \alpha_1 x_1 + \dots + \alpha_n x_n. \quad (7.28)$$

We apply $T - \lambda_n I$ and use $T(x_j) = \lambda_j x_j$:

$$(T - \lambda_n I)(x) = \alpha_1(\lambda_1 - \lambda_n)x_1 + \dots + \alpha_{n-1}(\lambda_{n-1} - \lambda_n)x_{n-1}.$$

We see that x_n no longer occurs on the right. Hence,

$$(T - \lambda_n I)(x) \in M_{n-1} \text{ for all } x \in M_n. \quad (7.29)$$

The M_n s are closed. By Riesz's lemma, there is a sequence (y_n) such that

$$y_n \in M_n, \quad \|y_n\| = 1, \quad \|y_n - x\| \geq \frac{1}{2} \text{ for all } x \in M_{n-1}.$$

We show that

$$\|T(y_n) - T(y_m)\| \geq \frac{1}{2}k_0 \text{ for all } n > m, \quad (7.30)$$

so that $(T(y_n))$ has no convergent subsequence because $k_0 > 0$. This contradicts the compactness of T since (y_n) is bounded.

By adding and subtracting a term, we can write

$$T(y_n) - T(y_m) = \lambda_n y_n - \tilde{x} \quad \text{where} \quad \tilde{x} = \lambda_n y_n - T(y_n) + T(y_m). \quad (7.31)$$

Let $m < n$. We show that $\tilde{x} \in M_{n-1}$. Since $m \leq n-1$, we see that $y_m \in M_m \subset M_{n-1} = \text{span}\{x_1, \dots, x_{n-1}\}$. Hence, $T(y_m) \in M_{n-1}$ since $T(x_j) = \lambda_j x_j$. By (7.29),

$$\lambda_n y_n - T(y_n) = -(T - \lambda_n I)(y_n) \in M_{n-1}.$$

Together, $\tilde{x} \in M_{n-1}$. Thus also $x = \lambda_n^{-1} \tilde{x} \in M_{n-1}$, so that

$$\|\lambda_n y_n - \tilde{x}\| = |\lambda_n| \|y_n - x\| \geq \frac{1}{2} |\lambda_n| \geq \frac{1}{2} k_0 \quad (7.32)$$

because $|\lambda_n| \geq k_0$. From this and (7.31), we have (7.30). Hence, the assumption that there are infinitely many eigenvalues satisfying $|\lambda_n| \geq k_0$ for some $k_0 > 0$ must be false and the proof is complete. ■

We said at the beginning of this section that the spectral theory of compact linear operators is almost as simple as that of linear operators on a finite-dimensional space (which is essentially eigenvalue

theory of finite matrices). An important property supporting that claim is as follows. For every non-zero eigenvalue that a compact linear operator may (or may not) have, the eigenspace is finite-dimensional. This is implied by the following theorem.

Theorem 7.4.2 Null Space of Compact Operators

Let $T : X \rightarrow X$ be a compact linear operator on a normed space X . Then, for every $\lambda \neq 0$, the null space $\mathcal{N}(T_\lambda)$ of $T_\lambda = T - \lambda I$ is finite-dimensional.

PROOF: It is enough to show that the closed unit ball M in $\mathcal{N}(T_\lambda)$ is compact.

Let (x_n) be in M . Then, (x_n) is bounded ($\|x_n\| \leq 1$), and $(T(x_n))$ has a convergent subsequence $(T(x_{n_k}))$ by definition of a compact linear operator. Now, $x_n \in M \subset \mathcal{N}(T_\lambda)$ implies that $T_\lambda(x_n) = T(x_n) - \lambda x_n = 0$, so that $x_n = \lambda^{-1}T(x_n)$ because $\lambda \neq 0$. Consequently, $(x_{n_k}) = (\lambda^{-1}T(x_{n_k}))$ also converges. The limit is in M since M is closed. Hence M is compact because (x_n) was arbitrary in M . This proves that $\dim(\mathcal{N}(T_\lambda)) < \infty$ by Theorem 5.2.11. ■

We shall now consider the ranges of $T_\lambda, T_\lambda^2, \dots$ for a compact linear operator T and any $\lambda \neq 0$. In this connection, we should first remember that for a bounded linear operator, the null space is always closed but the range need not be closed. However, if T is compact, then T_λ has a closed range for every $\lambda \neq 0$, and the same holds for T_λ^2, T_λ^3 , and so on.

Theorem 7.4.3 Range of a Compact Operator

Let $T : X \rightarrow X$ be a compact linear operator on a normed space X . Then, for every $\lambda \neq 0$ the range of $T_\lambda = T - \lambda I$ is closed.

Example 7.4.1 Let H be a Hilbert space, $T : H \rightarrow H$ a bounded linear operator, and T^* the Hilbert-adjoint operator of T .

1. Show that T is compact if and only if T^*T is compact.
2. If T is compact, show that T^* is compact.

SOLUTION: To be completed.

Theorem 7.4.4 Eigenvalues of Compact Operators

Let $T : X \rightarrow X$ be a compact linear operator on a Banach space X . Then, every spectral value $\lambda \neq 0$ of T (if it exists¹) is an eigenvalue of T .

¹A self-adjoint compact linear operator on a complex non-empty Hilbert space always has at least one eigenvalue, as we'll see shortly.

REMARK: The value $\lambda = 0$ was excluded in the above theorem as well as in many of the theorems encountered above, so it is natural to ask what we can say about $\lambda = 0$ in the case of a compact operator $T : X \rightarrow X$ on a complex normed space X . If X is finite-dimensional, then T has representations by matrices and it is clear that 0 may or may not belong to $\sigma(T) = \sigma_p(T)$, i.e., if $\dim(X) < \infty$, we may have $0 \notin \sigma(T)$. Then, $0 \in \rho(T)$. However, if $\dim(X) = \infty$, then we must have $0 \in \sigma(T)$. And all three cases,

$$0 \in \sigma_p(T), \quad 0 \in \sigma_c(T), \quad 0 \in \sigma_r(T)$$

are possible.

7.4.1 Operator Equations Involving Compact Linear Operators

Let us briefly consider a compact linear operator $T : X \rightarrow X$ on a normed space X , the adjoint operator $T^* : X' \rightarrow X'$, and the following equations:

$$T(x) - \lambda x = y \quad (y \in X \text{ given}, \lambda \neq 0) \quad (7.33)$$

$$T(x) - \lambda x = 0, \quad (\lambda \neq 0) \quad (7.34)$$

$$T^*(f) - \lambda f = g \quad (g \in X' \text{ given}, \lambda \neq 0) \quad (7.35)$$

$$T^*(f) - \lambda f = 0 \quad (\lambda \neq 0) \quad (7.36)$$

Here, $\lambda \in \mathbb{C}$ is arbitrary and fixed, not zero, and we shall state the existence of solutions x and f , respectively.

We have the following results:

1. (7.33) is **normally solvable**, i.e., (7.33) has a solution x if and only if $f(y) = 0$ for all solutions f of (7.36). Hence, if $f = 0$ is the only solution of (7.36), then for every y the equation (7.33) is solvable.
2. (7.35) has a solution if and only if $g(x) = 0$ for all solutions x of (7.34). Hence, if $x = 0$ is the only solution of (7.34), then for every g the equation (7.35) is solvable.
3. (7.33) has a solution x for every $y \in X$ if and only if $x = 0$ is the only solution of (7.34).
4. (7.35) has a solution f for every $g \in X'$ if and only if $f = 0$ is the only solution of (7.36).
5. (7.34) and (7.36) have the same number of linearly independent solutions.

7.5 Bounded Self-Adjoint Linear Operators on Hilbert Spaces

We now consider bounded self-adjoint linear operators that are defined on a complex Hilbert space H and map H into itself.

A bounded self-adjoint linear operator T may not have eigenvalues, but if T has eigenvalues, the following basic facts can readily be established.

Theorem 7.5.1 Eigenvalues, Eigenvectors

Let $T : H \rightarrow H$ be a bounded self-adjoint linear operator on a complex Hilbert space H . Then,

1. All the eigenvalues of T (if they exist) are real.
2. Eigenvectors corresponding to (numerically) different eigenvalues of T are orthogonal.

PROOF:

1. Let λ be any eigenvalue of T and x a corresponding eigenvector. Then, $x \neq 0$ and $T(x) = \lambda x$. Using the self-adjointness of T , we obtain

$$\lambda \langle x, x \rangle = \langle \lambda x, x \rangle = \langle T(x), x \rangle = \langle x, T(x) \rangle = \langle x, \lambda x \rangle = \bar{\lambda} \langle x, x \rangle.$$

Here, $\langle x, x \rangle = \|x\|^2 \neq 0$ since $x \neq 0$, and division by $\langle x, x \rangle$ gives $\lambda = \bar{\lambda}$, so λ is real.

2. Let λ and μ be eigenvalues of T , and let x and y be corresponding eigenvectors. Then, $T(x) = \lambda x$ and $T(y) = \mu y$. Since T is self-adjoint and μ is real,

$$\lambda \langle x, y \rangle = \langle \lambda x, y \rangle = \langle T(x), y \rangle = \langle x, T(y) \rangle = \langle x, \mu y \rangle = \mu \langle x, y \rangle.$$

Since $\lambda \neq \mu$, we must have $\langle x, y \rangle = 0$, which means orthogonality of x and y . ■

Theorem 7.5.2 Resolvent Set

Let $T : H \rightarrow H$ be a bounded self-adjoint linear operator on a complex Hilbert space H . Then, a number λ belongs to the resolvent set $\rho(T)$ of T if and only if there exists a $c > 0$ such that for every $x \in H$,

$$\|T_\lambda(x)\| \geq c \|x\|, \tag{7.37}$$

where recall $T_\lambda = T - \lambda I$.

From this theorem we immediately obtain the following.

Theorem 7.5.3 Spectrum of Bounded Self-Adjoint Operator

The spectrum $\sigma(T)$ of a bounded self-adjoint linear operator $T : H \rightarrow H$ on a complex Hilbert space H is real.

PROOF: Using the above proof, we show that a $\lambda = \alpha + i\beta$, for $\alpha, \beta \in \mathbb{R}$, $\beta \neq 0$, must belong to $\rho(T)$, so that $\sigma(T) \subset \mathbb{R}$.

For every $x \neq 0$ in H , we have

$$\langle T_\lambda(x), x \rangle = \langle T(x), x \rangle - \lambda \langle x, x \rangle,$$

and since $\langle x, x \rangle$ and $\langle T(x), x \rangle$ are real,

$$\overline{\langle T_\lambda(x), x \rangle} = \langle T(x), x \rangle - \bar{\lambda} \langle x, x \rangle.$$

Here, $\bar{\lambda} = \alpha - i\beta$. By subtraction,

$$\overline{\langle T_\lambda(x), x \rangle} - \langle T_\lambda(x), x \rangle = (\lambda - \bar{\lambda}) \langle x, x \rangle = 2i\beta \|x\|^2.$$

the left-hand side is $-2i\text{Im}(\langle T_\lambda(x), x \rangle)$. The latter cannot exceed the absolute value, so that, dividing by two, taking absolute values, and applying the Cauchy-Schwarz inequality, we obtain

$$|\beta| \|x\|^2 = |\text{Im}(\langle T_\lambda(x), x \rangle)| \leq |\langle T_\lambda(x), x \rangle| \leq \|T_\lambda(x)\| \|x\|.$$

Division by $\|x\| \neq 0$ gives $|\beta| \|x\| \leq \|T_\lambda(x)\|$. If $\beta \neq 0$, then $\lambda \in \rho(T)$ by Theorem 7.5.2. Hence, for $\lambda \in \sigma(T)$, we must have $\beta = 0$, that is, λ is real. ■

Example 7.5.1 Show that the operator $T : L_2[0, 1] \rightarrow L_2[0, 1]$ defined by

$$y(t) = T(x)(t) = tx(t)$$

is a bounded self-adjoint linear operator without eigenvalues.

SOLUTION: To be completed.

Theorem 7.5.4 Spectrum of Bounded Self-Adjoint Operators

The spectrum $\sigma(T)$ of a bounded self-adjoint linear operator $T : H \rightarrow H$ on a complex Hilbert space H lies in the closed interval $[m, M] \subset \mathbb{R}$, where

$$m = \inf_{\|x\|=1} \langle T(x), x \rangle, \quad \text{and} \quad M = \sup_{\|x\|=1} \langle T(x), x \rangle. \quad (7.38)$$

PROOF: $\sigma(T)$ lies on the real line, as we have seen in the previous theorem. We now show that any real $\lambda = M + c$ with $c > 0$ belongs to the resolvent set $\rho(T)$. For every $x \neq 0$ and $v = \|x\|^{-1}x$, we have $x = \|x\|v$ and

$$\langle T(x), x \rangle = \|x\|^2 \langle T(v), v \rangle \leq \|x\|^2 \sup_{\|\tilde{v}\|=1} \langle T(\tilde{v}), \tilde{v} \rangle = \langle x, x \rangle M.$$

Hence, $-\langle T(x), x \rangle \geq -\langle x, x \rangle M$, and by the Schwarz inequality, we obtain

$$\|T_\lambda(x)\| \|x\| \geq -\langle T_\lambda(x), x \rangle = -\langle T(x), x \rangle + \lambda \langle x, x \rangle \geq (-M + \lambda) \langle x, x \rangle = c \|x\|^2,$$

where $c = \lambda - M > 0$ by assumption. Division by $\|x\|$ yields the inequality $\|T_\lambda(x)\| \geq c \|x\|$. Hence, $\lambda \in \rho(T)$ by Theorem 7.5.2. For a real $\lambda < m$, the idea of proof is the same. ■

Example 7.5.2 What theorem about the eigenvalues of a Hermitian matrix do we obtain from the theorem above?

SOLUTION: To be completed.

Example 7.5.3 Find m and M (as in the above theorem) if T is the projection operator of a Hilbert space H onto a proper subspace $Y \neq \{0\}$ of H .

SOLUTION: To be completed.

Theorem 7.5.5 Norm

For any bounded self-adjoint linear operator T on a complex Hilbert space H , we have

$$\|T\| = \max\{|m|, |M|\} = \sup_{\|x\|=1} |\langle T(x), x \rangle|. \quad (7.39)$$

PROOF: By the Schwarz inequality,

$$\sup_{\|x\|=1} |\langle T(x), x \rangle| \leq \sup_{\|x\|=1} \|T(x)\| \|x\| = \|T\|,$$

that is, $K \leq \|T\|$, where K denotes the expression on the left. We show that $\|T\| \leq K$. If $T(z) = 0$ for all z of norm one, then $T = 0$ (why? because then $\|T\| = 0$) and we are done. Otherwise, for any z of norm one such that $T(z) \neq 0$, we set $v := \sqrt{\|T(z)\|}z$ and $w := \sqrt{\|T(z)\|}T(z)$. Then, $\|v\|^2 = \|w\|^2 = \|T(z)\|$. We now set $y_1 = v + w$ and $y_2 = v - w$. Then, by straightforward calculation, since a number of terms drop out and T is self-adjoint,

$$\langle T(y_1), y_1 \rangle - \langle T(y_2), y_2 \rangle = 2(\langle T(v), w \rangle + \langle T(w), v \rangle) = 2(\langle T(z), T(z) \rangle + \langle T^2(z), z \rangle) = 4\|T(z)\|^2. \quad (7.40)$$

Now, for every $y \neq 0$ and $x = \frac{y}{\|y\|}$, we have $y = \|y\|x$ and

$$|\langle T(y), y \rangle| = \|y\|^2 |\langle T(x), x \rangle| \leq \|y\|^2 \sup_{\|\tilde{x}\|=1} |\langle T(\tilde{x}), \tilde{x} \rangle| = K \|y\|^2,$$

so that by the triangle inequality and straightforward calculation we obtain

$$\begin{aligned} |\langle T(y_1), y_1 \rangle - \langle T(y_2), y_2 \rangle| &\leq |\langle T(y_1), y_1 \rangle| + |\langle T(y_2), y_2 \rangle| \\ &\leq K(\|y_1\|^2 + \|y_2\|^2) \\ &= 2K(\|v\|^2 + \|w\|^2) \\ &= 4K\|T(z)\|. \end{aligned}$$

From this and (7.40), we see that $4\|T(z)\|^2 \leq 4K\|T(z)\|$. Hence, $\|T(z)\| \leq K$. Taking the supremum over all z of norm one, we obtain $\|T\| \leq K$. Together with $K \leq \|T\|$, we have (7.39). ■

Actually, the bounds for $\sigma(T)$ in Theorem 7.5.4 cannot be tightened.

Theorem 7.5.6

Let H and T be as in Theorem 7.5.4 and $H \neq \{0\}$. Then, m and M defined in (7.38) are spectral values of T .

PROOF: We show that $M \in \sigma(T)$. By the spectral mapping theorem, the spectrum of $T + kI$ (for k a real constant) is obtained from that of T by a translation, and

$$M \in \sigma(T) \Leftrightarrow M + k \in \sigma(T + kI).$$

Hence, we may assume that $0 \leq m \leq M$ without loss of generality. Then, by the previous theorem, we have

$$M = \sup_{\|x\|=1} \langle T(x), x \rangle = \|T\|.$$

By the definition of a supremum, there is a sequence (x_n) such that

$$\|x_n\| = 1, \quad \langle T(x_n), x_n \rangle = M - \delta_n, \quad \delta_n \geq 0, \quad \delta_n \rightarrow 0.$$

Then, $\|T(x_n)\| \leq \|T\| \|x_n\| = \|T\| = M$, and since T is self-adjoint,

$$\begin{aligned} \|T(x_n) - Mx_n\|^2 &= \langle T(x_n) - Mx_n, T(x_n) - Mx_n \rangle \\ &= \|T(x_n)\|^2 - 2M \langle T(x_n), x_n \rangle + M^2 \|x_n\|^2 \\ &\leq M^2 - 2M(M - \delta_n) + M^2 \\ &= 2M\delta_n \rightarrow 0 \text{ as } n \rightarrow \infty. \end{aligned}$$

Hence, there is no positive c such that

$$\|T_M(x_n)\| = \|T(x_n) - Mx_n\| \geq c = c \|x_n\|, \quad \|x_n\| = 1.$$

Theorem 7.5.2 now shows that $\lambda = M$ cannot belong to the resolvent set of T . Hence, $M \in \sigma(T)$. For $\lambda = m$, the proof is similar. ■

Note that the above two theorems imply that for a bounded self-adjoint operator the absolute value of the smallest eigenvalue (if it exists) is precisely the operator norm of the operator. (See Proposition 4.8 in the Course Notes.)

Also observe that we can *define* the operator norm of a bounded self-adjoint operator L on an inner product space as

$$\|L\| = \sup_{\|x\|=1} |\langle L(x), x \rangle|. \quad (7.41)$$

Now, the subdivision of the spectrum of a linear operator into the point spectrum and another part seems natural since that “other part” is absent in finite-dimensional spaces, as is well known from matrix theory. A similar justification can now be given for the subdivision of that “other part” into the continuous and residual spectrum since the latter is absent for the large and important class of self-adjoint linear operators.

Theorem 7.5.7 Residual Spectrum

The residual spectrum $\sigma_r(T)$ of a bounded self-adjoint linear operator $T : H \rightarrow H$ on a complex Hilbert space H is empty.

PROOF: We show that the assumption $\sigma_r(T) \neq \emptyset$ leads to a contradiction. Let $\lambda \in \sigma_r(T)$. By the definition of $\sigma_r(T)$, the inverse of T_λ exists but its domain $\mathcal{D}(T_\lambda^{-1})$ is not dense in H . Hence, by the projection theorem, there is a $y \neq 0$ in H that is orthogonal to $\mathcal{D}(T_\lambda^{-1})$. But $\mathcal{D}(T_\lambda^{-1})$ is the range of T_λ , hence

$$\langle T_\lambda(x), y \rangle = 0 \text{ for all } x \in H.$$

Since λ is real (remember that for a self-adjoint operator $\langle T(x), x \rangle$ is real for all $x \in H$) and T is self-adjoint, we thus obtain $\langle x, T_\lambda(y) \rangle = 0$ for all x . Taking $x = T_\lambda(y)$, we get $\|T_\lambda(y)\|^2 = 0$, so that

$$T_\lambda(y) = T(y) - \lambda y = 0.$$

Since $y \neq 0$, this shows that λ is an eigenvalue of T . But this contradicts the assumption $\lambda \in \sigma_r(T)$. Hence, $\sigma_r(T) \neq \emptyset$ is impossible, so $\sigma_r(T) = \emptyset$ holds. ■

7.5.1 Compact Self-Adjoint Operators; The Spectral Theorem

We now focus specifically on bounded, linear, self-adjoint, compact operators on a Hilbert space.

Theorem 7.5.8 Eigenvalues of Compact Self-Adjoint Operator

A non-zero, linear, compact, self-adjoint operator L on a Hilbert space H has at least one non-zero eigenvalue λ .

PROOF: We will show that there is one non-zero eigenvector $\phi_1 \in H$ and that it satisfies $\|\phi_1\| = 1$ and the corresponding eigenvalue μ_1 satisfies $|\mu_1| = \|L\|$. In other words, ϕ_1 is the eigenvector corresponding to the smallest eigenvalue, whose absolute value is $\|L\|$.

In fact, we can search for such an eigenvector because, by definition of the supremum as present in the alternate definition (??) of the operator norm, there exists a sequence $(v_n) \subset H$ such that $\|v_n\| = 1 \ \forall n$ and $|\langle L(v_n), v_n \rangle| \rightarrow \|L\|$ as $n \rightarrow \infty$. Then, $\langle L(v_n), v_n \rangle$ converges, say to μ_1 , where $|\mu_1| = \|L\|$, i.e., $\mu_1 = \|L\|$ or $\mu_1 = -\|L\|$. Then,

$$0 \leq \|L(v_n) - \mu_1 v_n\|^2 = \|L(v_n)\|^2 - 2\mu_1 \langle L(v_n), v_n \rangle + \mu_1^2 \|v_n\|^2 \leq 2\mu_1^2 - 2\mu_1 \langle L(v_n), v_n \rangle \rightarrow 0,$$

where we used the fact that $\|L(v_n)\|^2 \leq \|L\|^2 = \mu_1^2$. Therefore, $L(v_n) - \mu_1 v_n \rightarrow 0$.

Now, (v_n) is bounded (why?), so by the compactness of L , the sequence $(L(v_n))$ has a convergent subsequence $(L(v_{n_k}))$. Suppose $L \neq 0$, which means that $\mu_1 \neq 0$. Then, (v_{n_k}) converges to an element, say ϕ_1 , i.e., $L(v_{n_k}) \rightarrow L(\phi_1)$ and $L(\phi_1) - \mu_1 \phi_1 = 0$, $\|\phi_1\| = \lim_{k \rightarrow \infty} \|v_{n_k}\| = 1$. Also, $|\langle L(\phi_1), \phi_1 \rangle| = |\langle \mu_1 \phi_1, \phi_1 \rangle| = |\mu_1| \|\phi_1\|^2 = |\mu_1| = \|L\|$. This completes the proof. ■

Continuing from the above proof, let us construct the next eigenfunction, call it ϕ_2 . Consider $H_1 = \{x \in H \mid \langle x, \phi_1 \rangle = 0\} = \text{span}\{\phi_1\}^\perp$. For $x \in H_1$, $\langle L(x), \phi_1 \rangle = \langle x, L(\phi_1) \rangle = \mu_1 \langle x, \phi_1 \rangle = 0$. This shows

that $L : H_1 \rightarrow H_1$ and that H_1 is a closed linear subspace of H . Because it is one-dimensional, we have that L is compact and self-adjoint on H_1 . This means that there exists $\phi_2 \in H_1$ such that $\|\phi_2\| = 1$ and $L(\phi_2) = \mu_2 \phi_2$. Furthermore,

$$\|L(v)\| \leq |\mu_2| \|v\| \quad \forall v \in H_1 \Rightarrow |\mu_2| = \sup_{v \in H_1, \|v\|=1} |\langle L(v), v \rangle|.$$

Note that $|\mu_2| \leq |\mu_1|$. This follows from the procedure used in the proof above: taking the supremum over the smaller set $H_1 \subset H$, there exists a sequence $(v_n) \subset H_1$, with $\|v_n\| = 1$ for all n , such that $\langle L(v_n), v_n \rangle \rightarrow \mu_2$. Then,

$$0 \leq \|L(v_n) - \mu_2 v_n\|^2 = \|L(v_n)\|^2 - 2\mu_2 \langle L(v_n), v_n \rangle - \mu_2^2 \leq 2\mu_2^2 - 2\mu_2 \underbrace{\langle L(v_n), v_n \rangle}_{\mu_2} = 0.$$

Continuing this procedure, when $\phi_1, \dots, \phi_n$ are determined, let $H_n := \{x \in H \mid \langle x, \phi_i \rangle = 0, i = 1, \dots, n\}$. Then, there exists $\phi_{n+1} \in H_n$ such that $\|\phi_{n+1}\| = 1$ and $L(\phi_{n+1}) = \mu_{n+1} \phi_{n+1}$; furthermore,

$$|\mu_{n+1}| = \sup_{v \in H_n, \|v\|=1} |\langle L(v), v \rangle|,$$

and we have $|\mu_1| \geq |\mu_2| \geq |\mu_3| \geq \dots$.

We then have that $\lim_{n \rightarrow \infty} \mu_n = 0$. Indeed, suppose not. Then there exists $\epsilon > 0$ and (n_k) such that $|\mu_{n_k}| \geq \epsilon \Rightarrow \left\{ \frac{\phi_{n_k}}{\mu_{n_k}} \right\}$ is bounded. Then,

$$\left(\left\| \frac{1}{\mu_{n_k}} \phi_{n_k} \right\| = \frac{1}{|\mu_{n_k}|} \leq \frac{1}{\epsilon} \right) \Rightarrow \left(L \left(\frac{1}{\mu_{n_k}} \phi_{n_k} \right) \right)$$

has a convergent subsequence because L is compact. But $\|\phi_{n_k} - \phi_{n_\ell}\|^2 = 2$ for $k \neq \ell$, so that (ϕ_{n_k}) cannot have a convergent subsequence.

Proposition 7.5.1

The $\{\phi_n\}$ constructed above is an orthonormal set of eigenvectors, the $\{\mu_n\}$ are all non-zero eigenvalues repeated according to multiplicity, and for any $v \in H$,

$$L(v) = \sum_{i=1}^{\infty} \langle L(v), \phi_i \rangle \phi_i = \sum_{i=1}^{\infty} \mu_i \langle v, \phi_i \rangle \phi_i.$$

PROOF: That $\{\phi_n\}$ is an orthonormal set of eigenvectors follows from the above construction.

Now, for $v \in H$, let $g_n := v - \sum_{i=1}^n \langle v, \phi_i \rangle \phi_i$. Then $g_n \in H_n$, which implies that $\|L(g_n)\| \leq |\mu_{n+1}| \|g_n\|$ since

$$\sup_{w \neq 0, w \in H_n} \frac{\|L(w)\|}{\|w\|} = |\mu_{n+1}|.$$

So

$$\|g_n\|^2 = \|v\|^2 - \sum_{i=1}^n |\langle v, \phi_i \rangle|^2 \leq \|v\|^2 \quad \text{and} \quad |\mu_{n+1}| \rightarrow 0,$$

which means that

$$L(g_n) = L(v) - \sum_{i=1}^n \langle v, \phi_i \rangle L(\phi_i) \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

or

$$\begin{aligned} L(v) &= \sum_{i=1}^{\infty} \langle v, \phi_i \rangle L(\phi_i) = \sum_{i=1}^{\infty} \mu_i \langle v, \phi_i \rangle \phi_i \\ &= \sum_{i=1}^{\infty} \langle v, \overline{\mu_i} \phi_i \rangle \phi_i = \sum_{i=1}^{\infty} \langle v, L(\phi_i) \rangle \phi_i = \sum_{i=1}^{\infty} \langle L(v), \phi_i \rangle \phi_i, \end{aligned}$$

as required. Now, suppose we “missed” an eigenvalue, i.e., suppose $L(\phi) = \mu\phi$, $\phi \neq 0$, $\mu \neq 0$, and $\mu \notin \{\mu_i\}$. Then $\langle \phi, \phi_i \rangle = 0$ since eigenvectors corresponding to different eigenvalues are orthogonal. Then

$$L(\phi) = \sum_{i=1}^{\infty} \langle L(\phi), \phi_i \rangle \phi_i = \sum_{i=1}^{\infty} \mu \langle \phi, \phi_i \rangle \phi_i = 0 \Rightarrow \mu\phi = 0,$$

which is a contradiction. So we haven’t “missed” any eigenvalues.

Finally, if $L(\phi) = \mu_j\phi$, $\phi \neq 0$, then

$$\mu_j\phi = L(\phi) = \sum_{i=1}^{\infty} \langle L(\phi), \phi_i \rangle \phi_i = \sum_{i=1}^{\infty} \mu_j \langle \phi, \phi_i \rangle \phi_i \Rightarrow \phi = \sum_{\{i \mid \mu_i = \mu_j\}} \langle \phi, \phi_i \rangle \phi_i \Rightarrow \phi \in \text{span}_{\{i \mid \mu_i = \mu_j\}} \{\phi_i\}.$$

Therefore, $\{\phi_i\}$ contains all linearly independent eigenvectors with non-zero eigenvalues, and $\{\mu_i\}$ are all the non-zero eigenvalues repeated according to multiplicity with each multiplicity finite. This completes the proof. ■

Proposition 7.5.2

A set $\{\phi_i\}$ of eigenvectors of a compact self-adjoint bounded linear operator on a Hilbert space H is an orthonormal basis for H if and only if $\mu = 0$ is not an eigenvalue of L (equivalently, if and only if $\mathcal{N}(L) = \{0\}$).

PROOF: Suppose $\{\phi_i\}$ is an orthonormal basis for H . Then, for any $v \in H$, the sum $\sum \langle v, \phi_i \rangle \phi_i$ converges, say, to an element w . This is because $\{\phi_i\}$ orthonormal implies $\sum |\langle v, \phi_i \rangle|^2$ converges, which implies that $\sum \langle v, \phi_i \rangle \phi_i$ converges in H . Now,

$$w = \sum \langle v, \phi_i \rangle \phi_i \Rightarrow L(w) = \sum \langle v, \phi_i \rangle L(\phi_i) = \sum \mu_i \langle v, \phi_i \rangle \phi_i.$$

Also, by the previous proposition

$$L(v) = \sum \langle L(v), \phi_i \rangle \phi_i = \sum \mu_i \langle v, \phi_i \rangle \phi_i \Rightarrow L(h) = 0 \text{ for } h = v - w.$$

Thus, $v = h + w + \sum \langle v, \phi_i \rangle \phi_i$. If $\mu = 0$ is not an eigenvalue of L , then $L(h) = 0 \Rightarrow h = 0 \Rightarrow v = \sum \langle v, \phi_i \rangle \phi_i$. Therefore, $\{\phi_i\}$ is an orthonormal basis for H .

Conversely, if $\{\phi_i\}$ is an orthonormal basis for H , then $L(v) = 0 \Rightarrow \sum \mu_i \langle v, \phi_i \rangle \phi_i = 0$ by the previous proposition, which implies that $\langle v, \phi_i \rangle = 0$ for all i , so that $v = 0$ since $\{\phi_i\}$ is an orthonormal basis. Therefore, $\mu = 0$ is not an eigenvalue of L , completing the proof. ■

Theorem 7.5.9 The Spectral Theorem

Let $L : H \rightarrow H$ be a compact and self-adjoint bounded linear operator on an infinite-dimensional Hilbert space. Then there exist orthonormal eigenvectors $\{\phi_i\}$ and eigenvalues $\{\mu_i\}$ such that $|\mu_1| \geq |\mu_2| \geq \cdots$, $\lim_{n \rightarrow \infty} \mu_n = 0$, and $L(v) = \sum_{i=1}^{\infty} \mu_i \langle v, \phi_i \rangle \phi_i$ for all $v \in H$. $\{\phi_i\}$ is an orthonormal basis for H if and only if $\mu = 0$ is not an eigenvector for L .

Example 7.5.4 Consider the following linear Fredholm integral operator on $L_2[0, 1]$:

$$L(x)(t) = \int_0^1 stx(s) \, ds.$$

The kernel $k(x, t) := st$ is symmetric, implying that L is self-adjoint. Since $k(s, t) \in L_2([0, 1]^2)$, L is compact. We now look for eigenvalues of L . Note that

$$L(x)(t) = t \int_0^1 sx(s) \, ds.$$

In other words, the range of L is the one-dimensional subspace $\text{span}\{t\} = \{at \mid a \in \mathbb{R}\}$. This implies that $v(t) = t$ is an eigenfunction of L . Substitution gives

$$\lambda t = t \int_0^1 s^2 \, ds = \frac{1}{3}t,$$

implying that $\lambda = \frac{1}{3}$. An independent calculation (do it!) shows that $\|L\| = \frac{1}{3}$.

In the subsequent sections, we will develop some more theory to write down the spectral theorem above in a different way.

7.5.2 Positive Operators

If T is self-adjoint, we have seen that $\langle T(x), x \rangle$ is real. Hence, we may consider the set of all bounded self-adjoint linear operators on a complex Hilbert space H and introduce on this set a *partial ordering* $\leq$ by defining

$$T_1 \leq T_2 \quad \text{if and only if} \quad \langle T_1(x), x \rangle \leq \langle T_2(x), x \rangle \quad \text{for all } x \in H. \quad (7.42)$$

Instead of $T_1 \leq T_2$, we also might write $T_2 \geq T_1$.

An important particular case is the following one.

Definition 7.5.1 Positive Operator

A bounded self-adjoint linear operator $T : H \rightarrow H$ is called **positive**, written

$$T \geq 0, \quad (7.43)$$

if and only if $\langle T(x), x \rangle \geq 0$ for all $x \in H$. Such an operator is more properly called “non-negative”, although “positive” is used more often.

Note that

$$T_1 \leq T_2 \Leftrightarrow 0 \leq T_2 - T_1,$$

that is, (7.42) holds if and only if $T_2 - T_1$ is positive.

Theorem 7.5.10 Basic Properties of Positive Operators

Let S and T be two bounded self-adjoint linear operator on a complex Hilbert space H that are positive. Then,

1. The sum $S + T$ is positive;
2. If S and T commute then the product ST is positive;
3. If $S \leq T$ and $T \leq S$, then $S = T$.

Also, let (T_n) be a sequence of bounded self-adjoint linear operator on a complex Hilbert space H such that

$$T_1 \leq T_2 \leq \cdots \leq T_n \leq \cdots \leq K, \quad (7.44)$$

where K is a bounded self-adjoint linear operator on H . Suppose that all T_j commute with K and with every T_m . Then, (T_n) is strongly operator convergent, i.e., $T_n(x) \rightarrow T(x)$ for all $x \in H$, and the limit operator T is linear, bounded and self-adjoint and satisfies $T \leq K$.

Proposition 7.5.3

If $T : H \rightarrow H$ is a bounded linear operator on a complex Hilbert space H , then TT^* and T^*T are self-adjoint and positive. In addition, the spectra of TT^* and T^*T are real and cannot contain negative values.

REMARK: What are the consequences of the second statement for a square matrix A ?

PROOF: To be completed. ■

Theorem 7.5.11 Spectra of Positive Operators

A bounded self-adjoint linear operator on a complex Hilbert space is positive if and only if its spectrum consists of non-negative real values only.

REMARK: What does this imply for a matrix?

PROOF: To be completed. ■

Proposition 7.5.4

Let $T : H \rightarrow H$ and $W : H \rightarrow H$ be bounded linear operators on a complex Hilbert space H and $S = W^*TW$. Then, if T is self-adjoint and positive, so is S .

PROOF: To be completed. ■

Proposition 7.5.5

If T is a bounded self-adjoint linear operator on a complex Hilbert space H , then T^2 is positive. In addition, the spectrum of T^2 cannot contain a negative value.

REMARK: What theorem on matrices do these statements generalise?

PROOF: To be completed. ■

Definition 7.5.2 Positive Square Root

Let $T : H \rightarrow H$ be a positive bounded self-adjoint linear operator on a complex Hilbert space H . Then, a bounded self-adjoint linear operator A is called a **square root** of T if

$$A^2 = T. \quad (7.45)$$

If, in addition, $A \geq 0$, then A is called a **positive square root** of T and is denoted by

$$A = T^{1/2}.$$

We first verify that the definition above makes sense.

Theorem 7.5.12 Positive Square Root

Every positive bounded self-adjoint linear operator $T : H \rightarrow H$ on a complex Hilbert space H has a positive square root A that is unique. This operator A commutes with every bounded linear operator on H that commutes with T .

Proposition 7.5.6

Let $T : H \rightarrow H$ be a positive bounded self-adjoint linear operator on a complex Hilbert space H . Then,

$$\|T^{1/2}\| = \|T\|^{1/2}.$$

PROOF: To be completed. ■

Example 7.5.5 Find operators $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T^2 = I$, the identity operator. Indicate which of the square roots is the positive square root of I .

SOLUTION: To be completed.

Example 7.5.6 Let $T : L_2[0, 1] \rightarrow L_2[0, 1]$ be defined by $T(x)(t) = tx(t)$. Show that T is self-adjoint and positive and find its positive square root.

SOLUTION: To be completed.

Example 7.5.7 Let $T : \ell_2 \rightarrow \ell_2$ be defined by $(\xi_1, \xi_2, \xi_3, \dots) \mapsto (0, 0, \xi_3, \xi_4, \dots)$. Is T bounded? Self-adjoint? Positive? Find a square root of T .

SOLUTION: To be completed.

7.6 Projection Operators

We saw briefly the projection operator in the context of the projection theorem, in which a Hilbert space H was represented as the direct sum of a closed subspace Y and its orthogonal complement $Y^\perp$:

$$\begin{aligned} H &= Y \oplus Y^\perp \\ x &= y + z, \quad y \in Y, z \in Y^\perp. \end{aligned} \tag{7.46}$$

Since the sum is direct, y is unique for any given $x \in H$. Hence, (7.46) defines a linear operator

$$P : H \rightarrow H, \quad x \mapsto y = P(x). \tag{7.47}$$

P is called an **orthogonal projection**, or simply **projection**, of H onto Y . Hence, a linear operator $P : H \rightarrow H$ is a projection on H if there is a closed subspace Y of H such that Y is the range of P and $Y^\perp$ is the null space of P and $P|_Y$ is the identity operator on Y .

Note now that in (7.46) we can now write

$$x = y + z = P(x) + (I - P)(x).$$

This shows that the projection of H onto $Y^\perp$ is $I - P$.

There is another characterisation of a projection on H , which is sometimes used as a definition.

Theorem 7.6.1 Projection

A bounded linear operator $P : H \rightarrow H$ on a Hilbert space H is a projection if and only if P is self-adjoint and idempotent, i.e., $P^2 = P$.

PROOF: Suppose that P is a projection on H and denote $P(H)$ by Y . Then, $P^2 = P$ because for every $x \in H$ and $P(x) = y \in Y$, we have

$$P^2(x) = P(y) = P(x).$$

Furthermore, let $x_1 = y_1 + z_1$ and $x_2 = y_2 + z_2$, where $y_1, y_2 \in Y$ and $z_1, z_2 \in Y^\perp$. Then, $\langle y_1, z_2 \rangle = \langle y_2, z_1 \rangle = 0$ because $Y \perp Y^\perp$, and self-adjointness of P is seen from

$$\langle P(x_1), x_2 \rangle = \langle y_1, y_2 + z_2 \rangle = \langle y_1, y_2 \rangle = \langle y_2 + z_1, y_2 \rangle = \langle x_1, P(x_2) \rangle.$$

Conversely, suppose that $P^2 = P = P^*$ and denote $P(H)$ by Y . Then, for every $x \in H$,

$$x = P(x) + (I - P)(x).$$

Orthogonality, $Y = P(H) \perp (I - P)(H)$, follows from

$$\langle P(x), (I - P)(v) \rangle = \langle x, P(I - P)(v) \rangle = \langle x, P(v) - P^2(v) \rangle = \langle x, 0 \rangle = 0.$$

Y is the null space $\mathcal{N}(I - P)$ of $I - P$ because $Y \subset \mathcal{N}(I - P)$ can be seen from

$$(I - P)(P(x)) = P(x) - P^2(x) = 0,$$

and $Y \supset \mathcal{N}(I - P)$ follows if we note that $(I - P)(x) = 0$ implies $x = P(x)$. Hence, H is closed by Corollary 5.5.1. Finally, $P|_Y$ is the identity operator on Y since, writing $y = P(x)$, we have $P(y) = P^2(x) = P(x) = y$. ■

Theorem 7.6.2 Positivity, Norm of Projections

For any projection P on a Hilbert space H ,

$$\langle P(x), x \rangle = \|P(x)\|^2 \tag{7.48}$$

$$P \geq 0 \tag{7.49}$$

$$\|P\| \leq 1 \quad \|P\| = 1 \text{ if } P(H) \neq \{0\} \tag{7.50}$$

PROOF: (7.48) and (7.49) follow from

$$\langle P(x), x \rangle = \langle P^2(x), x \rangle = \langle P(x), P(x) \rangle = \|P(x)\|^2 \geq 0.$$

By the Schwarz inequality,

$$\|P(x)\|^2 = \langle P(x), x \rangle \leq \|P(x)\| \|x\|,$$

so that $\frac{\|P(x)\|}{\|x\|} \leq 1$ for every $x \neq 0$, and $\|P\| \leq 1$. Also, $\frac{\|P(x)\|}{\|x\|} = 1$ if $x \in P(H)$ and $x \neq 0$. This proves (7.50). ■

The product of projections isn't necessarily a projection. But we do have the following result.

Theorem 7.6.3 Product of Projections

Let H be a Hilbert space.

1. $P = P_1 P_2$ is a projection on H if and only if the projections P_1 and P_2 commute, that is, $P_1 P_2 = P_2 P_1$. Then P projects H onto $Y = Y_1 \cap Y_2$, where $Y_j = P_j(H)$.
2. Two closed subspaces Y and V of H are orthogonal if and only if the corresponding projections satisfy $P_Y P_V = 0$.

PROOF: To be completed. ■

Similarly, a sum of projections need not be a projection, but we have

Theorem 7.6.4 Sum of Projections

Let P_1 and P_2 be projections on a Hilbert space H . Then,

1. The sum $P := P_1 + P_2$ is a projection on H if and only if $Y_1 = P_1(H)$ and $Y_2 = P_2(H)$ are orthogonal.
2. If $P = P_1 + P_2$ is a projection, then P projects H onto $Y := Y_1 \oplus Y_2$.

PROOF: To be completed. ■

Example 7.6.1 Show that a projection P on a Hilbert space H satisfies

$$0 \leq P \leq I.$$

Under what conditions will $P = 0$ and $P = I$?

SOLUTION: To be completed.

Example 7.6.2 Let $Q = S^{-1}PS : H \rightarrow H$, where S and P are bounded and linear. If P is a projection and S is unitary, show that Q is a projection.

SOLUTION: To be completed.

Theorem 7.6.5 Partial Ordering of Projections

Let P_1 and P_2 be projections defined on a Hilbert space H . Denote by $Y_1 = P_1(H)$ and $Y_2 = P_2(H)$ the subspaces onto which H is projected by P_1 and P_2 , and let $\mathcal{N}(P_1)$ and $\mathcal{N}(P_2)$ be the null spaces of these projections. Then the following conditions are equivalent:

$$P_2P_1 = P_1P_2 = P_1 \quad (7.51)$$

$$Y_1 \subset Y_2 \quad (7.52)$$

$$\mathcal{N}(P_1) \supset \mathcal{N}(P_2) \quad (7.53)$$

$$\|P_1(x)\| \leq \|P_2(x)\| \text{ for all } x \in H \quad (7.54)$$

$$P_1 \leq P_2. \quad (7.55)$$

Theorem 7.6.6 Difference of Projections

Let P_1 and P_2 be projections on a Hilbert space H . Then,

1. The difference $P = P_2 - P_1$ is a projection on H if and only if $Y_1 \subset Y_2$, where $Y_j = P_j(H)$.
2. If $P = P_2 - P_1$ is a projection, then P projects H onto Y , where Y is the orthogonal complement of Y_1 in Y_2 .

Theorem 7.6.7 Monotone Increasing Sequence

Let (P_n) be a monotone increasing sequence of projections P_n defined on a Hilbert space H . Then,

1. (P_n) is strongly operator convergent, say $P_n(x) \rightarrow P(x)$ for all $x \in H$, and the limit operator P is a projection defined on H .
2. P projects H onto

$$P(H) = \overline{\bigcup_{n=1}^{\infty} P_n(H)}.$$

3. P has the null space

$$\mathcal{N}(P) = \bigcap_{n=1}^{\infty} \mathcal{N}(P_n).$$

Theorem 7.6.8 Limit of Projections

If (P_n) is a sequence of projections defined on a Hilbert space H and $P_n \rightarrow P$, then P is a projection defined on H .

PROOF: To be completed. ■

7.7 Spectral Family

Our goal is to come up with a representation of bounded self-adjoint linear operator on a Hilbert space in terms of very simple operators, projections, called the *spectral representation* or *spectral decomposition*. We will do this by the use of a suitable family of projections called the *spectral family*.

Definition 7.7.1 Spectral Family/Decomposition of Unity

A real **spectral family**, or real **decomposition of unity**, is a one-parameter family $\mathcal{E} = (E_\lambda)_{\lambda \in \mathbb{R}}$ of projections E_λ defined on a Hilbert space H (of any dimension) that depends on a real parameter λ and is such that

$$E_\lambda \leq E_\mu \quad \text{hence} \quad E_\lambda E_\mu = E_\mu E_\lambda = E_\lambda, \quad \lambda < \mu \quad (7.56)$$

$$\lim_{\lambda \rightarrow -\infty} E_\lambda(x) = 0 \quad \forall x \in H \quad (7.57)$$

$$\lim_{\lambda \rightarrow \infty} E_\lambda(x) = x \quad \forall x \in H \quad (7.58)$$

$$E_{\lambda+0}(x) = \lim_{\mu \rightarrow \lambda+0} E_\mu(x) = E_\lambda(x) \quad \forall x \in H \quad (7.59)$$

REMARK: $\mu \rightarrow \lambda+0$ in (7.59) indicates that in this limit process we consider only values $\mu > \lambda$, and (7.59) means that $\lambda \mapsto E_\lambda$ is strongly operator continuous from the right. As a matter of fact, continuity from the left would do equally well.

From this definition, we see that a real spectral family can be regarded as a mapping

$$\mathbb{R} \rightarrow B(H), \quad \lambda \mapsto E_\lambda,$$

i.e., to each $\lambda \in \mathbb{R}$ there corresponds a projection $E_\lambda \in B(H)$, where recall that $B(H)$ is the space of all bounded linear operators from H into H .

$\mathcal{E}$ is called a **spectral family on an interval $[a, b]$** if

$$E_\lambda = 0 \quad \text{for} \quad \lambda < a, \quad E_\lambda = I \quad \text{for} \quad \lambda \geq b. \quad (7.60)$$

Such families will be of particular interest since the spectrum of a bounded self-adjoint linear operator lies in a finite interval on the real line.

We shall see in the next two sections that with any given bounded self-adjoint linear operator T on any Hilbert space we can associate a spectral family that may be used for representing T by a Riemann-Stieltjes integral. This is known as a *spectral representation*, as was mentioned before. Then we shall also see that in the finite-dimensional case, the integral representation reduces to a finite sum written in terms of the spectral family.

7.7.1 Bounded Self-Adjoint Linear Operators

7.8 Spectral Decomposition of Bounded Self-Adjoint Linear Operators

7.8.1 The Spectral Theorem for Continuous Functions

7.9 Properties of the Spectral Family of a Bounded Self-Adjoint Linear Operator

7.10 Sturm-Liouville Problems

7.11 Appendix: Banach Algebras

7.12 Appendix: C^* -Algebras

(Take from Marcoux notes and “Quantum Algebras...”)

8 Sobolev Spaces